

PANTANAL REDD+ PROGRAM



Project title	Pantanal REDD+ Program
Project ID	4747
Crediting period	14-October-2021 to 31-December-2070
Project lifetime	14-October-2021 to 31-December-2070; 50-years lifetime
(CCB) GHG accounting period	14-October-2021 to 31-December-2070; 50-years total period
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Version	1st version for public comment period

VCS Standard version	VCS Standard, v4.6 - 21 March 2024
CCB Standards version	Climate, Community & Biodiversity Standards, v3.1 - 21 June 2017
Project location	Brazil, State of Mato Grosso do Sul, Municipalities of Aquidauana, Miranda and Corumbá
Project proponent(s)	Biofílica Ambipar Environment Investments Plínio Ribeiro (plinio.ribeiro@ambipar.com), Soraya Pires (soraya.pires@ambipar.com) and Luana Cordeiro (luana.cordeiro@ambipar.com) Phone: +55 11 3073-0430
Validation/verification body	AENOR CONFIA, S.A.U.
History of CCB status	First validation
Gold Level criteria	GL1. Climate Change Adaptation Benefits: because of its significant support to help communities and/or biodiversity adapt to the impacts of climate change through strategies aligned with the activities planned by the Program. GL3. Exceptional Biodiversity Benefits: because the Project Zone includes a high priority area for biodiversity conservation that meets the criteria of vulnerability or uniqueness, identifying "trigger" species.
Expected verification schedule	1st VCS Verification - Period monitored: 14-October-2021 to 31-December-2023 (2 years and 79 days)
Prepared by	Biofílica Ambipar Environment Investments

CONTENTS

1	SUMMARY OF PROJECT BENEFITS	4
1.1	Unique Project Benefits	4
1.2	Standardized Benefit Metrics	5
2	PROJECT DETAILS	8
2.1	Project Goals, Design and Long-Term Viability	8
2.2	Without-project Land Use Scenario and Additionality	49
2.3	Safeguards and Stakeholder Engagement	51
2.4	Management Capacity	79
2.5	Legal Status and Property Rights	90
2.6	Additional Information Relevant to the Project	106
3	CLIMATE	108
3.1	Application of Methodology	108
3.2	Quantification of Estimated GHG Emission Reductions and Removals	144
3.3	Monitoring	154
3.4	Optional Criterion: Climate Change Adaptation Benefits	180
4	COMMUNITY	187
4.1	Without-Project Community Scenario	187
4.2	Net Positive Community Impacts	194
4.3	Other Stakeholder Impacts	197
4.4	Community Impact Monitoring	199
4.5	Optional Criterion: Exceptional Community Benefits	201
5	BIODIVERSITY	203
5.1	Without-Project Biodiversity Scenario	203
5.2	Net Positive Biodiversity Impacts	210
5.3	Offsite Biodiversity Impacts	214
5.4	Biodiversity Impact Monitoring	215
5.5	Optional Criterion: Exceptional Biodiversity Benefits	216
	APPENDIX 1: STAKEHOLDER DESCRIPTION TABLE	220
	APPENDIX 2: COMMERCIAL SENSITIVE INFORMATION	224

1 SUMMARY OF PROJECT BENEFITS

1.1 Unique Project Benefits

Under development and revision.

Outcome or impact estimated by the end of project lifetime	Section reference
1) The Pantanal REDD+ Program is expected to help mitigate climate change during its lifetime. For the first project activity instances validated, a total of 6,226,775 tCO ₂ e is expected to be avoided, through the avoided legal deforestation of 11,800 hectares of forests and forested savannahs on private properties in the Pantanal biome.	3.2
2) The Pantanal REDD+ Program aims to strengthen community resilience and biodiversity in the face of climate change, by maintaining the forest standing and consequently reducing greenhouse gas emissions. The program also seeks to reduce the vulnerability of local residents by encouraging the implementation of alternative sources of income and diversifying the Payment for Environmental Services (PES) models, as well as emphasizing the importance of education for climate adaptation. Climate benefits are also achieved through fire-fighting activities and the maintenance of ecosystem services.	3.4
3) The benefits expected for the people living in the region where the Pantanal REDD+ Program is located through its proposed activities include promoting an improvement in the quality of life and well-being for people through the promotion of security, improvement in the quality of food, improvement in communication processes, technical training for employees, greater access to health and education and diversification of the regional economy based on sustainable actions.	4.4
4) The activities implemented in the Pantanal REDD+ Program in the area of biodiversity aim to maintain continuous monitoring of the region's fauna and flora and of endangered and High Conservation Value species. In addition, the aim is to carry out awareness campaigns among farm employees and the local community about the negative impacts of illegal hunting and fishing activities on the region's biodiversity. These actions are expected to lead to greater conservation of the Pantanal biome's biodiversity, especially of species with some degree of threat in the region.	5.4

1.2 Standardized Benefit Metrics

Under development and revision.

Category	Metric	Estimated by the end of project lifetime	Section reference
GHG emission reductions or carbon dioxide	Net estimated removals in the project area, measured against the without-project scenario	Not applied	
	Net estimated reductions in the project area, measured against the without-project scenario	6,226,775	3
Forest ¹ cover	For REDD ² projects: Estimated number of hectares of reduced forest loss in the project area measured against the without-project scenario	11,800	3
	For ARR ³ projects: Estimated number of hectares of forest cover increased in the project area measured against the without-project scenario	Not applied	
Improved land management	Number of hectares of existing production forest land in which IFM ⁴ practices are expected to occur as a result of project activities, measured against the without-project scenario	Not applied	
	Number of hectares of non-forest land in which improved land management practices are expected to occur as a result of project activities, measured against the without-project scenario	Not applied	
Training	Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities	Potentially 137 people	4

¹ Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (VCS Program Definitions)

² Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (VCS Program Definitions)

³ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (VCS Program Definitions)

⁴ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (VCS Program Definitions)

Category	Metric	Estimated by the end of project lifetime	Section reference
	Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities	Potentially 48 women	4
Employment	Total number of people expected to be employed in project activities ⁵ , expressed as number of full-time employees ⁶	Not applied	
	Number of women expected to be employed as a result of project activities, expressed as number of full-time employees	Not applied	
Livelihoods	Total number of people expected to have improved livelihoods ⁷ or income generated as a result of project activities	Not applied	
	Number of women expected to have improved livelihoods or income generated as a result of project activities	Not applied	
Health	Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	Potentially 137 people	4
	Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	Potentially 48 women	4
Education	Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	Potentially 137 people	4
	Number of women and girls for whom access to, or quality of, education is expected to improve as result of	Potentially 48 women	4

⁵ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁶ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁷ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. The Sustainable Livelihood Approach to Poverty Reduction. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Estimated by the end of project lifetime	Section reference
	project activities, measured against the without-project scenario		
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Not applied	
	Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	Not applied	
Well-being	Total number of community members whose well-being ⁸ is expected to improve as a result of project activities	Potentially 137 people	4
	Number of women whose well-being is expected to improve as a result of project activities	Potentially 48 women	4
Biodiversity conservation	Expected change in the number of hectares managed significantly better by the project for biodiversity conservation ⁹ , measured against the without-project scenario	11,800	5
	Expected number of globally Critically Endangered or Endangered species ¹⁰ benefiting from reduced threats as a result of project activities ¹¹ , measured against the without-project scenario	4 species (2 fauna and 2 flora)	5

⁸ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

⁹ Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

¹⁰ Per IUCN's Red List of Threatened Species

¹¹ In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 PROJECT DETAILS

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

The Pantanal biome is located in the center-west of Brazil, covering portions of the states of Mato Grosso and Mato Grosso do Sul. Considered the largest continuous freshwater wetland on the planet, this biome oscillates between terrestrial and aquatic phases, forming a mosaic of landscapes that provide great biodiversity of flora and fauna. Analyzing the historical context of the region where the Pantanal is located, there has been a considerable expansion of agricultural activities (the main local economic activity, mainly focused on the breeding of Pantanal cattle) and this factor, together with the high rates of forest fires in this area, results in a scenario of unsustainable exploitation of the natural resources of this biome.

Considering all the applicable legal requirements, landowners who have a surplus of native vegetation on their property and choose to preserve it, would fulfill the first requirement for obtaining economic incentives from the voluntary carbon market, through Reducing Emissions from Deforestation and Forest Degradation (REDD) through Avoided Planned Deforestation (APD). Given all this context, the implementation of the REDD+ Carbon Program on a scale for the Pantanal biome is possible, providing a model of sustainable economic development that can help to redress the balance, reconciling the conservation of environmental services with traditional economic activities.

Biofílica Ambipar Environment, in partnership with rural landowners in the Pantanal of Mato Grosso do Sul who are committed to promoting dialogue that benefits both the environment and sustainable economic development, aims to develop economic activities linked to the conservation of environmental services and all their ecological functions through the implementation of the Pantanal REDD+ Program.

The first instance of the Pantanal REDD+ Program, made up of 11 private properties located in the municipalities of Aquidauana, Corumbá and Miranda, in the state of Mato Grosso Sul, has as its starting date the purchase of the Santa Sofia farm, which took place on October 14, 2021, an event that marks the beginning of conservation actions among the first properties participating in the Program in the Pantanal region.

The Pantanal REDD+ Program aims to mitigate climate change through the conservation of the Pantanal biome, and the implementation of the first instance of the Program, complying with all current laws and regulations, will avoid the legal deforestation of 11,800 hectares of forest and forested savannah phytophysiognomies and, in the first credit period of the validated instances (20 years), will reduce emissions by 6,226,775 tCO₂e, representing an average annual reduction in Greenhouse Gas (GHG) emissions of 311,339 tons of CO₂.

Given the vast richness of biodiversity in the Pantanal biome, it is important to implement additional activities under the REDD+ Program that adopt measures to protect the region's fauna and flora. In this sense, the project will carry out continuous monitoring of the fauna and flora and of species classified as being under some degree of threat of extinction and classified as being of High Conservation Value, with the aim of understanding the dynamics of the region's ecosystem. In addition, the implementation of scientific research on the farms will be encouraged and awareness campaigns will be carried out among employees and the local community about the negative impacts of illegal hunting and fishing. Finally, fire prevention and mitigation actions will be implemented/improved on the farms participating in the Program, which will have a direct impact on climate and biodiversity issues.

In addition, in terms of community benefits, the activities of the Pantanal REDD+ Program were designed following the main needs raised with the project's stakeholders, with the aim of promoting a better quality of life and well-being for the people who live in the region. The actions planned in this area are aimed at improving safety, food quality, communication processes, technical training for farm employees, greater access to health and education and diversification of the local economy.

The Pantanal REDD+ Program has a crediting period of 50 years (20 years per instance), with a lifetime and longevity of 50 years during which the project's activities will be maintained. It should be noted that this Program is not located in a region covered by REDD+ at the jurisdictional level, nor does it overlap any existing REDD+ Project areas.

2.1.2 Audit History (VCS, 4.1)

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	14-October-2021 to 31-December-2070	VCS & CCB	AENOR CONFIA, S.A.U.	50 years
Verification	14-October-2021 to 31-December-2023	VCS	AENOR CONFIA, S.A.U.	2 years and 79 days

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

Sectoral scope	14: Agriculture, forestry, and other land use
AFOLU project category ¹²	Reducing Emissions from Deforestation and Forest Degradation (REDD)
Project activity type	Avoided Planned Deforestation (APD)

¹² See Appendix 1 of the VCS Standard

2.1.4 Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

The Pantanal REDD+ Program meets the scope of the VCS Program, falling under the following eligibility criteria:

- 1) Reduces CO₂ emissions into the atmosphere, falling within the scope of the Greenhouse Gases stipulated in the Kyoto Protocol.
- 2) The activities of the Pantanal REDD+ Program follow the parameters established in an eligible methodology under the VCS Program, VM0007 (more details in section 3. Application of the Methodology).
- 3) The implementation of the activities of this Program does not result in the violation of any law applicable to its region of insertion (more details in section 2.5.1).
- 4) It is not located in a jurisdiction covered by a jurisdictional REDD+ program.
- 5) The activities of the Pantanal REDD+ Program work to avoid emissions resulting from planned deforestation and therefore do not fit within any of the items excluded from the scope of the VCS Program (presented in table 1 of item 2.1.3 of the VCS Standard v4.6):
 - Grid-connected electricity generation activities using hydroelectric power plants.
 - Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants.
 - Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.
 - Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.
 - Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.
 - Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).
 - Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.
 - Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions
- 6) It is established in the VCS Standard v4.6 (item 3.8.2) that AFOLU projects must start the pipeline listing process within a period of 3 years after the project start date. The Pantanal REDD+ Program's start date is 14/10/2021 and its registration was made on 19/12/2023. It is therefore understood that the Pantanal REDD+ Program meets this requirement.

- 7) The VCS Standard v4.6 (item 3.8.4) states that AFOLU projects with estimated ex-ante emission reductions of more than 20,000 tCO₂e per year must complete their validation within 5 years of the project start date. The Pantanal REDD+ Program has estimated ex-ante annual reductions of 311,339 tCO₂e and has a start date of 14/10/2021. Thus, this Program has until the date of 10/14/2026 to complete its validation, which is planned to take place in June 2024, thus meeting this criterion.
- 8) The draft of the Pantanal REDD+ Program was submitted for public consultation on April 12, 2024 and the audit will be carried out in the second half of June 2024.

AFOLU Project Eligibility

This is an eligible project in the AFOLU (Agriculture, Forestry and Other Land Use) category under the VCS Program by reducing emissions from deforestation and degradation (REDD) under avoided planned deforestation (APD).

According to the VCS Standard v4.6 (Appendix 1), eligible REDD+ activities are those that reduce greenhouse gas emissions by reducing deforestation (human-induced land use conversion from forest to non-forest land) and/or forest degradation. Among the eligible REDD+ activities are avoiding planned deforestation, which the Pantanal REDD+ Program fits into.

A project area must adhere to an internationally accepted definition of forest (FAO, UNFCCC) and must qualify as forest for at least 10 years prior to the project start date. According to data collected from the MapBiomas platform, Collection 8¹³, the area of the Pantanal REDD+ Program is eligible as forest or forested savannah (classified according to the FAO forest criteria, as agreed upon in the reference level submitted by Brazil to the UNFCCC¹⁴) for over 10 years, and therefore, there has been no ecosystem conversion with the purpose of generating carbon credits.

2.1.5 Transfer Project Eligibility (VCS, 3.23, Appendix 2)

Not applicable, as the Pantanal REDD+ Program is not a transfer project and CPAs seeking registration.

2.1.6 Project Design (VCS, 3.6)

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)

¹³ <https://brasil.mapbiomas.org/>

¹⁴ <https://forum.mapbiomas.org/t/area-da-formacao-florestal/163>

☒ Grouped project

2.1.6.1 Eligibility Criteria for Grouped Projects (VCS, 3.6; CCB, G1.14)

As it is a grouped project, the expansion of its scope is allowed through the inclusion of new instances after the initial validation of the project, occurring at each verification event, as long as all eligibility criteria are duly met.

This Program has been developed under a set of eligibility criteria specified in the CCB Standard v3.1 and VCS Standard v4.6, the latter relating to three main topics: baseline and additionality scenario; new instances of activity; and inclusion of new instances of activities in the project.

Based on the CCB Standard v3.1, the eligibility criteria to be adopted for the Program for the adoption of new instances are as follows:

- Adopt and apply project activities in the same manner specified in the project description.
- Be subject to the same climate, community, and biodiversity scenarios without project determined for the project.
- Have similar characteristics to the project regarding additionality.
- Be subject to the same stakeholder engagement processes described in G3 and respect for land, territories, and resources rights, including free, prior, and informed consent described in G5.
- Have similar monitoring elements.

The Pantanal REDD+ Program will comply with all eligibility criteria for grouped projects established by the CCB Standard when inserting a new instance. Additionally, the scalability limit of the Pantanal REDD+ Program is defined by the Project Zone, located in the states of Mato Grosso and Mato Grosso do Sul, as presented in item 2.1.16.

The project proponent has extensive experience in REDD+ projects in Brazil, as well as a qualified team and financial resources for such endeavors. Scaling up the project would bring greater environmental, social, cultural, and economic benefits, contributing to biodiversity conservation, climate regulation, water resource availability, local sustainable development, culture valorization, and generation and dissemination of scientific knowledge. Additionally, methodology VM0007 v1.6 used by the Program does not establish any limit on project size or number of activity instances. Therefore, the proponent does not acknowledge capacity limits, economic and managerial restrictions for the Program expansion. Since scalability limits are not applicable in the Program, there is no need to build risk mitigation actions for climate, community, and biodiversity benefits for this expansion.

Further detailed explanations can be found below, where the eligibility criteria applied to the Program will be presented, as provided in the VCS Standard v4.6.

Baseline Scenario and Additionality

- i. *Grouped projects shall specify one or more clearly defined geographic areas within which project activity instances may be developed:*

To delimit the geographical area of the Pantanal REDD+ Program, the Brazilian Pantanal biome as a whole was considered, excluding from this limit legally protected areas where legal deforestation would not be able to occur, these being: Conservation Units, Indigenous Lands and Quilombola Territories. In this way, the geographical area of this Program extends across the states of Mato Grosso and Mato Grosso do Sul in Brazil, covering around 16 municipalities where the Pantanal REDD+ Program can operate.

Eligible to become instances of the Pantanal REDD+ Program are legally established private properties with surplus vegetation classified as forest and/or forested savannah according to the Mapbiomas Collection 8¹⁵ classification, and which can be legally deforested according to the environmental legislation in force at the national, state and municipal levels.

To meet the minimum vegetation cover required by law, properties must conserve a percentage of 20% of Legal Reserve and, for properties that fall within the Restricted Use Area of the Pantanal Floodplain, 50% of vegetation cover classified as “Cerrado” formations with a high density of trees (Forested Savannah and Wooded Savannah) and 40% of vegetation cover classified as grassland formations (Park Savannah, Woody Grass Savannah and Steppic-Savannah) must be preserved.

Further details on the geographical area can be found in section 2.1.16 Project location, and a KML file containing the geodetic polygon clearly defining the boundaries of the geographical area into which this Program can expand, as well as the boundaries of the project area, will be provided together with its documentation.

- ii. *Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances. The initial project activity instances are those that are included in the project description at validation and shall include all project activity instances currently implemented on the issue date of the project description. The initial project activity instances may also include any instances of the project activity that have been planned and developed to a sufficient level of detail to enable their assessment at validation:*

In order to determine the baseline scenario and demonstrate the additionality of the Pantanal REDD+ Program, it was considered that all the properties within its geographical area are subject to the same legal framework (with peculiarities regarding legislation at state level) and have the same common practice scenario that dominates the Pantanal biome, which would be the conversion of land use from forested areas to livestock, mainly aimed at raising cattle.

More details on the baseline and additionality scenarios can be found in section 0.

¹⁵ <https://brasil.mapbiomas.org/>

- iii. *Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area:*

The Pantanal REDD+ Program has only one project activity, which is restricted to avoided planned deforestation (APD), following the guidelines stipulated by the methodology VM0007 v1.6.

In order to guarantee the conservation of the excess vegetation on the properties participating in this Program, a long-term contract has been signed between Biofílica Ambipar Environment and each of the landowners. In addition, social and biodiversity activities will be implemented, as well as activities aimed at mitigating leaks and monitoring activities, which are described in section 2.1.17.

- iv. *The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project. Where a single baseline scenario cannot be determined for a project activity over the entirety of a geographic area, the geographic area shall be redefined or divided such that a single baseline scenario can be determined for the revised geographic area or areas:*

The Pantanal REDD+ Program defines only one baseline that is applied to the entire extent of its geographical area and follows all the criteria stipulated by the VM0007 v1.6 methodology, as specified in item 3.2.1 Baseline Scenario.

- v. *The additionality of the initial project activity instances shall be demonstrated for each designated geographic area, in accordance with the methodology applied to the project. Where the additionality of the initial project activity instances within a particular geographic area cannot be demonstrated for the entirety of that geographic area, the geographic area shall be redefined or divided such that the additionality of the instances occurring in the revised geographic area or areas can be demonstrated:*

The Pantanal REDD+ Program has only one defined geographical area and its additionality is shown for its total extension, following all the criteria stipulated by the VM0007 v1.6 methodology. More details can be found in section 3.1.5 Additionality.

- vi. *Where factors relevant to the determination of the baseline scenario or demonstration of additionality require assessment across a given area, the area shall be, at a minimum, the grouped project geographic area. Examples of such factors include, inter alia, common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; determination of regional grid emission factors; and historical deforestation and degradation rates:*

All the relevant factors for determining the baseline and demonstrating the additionality of the Pantanal REDD+ Program were scaled for the entire geographical area of this Program.

New Project Activity Instance Eligibility Criteria

For instances to be included in a grouped project, it must meet one or more of the eligibility criteria listed below:

i. Meet the applicability conditions set out in the methodology applied to the project:

As presented in section 3.1.2 Applicability of the methodology, the Pantanal REDD+ Program follows all the requirements of the VM0007 v1.6 methodology, an eligible methodology under the VCS Program.

With regard to the instances included in this Program, all participating properties meet the following conditions of applicability:

- The project insertion area is not registered with the Clean Development Mechanism (CDM) or any other Greenhouse Gas (GHG) program;
- The implementation area of the project is classified as forest vegetation according to the Brazilian forest definition, and also includes areas classified as Savannah Forest, which are areas with a significant density of trees, but without the formation of a canopy. The main basis for delimiting these phytophysionomies in the project area was the MapBiomass platform, collection 8¹⁶;
- The baseline scenario for inserting the properties falls under the concept of avoided planned deforestation (APD);
- The baseline scenario is based on the legal conversion of land use from forested to non-forested areas, according to the environmental legislation in force at national, state and municipal level in the project area (Brazil, states of Mato Grosso and Mato Grosso do Sul);
- Leak prevention activities do not include: flooded agricultural land (such as areas for growing rice, for example); and areas where there is intensive livestock production through confinement, controlled diets to achieve rapid weight gain and/or the use of manure lagoons (a place where liquid manure produced by animals is stored).

ii. Use the technologies or measures specified in the project description:

The reduction in GHG emissions will be the effect of signing a long-term forest conservation agreement with the owners of the properties that are part of the current and future instances of this project. Complementary activities related to biodiversity conservation, social activities, as well as monitoring activities have been described in sections 4.4.1 e 5.4.1.

iii. Apply the technologies or measures in the same manner as specified in the project description:

Current and future instances of this project must apply the same technologies and measures specified in this document, and any adjustments/changes will be reported and duly described.

¹⁶ <https://brasil.mapbiomas.org/>

- iv. *Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area:*

The only geographical area of the Pantanal REDD+ Program is the Pantanal biome, which partially covers the states of Mato Grosso and Mato Grosso do Sul, including 16 municipalities. Item 2.1.16 Location of the project provides more details on this subject.

New instances must include legally constituted private properties that have forest cover that can be legally deforested, meeting the criteria for avoided planned deforestation (APD).

- v. *Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances:*

The new instances that will be part of this project should follow the same additionality approach presented in section 3.1.5.

Inclusion of New Project Activity Instances

Grouped projects provide for the inclusion of new project activity instances after the initial validation of the project. New project activity instances shall:

- *Occur within one of the designated geographic areas specified in the project description:*

New instances to be added will be restricted to within the previously established boundary of the only geographical area of the Pantanal REDD+ Program, which is based on the boundary of the Pantanal biome.

- *Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial conformance with multiple sets of eligibility criteria is insufficient:*

New instances that are added to the Pantanal REDD+ Program will meet the following eligibility criteria:

- Rural properties within the boundaries of the Pantanal biome, more specifically within the geographical area stipulated for the Pantanal REDD+ Program.
- Commitment to the long-term conservation of the surplus forest cover on the property, agreed through the signing of a long-term contract between Biofilica Ambipar Environment and the owner of the land where the project will be implemented.
- Rural Environmental Registry of the property with "Regular", "Active" or "Registered for analysis" status in the National Rural Environmental Registry System (SICAR) and the IMASUL System for Registration and Strategic Information on the Environment (SIRIEMA),

the latter only if the rural property in question is located in the state of Mato Grosso do Sul.

- *Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate conformance with the applicable set of eligibility criteria and enable evidence gathering by the validation/verification body:*

The monitoring report for new instances will be based on the CCB and VCS Monitoring Report Template, CCB v3.0, VCS v4.3 or the most up-to-date version available at the time, covering all the eligibility criteria stipulated in this item and providing evidence of all the activities planned for the Pantanal REDD+ Program.

- *Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions):*

The proponent of the Pantanal REDD+ Program (Biofílica Ambipar Environment) has a partnership agreement with the owners of the land where the project is being implemented. Since it is the landowners who have the right to use and possess the farms and their resources, a long-term contract has been signed between the parties, aimed at forest conservation in the areas targeted by this Program.

The current owners of the farms included in the first instance of this Program have remained the same since the start date stipulated for the project, October 14, 2021. All documents proving ownership of the project areas are kept under the management of the proponent and will be duly presented to the VVB during the validation/verification process.

- *Have a start date that is the same as or later than the grouped project start date:*

For future instances, the start date of the credit period will be determined based on proof of the owners' commitment not to remove vegetation in areas exceeding the legal reserve, which must be proven equal to or after the project start date.

- *Only be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the total project crediting period:*

The crediting period for new instances will be defined according to the baseline scenario stipulated for the project, which is based on the project area and the planned deforestation rate. Further details on the baseline can be found in section 3.1.4 Baseline Scenario.

- *Not be or have been enrolled in another VCS project:*

The Pantanal REDD+ Program has not been registered with any other GHG emissions program, nor have any of its current or future instances.

- *Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 – 3.6.9 of VCS Standard v4.6:*

Transcribing items 3.6.8 and 3.6.9 of the VCS Standard v4.6, we have:

3.6.8 The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other).

Capacity Limits

3.6.9 Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows:

- 1) Each project activity instance that exceeds one percent of the capacity limit shall be identified.
- 2) Such instances shall be divided into clusters, whereby each cluster is comprised any system of such instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters.
- 3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.

The capacity limit does not apply to the Pantanal REDD+ Program.

- *When the inclusion of a new project activity instance includes the addition of a new project proponent, such instances must be included in the description of the grouped project within five years (in the case of AFOLU projects) of the start date of the project activity instance:*

The Pantanal REDD+ Program has Bioflica Ambipar Environment Investments as its sole proponent and it is not intended to include another proponent in this Program.

Finally, to guarantee the management, monitoring and subsequent quality verification of the project activity instances (PAIs) that make up the Program, an identification code was developed for each property, which is structured as follows:

Signed contract number (C) + Number of property(ies) considered in the contract (P) + Instance validation year (Y) = **Cn-Pnn-Ynn**

Where:

- Signed contract number = C1, C2,...Cn;
- Number of property(ies) considered in the contract = P01, P02,...Pn;
- Instance validation year = Y24, Y23,...Y70.

2.1.7 Project Proponent (VCS, 3.7; CCB, G1.1)

Organization name	Biofílica Ambipar Environment Investments
Contact person	Plínio Ribeiro
Title	Chief Executive Officer (CEO)
Address	Av. Angelica, 2330 ed. New England, 5th floor – São Paulo – SP
Telephone	+55 11 3073-0430
Email	plinio.ribeiro@ambipar.com

2.1.8 Other Entities Involved in the Project

Organization name	Instituto Taquari Vivo - ITV
Role in the project	Provide support in logistics processes for Biofílica Ambipar Environment and contracted suppliers in strategic articulation in the region, in strategic articulation in the region together with public-private institutions of interest and acts as a communication channel with owners.
Contact person	Renato Roscoe
Title	Executive Director
Address	Avenida Afonso Pena, 5723 - Room 1504, Bairro Santa Fé – Campo Grande/MS
Telephone	-
Email	contato@taquarivivo.org

Organization name	Restaura Consultoria Ambiental e Treinamentos LTDA
Role in the project	Preparation of socio-economic and biodiversity diagnoses.
Contact person	Letícia Koutchin Reis
Title	Executive Director
Address	Rua Luis Gama, 121, Bairro Amambai – Campo Grande/MS
Telephone	+55 67 999175884

Email	contato@restauraconsultoria.com
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Organization name	STA Soluções Técnicas e Serviços de Engenharia Ambiental LTDA
Role in the project	Preparation of Carbon stock diagnosis.
Contact person	Raniery Vale Neri Branco
Title	Project Development Engineer
Address	Rua vinte e oito de setembro, 1226, Bairro Reduto – Belém/PA
Telephone	-
Email	raniery@ambientalsta.com.br

Organization name	Monteiro de Barros Sociedade de Advogados
Role in the project	Consultancy on legal and environmental issues for the Pantanal REDD+ Program.
Contact person	Theotonio Mauricio Monteiro de Barros
Title	Lawyer
Address	Rua Ferreira de Araújo, 221 – cj. 34, Bairro Alto de Pinheiros – São Paulo/SP
Telephone	+55 11 3039-1819
Email	theotonio@mdbadv.com.br

Organization name	Associação Aliança 5P
Role in the project	The Pantanal REDD+ Program includes properties that are members of the Associação Aliança 5P, an institution that supports the development of project activities.
Contact person	Ana Paulo Felício
Title	Executive Secretary
Address	Avenida Afonso Pena, 5723 – Sala 1504, Bairro Santa Fé – Campo Grande/MS

Telephone	-
Email	contato@aliança5p.com

2.1.9 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

The Program activities will be developed, as well as the Right to Use the Project Area is respected, in accordance with the criteria of VCS Standard v4.6 (pages 25-26):

- “1) Project ownership arising or granted under statute, regulation, or decree by a competent authority.
- 2) Project ownership arising under law.
- 3) Project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership).
- 4) Project ownership arising by virtue of a statutory, property or contractual right in the land, vegetation or conservational or management process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership).
- 5) An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent.
- 6) An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the land, vegetation or conservational or management process that generates GHG emission reductions or carbon dioxide removals which vests project ownership in the project proponent.
- 7) Project ownership arising from the implementation or enforcement of laws, statutes or regulatory frameworks that require activities be undertaken or incentivize activities that generate GHG emission reductions or carbon dioxide removals.

It is important to highlight that the right to property is guaranteed at a constitutional level, being enshrined as a fundamental right in the Brazilian legal system. In this sense, it should be clarified that, according to the Civil Code, an owner is the one who has the right to use, enjoy and dispose of the thing, and the right to recover it from anyone who unfairly possesses or holds it.

The Pantanal REDD+ Program area is located within the limits of private properties that hold the right to use and possess the farms and their resources, since they are private properties, and have all the documentation that proves their ownership. There are no title or rights disputes in the project area.

As it is a grouped project, the farms belonging to the Pantanal REDD+ Program have different owners. Thus, to establish responsibility and rights over the Program, contracts were signed between the farm owners and the project proponent Biofilica Ambipar Environment.

Documents proving ownership (land details of each property) and formalized contracts with each owner will be shared with the VBB on a confidential basis.

2.1.10 Project Start Date (VCS, 3.8)

Project start date	14-October-2021
Justification	The start date of the Pantanal REDD+ Program is aligned with the beginning of the Pantanal Conservation Initiative, marked by the acquisition of Fazenda Santa Sofia, component of C1-P01-Y24. Fazenda Santa Sofia was acquired by eight investors and its registration transcript was made official on 14/10/2021 in the name of Associação Onçafari, currently in charge of the management and maintenance of the property. The purchase of Fazenda Santa Sofia by those interested in the conservation of the Pantanal gave rise to the 5P Alliance (Pantanal, Preservation, Partnerships, Livestock – Pecuária in Portuguese – and Productivity), a non-profit association dedicated to promoting economic sustainability, environmental preservation and socioeconomic development in the Pantanal region and bordering areas. The 5P Alliance has a strategic role in the implementation of the Pantanal REDD+ Program activities. The consolidation of this purchase significantly contributes to maintaining the biodiversity corridor in the Pantanal, connecting properties in the regions of Miranda, Aquidauana, and Nhecolândia, where instances 02 to 06 are located. It is worth noting that the farms C7-P01-Y24 and C7-P02-Y24, although not part of Alliance 5P, are in the same Pantanal region, contributing to the conservation of the biodiversity corridor and therefore have the same start date. Given the context and aligned with the strategy for starting the Pantanal REDD+ Program, which aimed to connect the largest number of properties to make conservation actions viable, a commitment was formalized between the owners to avoid planned deforestation in the region.

2.1.11 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Crediting period	The project has a credit period of 50 years. This period is within that allowed by the VCS Standard, v4.6 (item 3.9.3) and by VM0007 (item 5.2.2) which say that for AFOLU projects the credit period must be between 20 and 100 years.
Start date of first or fixed	14 October 2021 to 31 Decembre 2070

crediting period	
CCB benefits assessment period	Changes in adaptive capacity to climate change and resilience, biodiversity and community well-being resulting from project activities are monitored over the project longevity period, which is 50 years, the same as the project credit period. The PAIs will generate reductions in GHG emissions eligible for emission as VCUs for 20 years, a period defined to guarantee the financial viability of the Program and, consequently, the implementation and its activities. Additionally, the partnership signed with the owners guarantees the continuity of management practices that will lead to the conservation of the project area throughout the duration of the contract. With the conclusion of the contractual period, in the years remaining until the completion of the project's longevity, the proponent will monitor the properties to ensure the permanence of the forest areas. For the next PAIs of the Program, the start date of the credit period will be determined based on proof of the owners' commitment not to carry out vegetation suppression in areas exceeding the legal reserve.

2.1.12 Differences in Assessment/Project Crediting Periods (CCB, G1.9)

There will be no difference between the assessment period and the credit period of the Pantanal REDD+ Program.

2.1.13 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

Indicate the estimated annual GHG emission reductions/carbon dioxide removals (ERRs) of the project:

☐ < 300,000 tCO₂e/year (project); ☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
14-October-21 to 31-December-21	466,852
1-January-2022 to 31-December-2022	478,803
1-January-2023 to 31-December-2023	490,753
1-January-2024 to 31-December-2024	502,704

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
1-January-2025 to 31-December-2025	514,654
1-January-2026 to 31-December-2026	526,605
1-January-2027 to 31-December-2027	538,556
1-January-2028 to 31-December-2028	550,506
1-January-2029 to 31-December-2029	562,457
1-January-2030 to 31-December-2030	574,407
1-January-2031 to 31-December-2031	464,555
1-January-2032 to 31-December-2032	142,628
1-January-2033 to 31-December-2033	93,489
1-January-2034 to 31-December-2034	81,538
1-January-2035 to 31-December-2035	69,588
1-January-2036 to 31-December-2036	57,637
1-January-2037 to 31-December-2037	45,687
1-January-2038 to 31-December-2038	33,736
1-January-2039 to 31-December-2039	21,785
1-January-2040 to 31-December-2040	9,835
Total estimated ERRs during the first or fixed crediting period	6,226,775
Total number of crediting years (PAI 01-07)	20
Average annual ERRs (first baseline period)	311,339

2.1.14 Physical Parameters (CCB, G1.3)

The history of land use in the region where the Pantanal REDD+ Program is located is mainly geared towards livestock practices, with areas with forest cover being converted to pasture for animals. This

deforestation dynamic in the Pantanal biome historically predates the date of implementation of the Pantanal REDD+ Program and, in this sense, this Program was not implemented with the aim of generating Greenhouse Gas emissions, but rather to transform the recurring dynamic that causes this fact, acting to prevent legalized deforestation inside private properties.

It is worth mentioning that the conditions prior to the start of the Pantanal REDD+ Program correspond to the baseline scenario. In this regard, below is a summary of the conditions prior to the start of the project, further details of which can be found in section 3.1.4 Baseline scenario.

- **Ecosystem type:** Flood and drought system, considered the largest floodplain in the world. It has a topography formed by mountains and plateaus, a tropical climate with rainfall well distributed throughout the year, fertile, sandy and clayey soil. It has the second largest biodiversity in Brazil, with the largest native vegetation cover among the country's biomes, and a concentration of endangered species.
- **Current and historical land use:** Historical local land use is predominantly dedicated to cattle breeding, which is a common practice in the Pantanal biome. Currently, land use in the area where the Pantanal REDD+ Program is being implemented is covered by forest and savannah.
- **Current and previous environmental conditions in the project area:**
- Considering the geographical area of the Pantanal REDD+ Program (which includes the Pantanal biome), we can observe the following physical characteristics:

Climate: The climate is hot and humid in the Pantanal region and, according to the Köppen classification system, the area falls into the Am, Af, and Aw classifications, each representing specific subtypes of the tropical climate with unique characteristics.

Hydrology: The geographical area falls within the Paraguay Hydrographic Region (RH).

Topography: The Pantanal region is an alluvial plain affected by rivers that drain the Upper Paraguay river basin, forming an extensive flooded area. Its plain is slightly undulating with rare isolated elevations and many shallow depressions.

Relevant historical conditions: The Pantanal biome is extremely important for the environment, being one of the most exuberant and diverse nature reserves on Earth. Because of its characteristics and importance, this biome is considered a World Natural Heritage Site and a Biosphere Reserve recognized by UNESCO.

Soils: The geographical area has predominantly sandy soils, comprising the following soil types: Agrisols, Cambisols, Chernosols, Gleisols, Latosol, Neosol Planosol, Plinthosol and Vertisol.

Vegetation: The vegetation typologies present in the geographical area of the Pantanal REDD+ Program are summarized as savannah typologies, but also include seasonal forests.

Ecosystems of the project area: The ecosystems of the Pantanal biome are characterized by savannahs and savannahs without periodic flooding, flooded fields and aquatic environments (freshwater or brackish lagoons, rivers, ebb and flow)¹⁷.

2.1.15 Social Parameters (VCS, 3.18; CCB, G1.3)

The Pantanal REDD+ Program has an area of activity planned for the entire Brazilian Pantanal biome and the properties of PAIs 1-7 (initial instances) of the project are located in the municipalities of Aquidauana, Miranda and Corumbá, in Mato Grosso do Sul.

Talking about the current and historical land use, the majority of vegetation in the region consists of native pasture, followed by savannah vegetation and forest on the latitudinal elevations of the terrain (capões e cordilheiras in Portuguese). Therefore, extensive livestock farming is the main economic activity in the plain. Currently, land use in the implementation area of the Pantanal REDD+ Program is vegetation cover represented by forest and forested savannah.

The socio-environmental diagnosis conducted indicated that there are no traditional communities such as indigenous, riverside, or quilombola communities in the project areas. Thus, as these are private properties, what is found in the project area are farm employees working in livestock farming, tourism, and other general services. These workers reside on the farms with their families in houses provided by the owner or in shared accommodations for employees. Socio-cultural information from the properties comprising the Program was collected in the field and is summarized in the following table:

Table 1. Socio-cultural characterization indicated in the Socio-environmental Diagnosis carried out

Declared Color/Race	%
Yellow	1%
White	21,17%
Indigenous	6,57%
Black	15,33%
Brown	56,20%
Gender	%
Female	35,04%
Male	64,96%
Sexual Orientation	%

¹⁷ <https://www.embrapa.br/pantanal/apresentacao/o-pantanal>

Bisexual	0,73%
Heterosexual	91,97%
Homosexual	7,30%
Marital Status	%
Married	20,44%
Divorced	3,65%
Single	50,36%
Common-law Marriage	24,82%
Widowed	0,73%
Place of Residence	%
Rural Area	4,38%
Urban Area	3,65%
Dependence on Property	90,51%
Not specified	1,46%
Education	%
Literate	8,03%
Elementary School	32,85%
Completed Elementary School	4,38%
Completed High School	18,98%
Attending High School	1,46%
Incomplete High School	16,06%
Completed College/University	7,30%
Attending College/University	2,92%
Incomplete College/University	2,19%
Illiterate	0,73%
Completed Postgraduate Studies	3,65%
Attending Postgraduate Studies	1,46%

Work Regime	%
CLT (Consolidação das Leis do Trabalho - Brazilian Labor Law)	79,56%
Direct Contract	3,65%
Outsourced Contract	2,92%
Not specified	1,46%
Temporary	12,41%
Length of Employment	%
Less than 6 months	20,44%
6 months to 1 year	27,74%
1 year to 2 years	19,71%
2 years to 5 years	13,14%
5 years to 8 years	2,19%
8 to 10 years	4,38%
More than 10 years	10,95%
Not specified	1,46%

Source: Data from field research carried out from October 2023 to January 2024 by the company Restaura.

More information about sociocultural characteristics can be found in Section 4.1.1 of this document or in the full socioeconomic report.

2.1.16 Project Zone Map and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

The region where the REDD+ Program is being implemented is the Pantanal biome, which is located in central-western Brazil, encompassing the states of Mato Grosso and Mato Grosso do Sul (Figure 1). This biome is subdivided into 11 regions: Porto Murtinho, Nabileque, Miranda, Aquidauana, Abobral, Nhecolândia, Paraguay, Paiguas, Barão de Melgaço, Paconé and Cáceres.

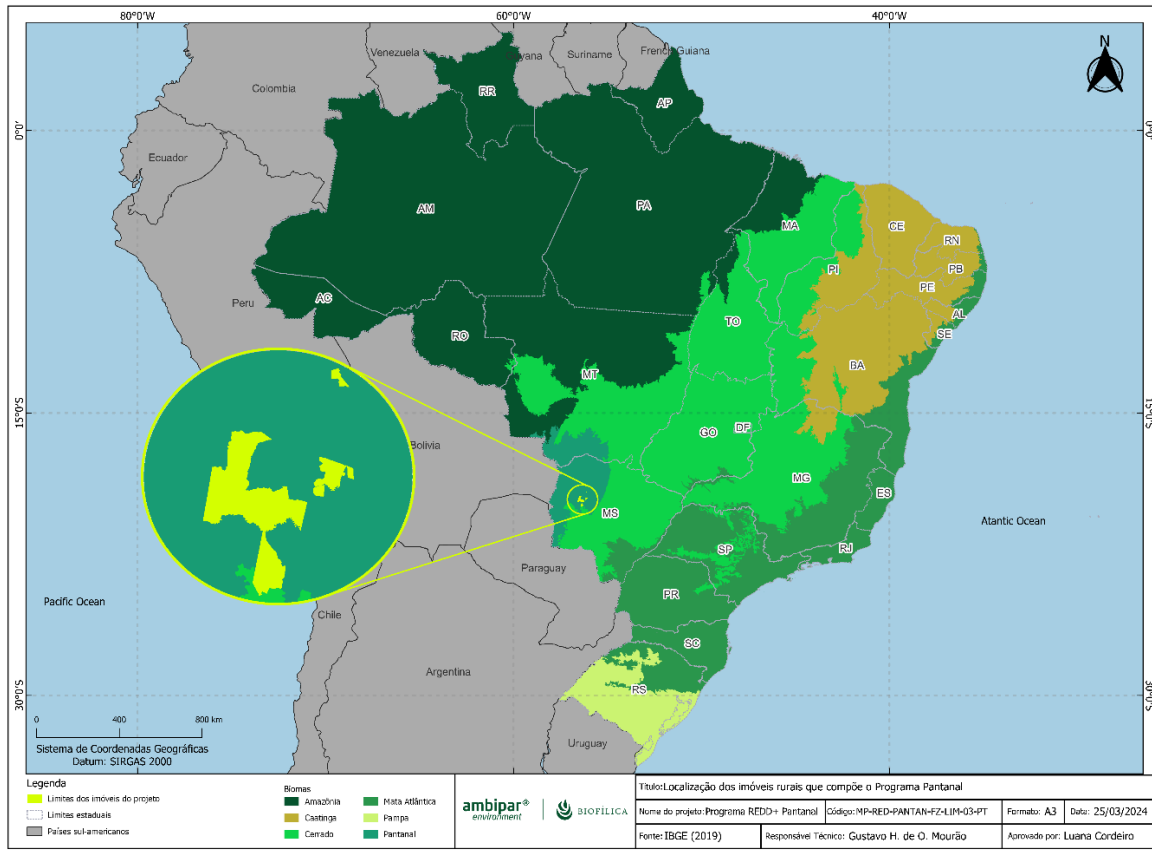


Figure 1. Location of the Pantanal biome, indicating the area where the Pantanal REDD+ Program is located

As this is a grouped project, the map of the Project Zone for the Program was constructed based on the boundaries of the Pantanal biome, taking into account the legislative and administrative characteristics of the region of interest. In addition, the following areas were excluded: (i) Conservation Units; (ii) Indigenous Lands; (iii) Quilombola Lands; (iv) Private Natural Heritage Reserves. For this, two official Brazilian government data sources were used: SIGEF¹⁸ and SNCI¹⁹. Criteria for vegetation, phytophysiology and current legislation were also considered to compose the map. More details on the construction of the boundaries of the Project Zone can be found in section 3.1.3.

PAIs 01-07 of the Pantanal REDD+ Program are located in Brazil, in the state of Mato Grosso do Sul, more specifically in the municipalities of Aquidauana, Miranda and Corumbá, and fall within the Pantanal sub-regions of Aquidauana, Miranda, Nhecolândia and Abobral. Its surroundings are characterized by the

¹⁸ SIGEF stands for Land Management System, an electronic system developed by Brazil's National Institute for Colonization and Agrarian Reform (INCRA). SIGEF was created to manage information on rural land in the country. It enables the electronic registration of rural land parcels, facilitating the land regularization process. The system also helps prevent land conflicts by providing a clear and up-to-date view of the country's land situation

¹⁹ SNCI, on the other hand, is the acronym for the National Rural Cadastre System, also developed by INCRA. The SNCI is a database that contains information on rural properties, their owners, the nature of land use, among other information. The SNCI is an important tool for land management and agrarian policies in Brazil. It is used for planning, monitoring and implementing public policies related to land and rural development.

presence of the Rio Negro Pantanal State Park and the Cachoeirinha Indigenous Land, located near the Rio Vermelho and Rio Negro, as illustrated in Figure 2.

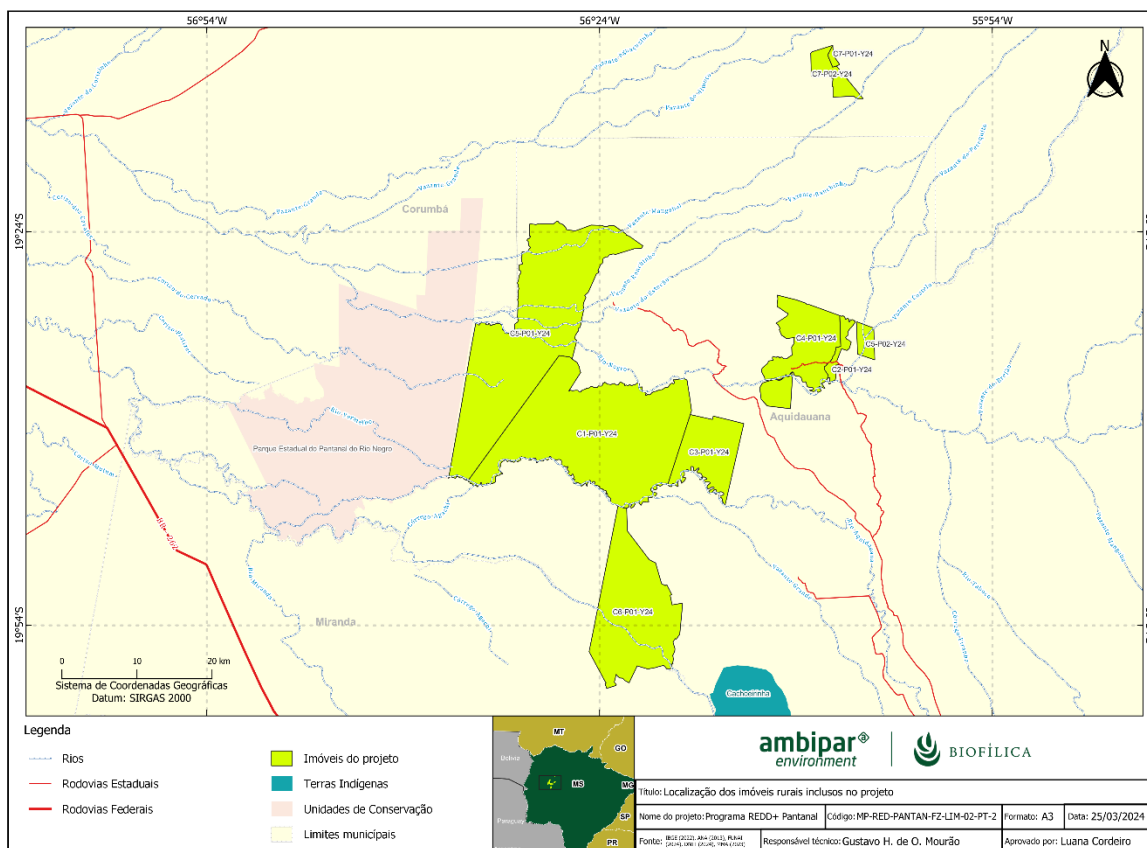


Figure 2. Location of the properties participating in the first instance of the Pantanal REDD+ Program

Table 2 details the reference coordinates (central point) of each of the 09 properties included in the Program's PAIs, as well as listing the numbers indicated on each property that appear in Figure 2. The coordinates of the vertices of each property will be provided via a KML file.

Table 2. Reference coordinates of the properties participating in the Pantanal REDD+ Program

Project Activity Instance	UTM coordinates	
	X	Y
C1-P01-Y24	562544.9	7826404.6
C2-P01-Y24	595015.0	7838415.5
C3-P01-Y24	577479.0	7823764.5
C4-P01-Y24	589882.9	7837925.6
C5-P01-Y24	553642.1	7840544.7

Project Activity Instance	UTM coordinates	
	X	Y
C5-P02-Y24	598474.4	7839314.5
C6-P01-Y24	567961.2	7803860.3
C7-P01-Y24	594330.2	7879131.7
C7-P02-Y24	594175.4	7876868.5

2.1.17 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

To gain a clearer understanding of how the Pantanal REDD+ Program intends to generate short-, medium- and long-term impacts in terms of climate, community and biodiversity, the Theory of Change was employed. This theory describes the logical sequence of events and expected results that should occur through the implementation of the planned activities.

The proposed activities were designed based on a comprehensive analysis of the context of the region of operation and the socio-economic and environmental reality present in the Pantanal. The focus was on addressing the main issues that can be mitigated through REDD+ Program actions. After defining the necessary activities, these were subdivided into major areas of action, described below.

Containing the expansion of deforestation: Deforestation causes various effects, such as climate change (water scarcity, rising temperatures) and the extinction of species of fauna and flora in the Pantanal. Focusing on implementing activities aimed at containing deforestation, actions are proposed such as carrying out frequent monitoring in the Project Area and the Leakage Belt by means of deforestation alerts using remote sensing data. This approach makes it possible to quickly identify legal and illegal deforestation activities, enabling effective intervention to contain them. In addition, it is necessary to implement and/or improve strategic property surveillance actions, integrating them with the activities already carried out on the farms. This implies reinforcing security measures to protect forest areas and prevent invasions and illegal deforestation activities.

Finally, thinking about the legal deforestation that can occur on properties, it is essential to monitor the expansion of pasture areas, ensuring that they do not advance into the Project Area or the Leakage Belt. This requires encouraging practices that promote sustainable land use and avoid converting forests into pastures, thus contributing to the conservation of the forest and forested savannah areas included in the Program.

Fire Prevention and Mitigation: Given the increasing visibility of the impacts of fire in the Pantanal region, it is extremely important to implement measures to prevent and mitigate the effects of fire in this region. In this sense, it is essential to focus on the installation of remote monitoring systems with wildfire alerts for properties, enabling early detection and swift response to fire outbreaks. Furthermore, it is crucial to encourage the formation and maintenance of fire brigades on farms, ensuring prompt intervention in

emergencies and strengthening response capabilities through the acquisition of appropriate equipment and machinery for firefighting.

In order to establish clear guidelines and effective strategies, the Project aims to support the development of the Integrated Fire Management Plan (IFMP, PMIF in Portuguese) for interested properties. Additionally, support in creating and improving communication channels among neighbors is essential to facilitate coordination of efforts and information exchange in emergencies, promoting a collaborative approach to preventing and combating wildfires.

These combined measures aim to significantly reduce the risks and impacts of wildfires, protecting vulnerable ecosystems and preserving lives and properties.

Promotion of Sustainable Actions in Rural Areas: Livestock farming is the main economic activity in the Pantanal and one of the main causes of the expansion of legally deforested areas in this biome. In light of this scenario, promoting sustainable practices in rural areas is crucial to ensure the sustainable development of the region.

Based on this, it is imperative that the Pantanal REDD+ Program seeks to expand its areas of operation, ensuring the commitment of properties not to engage in vegetation suppression in areas beyond what is legally permitted. Additionally, maintaining institutional partnerships and collaborations with public agencies and private entities is fundamental to strengthening the implementation of sustainable practices.

With the aim of perpetuating activities among properties involved in the program, it is essential to encourage the participation of partners in associations dedicated to promoting sustainable practices in the Pantanal rural areas, stimulating collaboration and exchange of experiences. Furthermore, diversifying models of Payments for Environmental Services (PES, PSA in Portuguese) in forest and forested savannah areas is necessary to further valorize environmental conservation.

Finally, in collaboration with partners, it is crucial to map and subsequently implement additional and alternative income sources, taking into account the peculiarities and objectives of each property, to ensure the economic viability of adopted sustainable practices. These combined actions aim to promote responsible use of natural resources, conserve biodiversity, and ensure the well-being of the local Pantanal population.

Sustainable Development and Quality of Life: Concern for the well-being and improvement of the quality of life of employees and property owners is a fundamental part of achieving the objectives proposed by the Pantanal REDD+ Program. Therefore, activities have been established to enhance safety, quality of food, access to healthcare and education, technical training, and communication processes among farm employees.

It is important to note that this is one of the most adaptable themes per property, as some demands identified during the process of assessing local "issues" were very specific to each property and region. Thus, some activities may be tailored according to the annual planning that will be developed between the proponent and partners. Among the aspects addressed are the improvement of local educational

infrastructure, the promotion of vegetable gardens, alternative productive systems, and the restoration of areas on farms, support for mobile medical care campaigns, implementation of higher-quality internet systems and alternatives for water treatment, as well as the development and execution of an emergency rescue plan, aiming to ensure the safety and well-being of employees.

Considering activities of broad application, regardless of specific contexts, training for employees and their families on fundamental topics aligned with the socio-environmental and economic objectives of properties is emphasized, with the aim of preserving the forest and Pantanal culture. Additionally, conversations and workshops on adaptation and preparation for possible future scenarios resulting from climate change will be promoted to increase community resilience. Investments in maintaining internal farm roads will also be carried out, contributing to mobility and safe access.

Finally, the processes of hiring new employees will be reviewed and improved with the aim of creating a safe and inclusive work environment. All these combined actions aim not only to promote sustainable development but also to significantly improve the quality of life of rural populations in the Pantanal.

Biodiversity Conservation: Considering the vast biodiversity present in the Pantanal biome, it is important to maintain activities within the scope of the Pantanal REDD+ Program that adopt specific measures to protect its unique fauna and flora.

Highlighted actions in this theme include continuous monitoring of fauna and flora and monitoring of endangered species, classified as Attributes of High Conservation Value (AHCVs), to understand ecosystem dynamics and identify any significant changes. Additionally, it is necessary to encourage local scientific research to expand knowledge and promote species conservation.

Simultaneously, looking at the context of each property, campaigns to raise awareness among employees and local communities about the impacts of illegal hunting and fishing on Pantanal biodiversity will be recommended. Similarly, campaigns will be supported to raise awareness about conflicts between livestock farming and jaguars, aiming to minimize human-wildlife conflicts. Furthermore, supporting research on these relationships will be important, seeking to better understand interactions and develop strategies for harmonious coexistence. These combined actions aim to ensure the preservation of the unique biodiversity of the Pantanal, protecting its species and ecosystems.

The table below details the activities designed for the Pantanal REDD+ Program separated by their corresponding categories (climate, biodiversity, and/or community) and grouped according to major themes of action within the context of the Pantanal REDD+ Program. Next, the expected results in the short, medium, and long term for the implementation of each activity are listed.

It is crucial to emphasize that the planning of activities was conceived with a broad vision, especially considering the expansion process of the Pantanal REDD+ Program. This implies that the specification of activities on each property under the program will occur during the annual strategic planning conducted by Biofilica Ambipar Environment and its partners for each involved property. The mentioned activities will serve as guidance for short, medium, and long-term decisions on properties participating in the program, outlining a clear path to achieve established objectives.

Table 3. Description of the activities and theory of change of the Pantanal REDD+ Program

Clim	Com.	Biod.	Activity Group	Outputs (short term)	Outcomes (medium term)	Impacts (long term)
			Containing the expansion of deforestation	Identification of deforested areas (legally or illegally), enabling rapid intervention if necessary	Mapping areas where there is a higher incidence of legal and illegal deforestation, making it possible to implement alternative actions in these places	Reducing deforestation rates in monitored areas
				Greater sense of security in the farms' areas	Identifying vulnerable points on the farms	Prevention of environmental damage, such as wildfires and illegal deforestation
				Identification of the purchase of new properties by Pantanal REDD+ Program participants and possible issuance of deforestation licenses in these locations	Preventing the use of financial resources from the Pantanal REDD+ Program for deforestation in other areas not included in the Program	Reduction in legal deforestation rates
				Alert to the expansion of pasture areas in the Pantanal REDD+ Program region	Protecting the Pantanal REDD+ Program area by taking measures to contain the expansion of cattle ranching in these areas	Mapping of the most suitable locations for pasture growth
			Fire Prevention and Mitigation	Immediate detection of forest fires, enabling rapid intervention for fire containment	Significant reduction in damage to forest areas, reducing the loss of natural resources and financial losses to owners	Establishing forest fire prevention plans based on the monitoring data collected
				Training of firefighting personnel for rapid response to fire containment emergencies	Reduction in the frequency and severity of fires on farms	Implementing a culture of fire prevention among farm employees and owners, encouraging safe and conscious practices to reduce the risk of fires
				Greater efficiency in emergency fire containment responses, extinguishing the fire more quickly	Reduction in the severity of forest fires	Reducing damage and losses due to fires on farms
				Implementation of actions to prevent, control and monitor fire on farms	Reduction in the number of forest fires	Reducing the negative impacts of fire on the environment

Clim	Com.	Biod.	Activity Group	Outputs (short term)	Outcomes (medium term)	Impacts (long term)
				Greater efficiency and speed in transmitting messages between neighboring properties, such as initial fire outbreaks	Coordination of actions between neighbors during emergencies, sharing resources, information and firefighting strategies	Strengthening collective efforts to prevent and fight fires, developing a sense of shared responsibility for protecting against forest fires in the region
			Promotion of Sustainable Actions in Rural Areas	Articulating partnerships with institutions that bring together rural landowners (unions and associations), prospectors, land brokers and local communities	Identifying local opportunities for implementing carbon projects	Development of carbon projects in the region, promoting economic activities that value the standing forest
				Identifying potential partners to exchange information and resources	Developing formal agreements and strategic partnerships for implementing carbon projects	Promoting economic activities that value the standing forest
				Promoting partnerships with associations that promote sustainable actions in the Pantanal region's rural areas	Promoting capacity building and training on ecological pastures and good livestock practices for partners in the Pantanal REDD+ Program	Reducing the negative impacts of cattle farming practices
				Generating extra income and encouraging environmental conservation	Ecosystem recovery and sustainable economic development	Valuing the standing forest and consequently conserving biodiversity
				Reducing dependence on just one economic activity to generate income on farms and create jobs	Increasing the resilience of farms to economic shocks (price fluctuations)	Diversification of the economy and an increase in the total revenue of farms, enhancing the value of the property
			Sustainable Development and Quality of Life	Basic understanding of how a carbon project works and what its relevant activities are	Implementation of Pantanal REDD+ Program activities in the day-to-day work of farm employees	Greater employee engagement and sense of responsibility in carrying out the project's activities, promoting the socio-environmental and economic benefits expected for the region

Clim	Com.	Biod.	Activity Group	Outputs (short term)	Outcomes (medium term)	Impacts (long term)
				Increased awareness among employees and farm owners about the impacts of climate change and available adaptation strategies	Implementation of measures to adapt to climate change	Increased resilience of farms to climate issues and reduction of their environmental impacts through the adoption of more sustainable practices
				Improved access to farms, facilitating the transportation of people, animals, and supplies	Stimulation of economic development, such as the promotion of tourist activities	Contribution to the sustainable development of the region, promoting socio-economic integration and improving the quality of life of local communities
				Improved teaching and learning conditions and easier access to education	Development of people with a higher level of training to work in the region	Community empowerment, reduction of poverty and inequality and promotion of sustainable development in the region
				Diversification of production	Increased income for farms, access to quality food and protection of natural resources	Improved quality of life for the rural population, promoting community resilience
				Improved access to health care for farm employees	Promotion of preventive health for employees and consequent reduction in health costs	Improved quality of life for employees
				Improved communication and easier access to information and culture	Internet access can boost economic development	Exchange of knowledge, culture and experiences, improving local quality of life
				Improved quality of water used on farms	Reduced risk of diseases associated with consumption of untreated water	Promotion of public health, improving the general well-being of the population
				Readiness for initial medical care in case of emergencies, such as minor accidents at work	Reducing the impact of accidents at work on farms	Creating a culture of safety on farms, preparing employees for an emergency
				Preparedness for more serious emergencies requiring hospital care, such as attacks by venomous animals and serious accidents at work	Reduction of more serious injuries and deaths due to accidents on farms	Creating a culture of safety at work, demonstrating concern for the safety and well-being of farm employees
				More efficient and faster hiring process for new employees	Improved farm reputation by guaranteeing and explaining all labor rights to employees	Improved quality and employee engagement on farms

Clim	Com.	Biod.	Activity Group	Outputs (short term)	Outcomes (medium term)	Impacts (long term)
			Biodiversity Conservation	Identifying species and population dynamics in the region (data on species diversity, distribution and abundance)	More accurate assessment of conservation status and identification of areas where populations are declining	Development of conservation strategies using the long-term monitoring data obtained
				Obtaining systematic data on endemic, threatened and venomous species, etc.	Knowledge of species diversity and environmental conditions in the project area	Assertive actions to conserve local biodiversity
				Identifying immediate threats to fauna species	Collection of data to direct biodiversity management activities	Protecting fauna species under some degree of threat
				Implementation of sustainable fishing practices and information on animals that are illegally hunted	Change in predatory cultural habits and awareness of the importance of conserving local fauna	Protection of fauna species vulnerable to hunting and fishing
				Implementing alternative measures to protect livestock (safer fences, for example)	Change in the perception about jaguars by livestock farmers, who may begin to see these animals not just as predators, but as part of the local ecosystem	Greater protection for jaguars in the Pantanal due to hunting by ranchers
				Understanding the dynamics of the relationship between humans and wildlife, including patterns of interaction, areas of conflict and what the influencing factors are	Development of strategies to protect fauna due to human interaction	Education campaigns on the importance of fauna and best practices to minimize identified human-fauna conflicts

2.1.18 Sustainable Development Contributions (VCS, 3.17)

Because it belongs to the Ambipar Group, the project proponent follows all the Group's policies and, consequently, all the activities carried out by the Pantanal REDD+ Program must follow the standards and rules established in the Group's Code of Conduct and Compliance (2023)²⁰, Sustainability Policy (2023)²¹ and Diversity and Inclusion Policy (2023)²².

In this sense, the Pantanal REDD+ Program 's planned activities are based on alignment with the Sustainable Development Goals (SDGs) and reinforce work on environmental, social and governance

²⁰ <https://ambipar.com/site2020/wp-content/uploads/2023/09/Codigo-de-Conduita-e-Compliance-2023.pdf>

²¹ <https://ambipar.com/site2020/wp-content/uploads/2023/09/Politica-de-Sustentabilidade-2023-portugues.pdf>

²² <https://ambipar.com/site2020/wp-content/uploads/2023/09/Politica-de-Diversidade-e-Inclusao-portugues.pdf>

aspects. Below are the activities planned for the Pantanal REDD+ Program in line with the UN Sustainable Development Goals (SDGs) and its National Targets²³:

Activity	SDG Target	ODS Indicator	Estimated contribution over the lifetime of the project	Net impact on ODS indicator
Encouraging the implementation of alternative means of natural food products on the properties	SDG 2.4	Ensure sustainable food production systems	Promote organic agriculture, which minimizes the use of pesticides. Raise awareness among farmers about safer alternatives and integrated pest control methods, as well as provide incentives for low-impact agriculture	Activities implemented to increase
Training sessions for farm employees and their families on topics aligned with the socio-environmental and economic objectives of the properties, aiming to preserve forests and Pantanal culture	SDG 4.7	Integration between the education of local schools and education for sustainable development for forest conservation	Increased engagement and sense of responsibility among project collaborators in carrying out project activities, promoting the social, environmental, and economic benefits envisaged for the area	Activities implemented to increase
Mapping and eventually implementing additional and alternative sources of income, aiming to combine economic viability and sustainable practices on the properties	SDG 8.4	Strive to decouple economic growth from environmental degradation	Diversification of the economy and promotion of sustainable actions in rural areas	Activities implemented to decrease

²³ <https://repositorio.ipea.gov.br/handle/11058/8636>

Activity	SDG Target	ODS Indicator	Estimated contribution over the lifetime of the project	Net impact on ODS indicator
Improving the processes for hiring new suppliers	SDG 8.8	Ensure the labor rights of all employees	Promotion of improved quality of life for collaborators on properties participating in the Pantanal REDD+ Program	Activities implemented to increase
Mapping on the properties and encouraging the implementation of a rescue plan for medical emergencies on the farms	SDG 8.8	Reduction of activity risks	Creation of a culture of workplace safety, demonstrating concern for the safety and well-being of farm collaborators	Activities implemented to decrease
Preventing planned deforestation on private properties in the Pantanal biome	SDG 13	Tons of greenhouse gas emissions avoided	The Pantanal REDD+ Program estimates to prevent the release of 6,226,775 tCO ₂ e into the atmosphere during the first reference period (10 years).	Activities implemented to increase
Conducting conversations/works hops on adaptation and/or preparation for future scenarios caused by climate change	SDG 13.3	Enhance education, increase awareness, and human and institutional capacity	Contribute to access to education and awareness about climate change and its risks	Activities implemented to increase
Frequent monitoring of the project area and Leak Belt through deforestation alerts using remote sensing data, implementation of surveillance actions	SDG 15.5.1br	on climate change, its risks, mitigation, adaptation, impacts, and early warning	Mapping of areas with the highest incidence of deforestation, enabling the implementation of alternative actions in these locations, aiming to reduce deforestation rates in monitored areas	Activities implemented to decrease

Activity	SDG Target	ODS Indicator	Estimated contribution over the lifetime of the project	Net impact on ODS indicator
on the properties, monitoring the expansion of pasture areas, and implementing a remote monitoring system with alerts for forest fires				
Continuous monitoring of fauna and flora and species threatened with extinction and classified as High Conservation Value Attributes (HCVAs)	SDG 15.5.2br	Reduce the rate of biodiversity loss and degradation and fragmentation	Develop strategies for biodiversity conservation through monitoring data and scientific research, thus enabling the application of urgent and significant measures to reduce habitat degradation and halt biodiversity loss	Activities implemented to decrease
Supporting and conducting (when possible) awareness campaigns on illegal hunting and fishing with farm employees	SDG 15.7	Index of threatened species of extinction	Biodiversity conservation	Activities implemented to increase
Encouraging local scientific research to expand knowledge and promote species conservation	SDG 15.a	Take urgent measures to end illegal hunting and fishing	Exchange of information and resources aimed at promoting economic activities that value standing forests	Activities implemented to increase
Implementation of Payment for Environmental Services (PES) models on the properties	SDG 15.a	Mobilize and increase financial resources for the conservation and sustainable use of biodiversity and ecosystems	Ecosystem restoration and sustainable economic development	Activities implemented to increase

Activity	SDG Target	ODS Indicator	Estimated contribution over the lifetime of the project	Net impact on ODS indicator
Formation of partnerships with public and private institutions	SDG 17.17	Official development assistance and public expenditures on conservation and	Promotion of economic activities that value standing forests through the implementation of carbon projects in the region	Activities implemented to increase

2.1.19 Implementation Schedule (CCB, G1.9)

Date	Milestone(s) in the project's development and implementation
Q1-2021	Consolidation of the implementation strategy for the Pantanal REDD+ Program
27-August-2021	Initiation of the prospecting process for the Pantanal REDD+ Program led by Biofílica Ambipar Environment
14-October-2021	Commencement of the Pantanal Conservation Initiative (completion of the acquisition of Santa Sofia Farm)
09-September-2022	Contract signing – C2-P01-Y24
13-September-2022	Contract signing – C3-P01-Y24
16-September-2022	Contract signing – C4-P01-Y24
16-September-2022	Contract signing – C5-P01-Y24 and C5-P02-Y24
21-September-2022	Contract signing – C1-P01-Y24
19-October-2022	Contract signing – C6-P01-Y24
01-November-2022	Conducting the 1st kick-off Roundtable of the Pantanal REDD+ Program with new owners

Date	Milestone(s) in the project's development and implementation
30-November-2022	1st Workshop on the Pantanal REDD+ Program at FAMASUL auditorium (Federation of Agriculture and Livestock of Mato Grosso do Sul)
19-April-2023	Hiring of Legal Consultancy
26-June-2023	Workshop on the Pantanal REDD+ Program with ABPO (Pantanal Association of Organic and Sustainable Livestock) and SEMADESC (Secretariat of Environment, Development, Science, Technology, and Innovation)
17-July-2023	Presentation of the Pantanal REDD+ Program at the "Pantanal Bridges" event
18-July-2023	Consolidation of the strategic partnership with Instituto Taquari Vivo (ITV)
19-July-2023	Meeting with IMASUL regarding communication of the Program implementation
August-2023	Contract signing – C7-P01-Y24 and C7-P02-Y24 and conducting the 2nd kick-off Roundtable of the Pantanal REDD+ Program with new owners
26-September-2023	Conducting the 1st Workshop with suppliers to present the work plan for socio-environmental and stock diagnostics
01-October-2023	Commencement of field diagnostics activities – Carbon Stock
09-October-2023	Commencement of field diagnostics activities – Socioeconomic and Biodiversity
19-December-2023	Submission of the draft PDD VCS on the VERRA registry platform for project listing
24-January-2024	1st Working Group Meeting led by Instituto Taquari Vivo, with the participation of Biofílica Ambipar Environment, Aliança 5P, FAMASUL, SEMADESC, and IMASUL, to discuss the effects of Law 6.160/2023 (Pantanal Law)
19-February-2024	2nd Working Group Meeting led by Instituto Taquari Vivo, with the participation of Biofílica Ambipar Environment, Aliança 5P, FAMASUL, SEMADESC, and IMASUL, to discuss the effects of Law 6.160/2023 (Pantanal Law)
01-February-2024	Hiring of VVB (AENOR)

Date	Milestone(s) in the project's development and implementation
February/March-2024	Conducting stakeholder feedback sessions
20-March-2024	3rd Working Group Meeting led by Instituto Taquari Vivo, with the participation of Biofílica Ambipar Environment, Aliança 5P, FAMASUL, SEMADESC, and IMASUL, to discuss the effects of Law 6.160/2023 (Pantanal Law)

2.1.20 Risks to the Project (CCB, G1.10)

The risk analysis for the Pantanal REDD+ Program was carried out using the Non-Permanence Risk Tool (AFOLU Non-Permanence Risk Tool, v4.2). This tool considers the internal risk of the project, the external risk, the natural risk, and the mitigation measures that help reduce the risks. The following table summarizes the results of the non-permanence risk analysis for the Program.

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Fire - Forest fires	Loss of carbon stocks (biomass) and biodiversity (fauna and flora)	The activities designed for the Pantanal REDD+ Program include the following: Implement monitoring with an alert system for forest fires on the properties; Encourage the creation and/or maintenance of fire brigades on the properties; Acquire fire-fighting equipment/machinery for the properties that need it; Support in drawing up the Integrated Fire Management Plan - PMIF for interested properties; and Support in drawing up more robust communication channels between neighbors.
Political risk	Failure to maintain activities due to political obstacles in the country	Although this is considered an external risk in which the Program's influence will be relatively low, Biofílica Ambipar Environment has an Advocacy and Engagement team that aims to create a favorable business environment for the implementation of the Pantanal REDD+ Program by establishing government, public and international relations. This team is responsible for establishing and maintaining strategic

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
		relationships with government bodies at local, regional, national and international level. Closely monitoring the development of policies related to NBS, identifying opportunities to influence the formulation of laws and regulations that encourage their implementation.
Project longevity	Failure to maintain the activities for the entire longevity period	The Pantanal REDD+ Program has an Adaptive Management Plan and a Management, Financial and Monitoring Plan that are applied throughout the life of the project. The first is a risk prevention or management response plan for the Program and consists of presenting actions and guidelines in order to minimize circumstances or events that are not fully expected or predictable, such as deviations in project performance or unforeseen environmental conditions. It includes a process for monitoring risks and the progress of adaptive actions applied to the project, documenting lessons learned and providing corrections that may be necessary, incorporating them into project decision-making. As for the Management, Financial and Monitoring Plan, the project proponents make public their intention to continue the management practices throughout the longevity of the project, as well as how the strategies for each scope of work (management, financial and monitoring) have been structured. The aim of this plan is to demonstrate the project's governance structure, the main resources used (financial, structural and labor), how these resources are managed and how the parties involved support each other and collaborate for efficient joint management and monitoring of the project.

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Lack of interest from stakeholders such as communities and public bodies	The lack of interest in participating in the Program's activities directly affects the benefits that would be generated at a social level. In addition, the lack of interest on the part of public bodies could jeopardize possible partnerships for the development of activities.	Activities are planned to strengthen and encourage greater involvement by all the parties involved in the elaboration and decision-making processes in relation to the Program's activities, through proposed actions that strengthen human capital (2.1.17). Another extremely important measure is linked to the improvement and dissemination of communication tools between the actors involved, such as consolidated communication channels (Section 2.3.12), procedures for redressing grievances (Section 2.3.15) and a continuous consultation process (2.3.11).
Market risk - Difficulty in selling verified carbon credits	Non-permanence of activities due to lack of access to planned financial resources	Bioflica Ambipar Environment has commercial, marketing, advocacy and market intelligence teams exclusively responsible for developing materials to publicize the project, participating in national and international events related to REDD+ and carbon markets in order to publicize the project, establishing and expanding the network of commercial contacts with potential buyers of the carbon credits to be made at the best possible prices that guarantee the financial sustainability of the project and Bioflica Ambipar Environment. The sales team is made up of professionals who are distributed among national and international clients, who deliver the credits to companies and institutions committed to the effective conservation of their areas and the co-benefits for communities and biodiversity. In addition, Bioflica Ambipar Environment is always looking for financing alternatives, such as donations and partnerships for the direct implementation of

Identified Risk	Potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
		project activities (not necessarily linked to the sale of credits).

In addition, the Program has several actions to mitigate the risks of non-permanence, such as:

- The management team is made up of experienced people in all areas.
- The project has available financial resources capable of maintaining the project for a period longer than the break-even point.
- The project is protected by a legally binding agreement to continue management practices that protect the carbon stocks credited during the crediting period of each property.
- Measures to prevent identified natural risks are implemented.

2.1.21 Benefit Permanence (CCB, G1.11)

The Pantanal REDD+ Program is scheduled to last until 2070, after which, if convenient, it could extend its period and continue with the revenue from the sale of VCUs, which would allow the benefits made possible by the credits to continue. However, another scenario is that of reaching a point where there is no need for this support, since the direction of the economy and society would have reached an "inflection point" where valuing the standing forest would be attractive, without depending on the implementation of projects.

All the activities carried out by the Pantanal REDD+ Program, indicated in Section 2.1.17, are based on the broad objective of allowing sustainable activities to be common practice, promoting the permanence of the social, environmental and climate benefits of the activities. Tools have been selected to maintain and improve these benefits, some of which are already in use and others will be implemented. The activities that have the potential to guarantee the permanence of the project's benefits are described below:

Structuring centralized governance for the Pantanal REDD+ Program: project governance guides teams and people in how projects should be conducted, and can encompass norms, rules, principles and policies that must be observed throughout the process. For the Pantanal REDD+ Program, this activity will bring benefits for the longevity of the project, as well-structured governance results in efficient actions, transparency, prevention of problems, compliance with requirements, standardization of information, structuring of decision-making and, ultimately, the success of the project. Governance is therefore an essential tool for guaranteeing the permanence of climate, community and biodiversity benefits.

Encouraging activities to prevent and mitigate fires: as one of the risks identified, fire is a threat to the permanence of the Program's benefits. Thus, the planned activities aimed at preventing, mitigating, and combating fire are fundamental to the Pantanal REDD+ Program in terms of protecting the forest and consequently enabling the climate, social and environmental benefits to be maintained.

Encouraging the implementation of different PES models on properties: Payment for Environmental Services has several advantages, such as encouraging conservation and reducing deforestation and forest degradation, as well as generating financial and social benefits. Some of the benefits can include receiving money to maintain environmental services, favors to obtain credit, subsidies for products, preferential provision of public services and tax exemptions²⁴. These benefits can help maintain and maintain the REDD+ Program and its environmental, social and climate activities.

Encouraging the implementation of alternative sources of income on farms in line with employee training: this initiative promotes a reduction in dependence on a single economic activity to generate farm income and jobs. The diversification of the economy and the increase in the total revenue of the farms increases the value of the property, so it is more likely that the Program's activities will continue for their entire longevity, guaranteeing the expected benefits. In addition, employee training is also part of this strategy for maintaining benefits.

Establishing strategic public-private partnerships in the region: Public-private partnerships bring several benefits to both sides, they will be signed to achieve common objectives aligned with the promotion of economic activities that value the standing forest. Such partnerships contribute to maintaining the benefits generated by the Pantanal REDD+ Program.

Encouraging the implementation of actions focused on biodiversity conservation: In addition to providing tools for sustainable socio-economic development, the aim of the project is to encourage scientific research. The Program, through partnerships with researchers and universities, will promote long-term research to monitor biodiversity and High Conservation Value Attributes. The aim of this monitoring is to assess impacts, implement mitigating actions and increase scientific knowledge about biodiversity in the region. Scientific knowledge is also a tool that promotes consistent and lasting benefits, and the research carried out in the area has the possibility of continuing even after the end of the project. In addition, as the project aims to promote partnerships with local universities, there will be an expansion of regional knowledge and the promotion of scientific benefits consistent with the reality and context of the region. Finally, it is also important to note that the conservation actions proposed by the Pantanal REDD+ Program (see Section 2.1.17), such as carrying out frequent monitoring of forest areas, implementing and/or improving asset surveillance actions and carrying out continuous monitoring of the Leakage Belt, for example, are essential actions for the CCB benefits to continue.

Establish a continuous channel of communication with stakeholders (directly and indirectly impacted): To ensure that the impacts of the project's actions are in fact aligned with the interests of the identified stakeholders, it is of great importance to periodically consult this public. The aim of the communication channels is to allow stakeholders to give their opinion and thus improve the project's activities, so that

²⁴ Payments for environmental services (PES): an effective instrument for environmental protection. Available at: <https://jus.com.br/artigos/36597/pagamentos-por-servicos-ambientais-psa-instrumento-eficaz-para-protecao-ambiental>

negative impacts are properly mitigated, and positive ones are improved. Thus, as presented in detail in Section 2.3.12, the Program will have channels for consultation and exchange of information between stakeholders, using e-mail, telephone calls, WhatsApp messages or even direct contact, if necessary. In addition, periodic meetings with the owners are planned throughout the life of the project, focusing on updating the parties on the progress of the Program. In addition to the meetings, monthly "Status Reports" are prepared and shared, which contain information on the progress of the Program's activities, important announcements, and other relevant information. All these actions were designed to ensure stakeholder engagement, focused on maintaining the climate, community and biodiversity benefits generated by the Program and maintaining its permanence throughout its life cycle.

2.1.22 Financial Sustainability (CCB, G1.12)

Biofílica Ambipar Environment has established a robust partnership instrument with each of the owners of the farms where the Pantanal REDD+ Program is located, which aims, among other things, to make investments in conservation on the properties viable in the long term, through the commercialization of environmental assets.

Under this instrument, the parties commit to jointly develop an Annual Activities Plan aimed at reflecting the activities and estimated costs for the development of the Pantanal REDD+ Program in the respective calendar year on each property. The implementation of this plan will only be possible through reinvestment of proceeds from the sale of generated VCUs. Any changes made to this Plan must be approved by the parties.

Furthermore, as a means of ensuring mutual transparency between the involved parties, both Biofílica Ambipar Environment and its partners must provide an annual accountability, demonstrating the investments made in the Program by each party. Within the governance structure designed for the Pantanal REDD+ Program, the management of resources for activity implementation shall be carried out through the joint work of the Operational Team with the Back Office Team (Administrative-Financial). With the implementation of this mechanism, it is understood that the financial sustainability of the Program will ensure the proposed climate, community, and biodiversity benefits.

Other information related to the financial analysis of the Pantanal REDD+ Program and the financial health statements of the proponent are considered commercially sensitive information and have been shared with the auditing team on a confidential basis.

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project (VCS, 3.13; CCB, G2.1)

The land use scenarios in the absence of the project were determined using the approved methodology VCS VM0007 version 1.1 and the VCS tool "VT0001 - Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities (T-ADD)", version 3.0. The likely scenarios, as well as the analysis of planned deforestation, agents, and vectors, were based on the reference scenario and are presented in more detail in Section 3.1.5.2.

2.2.2 Most-Likely Scenario Justification (CCB, G2.1)

According to the application of tool VT0001, the scenarios described below were identified for land use in the event of the Program not occurring.

Scenario 1 - Continuation of forest area on properties without financial incentives, such as the implementation of an AFOLU VCS-CCB project: In this scenario, the continuation of forest area on properties without financial incentives is considered unlikely due to several factors. Although properties maintain forest cover without changes, there are no economic incentives for its preservation. The direct costs associated with protecting the areas, combined with the opportunity cost related to the economic use of the properties, make it difficult to maintain forest cover. Additionally, legislation permits deforestation, which may bring economic benefits to landowners, and the maintenance of rural properties requires the payment of various fees and taxes, necessitating economic revenue generation to avoid financial losses or even property loss.

Faced with these challenges, it is unlikely that the continuation of land use as it existed before the project will be maintained without financial incentives to sustain surplus areas of the properties. The revenue sources available to maintain these areas are limited, making it difficult to maintain forest cover without financial incentives, such as carbon-derived financing or conversion of the area to other land uses.

Scenario 2 - Implementation of REDD+ activities without registration as an AFOLU VCS-CCB project: Although maintaining forest cover without financial incentives is unlikely, this scenario is feasible due to the profile of participants in the Pantanal REDD+ Program, who are committed to the conservation and sustainable use of their properties. While there are no financial incentives from the sale of verified carbon credits, these activities aim to protect the Project Area and bring benefits to local communities and biodiversity. Challenges such as frequent fires and lack of economic alternatives besides livestock highlight the importance of these activities. However, the economic viability of these actions is limited without additional revenue from registered VCS credits, which may hinder the generation of positive impacts in the region.

Scenario 3 - Legal deforestation for expansion of livestock by the landowner: This is the most likely scenario to occur when analyzing the land use history in the Pantanal, given the predominance of

livestock as the main economic activity in the region. Based on analyses of secondary databases, as well as field diagnostic surveys conducted by the Program, it can be observed that many properties contiguous to participating farms in the Program already have land use focused on livestock activities, with vegetation cover suppression occurring in the last 10 years. MapBiomass data show an expansion of areas classified as pasture when analyzing the last 10 years of land use and land cover in the project's insertion region, with a higher rate of land use conversion occurring in areas located south and east of the properties that are part of the Pantanal REDD+ Program. According to information from the State Agency of Animal and Plant Health Defense (IAGRO)²⁵, the State of Mato Grosso do Sul has the fourth largest cattle herd in Brazil with approximately 20 million head of cattle, with beef cattle production (predominantly based on pasture feeding) being one of the main economic activities in the region. News related to the Pantanal, such as that published in March 2023 on the UOL²⁶ website, indicate that the main cause of native vegetation loss in the Pantanal is due to conversion to pastures. Therefore, it is understood that livestock comprises the main factor responsible for the legal suppression demand of existing vegetation in the Pantanal biome, aiming for the conversion of forest cover to pasture areas.

Scenario 4 - Sale of the property to third parties and conversion of land use from forest cover to livestock:

This scenario is a significant possibility due to the growing demand and increasing prices of properties in the Project Zone. The sale may occur after the loss of forest cover (scenario 3) or with intact forest areas, which are subsequently converted to activities described in scenario 3. Land values disclosed by the Ministério da Fazenda²⁷ indicate the economic feasibility of this approach. The most plausible consideration is the approach where the property is sold with intact forest areas, leveraging the profile of participants in the Pantanal REDD+ Program, who are focused on conservation. Although this may reduce the initial sale price, it can generate real estate speculation in the region in the long term.

Detailed analyses of each alternative land use scenario to the Project Area can be found in Section 3.1.5.2.

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

Likely, community and biodiversity benefits would not occur in the absence of the Pantanal REDD+ Program, as many of the expected benefits would be derived from the investment resulting from the sale of verified project credits. Additionally, the project activities are not mandated by any law and probably would not have been implemented in the non-project scenario due to financial constraints. Land use in the absence of the project would likely be directed towards the expansion of extensive livestock farming. This context is better presented and detailed in Section 3.1.5 regarding additionality, the non-project scenario for the community in Section 4.1.4, and for biodiversity in Section 5.1.3.

²⁵<https://www.iagro.ms.gov.br/Geral/pecuaria/#:~:text=Com%20o%20quarto%20maior%20rebanho,predominantemente%20a%20base%20de%20pastagem>

²⁶<https://midiamax.uol.com.br/cotidiano/2023/biomass-em-risco-mato-grosso-do-sul-perde-67-milhoes-de-hectares-de-floresta-em-35-anos/>

²⁷<https://www.gov.br/receitafederal/pt-br/centrais-de-conteudo/publicacoes/documentos-tecnicos/vtn>

With the implementation of the Program, the generation of social benefits is associated with strengthening governance, with the consolidation of strategic partnerships in the region, aiming to monitor and operationalize the proposed activities. The social activities planned by the Program are related to improving quality of life, income diversification, communication and engagement, encouragement and dissemination of local culture (see Section 2.1.17).

Regarding biodiversity benefits, the project proposes activities related to the protection of forest areas, through satellite monitoring and enhancement of farm surveillance, aiming to protect the natural heritage in the area in question, in addition to conservation activities that are aligned with monitoring of fauna and flora, incentive for scientific research on different themes, and awareness-raising work with stakeholders through environmental education. These actions seek to ensure forest conservation and aim to monitor local fauna and flora and implement environmental education actions to engage stakeholders in supporting the biodiverse system where the Program will be implemented.

Thus, the Pantanal REDD+ Program, through a set of activities focused on environmental conservation and sustainable socioeconomic development, aims to ensure forest conservation, generating benefits for biodiversity, maintenance of ecosystem services, climate regulation, and local development. The positive impacts resulting from the project are, in essence, due to its proposed activities, generating net benefits for the community and biodiversity that would not occur in the absence of the project.

2.2.4 Benefits to be used as Offsets (CCB, G2.2)

In the Pantanal REDD+ Program, there are no distinct benefits for climate, community, or biodiversity intended to be used as compensation. The Program aims to produce compensations related to reduced emissions by avoiding planned deforestation, as described in Section 3 - Climate, following the premises of the VCS standard.

2.3 Safeguards and Stakeholder Engagement

2.3.1 Stakeholder Identification (VCS, 3.18, 3.19; CCB G1.5)

For the Pantanal REDD+ Program, a socioeconomic diagnosis was conducted aiming to identify and characterize the social and economic profile of populations that influence and can be influenced by the project activities, as well as agents, causes, and motivators of deforestation on farms and surrounding areas. The study was based on primary data (field surveys) as well as the systematization of studies and surveys already conducted in the area and surroundings and available information.

Various reliable sources were used to characterize the social and economic profile of the municipalities of Aquidauana, Corumbá, and Miranda, which are categorical elements to understand the local dynamics and guide the planning of the Pantanal REDD+ Program. Notable sources include the Brazilian Institute of Geography and Statistics (IBGE), the FIRJAN Municipal Development Index (IFDM), Municipal

Governments, State Secretariat of Environment and Economic Development, United Nations Development Program (UNDP), National Institute for Educational Studies and Research Anísio Teixeira (INEP)/Ministry of Education, State Department of Education, Higher Education Institutions, Datasus/Ministry of Health, and State Department of Health.

With this in mind, the diagnosis was conducted by Restaura Consultoria Ambiental e Treinamentos LTDA and involved a multidisciplinary team of specialized professionals. The process included:

(a) Characterization encompassing the population residing in urban and rural areas - for the municipalities of Aquidauana, Miranda, and Corumbá, the identification of social groups and actors such as communities, rural producers, and/or squatters under the project's influence, i.e., those who may be affected and/or benefited, and the assessment of their rights, interests, and relevance to the project.

(b) Characterization of the social profile - based on secondary data or primary surveys, the survey includes: Population density, number of social groups and actors; Historical and current occupation; Levels of education; Access to education and health services; Access to media; Population and migratory dynamics; Relationship and interaction among social groups and actors; Relationship and interaction of social groups and actors with entities/institutions (other stakeholders) that may affect or be affected by the project activities; Expectations of population growth; Cultural characteristics associated with gender; Relationship with natural resource use and traditional use; Mapping of social organizations, leadership figures, and educational/training institutions operating in the region, understanding the community dynamics with these figures; Current and potential public use in the areas covered, existence of attractions and historical-cultural heritage; Mapping of areas essential to supply the basic needs of local groups and social agents, such as subsistence, fuel, pasture for animals, medicines, and materials used in construction; Mapping of areas critical to the cultural identity of traditional communities (if any), such as areas and territories that communities occupy and from which they obtain natural resources as a condition for their cultural, religious, social, and economic reproduction.

(c) Characterization of the economic profile - was carried out through the collection of primary data: Socioeconomic indices; Infrastructure; Average family income; Productive structure: agriculture, livestock, forests, industry, services, among others; Current and historical land use; Volume and value generated by sector; Economy linked to forest products (timber and non-timber); Production system, local techniques, degree of product processing; Subsistence and local market, destination of production; Available labor and qualification, and availability of technical assistance.

In addition to the identification and diagnosis of communities, a survey of institutions, public bodies, and NGOs that may have an interest in the activities developed in this REDD+ Program was conducted. This survey aimed to bring stakeholders, project proponents, and partners closer together to maintain direct communication regarding the Program, as well as to listen to opinions and clarify external doubts.

Appendix 1 consists of the Stakeholder Identification Table, the presented list is dynamic, subject to updates as new PAIs are added to the Pantanal REDD+ Program.

2.3.2 Stakeholder Descriptions (VCS, 3.18, 3.19; CCB, G1.6, G1.13)

The research conducted by Restaura provides data that allows us to understand the social and environmental dynamics of the region where the Program is implemented, enabling the construction of a list of stakeholders that highlight the relationship between the project and the various actors involved.

The stakeholders of the Pantanal REDD+ Program are located in the municipalities of Aquidauana, Miranda, and Corumbá, in the state of Mato Grosso do Sul. Among the stakeholders who have some interest in the Pantanal REDD+ Program, the following can be mentioned:

- **Rural Landowners:** These are large private landowners whose main activities are livestock farming and/or ecotourism. They are directly impacted by the proposed forest conservation and can influence the dynamics of their properties.
- **Employees of partner farms:** These are employees who represent the human workforce of the properties and play essential roles in the success of tourism and/or livestock operations.
- **NGOs and other environmental movements:** These are institutions that advocate for the environment, socio-cultural resources, biodiversity conservation, and natural resources. They play a crucial role in defending ecosystems, providing support, resources, and expertise to the project. Examples include Onçafari Association and SOS Pantanal Institute.
- **Public Agencies:** Responsible for planning, implementing, and monitoring social, environmental, and economic actions and projects. Indispensable for the success of the project, they enhance positive impacts through public policies. This group includes mayors, councilors, deputies, senators, governors, state and municipal secretaries, IBAMA, fire departments, among others.
- **Research and Extension Institutions:** Various Brazilian teaching and research institutions, such as the Federal University of Mato Grosso do Sul - UFMS, State University of Mato Grosso do Sul - UEMS, and Federal Institute of MS - IFMS. They contribute scientific knowledge, essential for the effectiveness and continuous improvement of the project. They can collaborate in the implementation and monitoring of project activities, as well as stimulate research for the development and dissemination of scientific knowledge in the region.
- **Associations and unions:** These are fundamental actors that play roles in collective representation, community mobilization, negotiation, training, and oversight.

This mapping is a valuable resource for understanding the diversity of actors involved and their collective relevance for the success of the Pantanal REDD+ Program. It is worth noting that stakeholders were invited to participate in discussions of the Pantanal REDD+ Program to facilitate coordination and communication between the proponent, partners, and external actors involved. Details about each stakeholder can be found in Appendix 1: Stakeholder Description Table.

2.3.3 Stakeholder Access to Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

To ensure access to all information regarding the Pantanal REDD+ Program, project documents will be available in both Portuguese and English versions, both digitally and physically, to all stakeholders.

Before the audit takes place, documents related to the Program will be available virtually on the Verra registration platform for consultation by stakeholders. The link to the Pantanal REDD+ Program in the Verra Registry will be available on the Biofílica Ambipar Environment website, where summarized project information will also be described.

After the completion of the initial audit, in addition to updating the documents online on the platforms mentioned above, a printed version of each document related to the Program, such as the project design document, monitoring report, and validation and verification reports, will be made available for each landowner to keep at the headquarters of their respective farms. Additionally, copies will also be distributed to be kept at the offices of the 5P Alliance and ITV in Campo Grande, Mato Grosso do Sul.

Updates and news about the project will be published and disseminated via email and WhatsApp (to a predefined list of stakeholders' contacts), in the Biofílica Ambipar Environment Newsletter via blog²⁸, and on social media platforms (Instagram²⁹ and LinkedIn³⁰). The results obtained and the results of audits are also shared with stakeholders on these platforms. These tools will be used throughout the project's lifetime.

2.3.4 Dissemination of Summary Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

The summarized project documents, including the information required for G1.1-9 according to the CCB Standard v3.1, will be available on the Verra registration page and on the Biofílica Ambipar Environment website, where, in addition to the project information description, there will be a link to the Program documents page.

Information about the Program will also be disseminated via e-mail and WhatsApp (to a predefined list of stakeholders' contacts), and on social media platforms such as Instagram and LinkedIn, by the project proponent. For stakeholders who do not have internet access, a printed copy can be provided at strategic points (farm headquarters and partner offices).

In this way, it is expected that all necessary efforts to provide total transparency and distribute key information and documents of the Program will be achieved.

2.3.5 Informational Meetings with Stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

As initial informational meetings with stakeholders for the Pantanal REDD+ Program commence following the conclusion of negotiations with farm owners, which is the responsibility of Biofílica Ambipar

²⁸ Blog Biofílica Ambipar Environment. Available at: <https://www.biofilica.com.br/blog/>.

²⁹ Instagram Biofílica Ambipar Environment. Available at: https://www.instagram.com/biofilica_br/.

³⁰ LinkedIn Biofílica Ambipar Environment. Available at: <https://www.linkedin.com/company/biofilicabr>.

Environment's New Business Team. After the formal signing of contracts, the responsibility for communication is transferred to the Operations Team, which is responsible for conducting the "Kick off" meeting. This is an ongoing stage throughout the project lifecycle, as new properties join the Program. From this stage onward, the Operations Team is primarily responsible for communicating with Owners and their representatives regarding project advancements.

This meeting is structured as follows: in the first stage, the participating stakeholders of the Program are introduced, along with Biofilica Ambipar Environment's organization and team responsibilities, and the presentation of participating farms and Program partner institutions. The second stage involves presenting the technical and methodological concepts, as well as the data comprising the Program. In the final stage, the Program execution schedule is presented, along with its current status at the time of presentation, the activities, and communication methods to be undertaken for its development and implementation. Two Kick off meetings were held for validated stakeholders in November 2022 and August 2023.

Following the Kickoff presentation with the properties, periodic meetings with owners commenced, focusing on updating stakeholders on Program progress. Up to the point of document preparation, numerous meetings have been conducted with partners responsible for the initial Program Instances (PAIs 01-07). These interactions prioritize information exchange, feedback, and clarification of stakeholder project-related inquiries. Communication is transparent and tailored to meet the specific requirements of each group. In this way, it is understood that continuous involvement and dialogue with stakeholders ensure their concerns and needs are adequately addressed. Stakeholders are invited to these meetings via e-mail, phone calls, messaging apps (WhatsApp), or direct contact, as needed.

In addition to the meetings conducted and complementary communication methods with owners and their representatives, feedback meetings were held with stakeholders of the Pantanal REDD+ Program between February and March 2024. The feedback process included both in-person (field visits) and virtual (online meetings) agendas. The field agenda took place from February 25th to 28th and March 1st, 2024, involving in-person meetings with instances C2-P01-Y24, C4-P01-Y24, C5-P01-Y24, C5-P02-Y24, C6-P01-Y24, C7-P01-Y24, and C7-P02-Y24. The virtual agenda occurred on March 7th and 8th, 2024, with the feedback process with C1-P01-Y24 on March 7th and consultation process with other stakeholders of the Pantanal REDD+ Program on March 8th, including formal and traditional authorities of Mato Grosso do Sul such as Sebrae MS, IBAMA, Institute of Wildlife Conservation – ICAS, and Funai CR Campo Grande.

In this initial feedback stage, the meetings start with the presentation of general results obtained during the socioeconomic diagnosis survey conducted on farms (questionnaires administered to collaborators) from October 2023 to January 2024 by the Restaura company. Subsequently, Biofilica Ambipar Environment provided a presentation on the Pantanal REDD+ Program, explaining its general context, how carbon credits are obtained, where we stand in the project timeline, what the main expected outcomes of the feedback process were, and about the ongoing communication process that will occur throughout the existence of the Pantanal REDD+ Program. Upon the conclusion of the presentations, dynamic activities were conducted, tailored to each audience present, focusing primarily on collecting participants' views on the strengths and weaknesses of Program implementation for each farm, providing

an internal and external perspective given their strategic position as individuals and/or institutions in the Program.

Following the consolidation of all information gathered from these dynamics, Biofilica Ambipar Environment developed a proposal for the activities to be implemented by the Pantanal REDD+ Program (presented in Section 2.1.17). These activities were presented to farm owners on March 21st and 27th. Online meetings were conducted to present the themes and activities that will be developed by the Program, the responsibilities for executing each activity, the actions that are mandatory, and those that will be adaptable to each farm's reality, as well as the main communication channels (internal and external) to be developed by the proponent. Further details of these meetings can be found in Section 2.3.10.

2.3.6 Risks from the Project and No Net Harm (VCS, 3.18, 3.19)

	Identified risks	Preventive or mitigation measures adopted
Natural Risks	Fire	The activities under "Fire Prevention and Mitigation" were designed to prevent and mitigate this identified risk. Actions to be implemented include: Implementing a monitoring and alert system for forest fires on the properties; Encouraging the creation and/or maintenance of fire brigades on the farms; Acquiring fire-fighting equipment/machinery; among others (see Section 2.1.17).
Human-induced risks	No risk identified	The activities defined by the Program do not expect to cause any human-induced risks.
Risks for stakeholder participation	Lack of interest in participating in the program's activities	Preventive measures for this risk are focused on communication channels and stakeholder engagement. A structure has been set up so that stakeholders are actively involved in the project's activities and have an understanding of how a carbon project and its mechanisms work. Among the activities planned are periodic meetings to carry out annual planning and present the progress of the Pantanal REDD+ Program, among other actions (see Sections 2.3.11, 2.3.12 and 2.3.13).
Working conditions	No risk identified	The project activities are not expected to cause any risk to working conditions. On the other hand, the activities under the "Sustainable

Identified risks		Preventive or mitigation measures adopted
		Development and Quality of Life" theme are aimed at improving working conditions and processes.
Safety of women and girls	No risk identified	Project activities do not pose any risk to the safety of women and girls. This is reinforced by the Ambipar Group's Code of Conduct and Compliance (2023) ³¹ and Diversity and Inclusion Policy (2023) ³² .
Safety of minority and marginalized groups, including children	No risk identified	Project activities do not cause any risk to the safety of minority and marginalized groups, including children.
Pollutants (air, noise, water discharges, waste generation, release of dangerous materials)	No risk identified	The activities defined by the Program are not expected to cause any kind of pollution. Only activities that benefit the flora, fauna and community are planned.

2.3.7 Community Costs, Risks, and Benefits (CCB, G3.2)

As mentioned above in Section 2.3.5, participatory meetings were held with stakeholders (public and private institutions of interest, owners of the partner ranches and their employees). These meetings were used to present the results of the socio-economic diagnosis, to talk about the ongoing communication process and to propose a round table discussion about the Pantanal REDD+ Program and its benefits, as well as to clarify any doubts. An interactive process was also applied to encourage the participation of those present, with group dynamics aimed at exchanging knowledge and information. The activities carried out made it possible to collect the participants' perceptions of Strengths, Weaknesses, Opportunities and Threats (known as SWOT) in relation to the Program and the environment in which it operates, as well as identifying and prioritizing the problems encountered.

The SWOT analyses conducted with farm owners and collaborators highlighted several points, some specific to the individual reality of each farm and some contextual to the region of the Pantanal REDD+ Program 's operation. Among the main strengths identified were the high biological value of the farms (places where nature conservation and carbon project implementation are viable), engagement of owners and collaborators in the Pantanal REDD+ Program, the suitability of the area for eco-tourism, and the tradition that ensures the farm's continuity.

In contrast, the main weaknesses identified revolved around the costs of implementing the Program, poor quality of the region's roads, which hinders various aspects such as access to fresh food, education,

³¹ Code of Conduct and Compliance. Available at: <https://ambipar.com/site2020/wp-content/uploads/2023/09/Codigo-de-Conduto-e-Compliance-2023.pdf>

³² Diversity and Inclusion Policy. Available at: <https://ambipar.com/site2020/wp-content/uploads/2023/09/Politica-de-Diversidade-e-Inclusao-portugues.pdf>

and healthcare, and logistical challenges to reach the farms, poor internet quality, lack of knowledge and interest among some individuals regarding environmental issues, and climatic conditions, causing high temperatures and dry periods in the Pantanal.

Among the opportunities mentioned are actions to improve the technical skills of the employees, enabling more partnerships and projects through REDD+, job creation, economic diversification, courses/training, and improvement in quality of life. Finally, the main threats identified were wildfires in the biome, expansion of deforestation, prolonged periods of drought, and financing. Details of the scenario analysis (SWOT analysis) and the identification and prioritization of issues identified by stakeholders for the Pantanal REDD+ Program can be found in the Socioeconomic and Environmental Diagnosis Report, available to VVB.

It is also important to mention that the risks indicated in this PDD were presented to the owners prior to the submission of this documentation for public consultation and were based on diagnostic data, feedback, and the risk tool (AFOLU Non-Permanence Risk Tool, v4.2).

Regarding costs, it is important to emphasize that Biofílica Ambipar Environment establishes with each of the farm owners that comprise the Program a partnership agreement aimed at making investments in conservation on the properties viable in the long term through the commercialization of environmental assets. Thus, the proponent and the partner are committed to jointly developing an Annual Activity Plan containing the activities and estimated costs for the development of the Pantanal REDD+ Program. More details can be found in Section 2.1.22.

2.3.8 Information to Stakeholders on Validation and Verification Process (VCS, 3.18.6, 3.19; CCB, G3.3)

Stakeholders directly impacted by the project's activities, including farm owners, employees, and other stakeholders, were informed in advance about the CCB validation and verification process, as well as the visit of an independent auditor to their community and residence.

Communication with the farm owners has been ongoing since the conclusion of negotiations for entry into the Program and during the farm's "Kick off" meeting. Other stakeholders were informed during the Socioeconomic and Environmental Diagnostics conducted in 2023. Additionally, during the feedback meetings held in February and March 2024, stakeholders were also informed about the upcoming field visit for auditing purposes, scheduled for the second half of 2024.

For new instances entering the project, the audit process timeline will be communicated later via e-mail and WhatsApp. The project proponent's social media channels, such as Instagram and LinkedIn, will also be used to inform other stakeholders and the general public. Additionally, other communication methods may be adopted depending on demand and necessity.

2.3.9 Site Visit Information and Opportunities to Communicate with Auditor (VCS, 3.18.6; CCB, G3.3)

All stakeholders will always be notified in advance about the auditor's visit prior to the audit event, and the activity schedule will be sent to those involved through the designated communication channels. Stakeholders have already been informed about the first field audit scheduled for the second half of 2024, as described in Section 2.3.8.

The proponent, Biofilica Ambipar Environment, coordinates logistics with the VVB and informs farm owners, their representatives, suppliers involved in the program, and other stakeholders through the communication channels specified in the project. It is the responsibility of the farm owners to organize the participation of their employees to accommodate the dynamics of the field audit activity, ensuring that everyone has the right to communicate with the auditor freely without any coercion.

2.3.10 Stakeholder Consultations (VCS, 3.18; CCB, G3)

Date of stakeholder consultation	25-February-2024 to 28-February-2024 01-March-2024 07-March-2024 to 08-March-2024
Stakeholder engagement process	The initial feedback process with stakeholders included a face-to-face agenda (field visits) and a virtual agenda (meetings held online). The field agenda took place between 25-February-2024 to 28-February-2024 and 01-March-2024, where there were face-to-face meetings with the Barranco Alto, Vera Lúcia, Fazendinha, Estância Caiman, Havaí and Salinas farms. The virtual agenda took place on March 7 and 8, 2024. On March 7 there was a feedback process with the Santa Sofia ranch and on March 8 there was a consultation process with external parties interested in the Pantanal REDD+ Program, involving representatives from IBAMA, ICAS, Funai and SEBRAE. Starting the field agenda, on February 25 and 26, a visit was made to the Barranco Alto farm, where the Biofilica Ambipar Environment and Restaura team met on February 25 only with the owner of this farm and, on February 26 with some of the employees of this farm and of the Vera Lúcia and Fazendinha farms (the employees of the Vera Lúcia and Fazendinha farms were invited to Barranco Alto on the day of the meeting).

Both meetings began with a presentation of the general results obtained during the socioeconomic diagnostic survey carried out on the farms (questionnaires applied to employees), which took place between October 2023 and January 2024, by the company Restaura. Biofílica Ambipar Environment then gave a presentation on the Pantanal REDD+ Program, explaining its general context, how obtaining carbon credits works, where we are in the project's timeline, what the main expected results are from the feedback process that was being carried out and the ongoing communication process that will take place throughout the Pantanal REDD+ Program 's existence.

After the initial presentations by the companies Restaura and Biofílica Ambipar Environment, there were different dynamics depending on the audience present on each of the days.

The dynamic carried out with the owner of Barranco Alto focused on evaluating the internal environment using SWOT analysis to gather her opinion on the main strengths, weaknesses, opportunities and threats that exist on the Barranco Alto farm in relation to the Pantanal REDD+ Program. As for the external environment, the PESTEL tool was used to understand the main Political, Economic, Social, Environmental and Legal factors (laws) that could have an impact on the development of the Pantanal REDD+ Program on the farm in question, giving us a more accurate view of the general context of Barranco Alto in the region where it operates.

In an effort to get a closer look at the context of day-to-day work in the field, the dynamics with the employees of the Barranco Alto, Vera Lúcia and Fazendinha farms focused on finding out what the workers' view is of the main strengths, weaknesses, threats and opportunities that exist in the context of these farms in relation to the Pantanal REDD+ Program. These points were identified using the SWOT analysis tool, which helps the debate to identify Strengths, Opportunities, Weaknesses and Threats.

The main objective of these dynamics was to understand what the main demands of the people who live in the region are and to identify where the activities of the Pantanal REDD+ Program can act positively.

Continuing with the field agenda, on February 27 and 28 the feedback process took place at the Estância Caiman.

On the morning of February 27 there was a meeting with the owner and the main leaders of this farm, where the general results obtained during the socioeconomic diagnostic survey carried out on the ranches by the company Restaura were presented, as well as a presentation on the Pantanal REDD+ Program by Biofíllica Ambipar Environment. In the afternoon of the same day, there was a meeting only with the environmental manager of this farm, who presented us with the main activities underway at Estância Caiman, as well as the planning designed to be applied at IMARK (the focus of the Program). Afterwards, we used the SWOT analysis tool to identify the farm's strengths, weaknesses, threats and opportunities from the point of view of its environmental manager.

The results obtained during the SWOT analysis were presented to the owner and some of the farm's leaders on February 28, so that they could also give their views on the points previously raised. This compilation of ideas was extremely important for understanding what IMARK's main concerns and demands were in relation to the overall context of the Pantanal REDD+ Program.

Concluding the field agenda, on March 1st the feedback process was held with the owners of the Havaí and Salinas farms. Due to logistical difficulties in reaching the headquarters of these farms, it was decided to hold the meeting only with the owners, which took place at the headquarters of the Instituto Taquari Vivo in Campo Grande/MS, a partner institution of the Pantanal REDD+ Program.

The meeting's itinerary was similar to the others held on other farms, starting with a presentation on the general results obtained during the socioeconomic diagnosis survey carried out on the farms by the company Restaura, followed by a presentation on the Pantanal REDD+ Program by Biofíllica Ambipar Environment.

Subsequently, SWOT and PESTEL analyses were applied with the owners to identify the main points of attention from the external and internal viewpoints of these farms that should be taken into consideration when planning the activities of the Pantanal REDD+ Program.

Starting with the virtual activities, on March 7 the feedback process with the Santa Sofia farm took place following the same script as the other meetings (presentations by Biofíllica Ambipar Environment followed by Restaura). The SWOT analysis was then applied to

	identify the main strengths, weaknesses, threats and opportunities in the context of the Santa Sofia farm. Finishing the stakeholder consultation process, on March 8 there was interaction with external parties to the Pantanal REDD+ Program, where a letter of invitation was sent to various institutions that could have any kind of interest in the Pantanal REDD+ Program. A total of 34 institutions were invited to the meeting, 4 of which actually took part in the process. The participants included representatives from the Brazilian Institute for the Environment and Renewable Natural Resources (IBAMA), the Institute for the Conservation of Wild Animals (ICAS), the National Foundation for Indigenous People (FUNAI) and the Brazilian Micro and Small Business Support Service (SEBRAE). The script for this meeting differed from the others, with a presentation only by Biofílica Ambipar Environment to introduce the participants to the context of the Pantanal REDD+ Program. A PESTEL analysis was then carried out to identify the main external factors in the political, economic, social, environmental and legal spheres (laws) that could affect the Pantanal REDD+ Program in the view of the participants present at this meeting. Finally, it should be noted that at the end of all the meetings (both face-to-face and virtual) a questionnaire was administered on the understanding of the content that had been presented and an attendance list was kept so that there was a formal record of the people who took part in this process of feedback and consultation with the stakeholders of the Pantanal REDD+ Program.
Consultation outcome	The SWOT analyses carried out with the owners and employees of the farms highlighted several points, some of which are specific to the individual reality of each farm and some of which are part of the context of the region where the Pantanal REDD+ Program operates. Among the main strengths listed, we can mention the high biological value of the farms (places where there is nature conservation and viability for implementing carbon projects), the engagement of the owners and employees in the Pantanal REDD+ Program, a good place to live, ecological tourism and the tradition that perpetuates the continuity of the farm. On the other hand, the main weaknesses listed revolve around the costs of implementing the Program, the poor quality of the roads in

the region, which hinders various issues such as access to fresh food, access to education and health and makes logistics to the farms difficult, the poor quality of the internet, the lack of knowledge and interest of some people in environmental issues and, finally, climate issues, causing high temperatures and periods of drought in the Pantanal.

Among the opportunities mentioned were: actions to improve the technical skills of the employees, enabling more partnerships and projects through REDD+, job creation, diversifying the economy, courses/training and improving quality of life. Finally, the main threats identified were forest fires, the expansion of deforestation, long periods of drought and forms of financing for activities on the properties. The details of the scenario analysis (SWOT analysis) and the identification and prioritization of the problems identified by the stakeholders for the Pantanal REDD+ Program can be found in the Socioeconomic and Environmental Diagnosis Report, available to the VVB.

Analysis of the questionnaires provided to the participants, which covered questions relating to their understanding of the content presented during the feedback process, showed that the vast majority claimed to have understood that the Pantanal REDD+ Program will be implemented in the region, with only one person saying they had not understood.

Participants were asked if they thought that the activities proposed by the Pantanal REDD+ Program would be beneficial to them, and the answers were yes. There were two people who answered negatively, one of whom claimed that they would not be beneficial because this person would stop working on the farm.

When asked if the Program's activities could be harmful in any way, 3 people answered yes, while 21 answered no.

We also asked if the person was against the existence of the Pantanal REDD+ Program in the region. The answers indicated that 3 people were opposed to the implementation of the Program, 2 people claimed to be indifferent and 18 said they were in favor.

Finally, they were asked which would be the best means of communication to keep in touch with the participants and WhatsApp was the most popular, followed by radio, e-mail or in person.

Stakeholder input

In order to arrive at a general picture of the main problems facing the region where the Pantanal REDD+ Program is located in the context of its participating farms, a joint analysis was carried out of the results obtained during the feedback and consultation processes with stakeholders (SWOT and PESTEL dynamics applied) and the socio-environmental diagnosis carried out by Restaura on each of the farms.

The main points were listed in an Excel spreadsheet, grouped into the category of a macro problem and, based on this major problem, the subsequent effects and possible solutions for each point were listed.

The scenario obtained through the main problems listed together with the strengths and opportunities identified served as the basis for designing the activities planned for the Pantanal REDD+ Program and served as input for creating the Theory of Change.

The macro problems listed were:

- Concern about the expansion of deforestation and the impact on climate change (water scarcity, excessive heat), poor water quality and the extinction of species of fauna and flora in the Pantanal.
- Difficulty in accessing the farms due to roads with little or no infrastructure, causing difficulties in diversifying the local economy, given the logistical difficulties in getting products to and from the farms and limiting access to health, education and quality food.
- Attacks by venomous animals, such as snakes, spiders and bees.
- Lack of transmission of Pantanal traditions between generations, leading to a loss of regional culture.
- Possibility of disagreements between landowners and neighbors who have not joined the Pantanal REDD+ Program.
- Frustration of the expectations of external communities.
- Poor quality of the local internet, hindering communication with external parties.

	• High turnover of employees on the farms, causing various problems, such as a low rate of employee training, the occurrence of employees who have not received information about their labor rights, lack of engagement in the Pantanal REDD+ Program and high dependence on employees on the farms. • Expansion in the use of pesticides and the presence of exotic species of grass to create pasture areas. • Difficulty in investing in the diversification of the regional economy due to a lack of infrastructure and production standards and pricing of products from the region. • Difference between the properties participating in the Pantanal REDD+ Program in terms of available infrastructure, firefighting progress and area dedicated to the carbon project. • The occurrence of illegal hunting and fishing, which could lead to the extinction of some species of Pantanal fauna. • The occurrence of human-fauna conflict, generated by the predation of cattle by jaguars, causing economic losses for ranchers, which could lead to the illegal hunting of these animals.
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All the comments received during the consultation period with stakeholders described above are described below. Comments received after the public comment period on the Verra platform will be included in this section at a later date. The table indicates updates to the draft, and the insignificance or irrelevance of the comments, where appropriate.

Summary of comment received	When comment was received	Actions taken
“everything is right”	26/02/2024 – Stakeholder consultation meeting. Handwritten comment	No action necessary.
“Double or triple the selling price to China”	26/02/2024 – Stakeholder consultation	No action necessary.

Summary of comment received	When comment was received	Actions taken
	meeting. Handwritten comment	
"A good presentation"	26/02/2024 – Stakeholder consultation meeting. Handwritten comment	No action necessary.
"The project is great. Congratulations on the presentations and thank you"	26/02/2024 – Stakeholder consultation meeting. Handwritten comment	No action necessary.
"Congratulations, great communication, experts"	26/02/2024 – Stakeholder consultation meeting. Handwritten comment	No action necessary.
"I thought the lecture was great"	26/02/2024 – Stakeholder consultation meeting. Handwritten comment	No action necessary.
"You have to have the Program, it's valid"	26/02/2024 – Stakeholder consultation meeting. Handwritten comment	No action necessary.
"I'm still trying to understand it, so as soon as I understand the program, I'll come up with suggestions for improvement"	26/02/2024 – Stakeholder consultation meeting. Handwritten comment	A complementary explanation of the Program was given, as well as the communication channels that the Program plans to implement to maintain a continuous exchange between the stakeholders consulted and the proponent to collect suggestions, questions or criticisms.

Summary of comment received	When comment was received	Actions taken
"What is public consultation?"	21/03/2024 - Meeting to present activities of Pantanal REDD+ Program. Verbal comment	The necessary explanation was given so that the subject could be understood.

2.3.11 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

As an integral part of participatory and adaptive management of activities, permanent channels of communication and engagement are necessary, especially with regard to possible conflicts that may arise with stakeholders.

As indicated in Section 2.3.5, the first communication channel set up between the partners and the proponent are the "Kick off" meetings, after the official entry into the farms, periodic meetings with the owners begin, focusing on updating the parties on the progress of the Program. Communication at these meetings is transparent and adapted to meet the specific requirements of each group. In this way, it is understood that continuous involvement and dialog with stakeholders ensures that their concerns and needs are adequately addressed. For these meetings, stakeholders are invited by e-mail, phone call, WhatsApp message or even direct contact, if necessary.

In addition to the meetings, monthly "Status Reports" are prepared and shared, which contain information on the progress of the Program's activities, important announcements and other relevant information. These documents are sent to all interested parties via a WhatsApp channel and by e-mail.

The proponent understands that maintaining good communication between all the parties involved is essential for the maintenance and success of the activities, so, in addition to the actions carried out from the start of the Program, strategies were drawn up for communication and ongoing engagement to be implemented throughout the project's life cycle. The strategies are divided into external and internal communication actions.

The external communication strategies involve Biofíllica Ambipar Environment's Marketing Team, helping to convey the mission, activities and values of each PAI. For farms that have a well-established structure of communication channels, the work will be done jointly (the partner's team and the project proponent's team). The process of training the landowners who participate in the Program to become project promoters is also planned. In addition, another external communication and engagement strategy is active prospecting for new properties to join the Pantanal REDD+ Program by Biofíllica Ambipar Environment's New Business Team.

With regard to communication and internal engagement between the project proponent, landowners and other stakeholders, some problems were identified, such as: landowners and employees having doubts about the implementation of the Pantanal REDD+ Program and misalignment between the properties

participating in the Program. In view of these problems, it was possible to outline the following strategies to be implemented by the Program:

- Continue to hold regular meetings to present the progress of the Pantanal REDD+ Program.
- Implement annual planning for each property between Biofilica Ambipar Environment and the owner, based on the Program's activities and the property's objectives.
- Maintain a direct communication channel between Biofilica Ambipar Environment and the owners to answer questions about the Program and its progress.
- Hold periodic meetings with employees to present the property's annual planning and progress in carrying out the activities of the Pantanal REDD+ Program.
- Install information boards on the farms about the activities implemented by the Program.
- Articulate a space for exchange between properties, which can take place via meetings, workshops and more informal communication channels such as WhatsApp.

As an integral part of participatory and adaptive management for social activities, a permanent communication channel is necessary, especially regarding possible conflicts that may arise with stakeholders. In this regard, Biofilica Ambipar Environment has an official communication e-mail for stakeholders (internal or external) which is contato@biofilica.com.br. Through this channel, and also through Biofilica Ambipar Environment's social media, it is possible to send suggestions, questions and/or complaints which will be dealt with directly by the proponent. The main objective of these contacts is to build an open channel for conversations with the property's neighbors and external communities with the aim of maintaining transparency in the activities proposed for each property.

It's worth mentioning that some of the Program's partners have already established a means of internal and external communication, where they publicize their sustainability actions. The most widely used media tools are Instagram and LinkedIn, and printed media are also used, with distribution in guesthouses and communal areas. These communication mechanisms already in use could be used to disseminate information about the Pantanal REDD+ Program, opening up another means of contact with the project's stakeholders.

It is worth noting that the Pantanal REDD+ Program has an Adaptive Management Plan for the entire life of the project. The adaptive management plan identifies, evaluates and creates a mitigation plan for potential risks to the project and any other obstacles to project implementation. It includes a process for monitoring progress, documenting lessons learned or corrections that may be necessary and incorporating them into project decision-making in future monitoring periods.

The communication plan for the Pantanal REDD+ Program will be continuously monitored and updated to keep up with changes that may arise throughout the program's life cycle. All updates will be coordinated with partners and communicated to stakeholders, ensuring transparency and alignment.

2.3.12 Stakeholder Consultation Channels (CCB, G3.5)

As indicated in Section 2.3.5, the first communication channel set up between the partners and the proponent are the "Kick off" meetings. After the official entry into the farms, periodic meetings with the owners begin, focusing on updating the parties on the progress of the Program. Communication at these meetings is transparent and adapted to meet the specific requirements of each group. In this way, it is understood that continuous involvement and dialog with stakeholders ensures that their concerns and needs are adequately addressed. For these meetings, stakeholders are invited by e-mail, phone call, WhatsApp message or even direct contact, if necessary.

In addition to the meetings, monthly "Status Reports" are prepared and shared, which contain information on the progress of the Program's activities, important announcements and other relevant information. These documents are sent to all interested parties via a transmission channel on the WhatsApp app and by e-mail.

Contact with stakeholders was also made when the socioeconomic diagnosis was carried out, as well as meetings with public institutions in the state of Mato Grosso do Sul. Consultations and participatory processes were carried out directly with stakeholders or through their representatives, as described in Section 2.3.10.

In addition to these mechanisms, a debriefing was also held with stakeholders, where the results of the socioeconomic survey carried out on the farms by Restaura between October 2023 and January 2024 were presented. Biofílica Ambipar Environment then explained the Pantanal REDD+ Program, including its contextualization, the process of obtaining carbon credits and the expected results of the feedback process. Different dynamics were carried out for each audience present, with the aim of collecting feedback on the strengths and weaknesses of the program's implementation on each farm, considering both the internal and external perspectives of the participants.

After consolidating the information from the dynamics, Biofílica Ambipar Environment prepared a proposal for the activities to be implemented by the Pantanal REDD+ Program. This proposal was presented to the farm owners in online meetings held on March 21 and 27. Topics such as implementation responsibilities, mandatory and adaptable actions, as well as the program's internal and external means of communication were covered. More details on these activities can be found in Section 2.3.10.

As an integral part of participatory and adaptive management for social activities, a permanent communication channel is necessary, especially with regard to possible conflicts that may arise with stakeholders. In this sense, Biofílica Ambipar Environment has an official communication e-mail for stakeholders (internal or external) which is contato@biofilica.com.br. Through this channel, and also through Biofílica Ambipar Environment's social media, it is possible to send suggestions, questions and/or complaints which will be dealt with directly by the proponent. For landowners participating in the program, the direct channel of communication with the proponent is via e-mail and cell phone from the Pantanal REDD+ Program coordinator. The main objective of these contacts is to build an open channel

for conversations with the property's neighbors and external communities in order to maintain transparency in the activities proposed for each property.

It is worth mentioning that some of the Program's partners have already established a means of internal and external communication, where they publicize their sustainability actions. The most widely used media tools are Instagram and LinkedIn, and printed media are also used, distributed in the communal areas and in the Employees' Village. This already used communication mechanism can be used to disseminate information about the Pantanal REDD+ Program, opening up yet another means of communication.

2.3.13 Stakeholder Participation in Decision-Making and Implementation (VCS, 3.18, 3.19; CCB, G3.6)

The area in which the Pantanal REDD+ Program will be implemented does not, as yet, encompass any region where there are traditional communities living. It will be implemented entirely within private properties, whose owners have signed a long-term agreement with Bioflica Ambipar Environment to conserve the areas of surplus vegetation within their properties.

However, it was adopted as a good practice for the development and subsequent implementation of the activities related to the Program, to open up the participation of the owners together with their collaborators and other interested parties. In this way, the Program and the proponent follow measures and policies (Sections 2.3.14 and 2.3.16) to ensure that all stakeholders have access to communication channels, regardless of age, culture or gender.

Involvement in the design, implementation, monitoring and evaluation of the Program takes place through the available communication channels (Section 2.3.11) and information meetings, in which all stakeholders have the opportunity to participate, without suffering any kind of discrimination.

It's important to note that since the process of designing the project and defining the activities, all the stakeholders have actively participated by contributing to the discussion and exposing ideas, in order to contribute to the development of the Program, taking into account the local reality. To encourage the exchange of experiences, group dynamics were carried out in person and online with the stakeholders of the Pantanal REDD+ Program, as illustrated below.



Figure 3 Photos a) e b) C4-P01-Y24; Photo c) C6-P01-Y24.

2.3.14 Anti-Discrimination Assurance (VCS 3.19; CCB, G3.7)

As a member of the Ambipar Group, the project proponent follows all the Group's policies, which are well defined and structured, because over the years a solid culture has been built up around human rights and social responsibility policies. Ambipar is therefore working to improve its Compliance Program, constantly striving to achieve the highest standards of integrity, transparency and reliability in all its business dealings and relationships.

One of the main guiding documents is the Code of Conduct and Compliance³³, which applies to employees, suppliers, representatives, service providers and business partners. This policy contains ethical and moral principles that are non-negotiable for the Group, such as respect for human rights, diversity and inclusion. Below is some information from the Code of Conduct and Compliance that guarantees anti-discriminatory practices and is also applied to the Pantanal REDD+ Program.

- **ETHICAL PRINCIPLES AND VALUES:** Respect for people regardless of their hierarchical position, origin, color, ethnicity, culture, age, social level, physical ability, religion and sexual orientation, and any practice of discrimination is repudiated.
- **CONDUCT THAT MAY IMPLICATE TERMINATION OF THE EMPLOYMENT CONTRACT OR BREAKING OFF THE BUSINESS RELATIONSHIP:** Practicing acts of rejection motivated by prejudice, violence or any discriminatory attitude towards people on the grounds of gender, ethnicity, race, religion, social class, age, sexual orientation, physical disability or reduced mobility; Practicing threats, blackmail, false testimony, moral harassment, sexual harassment or any other act contrary to the principles and commitments of this Code.
- **RESPECT FOR DIVERSITY:** any type of discriminatory act is prohibited, with special emphasis on equality and diversity among its employees, with its own policy on the subject, including promoting the inclusion of minority groups.
- **HUMAN RIGHTS:** Respect for human rights is a central pillar of the company's activity and disrespect for human rights is considered a serious infraction that can eliminate any relationship with the company.

In addition to the Code of Conduct and Compliance, the Ambipar Group also has a Sustainability Policy (2023)³⁴ and a Diversity and Inclusion Policy (2023)³⁵, which provide transparency and reinforce commitments to sustainable development and encourage the provision of a favorable and healthy working environment for all professionals involved in its activities, valuing diversity and ensuring equal opportunities without any distinction as to the characteristics, history, religion or identity of individuals.

³³ Code of Conduct and Compliance. Available at: <https://ambipar.com/site2020/wp-content/uploads/2023/09/Codigo-de-Conduto-e-Compliance-2023.pdf>

³⁴ Sustainability Policy. Available at: <https://ambipar.com/site2020/wp-content/uploads/2023/09/Politica-de-Sustentabilidade-2023-portugues.pdf>

³⁵ Diversity and Inclusion Policy. Available at: <https://ambipar.com/site2020/wp-content/uploads/2023/09/Politica-de-Diversidade-e-Inclusao-portugues.pdf>

Other policies that help to combat discrimination and are pillars for building the management of the Pantanal REDD+ Program are: the Stakeholder Relations Policy and the Supplier Relations Policy. All these policies are the basis for building inclusive, ethical and beneficial relationships for all parties. All the Group's policies are available in full on the website: <https://ambipar.com/politicas-estatuto/>.

In this way, the Pantanal REDD+ Program is committed to inclusion and unconditional respect for all people, especially women, blacks, LGBTQIAPN+, people in situations of refuge, immigrants, people with disabilities and other minority groups.

In the development of the Program, discrimination of any kind will not be tolerated on the grounds of gender, race, religion, sexual orientation, political opinion, origin, class or social condition, nationality, place of birth, age, pregnancy, health status, disability, genetic predisposition, lifestyle, union membership, retirement, disability, or any other individual characteristics.

In addition, the Pantanal REDD+ Program expresses its rejection of any forced labor or labor in conditions analogous to slavery, any form of discrimination and harassment, both moral and sexual. Working conditions in the Program are ensured from the outset of the negotiation process, since before signing a contract with the owners of the farms participating in this Program, they must request a Labor Debt Clearance Certificate issued by the Labor Court, in order to confirm that the properties are in order in this regard.

To ensure compliance with these policies, Ambipar also has an online reporting channel, where complaints can be made about: Interpersonal relations; Physical aggression; Moral harassment; Discrimination; Sexual harassment; Conflict of interest; Violation of laws or non-compliance with policies/procedures; Relationship with the community, among other issues. This channel can also be used by the REDD+ Program as a way of ensuring compliance with these policies. Link to the reporting portal: <https://ambipar.com/denuncias/>.

2.3.15 Feedback and Grievance Redress Procedure (VCS, 3.18.4; CCB, G3.8)

Development process

Efficient management is fundamental to the success of the REDD+ Program. It is therefore essential to establish a well-defined procedure for receiving, hearing, responding to and resolving complaints related to the program in an effective manner. This procedure has been designed to take into account traditional methods of conflict resolution, such as attempt resolution, mediation and, ultimately, arbitration or litigation in the courts.

Thus, the Pantanal REDD+ Program provides for the implementation of a formal procedure for resolving conflicts with the actors involved, taking into account the local reality.

Conflict management will take place through a committee formed by Biofílica Ambipar Environment and the owners and/or representatives of the farms involved in the project, where any

	complaints and grievances received through the previously established means of communication will be analyzed and the appropriate measures will be taken. However, complaints focused on a specific property will be dealt with directly between Biofílica Ambipar Environment and the owner/representative. In addition, there is a plan to identify conflicts and take preventive measures in this area, through an ongoing relationship with social actors in the area surrounding the implementation of the Pantanal REDD+ Program.
Grievance redress procedure	The procedure presented here is based on Biofílica Ambipar Environment's experience in project management. These measures are applied to the applicant's projects that have already been certified and verified by the same certification standard. Here are the steps to be taken: 1. Receiving and Recording Feedback and Complaints The first step in the feedback and complaints procedure is to create a structured process for receiving and registering complaints. This defines the applicant's e-mail inbox and direct communication with the project team as the official channels for feedback and complaints. These channels will be easily accessible and well publicized to those involved in the Program. Any interested party can submit comments, suggestions and complaints about the Pantanal REDD+ Program by e-mail: contato@biofilica.com.br. It is the responsibility of Biofílica Ambipar Environment that all complaints received are stored including date, time, description of the complaint, contact information of the complainant and any relevant information. 2. Response and Resolution Process Once a complaint is received, it is essential to respond promptly. This involves a careful analysis of the problem and an initial attempt at resolution. Simple issues will be responded to within two weeks and for more complex issues the timeframe can extend to 30 days. In addition, interested parties will be informed of the progress of the complaint resolution. All responses to complaints, including the actions taken to resolve the problem, will be recorded, along with the full history of interactions with stakeholders. These measures

are aimed at finding solutions and ensuring that all parties involved are heard and valued.

3. Mediation stage

If the initial resolution attempt is unsuccessful or the complaint is of a more complex nature, the next stage is mediation. At this point, an impartial third party, such as a mediator or a panel of experts, can be involved to facilitate communication between the parties involved and assist in finding a mutually satisfactory solution. It is important to set reasonable deadlines for this stage in order to keep the process moving efficiently.

4. Arbitration or Court Trial

If mediation does not resolve the problem, the last stage of the procedure is arbitration or recourse to the courts. At this stage, an impartial third party, with the authority to make binding decisions, is involved to resolve the dispute definitively. Again, it is crucial to set reasonable deadlines for this stage, so that the parties involved can reach a final resolution effectively.

2.3.16 Accessibility of the Feedback and Grievance Redress Procedure (VCS, 3.19; CCB, G3.8)

Stakeholders will be informed about the feedback and grievance redress procedure by reading the Project Development Plan (PDD), as well as the updates that will be communicated in subsequent Monitoring Reports (MR). In addition, a verbal announcement will be made during scheduled meetings. To ensure even greater coverage, the project's virtual communication channels will be used, including e-mail and WhatsApp, using a pre-established list of stakeholder contacts. Dissemination will also take place via the Biofílica Ambipar Environment website and its social networks, such as Instagram and LinkedIn, ensuring effective transparency about the implementation of this process.

The registration of any complaint can be made in writing, via e-mail, telephone or other available channels. Responses will initially be provided to the complainant and, when necessary, will be made available to the public, especially in cases that require open justification. Such communications will be made in writing and/or through meetings, as appropriate.

Thus, the entire Pantanal REDD+ Program management team is committed to maintaining and updating open communication channels, aiming at the continuous improvement of the Program and ensuring effectiveness in receiving feedback from stakeholders.

2.3.17 Worker Training (VCS, 3.19; CCB, G3.9)

The measures needed to provide guidance and training to stakeholders through the project's activities are based on three main themes:

Promoting Sustainable Actions in the Rural Environment: with the aim of providing guidance and training to employees, as well as their families, with a view to implementing the project's activities, in line with the socio-environmental and economic objectives of the properties for forest conservation and the perpetuation of Pantanal culture, this action will be divided into the following stages:

1. Carry out a training needs assessment to determine which skills and knowledge will be essential for employees and their families to contribute effectively to the project. An initial assessment has already been carried out through the socio-environmental diagnosis and stakeholder feedback activities.
2. Develop embracing training programs that address all relevant aspects of the project, including forest conservation techniques, sustainable agricultural practices, natural resource management, preservation of Pantanal culture and socio-environmental awareness. Environmental and cultural education components will also be incorporated into the training programs.
3. Provide training in sustainable techniques in line with the objectives of each property, providing training specific to the roles and responsibilities of each individual within the project, ensuring that they clearly understand their tasks and how to carry them out effectively.
4. Promote stakeholder involvement in the training programs, encouraging active participation in the project and ensuring that their perspectives and knowledge are incorporated into the activities.
5. Establish evaluation and monitoring mechanisms, in line with the REDD+ Program's Monitoring Plan, to track employees' progress and identify areas that need strengthening or adjustments to the training programs.

Sustainable Development and Quality of Life: with the aim of providing guidance and training to owners, employees and their families on adapting and/or preparing for future scenarios caused by climate change, this action will be divided into the following stages:

1. Carry out a detailed assessment of the specific vulnerabilities faced by the community in relation to climate change, taking into account factors such as rising temperatures, extreme weather events, changes in rainfall patterns, among others. This stage has already been carried out in line with the socio-environmental diagnosis prepared by the project.
2. Promote sensitization and awareness-raising sessions for stakeholders on the impacts of climate change, highlighting the importance of adaptation and preparation to face these challenges.
3. Provide education on the risks and opportunities associated with climate change, highlighting how adaptation practices can help reduce negative impacts and take advantage of emerging opportunities.

4. Offer training in specific adaptation practices, aligned with the objectives of each property, such as sustainable management of natural resources, diversification of production, building resilient infrastructure, water and soil conservation, among others.

5. Assist owners, employees and their families in developing personalized adaptation plans, taking into account the needs and resources available on the farms, including identifying, assessing and responding to potential threats related to climate change, with the aim of increasing the resilience of communities.

External communication strategies: with the aim of providing guidance and training to owners, employees and their families to make them agents promoting the Pantanal REDD+ Program, this action will be divided into the following stages:

1. Train participants to develop communication and engagement skills to raise awareness about the importance of conserving the Pantanal biome through the Pantanal REDD+ Program, focusing on local mobilization for the expansion of the project.

2. Encourage the participation of landowners already involved in the Program in networks and partnerships with other organizations and institutions involved in the conservation of the Pantanal, enabling the exchange of experiences and cooperation in joint projects, as well as attracting participants to join the Pantanal REDD+ Program.

2.3.18 Community Employment Opportunities (VCS, 3.19.13; CCB, G3.10)

In accordance with the proponent's policies indicated in Section 2.3.14, the Pantanal REDD+ Program follows the premise that all actors who will be directly involved in the project's activities will have equal opportunities for hiring and promotion within the existing functions, as long as the job specifications are met, without any kind of discrimination, as well as encouraging equality, equity and universality of policies and initiatives to value diversity, with attention to minority groups.

Although each Pantanal REDD+ Program partner property has its own system for hiring employees, the project proponent encourages the adoption of good practices and the application of a fair and transparent process for selecting candidates, encouraging hiring, inclusion and unconditional respect for all people, especially women, young people, black people, LGBTQIAPN+ people, people in situations of refuge, immigrants, people with disabilities and other minority groups.

The Pantanal REDD+ Program will not allow any wage differentiation on the grounds of color, gender, religion, choice or personal characteristic that is not specifically related to the technical requirements for performing the job, observing, in particular, collective standards and current legislation.

2.3.19 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

Because it belongs to the Ambipar Group, the project proponent follows all the Group's policies and, consequently, the assessment of the activities carried out by the Pantanal REDD+ Program that may have

an impact on the health and safety of the workers involved in the Program will follow the Risk Management Policy³⁶, with the necessary adaptations to the local reality.

The main objective of adapting this Policy is to establish the principles, guidelines and responsibilities to be observed in the process of managing the risks to which the Program and the workers involved in the activities will be exposed, in such a way as to enable the correct identification, evaluation, chain of treatment priorities, constant monitoring and effective communication of the risks, always with a view to the perpetuity of the Program. The risk management process consists of the following stages:

(a) Identification of Risks and Events

The identification of risks and events will take place in a structured manner, in line with the Program's activities, through internal and external sources.

(b) Risk assessment

Risks will be identified and assessed in each operation according to their impact and probability of occurrence. In terms of impact, risks will be classified into four levels: Low, Medium, High and Extreme. In terms of probability, they should be classified according to the following scale: remote, possible, probable or very probable.

(c) Prioritization and treatment

Once the risks have been assessed, they can be compared relatively in terms of the levels of impact and probability assigned and their action plans prioritized. Risk treatment involves choosing one of the alternatives: eliminating the risk, reducing the risk, transferring the risk or accepting the risk.

(d) Monitoring

Continuously monitor and record the performance of risk indicators, as well as their limits, including the supervision, implementation and maintenance of action plans through ongoing management and independent internal or external evaluations, where applicable.

(e) Communication and consultation

The communication and consultation processes must permeate every activity of the Pantanal REDD+ Program, with the aim of sharing and providing information for the continuous management of Risks, such as the Risk Matrix, with the interested parties, especially those directly impacted.

(f) Types of Risks

Risk descriptions will follow a standardized and consistent format to facilitate their identification, assessment and monitoring, with Risks being classified by their nature. Risks can be categorized as: 1 - Strategic Risks; 2 - Financial Risks; 3 - Operational Risks; 4 - Legal, Regulatory and Compliance Risks; 5 - Political Risks; 6 - Technological Risks; and 7 - Social and Environmental Risks.

³⁶ Risk Management Policy. Available at: <https://ambipar.com/site2020/wp-content/uploads/2023/09/Politica-de-Gerenciamento-de-Riscos.pdf>

Additionally, to demonstrate that the Program respects rights related to work and labor, in addition to the points indicated in item 2.3.18, the actions related to the points below are indicated:

Discrimination and sexual harassment	As mentioned, the Pantanal REDD+ Program follows the rules stipulated by the Ambipar Group's Diversity and Inclusion Policy, which explicitly prohibits the practice of harassment, intimidation, mockery, threats, stigmatization or other attitudes of physical or psychological violence that offend a person's dignity. Likewise, the dissemination of material or behavior of any kind that promotes hatred, discrimination or the promotion of stereotypes is expressly prohibited. This position will be replicated in all activities promoted by the Program and will also be disseminated to partner owners.
Human trafficking, forced labor and child labor	The Pantanal REDD+ Program repudiates child labor, protecting the rights of children and adolescents, in line with the Ambipar Group's Diversity and Inclusion Policy. In addition, any activity related to this Program strictly prohibits forced labor and/or labor analogous to slavery.

2.4 Management Capacity

2.4.1 Project Governance Structures (CCB, G4.1)

For the design and implementation of the Pantanal REDD+ Program, the governance structure is divided as follows:

Biofílica Ambipar Environment (Project Proponent) - planning, design, implementation and monitoring:

Founded in 2008, Biofílica Ambipar Environment aims to be a global leader in Nature Based Solutions (NBS), generating value in the environmental asset market, mitigating climate change, conserving biodiversity and boosting well-being and social development. The company is dedicated to developing and co-managing projects aimed at reducing emissions through forest conservation and reforestation for carbon sequestration.

As the exclusive proponent of the Pantanal REDD+ Program, Biofílica Ambipar Environment plays a key role, being responsible for managing all partnerships with external properties and institutions, guaranteeing the implementation and sustainability of the program.

In addition, Biofílica Ambipar Environment believes it is crucial to count on the support of partners and other stakeholders, each assuming their specific responsibilities, to achieve the goals and objectives set by the Pantanal REDD+ Program. This collaboration is fundamental to guaranteeing tangible benefits for

the climate, the community and biodiversity, promoting continuous improvements in the actions and activities developed, while recording the changes and trajectory of the Program over time.

In order to achieve the expected results in the Program, all departments of Biofíllica Ambipar Environment are integrally involved from the initial planning stages to the implementation and monitoring phase of the activities. Below is a brief summary of each area's role in the Program:

New Business Team (NN): The mission of Biofíllica Ambipar Environment's New Business Team is to develop and implement prospecting strategies to identify investment opportunities, promote sustainable growth and strengthen the company's competitive position in the Nature Based Solutions (NBS) market. The team works in three main areas: Opportunity Identification, Feasibility Assessment and Negotiation.

In Opportunity Identification, the team uses various sources of information to identify areas for project development, as well as mapping out business opportunities, such as strategic partnerships. After identification, the Feasibility Assessment of the projects is carried out, considering aspects such as profit potential, associated risks and return on investment. The team conducts legal and financial due diligence to ensure relevant information before proceeding with the negotiation. In the Negotiation phase, the team conducts negotiations with owners and/or intermediaries, ensuring clear and comprehensive contracts to protect the interests of all parties involved.

The work of the NN team is crucial to the development of the Pantanal REDD+ Program, from identifying opportunities to concluding negotiations, through institutional partnerships, area prospectors and informal outreach.

NBS Operational Team: The Biofíllica Ambipar Environment Operational team is involved in all stages of carbon projects, coordinating activities such as baseline validation, development, monitoring and control of activities, as well as leading verification audits.

Its work involves planning and monitoring the Program's activities, coordinating with stakeholders, promoting institutional communication and supporting the geoprocessing team. The team is responsible for ensuring the efficiency of actions to mitigate greenhouse gas emissions, in accordance with Verra's methodologies, and leading project verification processes.

Geoprocessing team: An essential part of the operational and new business team, the Geoprocessing team uses advanced technologies to integrate and analyze geospatial data, providing effective and sustainable solutions. Its infrastructure includes geographic information systems, remote sensing, modeling, spatial analysis and data integration. This team plays a key role in the analysis and monitoring of the Program's properties.

Intelligence Team: The Operational Intelligence team focuses on knowledge management, researching and analyzing methodologies, best practices and trends for implementing NBS projects. It also develops practical and scalable operational solutions in line with Verra's methodologies and generates strategic information to be implemented in the Program.

Commercial Team: is responsible for the exclusive marketing of carbon credits, conducting technical and consultative negotiations. It works closely with technical teams and partners, providing support throughout the sales cycle, from lead qualification to after-sales and logistics for field visits.

Marketing Team: Biofílica Ambipar Environment's Marketing and Communications team follows the entire customer journey, from attraction to after-sales, using content, storytelling and automation strategies to educate, engage and convert leads into buyers. It also works closely with partners' marketing teams, providing technical support and planning joint actions.

Advocacy & Engagement Team: is responsible for creating a favorable environment for Nature Based Solutions, establishing and maintaining strategic relationships with government bodies and organizations, influencing policies and seeking partnerships to promote the implementation of NBS.

Back Office Team: this is the area that provides essential support for the Pantanal REDD+ Program 's financial and administrative operations, including accounting, administrative management, financial analysis and planning, compliance and risk management.

The detailed description of the REDD+ Program's operations, responsibilities and management tools is set out in the "Governance of the Pantanal REDD+ Program " document shared with the VVB on a confidential basis.

Suppliers – Initial Program Diagnostics Development:

Monteiro de Barros Sociedade de Advogados: The law firm Monteiro de Barros Sociedade de Advogados (MDB) dedicates special efforts to projects and initiatives aimed at overcoming inequalities and promoting actions for human development, culture, and the sustainability of the planet. In the Pantanal REDD+ Program, they provided legal and environmental consultancy services for the Pantanal REDD+ Program and also represent some partner property owners of the Program.

Restaura Consultoria Ambiental e Treinamentos LTDA: Restaura is an ecological consulting company focused on providing restoration/recovery services for degraded areas and technical and scientific training. In the Pantanal REDD+ Program, they worked on the development of socio-economic and environmental studies, such as the characterization of the physical environment and biodiversity of the region (flora and fauna).

Soluções Técnicas e Serviços de Engenharia Ambiental – STA: STA's mission is to provide efficient solutions for the sustainable use of natural resources, reconciling economic and social development with the maintenance of ecological processes. In the Pantanal REDD+ Program, they worked on the development of the carbon stock estimation study. Institutional Partnerships – Support to the proponent in the planning and implementation of the Program.

Instituto Taquari Vivo (ITV): Providing support in logistics processes for Biofílica Ambipar Environment and contracted suppliers, strategic articulation support in the region with public-private institutions of interest, and also serving as a communication channel with property owners.

Associação Aliança 5P: Providing support in field logistics; support in engagement and communication among the parties involved in the Program participating in the association; The Association's main purpose is to conduct, support, and articulate studies, projects, and initiatives associated with economic sustainability, nature preservation, and socio-economic and environmental development in the Pantanal region and surrounding areas.

2.4.2 Required Technical Skills (VCS, 3.19; CCB, G4.2)

The main technical skills required to implement the Pantanal REDD+ Program include knowledge of methodologies to prevent planned deforestation, understanding of development and management of forest conservation projects in Brazil, financial management, experience in initiatives to engage and mobilize landowners, and familiarity with territorial monitoring. It can be said that the project proponent possesses the necessary technical skills for the successful completion of the Project. Additionally, partnerships will be established to expand this expertise in order to enhance Program management.

Ambipar Group operates throughout the national territory offering integrated solutions for the entire business chain. In rapid global expansion, Ambipar respects compliance rules and socio-environmental responsibility, valuing ethics and prompt responsiveness to its clients' demands. As a crucial arm in the Group's strategies, Ambipar Environment focuses on environmental solutions, providing a comprehensive platform for the valorization of industrial waste, post-consumer waste, recycling, co-processing, reverse manufacturing, and other services related to waste management, all focused on the circular economy and ESG (Environmental, Social, and Governance) principles. Recently, forest maintenance and preservation have been added to the Environment area of the Group.

As part of the Ambipar Group, within the Environment pillar and aligned with its objectives, Biofílica Ambipar Environment is a Brazilian company that promotes the management of forest areas in the Amazon, Atlantic Forest, and now Pantanal biomes. The company has a specialized team and is a reference in the development of forest conservation projects, ensuring the quality and effectiveness of its activities.

Biofílica Ambipar Environment aims to make conservation economically attractive for forest owners, communities, and investors. Its goals include reducing deforestation and carbon emissions in the atmosphere, conserving biodiversity and water resources, and promoting social inclusion and community development in its operating regions through the sale of environmental services credits, development and financing of scientific research activities, and the development of sustainable business chains.

2.4.3 Management Team Experience (VCS, 3.19; CCB, G4.2)

Below is the experience of the members of the Pantanal REDD+ Program management team within Biofílica Ambipar Environment.

BOARD OF DIRECTORS:**Plínio Ribeiro – CEO**

Plínio Ribeiro holds a degree in Business Administration from the Instituto de Ensino e Pesquisa - INSPER and a Master's degree in Public Administration and Environment from Columbia University and the Earth Institute (USA). He has participated in several conservation projects on the lower Negro River, through the Institute of Ecological Research – IPÊ since 2005 and was one of the producers of the documentary "Retorno à Amazônia", by Jean Michel Cousteau. He has been working at Biofílica Ambipar Environment since 2008, where he has led Projects, Operations and Business Management. Currently, he is Executive Director and shareholder of the company.

Cláudio Pádua – Scientific Director

Cláudio Pádua holds a degree in Business Administration and Biology, a Master's in Latin American Studies and a PhD in Ecology from the University of Florida in Gainesville (USA). Retired professor at the University of Brasília, Pádua is currently dean of the Superior School of Conservation and Sustainability and vice-president of the Institute of Ecological Research (IPÊ). He is also a Senior Research Associate at the Center for Environmental Research and Conservation at Columbia University (USA) and Director of International Conservation at the Wildlife Trust Alliance, as well as a consultant to the Brazilian Biodiversity Fund (FUNBIO) and WWF Brazil. Pádua represents Brazil before the International Advisory Group (IAG) of the G7 Pilot Program. In 2003, together with his wife, Suzana Pádua, he was named "Hero of the Planet" by Time magazine for his activities in favor of biodiversity conservation. Between 1997 and 2007, he won six conservation awards, three national and three international, and, throughout his life, published two books and more than 30 articles in national and international scientific journals. Since 2008 he has directed Biofílica Ambipar Environment's involvement and scientific production as Scientific Director and advisor.

Soraya Dias Pires – Director of Operations and New Business

Soraya Dias has a degree in agronomy from the University of São Paulo (USP), two MBAs from USP, one in agribusiness and the other in strategic management, and an MBA from the Getulio Vargas Foundation (FGV) in business administration. He has more than 15 years' experience in agribusiness management and development, with a career developed in large multinationals in the sugar-energy sector: Adecoagro, BP Biocombustíveis and BP Bunge Bioenergia. She has experience in agro-industries in activities associated with agro-industrial planning and control, agricultural operations and negotiations. At Biofílica Ambipar Environment she is Director of Operations and New Business.

Alexandre Spinacé – Director of Sales & Marketing

Alexandre Spinacé has a degree in Business Administration and a post-graduate degree in Finance from FGV-SP. He has more than 15 years working in the commercial area of large national and international companies. He has worked at Biofílica Ambipar Environment since 2021, where he has led the Technology, Finance and Human Resources areas. He is currently responsible for the Sales and Marketing area, where he leads the commercial relationship in the global voluntary carbon market.

Annie Groth – Director of Advocacy & Engagement

Annie has a degree in International Relations (Maastricht University) and a Master's degree in International Public Administration from Sciences Po (Paris). Before joining the Biofílica Ambipar Environment team, she worked with the German Green Party Foundation in the areas of digital policy and infrastructure development.

Flavia Gillich - Financial Director

Graduated in Business Administration, with a post-graduate degree in Business Planning and Control and an MBA in Finance and Controllershship from USP. Professional with more than 20 years' experience in the Finance and Controllershship area, in national and multinational companies in a variety of sectors. Responsible for Strategic and Financial Planning, Budgeting, Accounting, Treasury, Accounts Payable, Accounts Receivable, Billing, Credit, Collection, Internal Auditing, Purchasing and Contracts.

Renan Rossi - Head of Technology

Professional with a degree from UNICAMP, with an extension in Leadership Management and Team Performance and further training in Complex Data Mining from the same institution. He has worked in the technology market since 2012, has experience in large software consultancies and has carried out various projects involving digital strategy, software development, digital transformation and business development.

OPERATIONS & NEW BUSINESS TEAM:**Caio Gallego – Intelligence Manager**

Caio Gallego is a Forest Engineer graduated from ESALQ-USP with an MBA in Corporate Sustainability from Fundação Getúlio Vargas. He specializes in geoprocessing and remote sensing for environmental conservation, mapping, and land use change analysis. He has knowledge in Sustainable Forest Management, environmental modeling, and alternative GIS use for forestry and agribusiness. He possesses advanced skills in GIS software and land use and cover change analysis such as ArcGIS, QuantumGIS, and DinâmicaEGO. He has experience in forest conservation and restoration, climate change, environmental modeling, and environmental project management. He serves as an operations manager in projects developed for the carbon market.

Marcio Sales – Spatial Statistics Specialist

Marcio Sales is a statistician, graduated from the Federal University of Pará, with a Master's in Geography from the University of California, Santa Barbara, and currently a Ph.D. candidate in Production Ecology and Conservation of Resources at Wageningen University in the Netherlands. He specializes in data analysis and conducts research in geostatistical modeling of processes distributed in space and time. At Biofílica Ambipar Environment, he produces projections of greenhouse gas emissions from future deforestation for project baselines and monitors deforestation via satellite.

Paula Conde – Administrative and Operational Support Manager

Paula Conde holds a degree in Business Administration from São Luís - PUC and a postgraduate degree in Accounting and Financial Management from FAAP. She has extensive experience, mostly with one of the largest media and education groups in Latin America - Editora Abril, where she worked with Financial Control and Reporting, Treasury, Financial and Accounting Reconciliation, Accounts Payable and Receivable, and Royalties. At Biofílica Ambipar Environment, she is responsible for administrative and financial activities, logistical support to the team and projects.

Raphael Ramiro – New Business Manager

Graduated in Financial Management, with an MBA in Corporate Finance from UFSCar. Currently pursuing Accounting at FIPECAFI (Foundation Institute of Accounting, Actuarial, and Financial Research), and an MBA in Ecological Restoration from UFSCar. A professional with over 15 years of experience in Financial Planning, Financial Modeling, Project Evaluation, Project Finance, Strategic Planning, and Corporate Finance.

Carla Paulino – New Business Analyst

Graduated in Agronomic Engineering and Business Administration from ILES/ULBRA Itumbiara, with a specialization in Compliance and Project Management. She has been involved in the development and execution of Continuous Improvement projects for the origination processes and procedures. With 16 years of experience in land analysis for partnership and supply contracts, she currently serves as a New Business Analyst III, focusing on prospecting and customer service for the implementation of Carbon Projects at Biofílica Ambipar Environment, one of the pioneering companies in Nature-Based Solutions (NBS). She has had the opportunity to develop teams and conduct intercompany training sessions.

Luana Cordeiro – Project Manager

Luana Cordeiro is a forest engineer graduated from the University of São Paulo – ESALQ/USP, with a postgraduate degree in Project Management (PMI) from SENAC, and a technical degree in environment from the São Paulo State Technical School. She has additional courses in Low Impact Forest Management in the Amazon by INPA and IFT. With over 6 years of experience at Biofílica Ambipar Environment, she has gained expertise in project development, monitoring, and management, preparing documentation audited by international certifiers (Verra), budget management, co-management with partners from the private sector to the third sector, contracting and managing suppliers, and participating in the consolidation of methodological intelligence and certification standards for carbon projects. She also provides technical support to the commercial and communication areas in the preparation of project-related and institutional materials.

Lygia Miolaro – Project Analyst

Lygia Miolaro is a Forest Engineer graduated from the University of São Paulo – ESALQ/USP, with a postgraduate degree in Project Management. She has experience in environmental licensing processes, projects for the recovery of degraded areas, environmental feasibility analysis, and environmental impact assessments. She is proficient in geoprocessing and various mapping techniques. Currently, she works on the development and implementation of AFOLU projects for voluntary carbon markets.

Caroline Pereira – Project Analyst

Caroline Pereira is an Environmental and Sanitary Engineer from the Federal University of Goiás, with professional experience in voluntary and regulated carbon markets, in AFOLU and Renewable Energy Generation projects. She is part of the analyst team of the Pantanal REDD+ Program, working on project development, monitoring, and management, preparing documentation audited by international certifiers. Additionally, she participates in carbon market intelligence analyses focusing on methodologies and certification standards for REDD+ projects.

Fábio Souza – Geotechnologies Coordinator

Fábio Souza is a Geographer with a master's degree in the use and conservation of natural resources, specializing in climate analysis, and a technologist in big data and analytical intelligence. He has worked on environmental diagnostics, development of geotechnology services, and management of geoprocessing teams. He is an expert in PostGIS database modeling, webGIS platform development, business intelligence, and geoprocessing process automation. Currently, he is part of Biofílica Ambipar Environment's operational team as the geotechnologies coordinator.

Gustavo Mourão – Geotechnologies Analyst

Gustavo Mourão is a Forest Engineer graduated from the Federal University of Vales do Jequitinhonha and Mucuri (UFVJM), with a master's degree and currently pursuing a Ph.D. in Forest Science from the same institution. He demonstrates deep knowledge and skills in geoprocessing and programming, including LiDAR point cloud processing and advanced use of R and Python languages for geostatistics. He is proficient in PostgreSQL query languages (PostGIS/SQL), online map publication using Geoserver and Mapstore, and data manipulation with Google Earth Engine. His professional experience at Biofílica Ambipar Environment as a Geoprocessing Analyst III includes analysis of deforestation and degradation data using information from PRODES, DETER, and land use and land cover classification provided by Mapbiomas, aiming to quantify deforestation rates in conservation units and rural properties. He has a track record of consultancy and projects, such as digital vegetation classification for APERAM, LiDAR analysis for the Brazilian Forest System, as well as autonomous services for rural producers in Itamarandiba and previous experience at Agroita Engenharia, significantly contributing to environmental conservation and sustainable management of natural resources. His work also includes the creation of process automation scripts, seeking new data processing tools, and contributing to carbon credit estimates in REDD+, Afforestation, Reforestation and Revegetation (ARR), and Agricultural Land Management (ALM) projects.

SALES & MARKETING:**Laion Pazian – Sales Manager**

Laion Pazian is an economist from USP (ESALQ Campus) and holds an MBA in Sales Management from Fundação Getúlio Vargas. He manages the carbon credit sales team, serving national key-accounts and supporting relevant international sales. He leads part of the commercial strategies and manages the

carbon credit portfolio of Biofílica Ambipar Environment, reporting results to the company's Board of Directors.

Marcos Almeida – Sales Manager (International Channel)

Marcos Almeida is the Head of International Sales and Business Development at Biofílica Ambipar Environment, with over 10 years of experience in international business. He holds a degree in Business Administration from the Federal University of Itajubá (UNIFEI) and a Postgraduate degree in Business Administration from Insper. He leads sales operations, sourcing, brokerage, and trading at Biofílica Ambipar Environment.

Ricardo Cordeiro – Marketing and Communications Coordinator

Ricardo Cordeiro holds a degree in digital marketing and an MBA in Marketing, Branding & Growth from PUC Rio Grande do Sul. With over 16 years of experience in art direction (online and offline) and project management, he is a specialist in digital marketing strategies, SEO, Google Adwords, and Social Ads. Throughout his career, he has worked in digital agencies, trade, and live marketing. At Biofílica Ambipar Environment, he leads the marketing team, planning and executing actions focused on Branding and Growth.

ADVOCACY & ENGAGEMENT:**Luisa Cotrim – Carbon Market Specialist**

Luisa Cotrim graduated in Economics from the State University of Campinas (UNICAMP) and has professional experience in the voluntary carbon market and in strategic and management consulting. At Biofílica Ambipar Environment, she leads market intelligence initiatives, including price and competition monitoring, market research and reports, and tracking trends and opportunities related to carbon markets.

Gabriel Trivelino – Senior Advocacy Analyst

Gabriel Trivelino is a Senior Advocacy Analyst at Biofílica Ambipar Environment, focusing on the company's relationship with the public sector and representative organizations. He holds a degree in Political Science from the University of Brasília (UnB) and an MBA in Government Relations from Fundação Getúlio Vargas (FGV), with over 10 years of professional experience representing interest groups in specific legislation or regulations in decision-making bodies such as the Brazilian Executive and Legislative branches.

2.4.4 Project Management Partnerships and Team Development (VCS, 3.19; CCB, G4.2)

The Pantanal REDD+ Program has all the internal expertise and necessary partnerships for the construction and implementation of the proposed activities. Currently, the partner institutions, mentioned in Section 2.4.1, are responsible for the elaboration of Socioeconomic and Environmental

Diagnostics, Carbon Stock, and Baseline, which compose the Program's PDD (Project Design Document), through service provision contracts. Additionally, institutional partners have provided all necessary support for the development, implementation, and subsequent expansion and monitoring of the Program.

When other initiatives during the Project's development require new technical knowledge, the Program proponent is responsible for coordinating with governmental, non-governmental, and private sector organizations, enabling the generation of net positive impacts on society and biodiversity.

2.4.5 Financial Health of Implementing Organization(s) (CCB, G4.3)

The Biofílica Ambipar Environment Investments S.A is a closely held corporation domiciled at Av. Angelica, 2330 ed. New England, 5th floor – São Paulo – SP, primarily engaged in the development and commercialization of environmental services credits. The Company manages and conserves forests through the commercialization of environmental services, investments in research, and socio-economic development of people and communities inhabiting the managed areas. With the aim of contributing to the creation and development of a solid and reliable environmental credits market, the Company invests in an innovative business model, allowing for the reduction of deforestation, appreciation of standing forests and the environmental services they provide, biodiversity protection, and carbon emissions reduction, making environmental conservation an economically viable activity for forest owners, communities, governments, and investors.

In July 2021, Ambipar Participações e Empreendimentos S.A. concluded the acquisition of 53.6% of Biofílica Ambipar Environment's share capital. The main objective of this transaction was to accelerate the company's growth, leveraging the Ambipar Group, resulting in the expansion of its portfolio of environmental services offered. As a consequence of this acquisition, the company underwent a nominal change, resolved in an extraordinary general assembly, renaming itself Biofílica Ambipar Environment Investments S.A. The entry of the shareholder (Ambipar) into Biofílica Ambipar Environment did not alter the Company's Objectives, which remain strongly related to the promotion and financing of activities related to the environment and environmental conservation, with a detailed description available in Article 3 of the company's updated bylaws, provided to VVB.

The entry of a new shareholder had a positive effect, as the influx of resources, expertise, and economies of scale allows Biofílica Ambipar Environment to advance more consistently and rapidly in its activities related to the environment and environmental conservation. Additionally, there are no contractual implications in any project implemented by Biofílica Ambipar Environment with the entry of a new shareholder. The main objectives of the investments in this transaction are technological development and expertise in Nature-Based Solutions (NBS) techniques.

In this context of accelerated expansion, Biofílica Ambipar Environment currently has several active projects and various opportunities in its pipeline focusing on the generation of forest carbon credits of the REDD+ type (Reducing Emissions from Deforestation and Forest Degradation). Additionally, the Company is also beginning to operate in other NBS (Nature Based Solutions) fronts:

Afforestation/Reforestation (AR), Agricultural Land Management (ALM), and Blue Carbon (mangrove carbons).

Thus, it can be said that the project proponent is financially healthy to ensure financial support throughout the Program period as needed. The documentary evidence of the proponent's financial health is classified as Commercially Sensitive Information and has been shared with the audit team on a confidential basis.

2.4.6 Avoidance of Corruption and Other Unethical Behavior (VCS, 3.19; CCB, G4.3)

Biofíllica Ambipar Environment maintains a legal team responsible for implementing and monitoring the best practices established in the "Ambipar Group Code of Conduct and Compliance," including the "Anti-Corruption and Anti-Money Laundering Policy," which is in perfect alignment with the Brazilian Anti-Corruption Law (Law No. 12,846/13), Law No. 9613/98 (Money Laundering Law), as well as with International Anti-Corruption Conventions (UN, OECD, Transnational Convention on Combating Bribery of Foreign Public Officials in International Business Transactions, Mérida Convention, Palermo Convention, Convention against Transnational Organized Crime), and observes the principles outlined in the FCPA (Foreign Corrupt Practices Act) and the UK Bribery Act.

Thus, Biofíllica Ambipar Environment clearly expresses its opposition to the practice of corruption and/or conduct aimed at personal gain at the expense of the company, society, or the government, strictly adhering to national and international anti-corruption principles, as well as the Ambipar Group's main guidelines and positions regarding the fight against money laundering, terrorism financing, and all forms of corrupt conduct, such as bribery (active and passive corruption), deviations, and granting undue advantages (administrative impropriety), and prohibiting acts by its employees that hinder investigation and oversight activities.

In this sense, ethical principles and values applied to the Pantanal REDD+ Program include:

- Proper and transparent relationships with employees, third parties, customers, and society in general.
- Integrity and honesty in conducting business.
- Prohibition and zero tolerance for acts of corruption.
- Good corporate governance practices, good accounting and management principles and practices, clear, objective, and timely communication to investors and the capital markets.

It is also emphasized that Biofíllica Ambipar Environment undergoes annual financial audits to ensure that its resources are allocated responsibly and free from corruption.

Finally, to promote compliance with these policies, Ambipar has an online reporting channel that can be used by stakeholders of the Program, where complaints can be made about: Conflict of interest; Corruption involving public officials; Fraud/theft or misappropriation of funds; Favoritism towards

suppliers or customers; Damage to company property; Misuse of resources; Leakage/misuse of information; Violation of laws or non-compliance with policies/procedures, among others. Complaints should be made, whenever possible, through the existing channel on the company's website, at the link <https://ambipar.com/denuncias/> or by email at canaldeetica@ambipar.com.

All Ambipar Group policies are available on its website: <https://ambipar.com/politicas-estatuto/>

2.4.7 Commercially Sensitive Information (VCS, 3.5.2 – 3.5.4; CCB Rules, 3.5.13 – 3.5.14).

Some information required by the VCS and/or CCB standards is considered confidential or commercially sensitive and cannot be publicly disclosed by the project proponents. This information was provided in full to the audit team during the validation process attached to this document, but was not included in the public version. Below is a list of the information that was made available:

- Land documents and legal status of the properties participating in the Program.
- Agreements signed between the parties involved.
- Project Financial Performance Spreadsheet and other related documents
- Biofílica Ambipar Environment's financial statements.
- Minutes of meetings between Biofílica Ambipar Environment and the parties involved in the Program.
- Attendance list with personal information of Program participants.
- Agreements and contracts signed between the parties involved.

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

With regard to REDD+ activities in Brazil, there is a history of initiatives, despite the construction and negotiation of this concept through agreements and meetings in the United Nations Framework Convention on Climate Change (UNFCCC).

In December 2015, Brazil established the National Strategy for REDD+ (ENREDD+) through MMA Ordinance No. 370. This document formalizes to Brazilian society and UNFCCC signatory countries how the Brazilian government has structured its efforts, contributing to mitigating climate change through forest deforestation and degradation control, promoting forest recovery, and sustainable development. In this context, Brazil's Decree No. 10,144 of November 28, 2019, established the National Commission for REDD+ (CONAREDD+), aimed at coordinating, monitoring, and revising the National Strategy for REDD+ and guiding the development of requirements for accessing payments for REDD+ policies and

actions results in the country. The CONAREDD+ internal regulations were published the following year, through Ordinance No. 544 of October 26, 2020.

At the same time, the analysis of Bill No. 572/2020 is of widely relevant nature. This bill "establishes the national system for reducing emissions from deforestation and degradation, conservation, sustainable forest management, maintenance, and increase of forest carbon stocks (REDD+) and provides other measures." So far, the text is under consideration in the Chamber of Deputies.

Regarding the carbon market, there are several bills under consideration in the National Congress aimed at establishing the Brazilian Regulated Carbon Market and regulating points related to the purchase and sale of carbon credits in the country from activities such as Reduction of Emissions from Deforestation and Forest Degradation (REDD+). The promotion of a Brazilian Emission Reduction Market is provided for in the Law that established the National Policy on Climate Change (Law No. 12,187, of 12/29/2009).

Topic: Constituição Federal Brasileira de 1988

It defines and regulates how laws should be created in the country, how the three branches of government (Executive, Legislative and Judiciary) should work and the bodies that act in conjunction with them. The Constitution's fundamental principles are:

I - sovereignty.

II - citizenship.

III - the dignity of the human person.

IV - the social values of work and free enterprise.

V - political pluralism.

Applicability to the project: Regulates and directs all national legislation that influences the project.

Topic: Land Regulation

Related Legislation:

Law 601/1850 - Law of Devolved Lands: Provides for devolved lands in the Empire, and those possessed by title of sesmaria without meeting legal conditions, as well as by simple title of peaceful and quiet possession; and determines that, once measured and demarcated, they shall be ceded on an onerous basis, both to private enterprises and for the establishment of colonies of nationals and foreigners, with the Government authorized to promote foreign colonization in the manner described.

Law 4.504/1964 - Land Statute: Provides for the land statute and other provisions. This Law regulates the rights and obligations concerning rural real estate, for the purposes of implementing Agrarian Reform and promoting Agricultural Policy.

Law 9.393/1996 - ITR Law - Rural Property Tax: Provides for the ITR Tax, payment of the debt represented by Agrarian Debt Titles and other provisions.

Law 10.406/2002 - Brazilian Civil Code - Concerning rights and duties in private life, of persons, goods and legal facts. Divided into two parts, (1) General Part, concerning persons, goods and legal facts and (2) Special Part, concerning the law of obligations, companies, things, family and successions.

Law No. 8.629/1993 - Law on Agrarian Reform - regulates and disciplines provisions relating to agrarian reform, as provided for in the Federal Constitution.

Applicability to the project: The project complies with the listed laws, which ensures the right to property including the Project Area and compliance with taxes related to rural properties.

Topic: Environment Related Legislation

Related Legislation:

Law 12.651/2012 - Brazilian Forest Code: Establishes general norms on the protection of vegetation, Permanent Preservation Areas, Legal Reserves; forest exploitation, supply of forest raw materials, control of the origin of forest products, and control and prevention of forest fires, and provides economic and financial instruments to achieve its objectives.

Law 9.605/1998 - Federal Law on Environmental Crimes: Provides for criminal and administrative sanctions arising from conduct and activities harmful to the environment, and provides other provisions.

MMA Ordinance No. 148, of June 7, 2022 - Recognizes the following official lists: "National Official List of Threatened Species of Flora", "Official List of Threatened Species of Brazilian Fauna" and "Official List of Extinct Species of Brazilian Fauna".

Law 7.347/1985 - Law on Public Civil Action: Regulates public civil action for liability for damage caused to the environment, consumers, assets and rights of artistic, aesthetic, historical, tourist value and provides other provisions.

Law No. 3,839, of December 28, 2009: Establishes the Territorial Management Program of the State of Mato Grosso do Sul (PGT/MS); approves the First Approach of the Ecological-Economic Zoning of the State of Mato Grosso do Sul (ZEE/MS), and provides other provisions.

Law No. 5,235, of July 16, 2018: Provides for the State Policy for Preservation of Environmental Services, creates the State Program for Payment for Environmental Services (PESA), and establishes a Management System for this Program.

Law No. 4,555, of July 15, 2014: Establishes the State Policy on Climate Change - PEMC, in the Territory of the State of Mato Grosso do Sul.

Decree No. 15,798, of November 3, 2021: Regulates the Voluntary Public Registry of Annual Greenhouse Gas Emissions and the State Communication, provided for in the State Climate Change Policy, provided for in State Law No. 4,555, of July 15, 2014.

Law No. 4,163, of January 2, 2012: Regulates, in the State of Mato Grosso do Sul, the exploitation of forests and other forms of native vegetation, the use of forest raw materials, the obligation of reforestation.

Law No. 6,160, of December 18, 2023: Provides for the conservation, protection, restoration and ecologically sustainable exploitation of the Restricted Use Area of the Pantanal Plain (AUR-Pantanal), in the State of Mato Grosso do Sul, and creates the State Fund for Sustainable Development of the Pantanal Biome.

Decree No. 14,755, of June 12, 2017: Provides for the establishment and recognition of Private Reserves of Natural Heritage, in the State of Mato Grosso do Sul.

Decree No. 13,977, of June 5, 2014: Provides for the Rural Environmental Registry of Mato Grosso do Sul and the MS More Sustainable Program.

Decree No. 16,388, of February 16, 2024: Regulates provisions of Law No. 6,160, of December 18, 2023, which provides for the conservation, protection, restoration and ecologically sustainable exploitation of the Restricted Use Area of the Pantanal Plain (AUR-Pantanal), in the State of Mato Grosso do Sul.

SEMAG Resolution No. 11 of July 15, 2014: Implements and disciplines procedures related to the Rural Environmental Registry and the MS More Sustainable Program referred to in State Decree No. 13,977, of June 5, 2014 (consolidated).

Ordinary Law No. 2,548/2017: Provides for the Environmental Policy for Protection, Control, Conservation and Recovery of the Environment of the Municipality of Aquidauana.

Organic Law of the Municipality of Aquidauana-MS: Presents rules that regulate public life in the city, always respecting the Federal Constitution and the State Constitution.

Organic Law of the Municipality of Miranda-MS: Presents rules that regulate public life in the city, always respecting the Federal Constitution and the State Constitution.

Organic Law of the Municipality of Corumbá-MS: Presents rules that regulate public life in the city, always respecting the Federal Constitution and the State Constitution.

Applicability to the project: The farms where the project is located have Permanent Preservation Areas, Legal Reserves, and threatened species of Brazilian fauna and flora, making the area owner susceptible to legal consequences if they fail to fulfill their responsibilities. Additionally, the project affects the Restricted Use Area of the Pantanal Plain, which has specific restrictions at the state level. It is worth mentioning that Ambipar Group is committed to the environment, and its Code of Conduct and Compliance explicitly states that all its activities and those of its employees must strictly comply with environmental legislation and standards, seeking the optimization of natural resources.

Topic: Anti-corruption

Related Legislation:

Law 12,846/2013 - Anti-Corruption Law: Provides for the administrative and civil liability of legal entities for acts against the public administration, whether national or foreign, and provides other provisions.

Applicability to the project: It is important to emphasize that any action of this Program is contrary to practices of corruption or conduct aimed at personal benefit to the detriment of any of the companies

involved in the Program, society, or the Government, and it is incumbent upon all involved to refrain from such practices.

Topic: International agreements

FCCC/CP/2005/Misc.1: Reducing emissions from deforestation in developing countries: approaches to stimulate action. Submission from Parties.

FCCC/CP/2007/6/add.1: Report of the Conference of the Parties on its thirteenth session, held in Bali from 3 to 15 December 2007. Addendum. Part two: Action taken by the Conference of the Parties at its thirteenth session.

FCCC/CP/2009/Add.1: Report of the Conference of the Parties on its fifteenth session, held in Copenhagen from 7 to 19 December 2009. Addendum. Part Two: Action taken by the Conference of the Parties at its fifteenth session.

FCCC/CP/2010/7/Add. 1: Report of the Conference of the Parties on its sixteenth session, held in Cancun from 29 November to 10 December 2010. Addendum. Part Two: Action taken by the Conference of the Parties at its sixteenth session.

FCCC/CP/2011/9/Add. 1: Report of the Conference of the Parties on its seventeenth session, held in Durban from 28 November to 11 December 2011. Addendum. Part Two: Action taken by the Conference of the Parties at its seventeenth session.

FCCC/CP/2012/8/Add.1: Report of the Conference of the Parties on its eighteenth session, held in Doha from 26 November to 8 December 2012. Addendum. Part two: Action taken by the Conference of the Parties at its eighteenth session.

FCCC/CP/2013/Add.1: Warsaw Framework for REDD-plus, held in Warsaw, Poland, from 11 to 22 November 2013, in particular the following decisions:

Decision9/CP.19: Work programme on results-based finance to progress the full implementation of the activities referred to in decision 1/CP. 16, paragraph 70.

Decision10/CP.19: Coordination of support for the implementation of activities in relation to mitigation actions in the forest sector by developing countries, including institutional arrangements.

Decision12/CP.19: The timing and the frequency of presentations of the summary of information on how all the safeguards referred to in decision1/CP.16, appendix I, are being addressed and respected.

Decision13/CP.19: Guidelines and procedures for the technical assessment of submissions from Parties on proposed forest reference emission levels and/or forest reference levels.

Decision14/CP.19: Modalities for measuring, reporting and verifying.

Decision15/CP.19: Addressing the drivers of deforestation and forest degradation.

FCCC/CP/2015/Add.1: Report of the Conference of the Parties on its twenty-first session, held in Paris from 30 November to 13 December 2015. Addendum. Part two: Action taken by the Conference of the Parties at its twenty-first session.

FCCC/CP/2015 Paris Agreement: Global, legally-binding agreement that sets out a global framework to avoid dangerous climate change by limiting global warming to well below 2°C and pursuing efforts to limit it to 1.5°C. Entry into force on 4 November 2016.

FCCC/CP/2016 Decisions adopted by the Conference of the Parties (COP): Especially decisions 1 (preparation into force of the Paris Agreement), 3 (Warsaw International Mechanism for Loss and Damage associated with Climate Change Impacts), 6 (National adaptation plans) and 7 (Long-term climate finance).

FCCC/CP/2017, FCCC/CP/2018, FCCC/CP/2019 Decisions adopted by the COP: Especially decision 1 reporting on developments of the implementation of the Paris Agreement.

CITES, de 03/03/1973: “Convention on International Trade in Endangered Species of Wild Fauna and Flora”

Article 6 of the Paris Agreement (2021): Decision 1/CP.21 mandated the SBSTA to operationalize the provisions of this Article through recommending a set of decisions to the COP serving as the meeting of the Parties to the Paris Agreement at its first session. At COP26, the Parties to the Paris Agreement at its third session (CMA 3) adopted three main decisions related to Article 6: decision 2 (on Article 6.2), decision 3 (on Article 6.4) and decision 4 (on Article 6.8).

Glasgow Leaders' Declaration on Forests and Land Use (2021): Signatories (including Brazil) promise to reverse and end deforestation by 2030.

Brazilian Nationally Determined Contribution (NDC): First Brazilian NDC submitted in in September 2015 to the UN Framework Convention on Climate Change for mitigation, adaptation and means of implementation, in a manner consistent with the purpose of contributions to achieve the ultimate objective of the Convention, pursuant to Decision 1/CP.20, paragraph 9. The updated Brazilian NDC was presented at the COP26 on December 8th, 2022.

2.5.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18.2; CCB, G3.11)

We guarantee that all employees belonging to Bioflica Ambipar Environment and the companies that provide services are legally employed in compliance with Brazilian labor legislation. In addition, international agreements ratified by Brazil and issues relating to worker welfare are respected.

Every year, Bioflica Ambipar Environment's compliance with labor standards and laws is checked by an audit, which is due to the fact that it is an S.A. company. Its financial statements are published on the Jus Brasil website, the largest open legal community in Latin America.

After the employee is hired and before they start working, they receive training on technical procedures and are empowered about their rights and the applicable legislation. In addition, employees are instructed to join the institution responsible for defending their rights, the unions for their area of work.

The Pantanal REDD+ Program is committed to complying with current labor laws and providing a quality of life for its employees. Listed below are the relevant laws and regulations that protect workers' rights in Brazil, as well as the international agreements ratified by Brazil on labor issues.

Federal legislation and regulations:

Decree-Law No. 5,452 of May 1, 1943: Approves the Consolidation of Labor Laws.

Law No. 6.514, of December 22, 1977: Amends Chapter V of Title II of the Consolidation of Labor Laws, relating to occupational safety and medicine, and makes other provisions.

International Agreements Ratified by Brazil:

International Labor Organization Convention No. 29 of 1930, ratified by Brazil on 25/04/1957: Abolition of forced labor.

International Labor Organization Convention No. 87 of 1940: Freedom of association and protection of the right to organize.

International Labor Organization Convention No. 97 of 1949, ratified by Brazil on 18/06/1965: Migration for Employment Convention.

International Labor Organization Convention No. 98 of 1949, ratified by Brazil on 18/11/1952: Right to Organize and Collective Bargaining Convention.

International Labor Organization Convention No. 100 of 1951, ratified by Brazil on 25/04/1957: Equal Remuneration Convention.

International Labor Organization Convention No. 105, ratified by Brazil on 18/06/1965: Abolition of Forced Labor Convention.

International Labor Organization Convention No. 111 of 1958, ratified by Brazil on 01/03/1965: Discrimination (Employment and Occupation) Convention.

International Labor Organization Convention No. 131 of 1970, ratified by Brazil on 04/05/1983: Minimum Wage Fixing Convention.

International Labor Organization Convention No. 138 of 1973, ratified by Brazil on 28/06/2001: Minimum Age Convention.

International Labor Organization Convention No. 142 of 1975, ratified by Brazil on 24/11/1981: Human Resources Development Convention.

International Labor Organization Convention No. 143 of 1975: Migrant Workers (Supplementary Provisions) Convention.

International Labor Organization Convention No. 155 of 1981, ratified by Brazil on 18/05/1992: Occupational Safety and Health Convention.

International Labor Organization Convention No. 169 of 1989, ratified by Brazil on 25/07/2002: Indigenous and Tribal Peoples Convention.

International Labor Organization Convention No. 182, ratified by Brazil on 02/02/2000: Worst Forms of Child Labor Convention.

In addition to international legislation and agreements, the Pantanal REDD+ Program is based on well-structured policies that guide program management practices. For example, the Ambipar Group's Code of Conduct and Compliance aims to guide all involved parties in adopting best practices in relationships and business conduct based on ethical and moral principles. The document provides guidelines on hiring, remuneration and benefits, training, labor relations and unions, modern slavery, harassment, and workplace quality, among others.

Another point demonstrating the Pantanal REDD+ Program 's commitment to labor laws and regulations is outlined in the Social Statute of the Aliança 5P Association. This is relevant because some properties involved in the program are part of this association and therefore adhere to its statute. The document states that a requirement for joining the association is compliance with and respect for Brazilian labor laws on owned or leased properties located in the Pantanal region, as well as treating employees and collaborators, whether direct or indirect, with respect and dignity.

Lastly, it's worth noting that before entering into contracts with the participating farm owners, a Negative Certificate of Labor Debts issued by the Labor Court was requested to confirm that the properties are compliant with labor laws.

2.5.3 Human Rights (VCS, 3.19)

The Pantanal REDD+ Program is currently implemented entirely within privately owned properties, with no traditional communities residing in the area. These properties' owners have signed a long-term agreement with Biofílica Ambipar Environment for the conservation of surplus vegetation areas within their properties. Therefore, it is understood that applicable international legislation on human rights, as well as the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention No. 169 on Indigenous and Tribal Peoples, do not apply in the context of the program at this time.

However, regardless of this fact, the Pantanal REDD+ Program is guided by the Ambipar Group's Code of Conduct, as described in Section 2.3.14, and commits to stakeholders in its vicinity by acting on the following premises:

- Encouraging a respectful attitude towards people, their traditions, and values.
- Adopting and maintaining a transparent process for defining social actions.
- Respecting the interests and needs of the communities where it operates.

- Collaborating with knowledge and technical information that can bring public benefits.
- Preserving and ensuring the political freedom and freedom of expression of collaborators.

Guiding the relationship of Pantanal REDD+ Program representatives with authorities, politicians, and public officials by transparent, professional, and correct attitudes. Any form of pressure or request from public agents that does not correspond to this definition must be immediately rejected and reported to the management of Biofílica Ambipar Environment.

Guiding the relationship based on respect for laws and conventions that address fundamental human rights and sustainability protection. In all its relationships, representatives of the Pantanal REDD+ Program must observe ethical standards, health, safety, and respect for human rights and socio-environmental responsibility.

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

The current scope of the Pantanal REDD+ Program so far does not include any areas inhabited by traditional communities, and there are no communities that are directly dependent on the project area (forest area), either for subsistence or other activities.

However, it is recognized that the Pantanal REDD+ Program can play a crucial role in preserving and protecting the Pantanal's rich cultural heritage, maintained by landowners and the families who work on the farms. In this context, the project aims to implement activities aimed at employees and their families, aligned with the socio-environmental and economic objectives of the properties, aimed at forest conservation, preserving and protecting Pantanal culture in the region.

2.5.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

The Pantanal REDD+ Program area is located within the boundaries of private properties that hold the right to use and possess the farms and their resources. These private properties have all the necessary documentation proving their ownership, and there are no disputes over titles or rights in the project area. In addition, the Right of Use of the Project Area is strictly respected, following the criteria established by the VCS Standard v4.6 (pages 25-26), as indicated in Section 2.1.9.

It is crucial to note that the right to property is constitutionally guaranteed and is considered a fundamental right in the Brazilian legal system. According to the Civil Code, the owner is the person who has the right to use, enjoy and dispose of the property, as well as the right to recover it from anyone who unjustly possesses or detains it.

Due to the grouped nature of the project, the farms that make up the Pantanal REDD+ Program have different owners. Contracts have therefore been drawn up between these owners and the project

proponent, Biofilica Ambipar, Environment to clearly define responsibilities and rights related to the program.

The documents proving ownership (including the land data for each property) and the contracts formalized with each owner will be shared with the audit team on a confidential basis.

2.5.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

The Pantanal REDD+ Program recognizes and respects all property rights, complying with significant statutory and regulatory requirements, as well as obtaining the necessary approvals from state and local authorities.

All owners involved in the Program have property titles and registrations for rural properties that certify and ensure rights over the land. All rights have been secured through the execution of contracts, complying with relevant legal requirements, and observing the free, prior, and informed consent of all identified stakeholders. It is also worth noting that in selecting contractual partners, areas were chosen where there are no collective conflicts over land ownership or pending related actions.

The documents proving ownership (including land data for each property) and the contracts formalized with each owner will be shared with the auditing team confidentially.

2.5.7 Free, Prior and Informed Consent (VCS, 3.18; CCB, G5.2)

Description of process for obtaining consent

The Pantanal REDD+ Program, as of the development of this documentation, does not encompass any properties where there are traditional or indigenous communities or government lands to carry out its activities. The project operates within privately-owned properties, and thus, the necessary consent for the development of the Pantanal REDD+ Program was obtained from the owners of the areas comprising the project. The contracts were freely negotiated, accepted, and signed by those with the legal authority to do so, as evidenced by the partnership contracts executed and shared with VVB under confidentiality agreements.

Nevertheless, the Program adopted as a best practice for the feedback process to conduct a consultation with all identified stakeholders to understand their stance regarding the implementation of the Pantanal REDD+ Program, allowing them to express support or opposition. For this purpose, a questionnaire was administered at the end of each meeting, covering questions about the understanding of the presented information and opinions

	regarding the implementation of the Pantanal REDD+ Program in the region. The farms participating in the Program have employees who reside on the project's properties. Therefore, both in-person and remote activities were conducted to carry out consultations, communication, project dissemination, presentation of socio-economic and environmental diagnostic results, and clarification of doubts. As a result, all stakeholders who could potentially be impacted in any way, either directly or indirectly, by the Pantanal REDD+ Program were informed and consulted. Further information about the consultation process can be found in Sections 2.3.5 and 2.3.10.
Outcome of FPIC process	The stakeholder consultation process took place through participatory meetings, group dynamics and the application of a questionnaire. Participants were able to express their opinions openly at the meetings or anonymously through the established communication channels. The questionnaire applied at the end of the feedback meeting served to enable the project proponent to map out the participants' (farm employees and landowners) understanding of the REDD+ Program. The questions were related to the implementation of the Program, impacts and benefits, communication channels used, suggestions and criticisms, as well as declaring support or opposition to the Program. When analyzing all the results of these dynamics, no opposition to the implementation of the Pantanal REDD+ Program was identified. It is essential to point out that the Program's activities do not encourage land invasion, the displacement of people without consent and do not force the physical or economic displacement of the parties.

2.5.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

Process used to design the benefit sharing plan	As mentioned in Section 2.5.7 above, as of the date of development of this documentation, the Project Area does not encompass properties inhabited by traditional or indigenous communities or by the government for the execution of its activities. However, Biofílica Ambipar Environment, as the sole proponent of the REDD+ Program, plays a crucial role in managing the distribution of the benefits generated, ensuring the participation of both owners and collaborators identified as stakeholders who will be directly affected by the project. This aims to
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	guarantee transparency with regard to the financing and costs of the project, as well as the distribution of benefits. In this sense, the proponent is in the final stages of drawing up its benefit-sharing plan, in line with the contractual agreements established with the Program's owners. This description will be contained in the document "Governance of the REDD+ Program in the Pantanal", which will be shared with the VVB on a confidential basis.
Summary of the benefit sharing plan	The proponent is in the final stages of drawing up its benefit-sharing plan, in line with the contractual agreements established with the Program's owners. This description will be contained in the document "Governance of the REDD+ Program in the Pantanal", which will be shared with the VVB on a confidential basis as it deals with the parties' financial information, which is considered commercially sensitive.
Approval and dissemination of benefit sharing plan	Once the structure of this plan has been completed, it will be presented to the partners (owners) in joint meetings, with the aim of capturing suggestions, clarifying doubts, addressing criticisms and identifying points that can be improved in the outlined strategy. Once approved, the documentation will be shared with the Program's current partners, and the proponent will take responsibility for storing and updating the mechanism as necessary. Each time a new partner joins the project, it will be presented to the stakeholders.

2.5.9 Property Rights Protection (VCS, 3.18, 3.19; CCB, G5.3)

The partners of the Pantanal REDD+ Program have the right to use their respective farms, as they are private properties, and they possess all the documentation proving land ownership. No activity related to the Project will result in the removal or involuntary relocation of Rights Holders from their lands or territories, nor will it force them to relocate activities important to their culture or livelihoods.

Additionally, as identified in the socio-economic diagnosis, although there are resident employees on the farms, these populations do not inhabit the forest or forested savannah areas that are determined as the project area. Thus, it is understood that there is no direct dependence on these workers for their subsistence on the project areas. In any case, there has been no plan or intention to remove or relocate these employees at any instance of the project.

2.5.10 Illegal Activity Identification (VCS, 3.19; CCB, G5.4)

According to the stakeholder survey, the most likely illegal activities that could occur in the Project Area are: Illegal deforestation and degradation; Trespassing; Unnatural fires; Illegal hunting and fishing. In

order to prevent and mitigate these actions, the project proposes the following measures in line with the proposed activities (detailed in Section 2.1.17):

Remote sensing aligned with physical presence on the properties: using remote sensing technologies, such as satellite images, it is possible to quickly detect any suspicious deforestation activity, allowing an immediate response by the team present in the field on the properties. This action not only allows for the early identification of illegal activities, but also, by being carried out by trained and equipped teams, it is possible to increase the effectiveness in detecting and responding to any suspicious activity. This strategic approach allows for wider coverage of the property, ensuring that vulnerable areas are proactively protected, while any invasions are promptly identified and contained, thus preventing the expansion of illegal deforestation and guaranteeing the security of the farm's assets.

Strengthening the capacity for prevention and response to non-natural fires on the properties: Through the use of remote sensing tools, such as fire hotspot alerts, it is possible to detect almost real-time indications of fire, allowing for a rapid and efficient response. Additionally, encouraging the creation and/or maintenance of fire brigades on farms is crucial to ensure an immediate and coordinated response in case of emergency, keeping trained personnel equipped to effectively combat the fire. Additionally, supporting the development of more robust communication channels between neighbors is equally important, as it allows for quick exchange of information and mutual support in crisis situations, facilitating the coordination of efforts to contain and control fires. It also serves as a valuable ally in disseminating environmental education and awareness actions and serves as a channel for reporting illegal or harmful activities to properties in the region. Together, these actions not only strengthen the prevention and response capacity of properties in case of non-natural fires but also contribute to reducing environmental and economic damages resulting from these events.

Promotion of environmental education and appreciation of Pantanal fauna: Although not a common practice on Program properties, for the prevention and mitigation of illegal hunting and fishing, the Program plans to conduct awareness campaigns with employees and the surrounding community about the negative impacts of illegal hunting and fishing. The goal is not only to inform about the environmental and legal consequences of these activities but also to raise awareness about the importance of conserving local biodiversity. By involving property employees and the neighboring community, these actions promote a culture of respect for environmental laws and property rights, hoping to reduce the incidence of illegal activities and protect natural resources in the areas.

Establishing solid partnerships with environmental enforcement agencies: These partnerships can vary in their forms and levels of collaboration between properties, being essential for preventing all risks identified by the project. A fundamental measure is the sharing of information, facilitating the early detection of suspicious activities such as deforestation, invasions, and illegal hunting. In other words, when illegal activity is identified in the Project Area, enforcement agencies should be notified immediately so that appropriate actions can be taken. Additionally, training and capacity-building activities on the properties can strengthen the skills in identifying and reporting illegal practices, so that individuals will be able to identify, document, and report illegal activities more efficiently. The establishment of these partnerships benefits not only the property owners but also the entire region where the properties are

located, strengthening law enforcement and significantly improving the capacity to prevent and combat a variety of illegal activities threatening the region.

As indicated in Section 2.3.19, the Pantanal REDD+ Program condemns child labor, protecting the rights of children and adolescents, in line with the Diversity and Inclusion Policy of the Ambipar Group. Furthermore, concerning any activity related to this Program, forced labor and/or slavery-like practices are strictly prohibited on any property participating in the defined activities.

2.5.11 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

In the area designated for the Pantanal REDD+ Program, no ongoing or unresolved conflicts or disputes over land, territories, and resources have been registered, nor have there been any disputes resolved in the past twenty years.

Therefore, all partner properties have legitimate owners and possessors as described in the relevant documents. As such, there is no impact of this Program regarding conflicts or disputes, as they do not exist.

In the event of future disputes over land ownership, the project will implement activities to assist in their resolution and to clarify emerging claims, in order to pacify the situation and allow for the consolidation of the developments. In other words, a conflict management procedure will be implemented, as described in Section 2.3.15 Feedback and Grievance Redress Procedure.

2.5.12 Approvals (CCB, G5.7)

Currently, in Brazil, there are no national or jurisdictional REDD+ policies or a competent authority to address projects under Article 6 of the Paris Agreement. Therefore, in the country, the National Designated Authority (DNA), represented by the Interministerial Commission on Global Climate Change (CIMGC), only deals with approvals for projects registered under the Clean Development Mechanism (CDM), so projects in the voluntary market are outside its scope. Additionally, the properties included in the Pantanal REDD+ Program are private lands and, as such, do not require approval from the host country or local governments to be developed as a conservation project.

However, the project proponent is always attentive to new information, actively participating in discussions in federal and state government forums to contribute to the formulation of these policies and regulations, being readily available to adapt the project to newly established rules.

Nevertheless, the Pantanal REDD+ Program sought approval from stakeholders, organizing presentations and consultations with formal and traditional authorities in Mato Grosso do Sul, such as SEMADESC (State Secretariat for Environment, Development, Science, Technology, and Innovation) and IMASUL (Mato Grosso do Sul Institute of Environment), as well as consulting with other state and federal institutions such as Sebrae MS, Ibama, Institute for the Conservation of Wild Animals – ICAS, and FUNAI

CR Campo Grande. All meetings, gatherings, and events were of paramount importance for the validation and approval of the conception, development, and implementation of the Program by the proponent, communities, and other stakeholders; ensuring transparency and active participation of all mentioned actors. It is worth mentioning that all information related to these events is formally recorded.

2.5.13 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB G5.9)

2.5.13.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes ☒ No

The Pantanal REDD+ Program has not received and does not seek any other form of environmental or social credit, climate credit, community credit, or biodiversity unit other than submission to validation and verification standards under VCS (Verified Carbon Standard) and CCBS (Climate, Community and Biodiversity Standard). Currently, there is no national or international REDD+ regulatory regime applicable to the REDD+ Pantanal Program.

2.5.13.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

The Pantanal REDD+ Program has not been submitted for validation/verification under any other greenhouse gas (GHG) program.

2.5.13.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

The Pantanal REDD+ Program has not been submitted for validation/verification under any other GHG program. Therefore, it has not been rejected by any other GHG program.

2.5.14 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.5.14.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

Brazil is not a country that is part of Annex 1 of the Kyoto Protocol and, therefore, does not have commitments to reduce greenhouse gas emissions under the Convention. Furthermore, the Pantanal REDD+ Program has no projects related to the generation of carbon credits under the CDM or any other regulatory scheme within the project area.

2.5.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

The Pantanal REDD+ Program does not hold or intend to generate any type of environmental credit related to the reduction of greenhouse gas emissions or claimed removals beyond the VCS Program. The Program aims to generate benefits for the climate, communities, and biodiversity, but only the net reductions and removals of greenhouse gases, i.e., VCUs, will be commercialized after being properly registered through the Verra registry platform.

In terms of double counting at the national level, current Brazilian regulations do not require any voluntary carbon market project to be registered with local or federal authorities. However, there are currently several bills under consideration in the National Congress that aim to establish the Brazilian Regulated Carbon Market and regulate issues related to the purchase and sale of carbon credits in the country. The promotion of a Brazilian Emission Reduction Market is provided for in the Law that established the National Policy on Climate Change (Law No. 12,187, of 12/29/2009).

The proponent will follow the necessary rules and processes to ensure that the Program's credits are not double counted.

2.5.14.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes☒ No

It is not planned for Pantanal REDD+ Program activities to impact emissions from goods and services in any supply chain.

2.6 Additional Information Relevant to the Project

2.6.1 Leakage Management (VCS, 3.11, 3.15)

Considering that the project activities will focus on reducing GHG emissions caused by planned deforestation, the Leakage Management Plan will consider monitoring other private areas within the same property of the participants in the REDD+ Pantanal Program. This activity is part of the "Containment of deforestation expansion" theme described in Section 2.1.17. Below is the planning for this action:

Implementation of Remote Sensing Technologies:

- Use remote sensing systems to effectively monitor the areas of interest in the Leakage Belt.
- Establish deforestation alerts to quickly identify any suspicious deforestation activity, allowing for an immediate response with the landowner.
- Regularly monitor the expansion of pasture areas on the properties within the Leakage Belt.

Diversification of Payment for Environmental Services (PES) Models (PSA in Portuguese):

- Encourage the participation of properties located in the Leakage Belt in the REDD+ Pantanal Program.
- Stimulate the implementation of different PES models for forest and forest savannah areas within the Leakage Belt, aiming to value environmental conservation and incentivize program participants to adopt sustainable practices.

Stakeholder Engagement:

- Actively involve landowners in the leakage management process, ensuring their participation and support.
- Promote transparency by providing updated information on activities carried out on the properties, ensuring the conservation of leakage areas.

Leakage Response:

- Establish, with landowners, response procedures to deal with leakage occurrences, as well as communication regarding the purchase of new properties to open forest and forest savannah areas.

By implementing this leakage management plan, the main goal is to prevent the misuse of financial resources and contribute to reducing legal deforestation rates in the Pantanal biome. Additionally, it is not expected that leakage management activities will include activities that increase carbon or GHG reserves compared to reference scenarios in leakage areas. However, if these activities are implemented during the project period, their emissions will be monitored and accounted for as described in Section 3.3.3.

2.6.2 Further Information

Not applicable.

At the time of completing this documentation, there is no additional information beyond what is described in this document that is relevant in terms of legislative, technical, economic, sectoral, social, environmental, geographic, site-specific, and/or temporal aspects that could influence the project's eligibility, greenhouse gas emissions reductions or carbon dioxide removals, or the quantification of the project's reductions or removals.

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology (VCS, 3.1)

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF)	1.6
Module	VMD0001	Estimation of carbon stocks in the above and belowground biomass in live tree and non-tree pools (CP-AB)	1.1
Module	VMD0002	Estimation of carbon stocks in the dead wood pool (CP-D)	1.1
Module	VMD0006	Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL)	1.3
Module	VMD0009	Estimation of emissions from activity shifting for avoiding planned deforestation/forest degradation and avoiding planned wetland degradation (LK-ASP)	1.3
Module	VMD0013	Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB)	1.2
Module	VMD0015	Methods for monitoring of greenhouse gas emissions and removals (M-REDD)	2.2

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Module	VMD0016	Methods for stratification of the project area (X-STR)	1.2
Module	VMD0017	Estimation of uncertainty for REDD+ project activities (X-UNC)	2.2
Tool	-	CDM Tool for testing significance of GHG emissions in A/R CDM project activities (T-SIG)	1.0
Tool	VT0001	Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities (T-ADD)	3.0
Tool	AFOLU Non-Permanence Risk Tool	Tool for AFOLU non-permanence risk analysis and buffer determination (T-BAR)	4.2

3.1.2 Applicability of Methodology (VCS, 3.1)

Reference ID/Title	Applicability condition	Justification of conformance
VM0007 - All REDD Activity Types	Land in the project area must qualify as forest (as defined by the VCS) for at least 10 years prior to the project start date.	According to satellite images analysis as well as information from the Program farms owners, the Project Area has been classified as forest for more than 10 years.
VM0007 - All REDD Activity Types	Where land within the project area is peatland or tidal wetlands and emissions from the SOC pool are deemed significant, the relevant WRC modules are applied	As shown in the next sections, there are no organic soils (or peatlands) in the Project Area. Furthermore, soil carbon stock was conservatively excluded. Therefore, the project is not subject to VCS WRC requirements.

Reference ID/Title	Applicability condition	Justification of conformance
	alongside other relevant methods	
VM0007 - All REDD Activity Types	Baseline deforestation in the project area falls within either of the following categories: - Unplanned deforestation (VCS category AUDef) - Planned deforestation (VCS category APDef)	The Pantanal REDD+ Program falls into the category of avoided planned deforestation (VCS APD Category).
VM0007 - All REDD Activity Types	Leakage prevent activities must not include: - Flooding agricultural lands to increase production (e.g. rice paddies) and/or; - Intensifying livestock production through use of feed-lots and/or manure lagoons	None of the Program's containment and leakage activities include actions to flood agricultural land or create containments.
VM0007 - Avoiding Planned Deforestation/ Degradation	Avoiding planned deforestation activities area applicable under the following condition: Where conversion of forest lands to deforested condition is legally permitted.	According to Brazil and state of Mato Grosso do Sul current legislation, the Pantanal REDD+ Program area meets all applicable legal requirements and can be legally deforested.

3.1.3 Project Boundary (VCS, 3.12)

Source		Gas	Included?	Justification/explanation
Baseline	Burning of woody biomass	CO ₂	Included	Carbon stock decreases due to burning are accounted as a carbon stock change
		CH ₄	Excluded	According to REDD-MF is conservative to exclude Non-CO ₂ gases emissions

Source	Gas	Included?	Justification/explanation
Project	N ₂ O	Excluded	According to REDD-MF is conservative to exclude Non-CO ₂ gases emissions
	Combustion of fossil fuels	CO ₂	According to REDD-MF is conservative to exclude CO ₂ emissions from fossil fuels combustions and potential emissions from CH ₄ and N ₂ O are negligible. In addition, E-FFC module confirms that.
		CH ₄	
		N ₂ O	
	Use of fertilizers	CO ₂	There is no use of fertilizers in the baseline. In addition, according to REDD-MF potential emissions are negligible and/or is conservative to exclude.
		CH ₄	
		N ₂ O	
	Burning of woody biomass	CO ₂	Justification similar to that presented in the baseline
		CH ₄	Must be included if fire occurs
		N ₂ O	Must be included if fire occurs
Project	Combustion of fossil fuels	CO ₂	Justification similar to that presented in the baseline
		CH ₄	
		N ₂ O	
	Use of fertilizers	CO ₂	Justification similar to that presented in the baseline
		CH ₄	
		N ₂ O	

Geographical Boundaries - Property Areas

The Pantanal REDD+ Program covers an extensive area of 105,936 hectares, comprising nine rural properties (Table 4). The project is characterized by a fragmented land ownership structure, consisting of a mosaic of rural land parcels. These parcels of land, although under different management, share a common vision of conservation and sustainable use of natural resources.

The Rural Environmental Registry (CAR) and the Land Management System (SIGEF) are two crucial tools in the context of environmental and land management of rural properties in Brazil. While the CAR is a mandatory tool aimed at the environmental regularization of rural properties, the SIGEF plays a complementary role by integrating information from the rural cadaster administered by the National

Institute for Colonization and Agrarian Reform (INCRA) with the rural registry maintained by the notary system.

A crucial aspect in understanding the interaction between the Rural Environmental Registry (CAR) and the Land Management System (SIGEF) is the possibility of a single property registered in the CAR covering several registrations recorded in the SIGEF. This feature reflects the complexity and diversity of rural land structures in Brazil, where a single property can be made up of several parcels of land, each with its own registration in the land registry.

The importance of this multifaceted relationship between CAR (Rural Environmental Registry) and SIGEF (Georeferenced Rural Properties System) stands out primarily in environmental management and the regularization of rural properties. While CAR focuses on identifying and registering the environmental characteristics of a property, SIGEF ensures the accuracy of information regarding ownership and physical delineation of these properties through their registrations. Thus, a rural property registered in CAR, representing a unit of environmental management, can be legally divided into several registrations in SIGEF, reflecting its land composition.

CAR aims to map and regularize areas of rural properties, focusing especially on environmental aspects such as the delimitation of Permanent Preservation Areas (APPs), Legal Reserves (RL), and other areas of environmental interest. This regularization is essential for the sustainable management of natural resources and is a prerequisite for obtaining a license for planned deforestation.

In the selection process, some properties were delineated with the entirety of the rural property as registered in CAR, while others were based only on certain plots identified by SIGEF. However, it is important to note that, even in properties whose delineation was based on SIGEF registrations, it was necessary to obtain the respective CAR for each of them. This procedure was essential for the precise delineation of Permanent Preservation Areas and Legal Reserves, fundamental for the advancement and environmental compliance of the project. Properties C7-P01-Y24 and C6-P01-Y24 were delineated according to SIGEF, with the former having only one included registration and the latter having three. The remaining properties were delineated with CAR. However, it should be noted that for C4-P01-Y24, the entire property was delineated considering four CAR registrations.

Table 4. Property code, property area and delimitation source

ID	Area (ha)	Delimitation source	Quantity
C1-P01-Y24	33,474	CAR	1
C2-P01-Y24	1,069	CAR	1
C3-P01-Y24	7,144	CAR	1
C4-P01-Y24	9,629	CAR	4
C5-P01-Y24	33,539	CAR	1
C5-P02-Y24	1,027	CAR	1
C6-P01-Y24	17,875	SIGEF	3

C7-P01-Y24	219	SIGEF	1
C7-P02-Y24	1,960	CAR	1
Total	105,936		

The properties belonging to the Pantanal REDD+ Program are distributed between two Brazilian biomes of immense ecological importance: mostly (10,038 ha) in the Pantanal and to a lesser extent in the Cerrado (3,897 ha), in a transition zone to the Pantanal. The Pantanal, representing 96.32 % of the extension of rural properties. The Pantanal is one of the largest continuous wetlands on the planet and is known for its biodiversity and concentration of aquatic fauna³⁷. The Cerrado, on the other hand, makes up the remaining 3.68% of properties. The Cerrado is a tropical savannah biome, known for its diversity of flora and fauna, as well as being one of the most biodiverse savannah regions in the world³⁸.

Project Area (VM0007)

The methodology applied to define the Project Area (AP) involves a detailed and rigorous approach, in line with VERRA standards. The definition of forests and savannahs follows the FAO classification, which considers forests to be areas of more than 0.5 hectares, with trees greater than 5 meters in height and a canopy cover of more than 10%, excluding areas predominantly under agricultural or urban use.

MapBiomass³⁹ is a collaborative initiative involving organizations from the science, technology and academic sectors, with the aim of mapping land cover and land use in Brazil. Using satellite images and advanced data processing algorithms, MapBiomass offers annual historical series of land use and land cover maps, from 1985 to 2022. This tool provides valuable data for monitoring changes in the Brazilian landscape, including deforestation, forest regeneration, agricultural expansion, and urbanization.

The initiative produces data categorized into various land use and land cover classes, such as forests, savannahs, agricultural areas, pasture areas, bodies of water, urban areas, among others, allowing for a detailed and temporal analysis of the transformations of the territory. The spatial resolution of these maps makes it possible to identify patterns and trends of change on a national, regional and local scale.

The methodology adopted by Mapbiomas to map land use and land cover in the Pantanal biome from 1985 to 2022 involves a series of sophisticated steps. For mapping, Mapbiomas uses decision trees to classify predominant types of native vegetation, such as forests, savannahs, and grasslands. Subsequently, the Random Forest algorithm is employed on stable samples for training and temporal series classification. For annual mosaics, Mapbiomas utilizes surface reflectance images from Landsat satellites, mainly selected during the dry season to minimize confusion in identifying flooded areas and different vegetation classes. To achieve this, the Pantanal is divided into seven regions based on drought patterns, sub-basins, and vegetation distribution, aiming to reduce noise in classification. Digital

³⁷ <https://www.embrapa.br/pantanal/apresentacao/o-pantanal#:~:text=O%20Pantanal%20%C3%A9%20uma%20das,bacia%20hidrogr%C3%A1fica%20do%20Alto%20Paraguai>

³⁸ <https://educa.ibge.gov.br/jovens/conheca-o-brasil/territorio/18307-biomas-brasileiros.html>

³⁹ <https://brasil.mapbiomas.org/o-projeto/>

classification is then conducted into subsets of 8 main classes, using stable and complementary samples for training. A set of 38 variables is defined for classification in the Pantanal, utilizing the Random Forest algorithm. Temporal and spatial filters are applied to correct classification errors and invalid class transitions, based on temporal series analysis. Finally, the resulting map is integrated with cross-cutting theme maps, resulting in final biome maps with 13 distinct classes. The overall accuracy of Mapbiomas land use class maps is evaluated at 85.9%.

MapBiomas methodology identifies various land use and land cover classes, among which some can be classified as forests or partially forests according to the FAO definition. Among these classes, forests and savannahs stand out for the Pantanal. These classes are on non-floodable lands. Additionally, areas classified as forests are characterized by a significant tree canopy density, generally forming a continuous canopy. Forests typically have denser biomass with larger trees. On the other hand, savannahs are marked by the presence of scattered trees, combined with grasses and shrubs, resulting in a biomass density usually lower than that found in forests. Vegetation stratification is crucial as different vegetation compartments present distinct carbon stocks.

Different stratifications were necessary to achieve the ideal accuracy of greenhouse gas emissions or net removal estimates for the project. Module VMD0016 (X-STR) was employed to enhance the accuracy of carbon stock estimates and greenhouse gas emissions (GHG). This methodology allowed the stratification of the project area into discrete units, reflecting intrinsic variations in vegetation and biomass between savannah and forest. Stratification categorization within the project was carried out with the support of MapBiomas data (collection 8).

MapBiomas from the year 2021 was used to exclusively select areas classified as forest and savannah, excluding all other land use and land cover categories for the same year. This careful selection allowed the creation of two strata for the project area, initially referred to as Forest and Savannah. Mapbiomas was adopted for its data due to its ability to discern the subtleties of vegetation cover, being proven accurate and widely used in literature (PEDREIRA; CRUZ, 2023⁴⁰; SANTOS, 2023⁴¹; FREITAS; CAVALCANTI; BRAZ, 2023⁴²).

⁴⁰PEDREIRA¹, João Pedro das Neves Cardoso; CRUZ¹, Carla Bernadete Madureira. AVALIAÇÃO DA ACURÁCIA TEMÁTICA DO MAPBIOMAS PARA A CLASSIFICAÇÃO DE ÁREAS VERDES INTRAURBANAS. 2023.

⁴¹ SANTOS, Rosângela Leal et al. ANÁLISE DO USO E OCUPAÇÃO DO SOLO NA REGIÃO CACAUEIRA PÓS-VASSOURA DE BRUXA: USO APLICADO DO MAPBIOMAS. 2023.

⁴² FREITAS, Leonardo Cristiano da Silva; CAVALCANTI, Lucas Costa de Souza; BRAZ, Adalto Moreira. AVALIAÇÃO DA ACURÁCIA DAS CLASSES DE USO E COBERTURA DA TERRA DO MAPBIOMAS (COLEÇÃO 6) PARA O MUNICÍPIO DE CARPINA (PE). Caderno Prudentino de Geografia, v. 45, n. 2, p. 38-52, 2023.

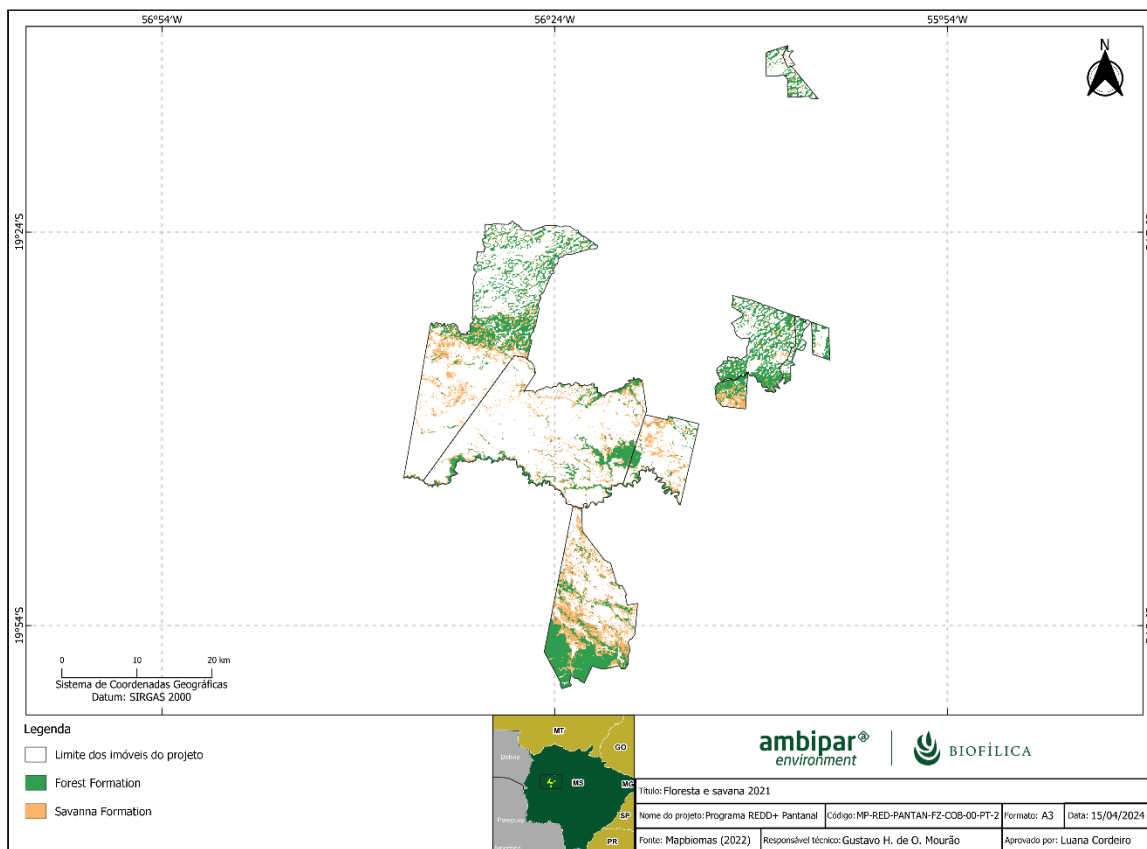


Figure 4. Areas of Forest and Forested Savannah identified within the properties participating in the Pantanal REDD+ Program in 2021

APD projects require the stability of areas with tree cover. It is essential that these areas have remained as forest for at least a decade, with no deforestation occurring during this period. To verify this stability, we used the 2011 land cover maps from Mapbiomas to compare with the delimited areas of 2021. The analysis focused on the forest and savannah classes and consisted of comparing the evolution of these classes over time. An intersection technique was used, keeping only areas where the classification remained unchanged between the two moments, remaining consistently as forest or savannah. Areas that met this stability criterion were then considered "stable forests" and qualified for inclusion in the project.

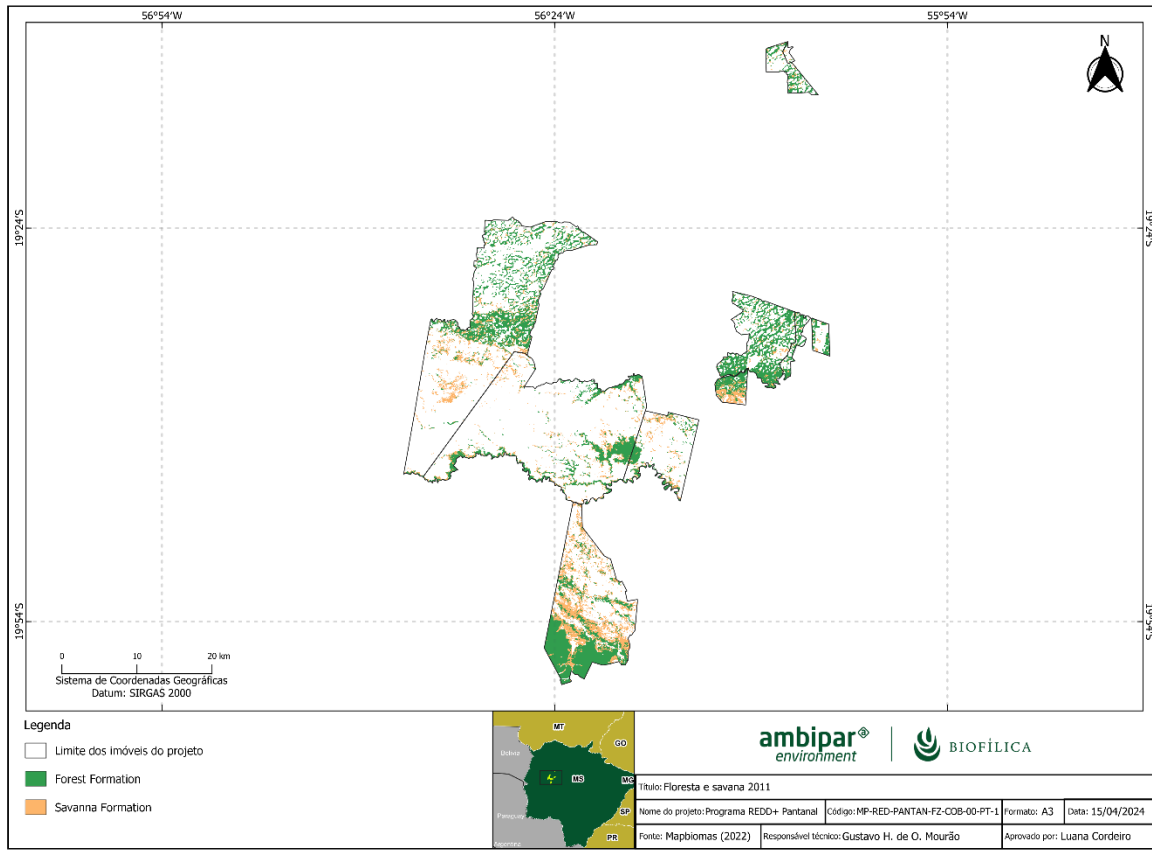


Figure 1. Areas of Forest and Forested Savannah identified within the properties participating in the Pantanal REDD+ Program in 2011

The savannah stratum presents variations in its floristic composition, not always being considered as forests. To select suitable savannah areas for the project, we established an inclusion criterion based on a minimum tree cover of 10% within a 100 m² analysis window.

The Tropical Tree Cover dataset provides detailed information on the extent and tree cover in the tropics, covering an area of 4.3 billion hectares with a spatial resolution of 10 meters and tree cover data at a scale of 100 square meters⁴³. Developed by the World Resources Institute (Brandt et al., 2023⁴⁴), this dataset generates a binary layer distinguishing areas with and without trees. The methodology employs convolutional neural network models on Sentinel images, with annual updates from 2017 to 2020. This resource adopts specific definitions for "tree" and "tree cover," classifying woody vegetation over 5 meters in height, or between 3 and 5 meters with a minimum canopy diameter of 5 meters, as trees. The binary layer maps the presence of trees, reflecting local forest cover.

⁴³ Dados disponíveis em <https://data.globalforestwatch.org/datasets/tropical-tree-cover/explore>

⁴⁴ Brandt, J., Ertel, J., Spore, J., & Stolle, F. (2023). WALL-to-wall mapping of tree extent in the tropics with sentinel-1 and sentinel-2. *Remote Sensing of Environment*, 292, 113574. <https://doi.org/10.1016/j.rse.2023.11357>

We used the Tropical Tree Cover classification to filter eligible savannahs for the project. The "stable forests" data related to savannahs were intersected with information from the Tropical Tree Cover to refine the selection of areas for the APD project. This data intersection process aimed to identify and retain only those savannah areas with a tree cover exceeding 10%. The use of these two robust and complementary datasets ensured a more precise and reliable selection of savannah areas suitable for inclusion in the APD projects, ensuring that only regions with significant tree cover are considered in the project. This stratum is then considered as Forested Savannah.

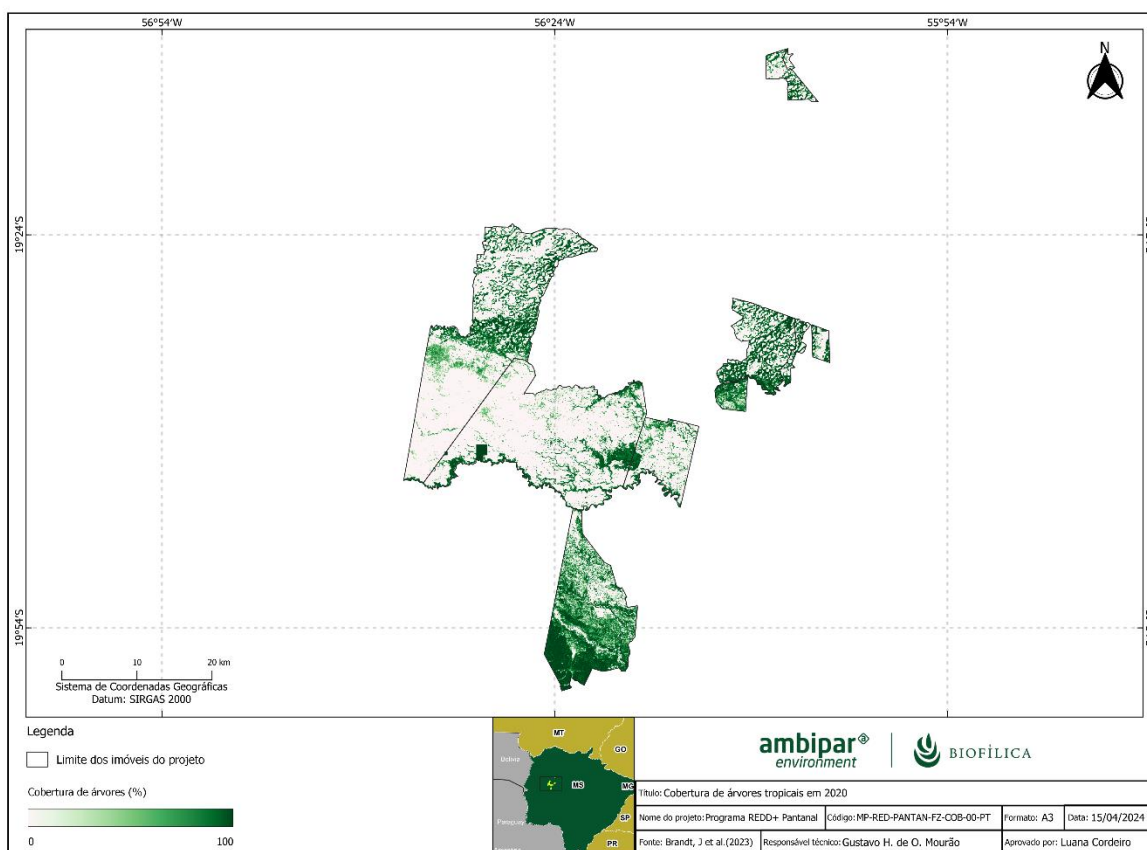


Figure 6. Tropical tree cover in 2020 on properties participating in the Pantanal REDD+ Program

To ensure that forests can be cleared legally, following all Brazilian regulations, areas delimited as protected must be removed from the project. All the exclusions of forests that cannot be legally cleared will be detailed below.

The Legal Reserve is a protected area, required by law, within a rural property, intended for the preservation and restoration of native vegetation. These areas have significant restrictions on their use, with the aim of protecting natural resources, contributing to the conservation of biodiversity and helping to create balance. Due to these restrictions and their environmental conservation purpose, Legal Reserve areas are not suitable for APD projects.

Permanent Preservation Area (APP) is defined by Law 12.651/2012 as a protected area, which may or may not be covered by native vegetation, designed to preserve water resources, the landscape, geological

stability and biodiversity. Its additional environmental functions are to facilitate the gene flow of fauna and flora, protect the soil and ensure the well-being of human populations. APPs are fundamental to environmental conservation, playing a crucial role in maintaining natural ecosystems and supporting life. Deforestation permits are prohibited for these areas.

The Legal Reserves (RL) and Permanent Preservation Areas (APP) of the properties involved in the project were identified using the records in the Rural Environmental Registry (CAR). These areas were used as exclusion criteria in the process of delimiting the project area. Consequently, forests and wooded savannahs contained within the RL and APP were excluded from the project area.

Private Natural Heritage Reserves (RPPNs) are categorized as Conservation Units (UC) and are located on private land. These reserves are established through the voluntary initiative of rural landowners, whether they are individuals or companies. RPPNs are recognized and validated by the government, in a process that demonstrates the owners' commitment to environmental preservation. RPPNs play a crucial role in conserving biodiversity and maintaining natural ecosystems, contributing significantly to environmental conservation efforts at a national level.

Although they often share boundaries with Legal Reserves, RPPNs have their own characteristics and exclusive areas that needed to be identified and removed from the study. For this reason, the forests and wooded savannahs contained in the Private Natural Heritage Reserves (RPPNs) were removed from the project.

An area located in C6-P01-Y24 was excluded from the analyses related to the REDD+ Pantanal Program's APD project. This exclusion is justified by the contractual agreement signed between the parties. Removing this region from the scope of the project was a critical decision in order to maintain the integrity and validity of the data.

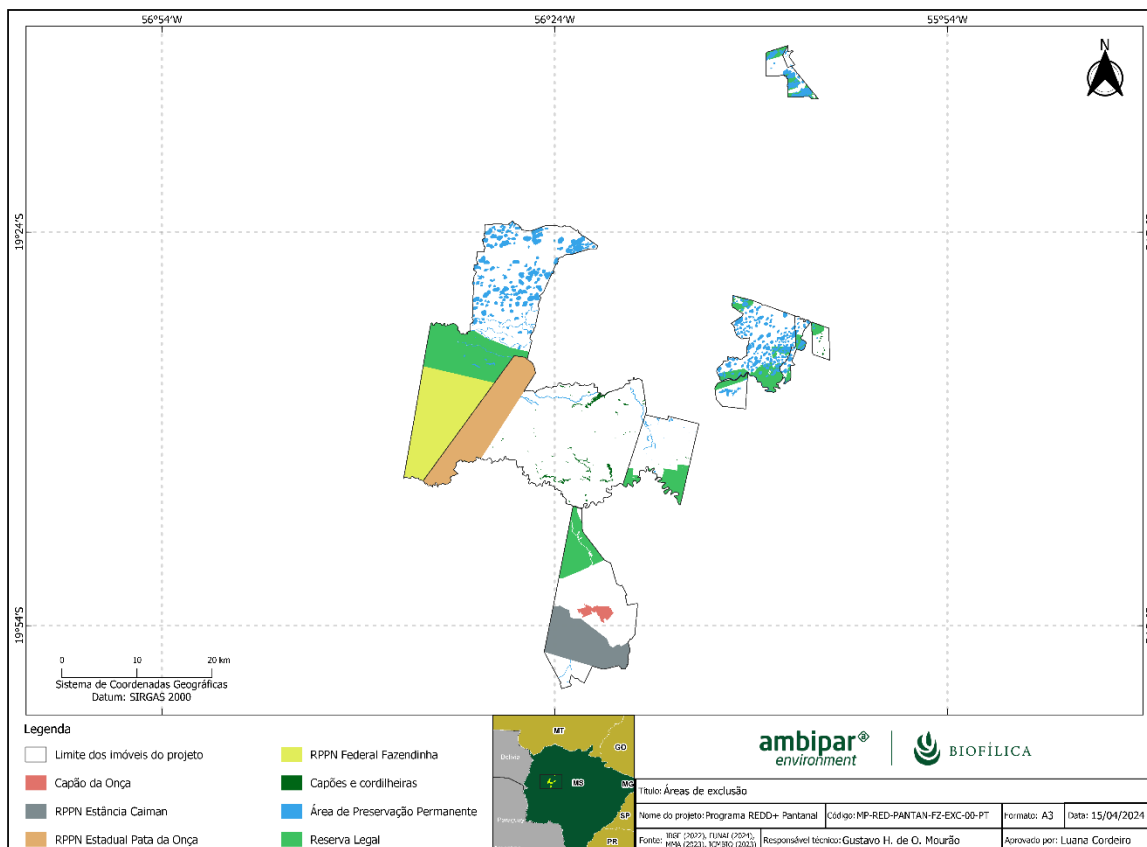


Figure 7. Areas excluded in the Project Area design process in accordance with current legislation

Ecological corridors are essential for maintaining biodiversity and ecological integrity in development projects involving land use. These areas act as vital connections between fragmented habitats, allowing for the migration and dispersal of species and the maintenance of essential ecological processes. This ensures connectivity between protected areas such as Legal Reserves, Permanent Preservation Areas and Private Natural Heritage Reserves (RPPNs).

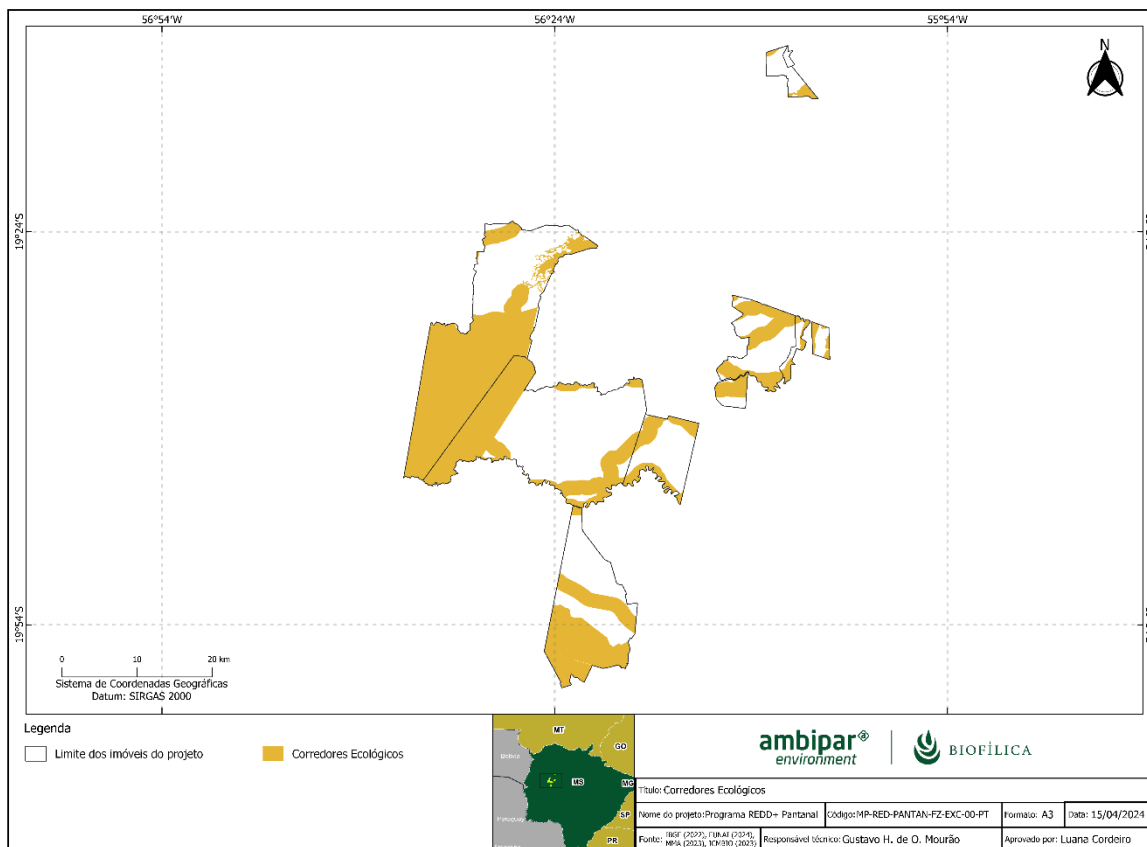


Figure 8. Ecological corridors defined by Law N° 6,160 of December 18, 2023 within the properties participating in the Pantanal REDD+ Program.

In implementing the project areas, current legislation was respected, excluding areas with the presence of “capões”, “cordilheiras”, ecological corridors and other limits established by Law N° 6.160 of December 18, 2023, which determines the protection of these structures in the Pantanal Plain Restricted Use Area (AUR-Pantanal) of the State of Mato Grosso do Sul.

These areas are considered to be specially protected and are not subject to the suppression of native vegetation, except in the cases provided for by law. However, on properties where the protected areas covered 60% or more of the property, the same law allows for a review of these restrictions, granting a soil alternative of up to 40% (forty percent) of the property's area. Thus, according to Decree No. 16.388 of February 16, 2024, some areas of the properties could be reconsidered for inclusion in the project. This decree regulated the use of up to 40% of the areas of mountain ranges, portions of biodiversity corridors and capões with an area of less than 3 hectares, as long as the criteria and limits imposed were respected, such as the preservation of a minimum width of 500 meters for ecological corridors and the maintenance of APP areas.

In the project, the configuration of the ecological corridors was adjusted to allow properties to have at least 40% of their territory available for alternative land use. This measure is in line with Decree N° 16.388, which authorizes the reconsideration of previously restricted areas, such as part of biodiversity

corridors, capons smaller than 3 hectares and mountain range areas. The application of this rule guarantees compliance with legal requirements, including the preservation of a minimum width of 500 meters for ecological corridors and the maintenance of Permanent Preservation Areas (APPs), promoting balanced land use management.

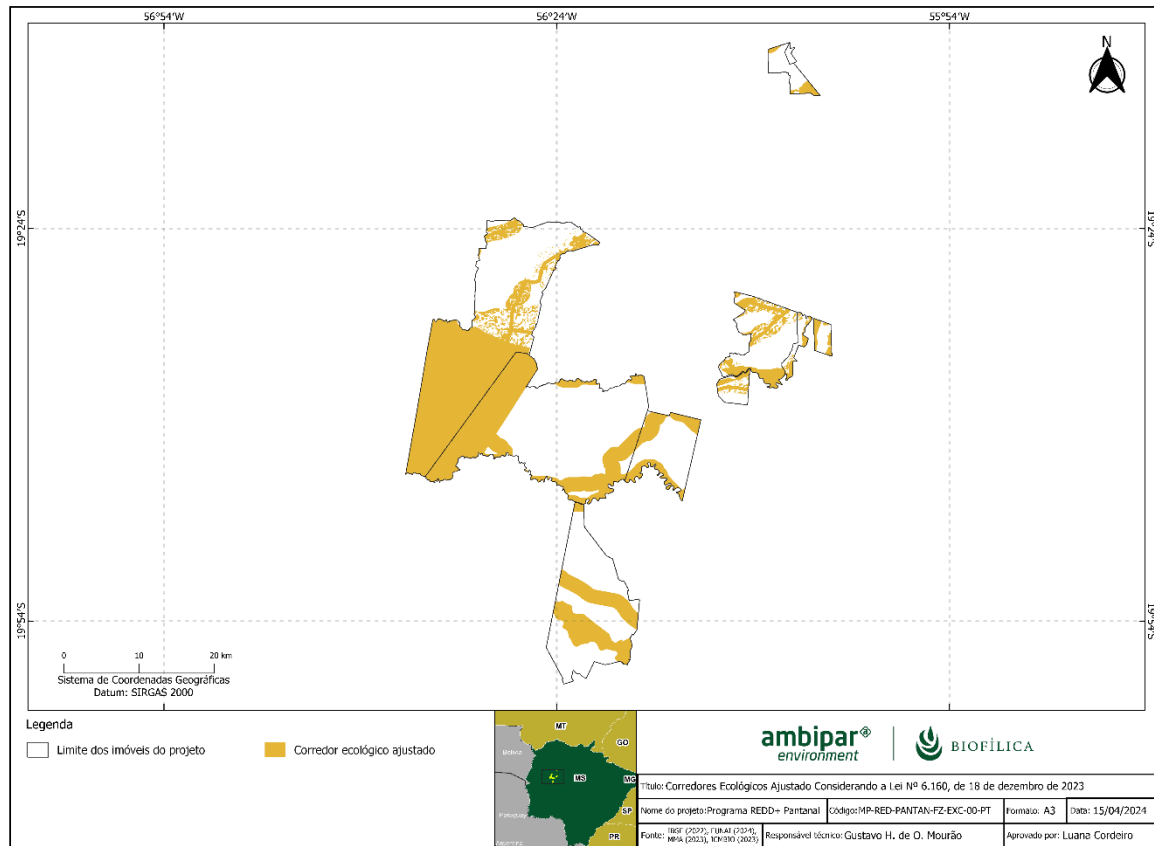


Figure 9. Ecological corridors adjusted according to the guidelines of Law No. 6,160 of December 18, 2023 within the properties participating in the Pantanal REDD+ Program.

After careful analysis and following all the legal regulations, an effective delimitation of protected areas where the suppression of vegetation is prohibited was achieved. Respecting Brazilian environmental laws and the specific guidelines for the Pantanal Plain, areas such as Legal Reserves, Permanent Preservation Areas and Private Natural Heritage Reserves were clearly demarcated and all vegetation contained within them was excluded from the project.

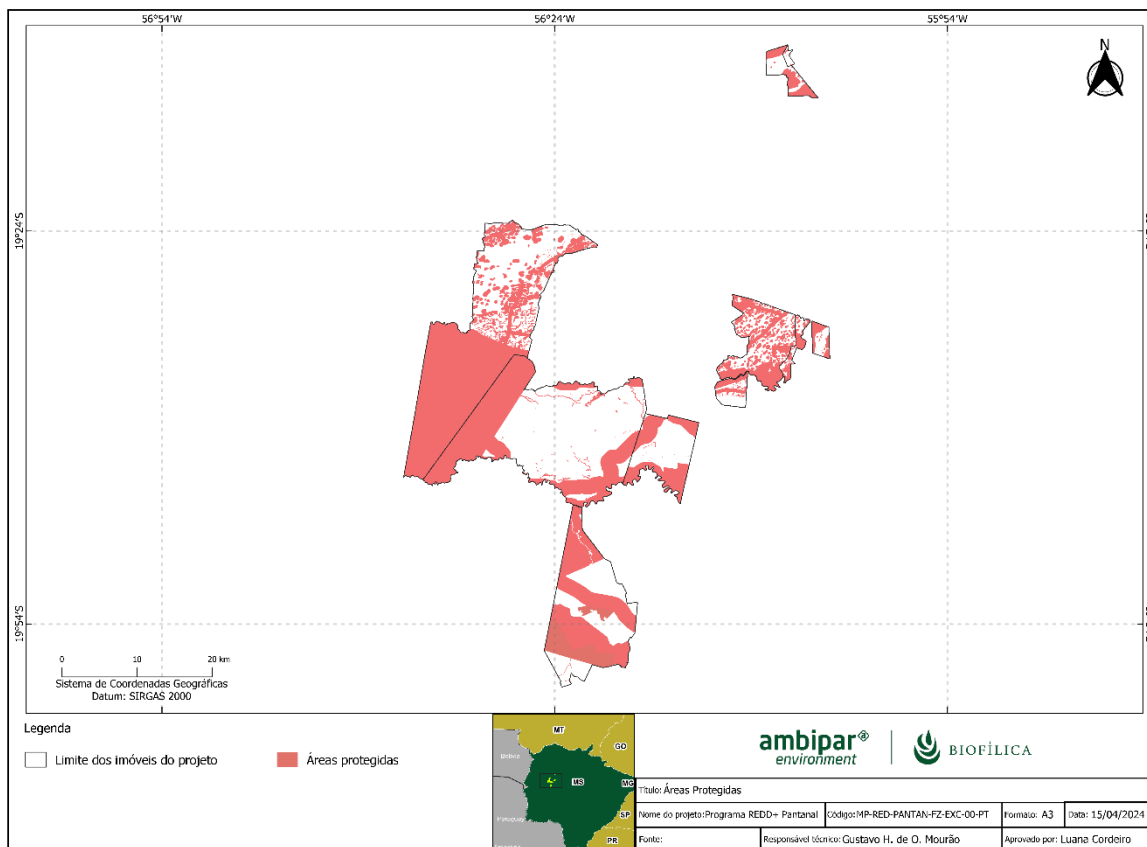


Figure 10. Total area identified as protected within the properties participating in the Pantanal REDD+ Program

After a meticulous process of analysis and exclusion, the REDD+ Pantanal Program project area was effectively delimited. This delimitation resulted in the selection of remaining forest and savannah plots that strictly met the established criteria: all forest and forested savannah regions that had remained stable for a decade, without deforestation, and that were not located in Legal Reserves, Permanent Preservation Areas, Private Natural Heritage Reserves (RPPNs), Capões, Cordilheiras, Ecological Corridors, or in regions planned for future deforestation. In addition, for savannah areas, only those that met the criterion of having more than 10% tree cover were considered in the project. This careful process ensured that the project area encompassed only those regions that effectively aligned with the project's focus on regions with real potential to prevent planned deforestation, excluding sites that could not legally be deforested or where future deforestation was planned.

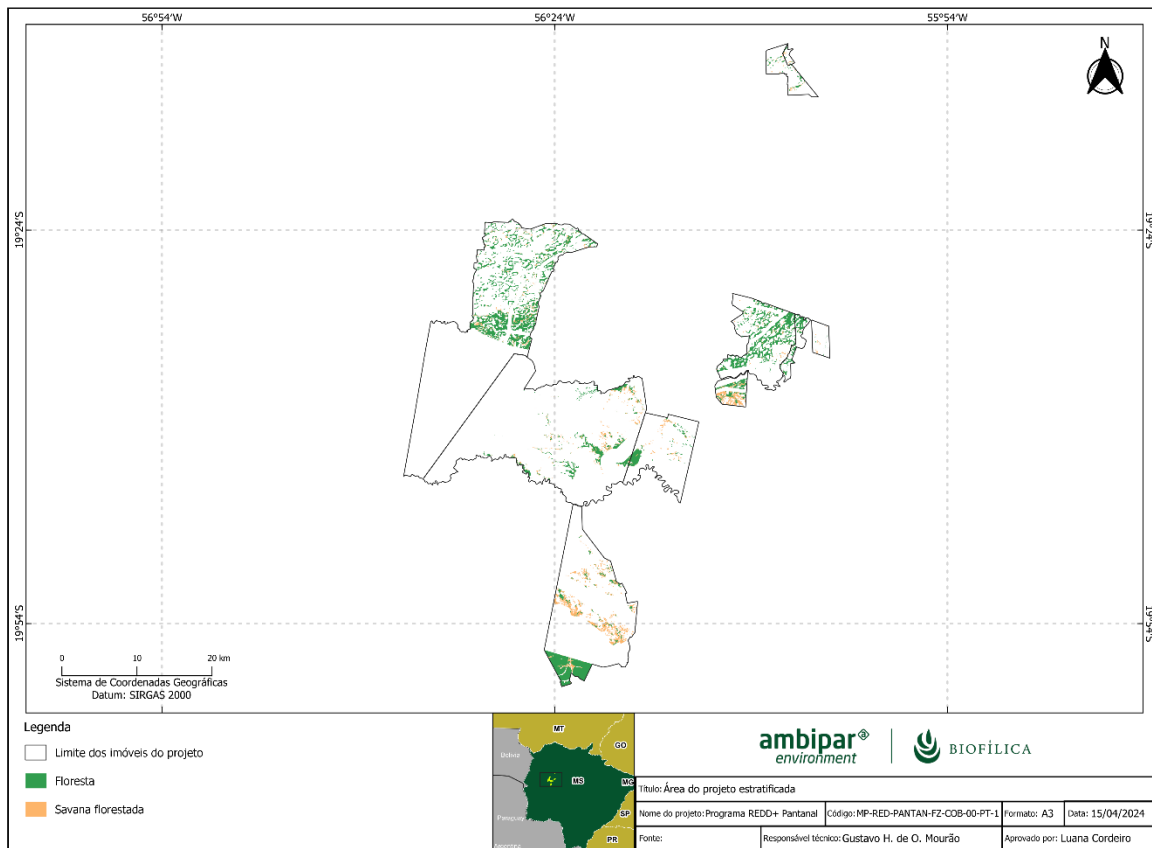


Figure 11. Project area of the REDD+ Pantanal Program within the participating properties stratified as forest and forested savannah.

Table 5. Total Project Area identified by stratum within the Pantanal REDD+ Program in PAIs

Extrato	Área (ha)
Floresta	9,023
Savana Florestada	2,778
Total	11,800

3.1.4 Baseline Scenario (VCS, 3.13)

Stratification of the project area and identification of planned deforestation agent in each stratum

Module VMD0006 v1.3 stipulates that the Project Area be stratified based on the spatial distribution of carbon stocks, reflecting parameters related to forest productivity and activity parameters. Since land use activities are nearly homogeneous in the domain, we stratified the project area based on the type of forest. Therefore, two forest strata were identified: forest and forested savannah.

The classification of land cover was performed based on Mapbiomas Collection 8 data. The inclusion of savannah was refined using degradation and deforestation data from the Global Forest Watch⁴⁵. To include only savannahs with forest cover, a mask was first applied to identify deforested areas in the vicinity of the project area over the past 10 years. Next, we checked which of these pixels had experienced loss of tree cover until 2021, to identify forested savannah areas with past forest cover loss, which we assumed to be biomass loss due to human action. The logic is that non-forested areas do not undergo significant tree cover loss. We then filtered, for the entire project area, only pixels with tree cover in 2021 of 35% or more. For the forest class, all pixels were included.

The REDD+ Pantanal Program has the "simplest scenario" mentioned by Module VMD0006 v1.3, where the agent is an individual, an organization, or a predefined corporation.

In this case, the identified deforestation agent within the strata stipulated in the baseline scenario, indicated in section 3.1.3, are the owners of the farms themselves. This scenario applies to all REDD+ Pantanal Program Implementation Areas (PAIs).

Deforestation area

The area of planned deforestation that will be avoided, i.e. the area where the REDD+ Pantanal Program will be implemented, was defined based on the following factors:

Legal permission for deforestation: The environmental laws in force at the location of the REDD+ Pantanal Program, i.e. in Brazil, the state of Mato Grosso do Sul, in the municipalities of Aquidauana, Corumbá and Miranda, were surveyed and analyzed in order to precisely define the places that can be legally deforested. The laws that provide guidelines in this area are listed below:

Federal Law 12.651/2012 - Brazilian Forest Code:

The Brazilian Forest Code defines, among other factors, areas that must be protected and criteria for requests to suppress vegetation, these being:

Article 12 of this law stipulates that, outside the boundaries of the Legal Amazon (which includes areas within the Pantanal biome), the minimum Legal Reserve percentage that must be maintained by landowners is 20% of the total area of their property.

Article 4 of this law stipulates that a Permanent Preservation Area (APP) is considered to be the marginal strips of any natural perennial or intermittent watercourse, from the edge of the channel of the regular bed, with widths that vary according to the width of these watercourses, as stipulated by law.

This law establishes the "Restricted Use Areas" (Art. 10), which state that ecologically sustainable exploitation is permitted in the Pantanal and Pantanal plains, taking into account the technical recommendations of official research bodies, with new suppressions of native vegetation for alternative land use subject to authorization from the state environmental agency.

⁴⁵ <https://www.globalforestwatch.org/>

Article 26 of this law also states that the suppression of native vegetation for alternative land use will depend on the property's registration with the CAR (Rural Environmental Registry) and prior authorization from the competent state agency of the Sisnama (National Environment System).

Law N. 6.160/2023 – Provides for the conservation, protection, restoration, and ecologically sustainable exploitation of the Restricted Use Area of the Pantanal Plain (AUR-Pantanal), within the scope of the State of Mato Grosso do Sul:

Considering that the properties participating in this Program are located within the boundaries established by the AUR-Pantanal, this law is of fundamental importance for determining the quantities of vegetation that can legally be deforested.

Article 12 of this law establishes that the following areas are specially protected and not subject to projects for the suppression of native vegetation:

I - in the "veredas": in addition to the watercourse, all wet areas and vegetation existing up to 50 (fifty) meters from the upper limit of the wet field, regardless of the type of vegetation existing in this range;

II - in the "landis": all arboreal vegetation covering the watercourse or bordering it, up to its outer limit with the grassland or savannah vegetation;

III - in the "salinas": in addition to the surrounding beach, a marginal strip of 100 (one hundred) meters;

IV - in the "capões" and ridges: in 80% (eighty percent) of the area covered with arboreal-shrub vegetation, as regulated;

V - in low areas, such as bay edges, floodplains, temporary bays, lowlands, marshes, comprised of native pastures with quality forage, identified by the Brazilian Agricultural Research Corporation (Embrapa), as regulated;

VI - the "murundus", present in the murundus fields;

VII - in identified ecological corridors.

Article 14 of this same law states that for the granting of environmental authorizations for the suppression of vegetation or the conversion of native pastures in the AUR-Pantanal, the following prerequisites must be proven:

I - the rural property is regularly enrolled and approved in the Rural Environmental Registry of Mato Grosso do Sul (CAR-MS);

II - the competent environmental agency has not recorded administrative infractions, finally adjudicated within the last 3 (three) years, regarding irregular suppression of native vegetation on the respective property;

III - the management of livestock in native pastures is being conducted in compliance with technical recommendations, aiming at their best performance and to conserve the quality, availability, diversity, and capacity for recovery of these pastures;

IV - the management of livestock in cultivated pastures is being conducted in compliance with technical recommendations, aiming at their best performance and to avoid or minimize their degradation;

V - the cleaning of native and cultivated pastures and the use of fire for grassland vegetation management are being conducted in accordance with criteria established in the environmental licensing;

VI - the legal reserve areas and permanent preservation areas and, if existing, the areas of the Atlantic Forest, as provided for in article 14 of Federal Law No. 11,428, of 2006, are being protected and used in accordance with legal and regulatory provisions.

Finally, article 15 of this law states that for the suppression of native vegetation, ecological relevance must be considered in order to maintain representative samples of the diversity of vegetation types (phytosociologies) existing on the rural property inserted in the AUR-Pantanal.

§ 1. Representative samples of the diversity of vegetation types (phytosociologies) are considered to be maintained when:

I - the native vegetation cover of the phytosociologies (landscape units), represented by areas of cerrado formations and forest formations, is equal to or greater than 50% (fifty percent) of the total of these areas existing on the property;

II - the native vegetation cover of the phytosociologies (landscape units), represented by areas of grassland formations, is equal to or greater than 40% (forty percent) of the total of these areas existing on the property.

§ 2. When the application of the provisions of items I and II of § 1 of this article and the prohibitions provided for in article 12 of this Law affect 60% (sixty percent) or more of the property area, alternative land use may be authorized for up to 40% (forty percent) of the property area, in accordance with the regulations, observing the technical recommendations of EMBRAPA Pantanal.

State Decree No. 13.977/2014:

This decree stipulates that Restricted Use Areas, within the scope of the State of Mato Grosso do Sul, are considered to be areas with a slope between 25° and 45° and according to the limits stipulated by the Pantanal floodplain.

Decree No. 16.388/2024 - Regulates provisions of Law No. 6.160, of December 18, 2023:

Provides for the conservation, protection, restoration and ecologically sustainable exploitation of the Pantanal Plain Restricted Use Area (AUR-Pantanal), within the scope of the State of Mato Grosso do Sul.

It is worth mentioning here that Decree N. 14.273, of October 8, 2015, was revoked by Decree No. 16.388, of February 16, 2024.

Decree No. 16.388/2024 regulates the provisions of Law No. 6.160/2023, serving as a guide to the applicability of this law.

Suitability of the project area for conversion to alternative non-forest use: This item is detailed in Section 3.1.3, in the description of the delimitation of the boundaries of the Project Area.

Approval from the competent environmental agency: The state environmental agency responsible for issuing environmental permits in the state of Mato Grosso do Sul is IMASUL (Institute of Environment of Mato Grosso do Sul). As detailed in the section regarding legal permission for deforestation, the proper registration of properties in CAR, as well as their analysis and approval by the competent state agency (in this case, IMASUL), is essential to prove that the property complies with all current environmental regulations. It is also a way to verify if the property has remaining native vegetation that can be legally deforested.

Furthermore, registration of properties in CAR-MS is a prerequisite for environmental regularization and the issuance of authorizations or environmental licenses for activities located on rural properties. The property's compliance with CAR-MS will be characterized by the digital issuance of the Certificate of Regularity after the validation of the property's registration. This certificate includes the following information: total documented area of the property (ha); total calculated area of the property (ha); remaining native vegetation (ha); permanent preservation area (ha); restricted use area (ha); required legal reserve area (ha); existing legal reserve area (ha); and proposed area for legal reserve (ha).

For registration in CAR-MS, the owner of the rural property must provide: identification of the owner and possessor, if different; proof of ownership or possession; indication of the property's perimeter and, if applicable, the remaining areas of native vegetation, permanent preservation areas, administrative and environmental easements, private natural heritage reserves (RPPN), restricted use areas, consolidated areas, legal reserve, and the location of water resources.

Biofílica Ambipar Environment had a meeting with the director-president of IMASUL in July 2023, where it was confirmed that the CAR of the properties is the main basis used by the environmental agency to evaluate if the property complies with all applicable legal requirements and, thus, define the amount of surplus forest that can be legally deforested within each property. Additionally, it was informed that, in the absence of a request for vegetation suppression, CAR would serve as a means to prove that the property has surplus vegetation that can be legally deforested.

In this regard, we attach to this document the Certificate of Regularity of CAR for the properties comprising the first instance of the REDD+ Pantanal Program, in order to prove that they have remaining native vegetation that, in the absence of the implementation of this Program, would be eligible for legal deforestation.

Compiling all the aforementioned factors and applying them to the project's reality, the total area for planned deforestation in the properties of the first instance of the project is 11,800 ha. The table below shows the planned deforestation area for each stratum ($A_{(planned,i)}$), for each property that comprises the PAIs of the REDD+ Pantanal Program.

Table 6. Planned deforestation area for each stratum per property in the PAI's of the REDD+ Pantanal Program

ID PAI's	A _{planned,i}	Forest Area	Forested Savannah Area
C1-P01-Y24	1.341	903	438
C2-P01-Y24	307	293	14
C3-P01-Y24	452	331	121
C4-P01-Y24	2.591	2.102	489
C5-P01-Y24	4.160	3.650	511
C5-P02-Y24	24	8	17
C6-P01-Y24	2.743	1.613	1.130
C7-P01-Y24	33	18	16
C7-P02-Y24	147	105	42
Total	11.800	9.023	2.778

Relative deforestation rate

Based on module VMD0006, which deals with estimating changes in carbon stocks and greenhouse gas emissions from planned deforestation and degradation of forests and wetlands, we have adopted a specific approach to determine the deforestation rate in different strata. This approach aligns with item 5 (Procedures) and sub-items 1.2 and 1.3 of the document, which emphasize the need for a concrete and imminent threat of deforestation.

When addressing planned deforestation, module VMD0006 highlights the importance of having concrete documentary evidence to prove the imminent threat of deforestation. However, there are situations where such specific deforestation documents may not be available. In these cases, the methodology allows the use of proxy areas to estimate the deforestation rate.

The criteria established in methodology VMD0006 for choosing proxy areas are crucial to ensure accurate estimates of the planned deforestation rate, ensuring that these are representative of the project's reality. These criteria aim to ensure that the selected proxy areas have conditions similar to the project area in essential aspects, such as land management practices, post-deforestation use, and environmental characteristics, thus ensuring a correspondence between the areas.

Initially, a Geographic Area (GA) was defined to identify possible proxy areas that met all the criteria established by VMD0006. When defining the GA for the analysis of proxy areas, we chose to encompass the entire Pantanal biome. This decision ensures that the GA reflects the typical environmental and land use characteristics of the Pantanal. To ensure a comprehensive and representative analysis, the GA was expanded to also include rural properties bordering the Pantanal, thus incorporating areas that, although not strictly within the limits of the biome, share similar environmental characteristics.

To ensure a precise analysis focused on the project's objectives, indigenous lands and conservation units were excluded from the GA. This exclusion is crucial because indigenous lands and conservation units have distinct land use management and environmental conservation regimes from rural private areas.

These areas are protected by specific legislation that restricts activities such as deforestation and land conversion, resulting in significantly different land use dynamics from those observed on private properties. By excluding indigenous lands and conservation units, the GA focuses on areas where land use practices and deforestation threats align with the characteristics of the project area, providing a more appropriate basis for the selection and analysis of proxy areas.

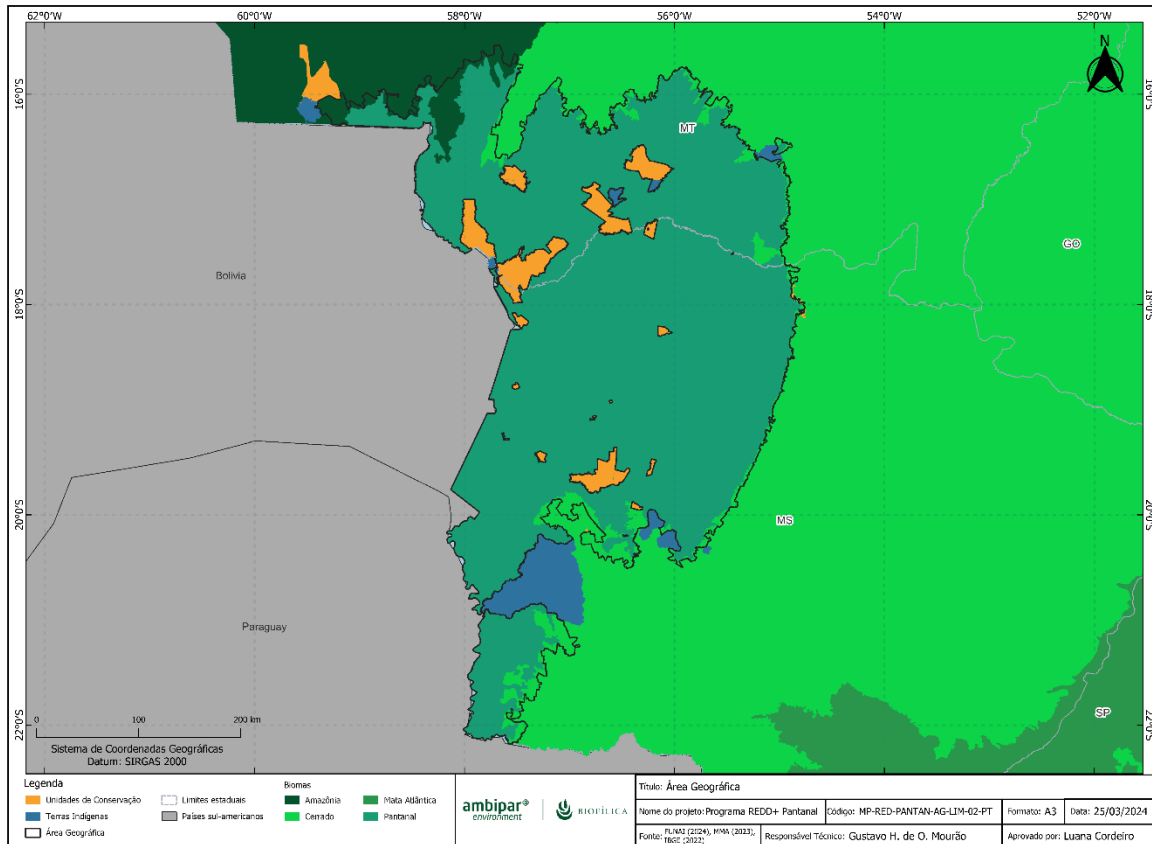


Figure 12. Delimitation of the Geographical Area selected for the analysis of proxies

VMD0006 requires that the proportion of terrain slope classes in proxy areas must be similar to that of the project area, with a permitted variation of $\pm 20\%$. Terrain slope is divided into two main categories: "gentle" (less than 15% slope) and "steep" (15% or more slope). This ratio is crucial because the slope of the terrain can influence both the ease of clearing and the type of land use after clearing.

The project area is predominantly flat, with a slope of less than 15%. The Geographical Region defined for the analysis and selection of proxy areas has similar characteristics, with a predominance of flat areas. This aspect is essential, as it ensures that the majority of properties that meet the other requirements and are located within this geographical region are potentially eligible as proxy areas. Initially, the entire geographical region was kept as a potential area for the selection of proxy areas, thus ensuring a broad spectrum of options for analysis and comparison with the project area.

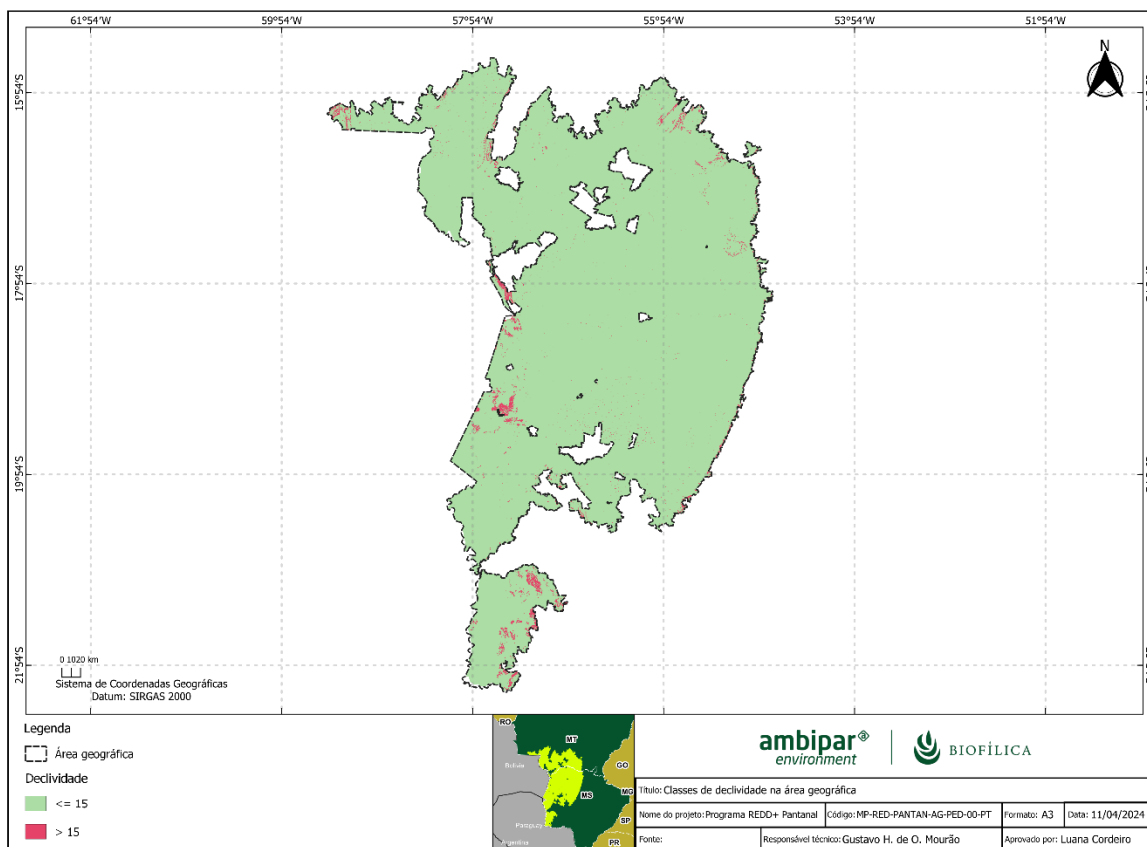


Figure 13. Slope classes within the delimited geographical area

Elevation classes in the proxy areas, grouped in 500-meter intervals, must be present in proportions that do not vary by more than $\pm 20\%$ in relation to the proportions in the project area. Elevation can impact both the local microclimate and biodiversity, affecting the feasibility and impact of deforestation.

The project area has an elevation of less than 500 meters. The Geographical Region chosen for analysis of the proxy areas predominantly has an elevation below 500 meters. This uniform elevation characteristic makes it easier to identify proxy areas that meet this elevation criterion. Thus, most of the Geographical Region qualifies for selection as a proxy area, as long as it meets the other requirements necessary for the analysis.

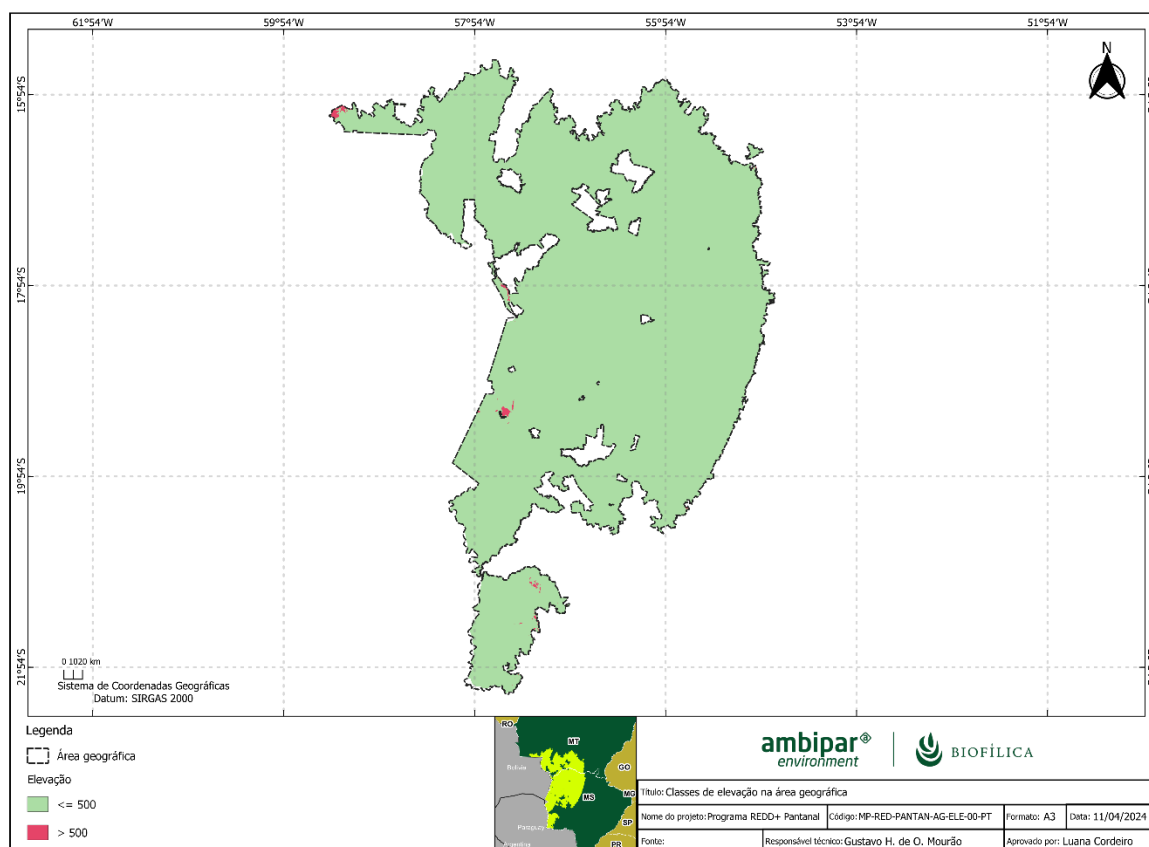


Figure 14. Elevation classes within the delimited geographical area

In the Geographic Area, a diversified soil composition is observed, reflecting the complexity and heterogeneity of the territory. Among this variety of soil orders, Neossolos, Planossolos, Plintossolos and Gleissolos stand out as the predominant types. This predominance not only highlights the specific formation conditions and underlying evolutionary processes of the region but also underscores the importance of these soil classes in the environmental and agricultural context.

Neossolos are primary soils with characteristics predominantly derived from the parent material, showing little or no pedological evolution due to the resistance of the original material or limiting factors such as climate, relief, or time. They are characterized by the absence of diagnostic horizons indicating a more advanced formation process. According to the Brazilian Soil Classification System (SiBCS) by Embrapa (2006), they are classified as incipient soils, reflecting an early stage of development. Neossolos vary significantly in terms of depth, fertility, acidity, and permeability, with some areas having soils rich in bases (eutrophic) and others with acidic soils poor in nutrients (dystrophic).

Planossolos are mainly found in flat or gently undulating terrain, often in regions of low altitude. These soils have a clear and sandy surface horizon, followed by a dense and clayey B horizon (plinthic), which hinders perfect drainage, leading to the temporary formation of suspended groundwater tables. Although they generally have good fertility, with high cation exchange capacity and base saturation, imperfect drainage can limit their agricultural use without proper soil management. They are common in the Agreste

region of Pernambuco and in areas with similar climates in states of the Brazilian Northeast, covering extensive areas that represent a significant portion of the semiarid region.

Plintossolos, as defined by the SiBCS, are soils that contain a plinthic horizon, characterized by the concentration of iron, which acts as natural cement. These soils are typically acidic and can be found with low or high base saturation. They may exhibit sodic and sodic-like characteristics, indicating variations in their chemical composition and structure. The plinthic horizon gives these soils particular physical and chemical properties, notable for their ability to retain water and nutrients under certain conditions.

Gleissolos are hydromorphic soils, characterized by prolonged or permanent water saturation, either due to internal stagnation or lateral flow, leading to a reducing environment practically devoid of dissolved oxygen. This moisture condition, present for much of the year or continuously, facilitates the process of gleization, marked by chemical reduction due to biological oxygen demand. Identified by the Brazilian Soil Classification System (SiBCS) by Embrapa in 2006, these soils present a gleic horizon, which can be either subsurface (C, B, or E) or surface (A), distinguished by colors ranging from gray to black and a thickness that typically varies between 10 and 50 cm, with average to high organic carbon contents. Due to their particular characteristics, Gleissolos are marked by a specific moisture regime, significantly influencing their morphology, chemistry, and biology.

The overlay of Mapbiomas' land use and land cover classification with the soil map of the Geographic Region reveals that soils classified as Planossolos, Neossolos, and Plintossolos are extensively used for livestock activities in the Pantanal biome. This analysis corroborates the suitability of these soils for land use practices carried out by deforestation agents in the project area, in accordance with the requirements of methodology VMD0006. Both soils are recognized as suitable for supporting anthropized pastures, reflecting the dominant livestock activities in the region.

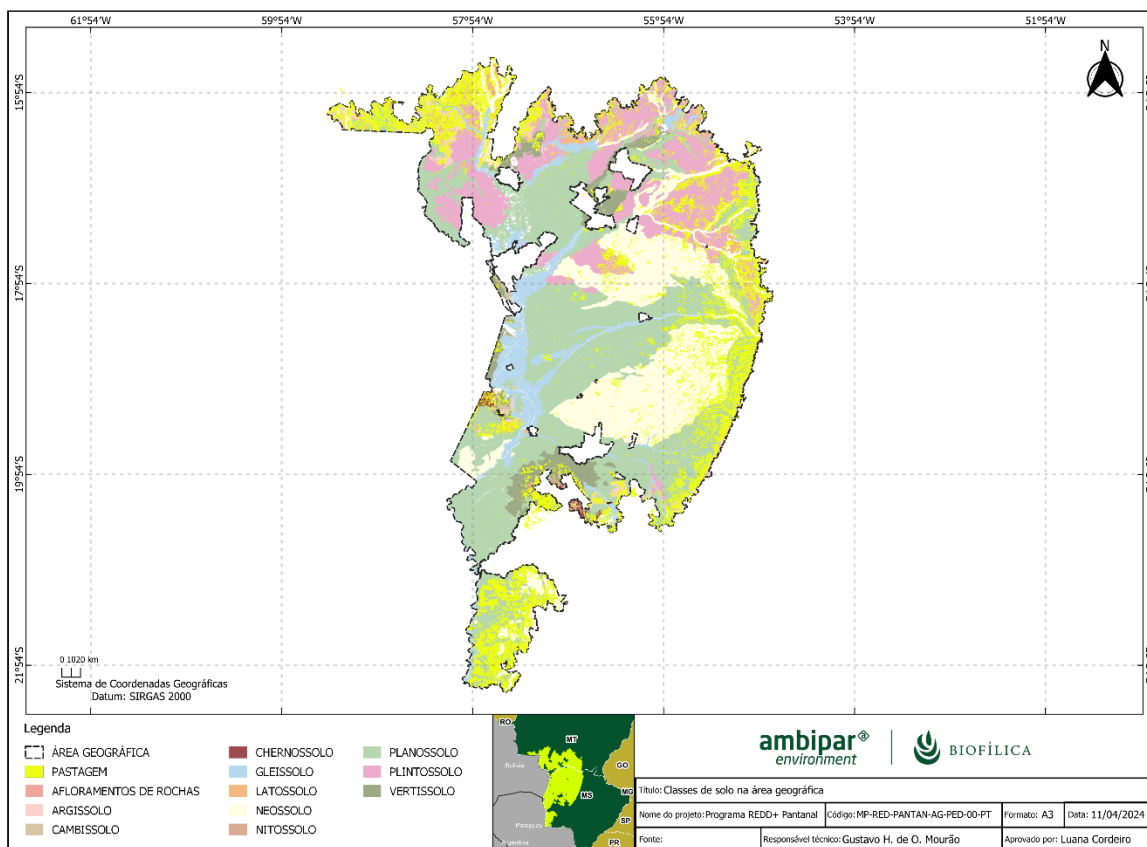


Figure 15. Soil classes in the defined geographical area

The properties where the project area is located are predominantly situated in two soil categories: Neossolos and Planossolos. Consequently, when defining possible locations for the allocation of proxy areas, we prioritized those where Neossolos and Planossolos predominate. This choice aligns with the land use practices observed in the project area and reinforces the representativeness and relevance of the selected proxy areas.

The other soil types, although capable of supporting pastures, were categorized as less suitable for the land use practices practiced by the deforestation agent, as they are not as representative in the properties within the project area. Thus, the selection of proxy areas was directed to locations where these two soil categories are predominant, ensuring the accuracy and relevance of deforestation estimates for the project.

The types of forest around the proxy area or in the proxy area before deforestation are in the same proportion as in the project area, within a variation of $\pm 20\%$. To achieve this, we used the same methodology applied for stratifying the project area; however, some modifications were made: the vegetation prior to deforestation was delimited from the land use and land cover map produced by Mapbiomas (2023) for the year 2011, ensuring that ten years before the start of the project, the vegetation was characteristically similar to the project area. For this, as well as for the project area (see Section 3.1.3), we stratified the vegetation into forested areas and forested savannah, where the

additional criterion of savannah with tree cover greater than 10% should also be met. The Mapbiomas classification, after stratification, ensured total similarity between the different vegetation strata and the possible proxy areas.

The intersection of these four essential requirements - forest types, soil composition, terrain slope, and elevation classes - delineated a specific area for the selection of proxy areas. This meticulous process ensured that the identified regions were aligned with the environmental and topographic conditions of the project area, guaranteeing an accurate representation of deforestation potential. However, other criteria were used before the definitive selection of areas.

The land conversion practices in the proxy areas must be identical to those used in the project area, ensuring that both the activities and conversion methods are the same to ensure accurate deforestation estimates. Similarly, the land use after deforestation in the proxy areas must reflect the intended use for the project area, maintaining consistency with expectations of future use under normal operating conditions.

Data derived from Mapbiomas shows that due to the peculiarity of the Pantanal biome, with its marked flood and dry seasons, 16.2% of the land is dedicated to agriculture, 99% of which is pasture and only 1% to agriculture. This distribution is due to the fact that, during the floods, a large part of the Pantanal is inundated, making agriculture a risky and unviable activity. On the other hand, livestock farming benefits from the mobility of cattle, which can be moved to safe areas during periods of intense flooding. This scenario is reflected in the growth of pastures, which increased from 0.7 million hectares in 1985 to 2.4 million hectares in 2019, a significant increase of 263%, demonstrating an adaptation to the Pantanal's natural dynamics⁴⁶.

In selecting proxy areas in the Pantanal, rural properties with land use practices similar to those of the project area, primarily the conversion of forests into pastures for livestock, were chosen. This choice reflects the predominant activity in the region, adapted to the natural characteristics of the biome. Following deforestation, these areas remained as pastures, aligned with the intended use for livestock, similar to what was planned for the project area in the absence of an APD intervention.

Proxy areas must have types of land management and land use rights similar to those of the project area. This ensures that the legal and administrative factors influencing deforestation are comparable. Additionally, deforestation agents in the proxy areas must have histories and practices of deforestation similar to those anticipated for the project area. This includes compliance with laws and regulations, as well as common land use practices. The proxy areas were selected considering only private properties, ensuring that land management and rights of land use are similar to those of the project area, ensuring that deforestation and land use practices are legally regulated and administratively comparable, as the project area is also composed only of private properties. The agents responsible for deforestation in the proxy areas have practices and histories similar to those of the project area, including adherence to environmental laws and regulations, as well as land conversion to pasture, a common and legal practice in the region.

⁴⁶ <https://www.sospantanal.org.br/cobertura-do-solo-e-uso-de-terra-no-pantanal/>

Ideally, proxy areas should be located in the immediate region of the project. If this is not possible, areas should be selected in other parts of the same country or, if necessary, in neighboring countries, always maintaining similarity with the project conditions. With this in mind, we prioritize proxy areas located in the same region of the Pantanal, in municipalities where the project area is located or in neighboring municipalities.

Deforestation agents in proxy areas must meet specific criteria, including legal compliance and suitability for conversion. In Brazil, the lack of details on legal deforestation processes requires the use of public tools to assess the legality of deforestation.

The Rural Environmental Registry (CAR) is one of these instruments. As a mandatory electronic register for rural properties, the CAR integrates relevant environmental information and is vital for monitoring and controlling deforestation, as well as being crucial for environmental conservation. It provides details of native vegetation, preservation areas, legal reserves (LR) and other restricted use zones⁴⁷. Another important tool is the embargoes on properties, applied in response to environmental infractions. The public list of embargoes helps identify areas with illegal deforestation⁴⁸.

In the context of the Pantanal, the Brazilian Forest Code (Law No. 12,651, of May 25, 2012), in its Art. 12, allows deforestation of up to 80% of the total area of properties in the Pantanal, requiring the maintenance of 20% as Legal Reserve. Thus, deforestation is legally permitted outside the Legal Reserve areas, respecting this limit and the laws and regulations governing this activity.

For the selection of proxy areas, properties with embargoes or without registration in the CAR were excluded from the analysis. Furthermore, to compose the proxy plots, portions of the properties located within the Legal Reserve were also excluded. This exclusion ensures that the analysis focuses on areas with a higher likelihood of legalized deforestation and compatible with the objectives of the REDD+ Pantanal Program. This care is essential to ensure the accuracy and legality of the estimated deforestation rates, thus aligning with the same methodology used for defining the project area.

After passing through all these criteria, we sought rural properties that were eligible to become proxy areas. The final selection focused on rural properties registered in Biofísica Ambipar Environment's local database, thus ensuring that the most accurate information regarding the limits of rural properties was available.

Considering all four established criteria (vegetation, elevation, slope, and soils), eight proxy areas that satisfy the necessary conditions for inclusion in the project were selected. These areas demonstrated in 2011 the minimum required similarity to the project area. It is also worth noting that, similar to the project areas, in each property, the vegetation was stratified into forest and forested savannah, ensuring that vegetation stratification reflects the conditions of the proxy areas.

⁴⁷ <https://www.fundoamazonia.gov.br/pt/projeto/Amazonia-Regularizacao-Ambiental/>

⁴⁸ <https://servicos.ibama.gov.br/ctf/publico/areasembargadas/ConsultaPublicaAreasEmbargadas.php>

Table 7. Similarity analysis between the defined Proxy Areas and the Project Area

Similaridade Áreas Proxy x Área do Projeto							
Savana				Floresta			
Proxy	Declividade	Altitude	Solo	Proxy	Declividade	Altitude	Solo
px_1	99,71%	100,00%	94,26%	px_1	99,01%	100,00%	93,15%
px_2	99,71%	100,00%	93,71%	px_2	99,01%	100,00%	93,15%
px_3	99,71%	100,00%	94,26%	px_3	99,01%	100,00%	93,15%
px_4	99,77%	100,00%	94,26%	px_4	99,20%	100,00%	93,15%
px_5	99,71%	100,00%	94,26%	px_5	99,01%	100,00%	93,15%
px_6	99,71%	100,00%	94,26%	px_6	99,03%	100,00%	93,15%
px_7	99,71%	100,00%	94,26%	px_7	99,01%	100,00%	93,15%
px_8	99,71%	100,00%	94,26%	px_8	99,01%	100,00%	93,15%

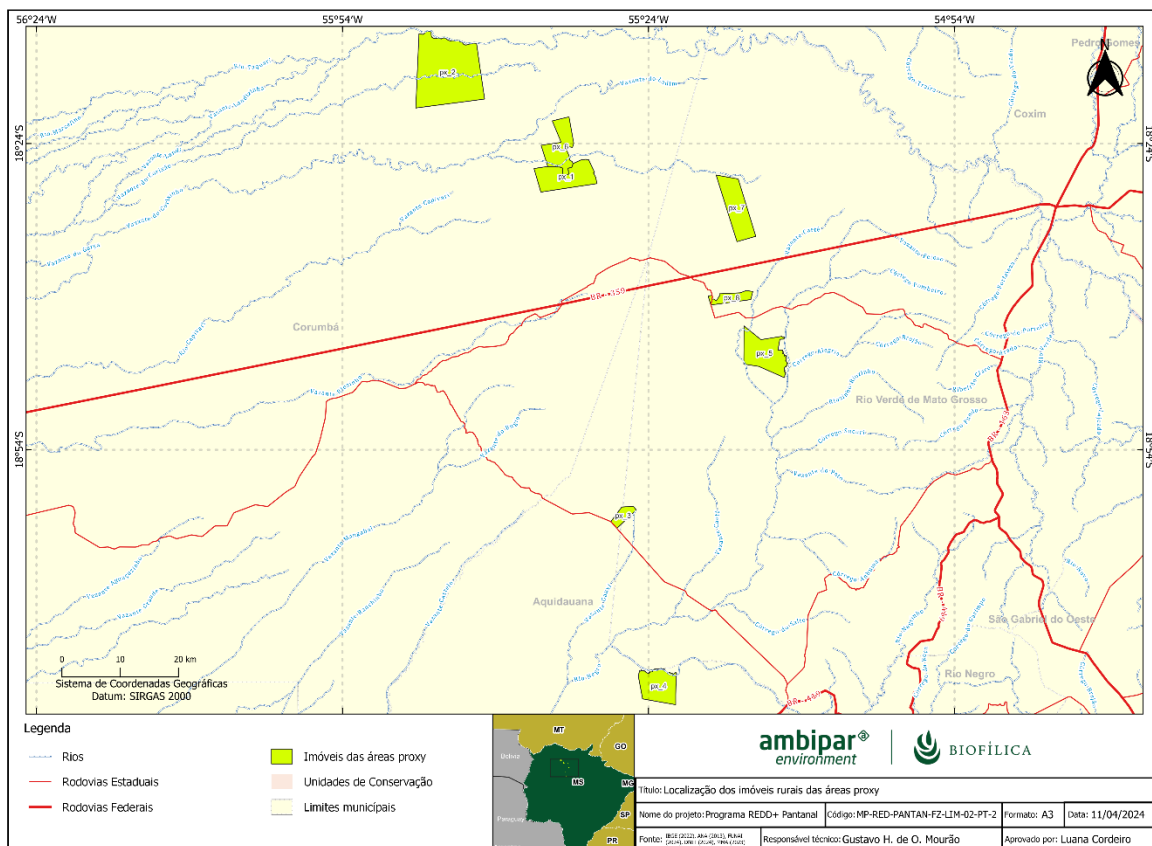


Figure 16. Location of the defined Proxy Areas

For the analysis of vegetation deforestation on the properties within the proxy areas, we utilized raster layers from MapBiomas representing land use and land cover classes over the years. Based on the nine selected properties, we focused on analyzing changes in land use between 2011 and 2021. The year 2011 was chosen as the reference point to ensure a historical record of at least a decade before the

project's initiation. Areas identified as forest or savannah in 2011 were retained in the raster, excluding other land uses that are not eligible to comprise the proxy areas. This initial marking ensures that only areas that were not deforested before the project's initiation are considered, which is crucial for determining the similarity of the areas to the project area.

After defining the year 2011 as the reference point in the MapBiomass land use and land cover raster, we proceeded with the application of a filter to identify changes in land use from forest and savannah to pasture between the years 2012 and 2021. In the MapBiomass land use and land cover raster for the reference year, 2011, forests and savannahs were classified with the value zero, indicating areas without alterations up to that point. Areas that remained as forest vegetation and savannah in subsequent years were kept with the value 0 in the raster, while those converted to pasture were reclassified numerically from 1 to 10, corresponding to the specific year of conversion relative to 2011. Next, we created a mask in the reference raster to record the specific changes: forest and savannah areas continued with the value 0, and those converted to pasture were marked with values from 1 to 10, indicating the year of the first conversion. For example, if a forest area was converted to pasture in 2012, the corresponding pixel in the mask was marked with the value 1; if the conversion occurred in 2015, the pixel was marked with the value 4. This process details not only if the conversion occurred but also the exact timing of this change, providing a clear and quantifiable view of deforestation.

The 2011 raster was reclassified to differentiate forest and savannah, and then combined with the results from the previous steps to create a detailed map indicating deforested areas. This map not only distinguishes areas without deforestation, retaining the base value of the vegetation type, but also highlights areas where deforestation occurred, specifying the year of the first occurrence of conversion to other uses, such as pasture. This temporal mask provides a clear and quantifiable view of deforestation. Continuing the process, this same raster was used to cross-reference the location and year of deforestation data with the 2011 land use and land cover information. This cross-referencing was crucial for defining the type of vegetation originally present and understanding the original characteristics of the affected areas, ensuring an accurate analysis of the impact of deforestation over time on the savannah and forest.

In the specific case of savannah, we adopted an additional procedure to ensure the accuracy of the analysis. In addition to cross-referencing the 2011 Mapbiomas data, the areas classified as savannah were filtered using an additional tree cover criterion. To do this, we used the Global Tree Cover dataset for the year 2010 on tree cover, applying the criterion that the savannah should have a tree cover of more than 10%. The Global Tree Cover⁴⁹ from the University of Maryland, in partnership with Google, USGS and NASA, provides a detailed estimate of global tree cover for the years 2000 and 2010. With a special resolution of 30 meters, tree cover is defined as vegetation over 5 meters high, covering both natural forests and plantations. The 2010 dataset was used because it was closer to the reference year when monitoring began (2011). This approach ensures that the savannah areas (from ten years before the start of the project) selected for the study have similar characteristics to those in the project area, thus fulfilling the requirements established by the VMD0006 module for choosing proxy areas.

⁴⁹ <https://glad.umd.edu/dataset/global-2010-tree-cover-30-m>

To finalize the analysis, the data in raster format was converted into vectors and delimited for each proxy area, enabling precise quantification of the extent of deforestation in each year and for the different strata within the proxy areas. The resulting data was then used to calculate deforestation rates.

These data cross-referencing and filtering methods provided a detailed and accurate view of the type of vegetation that existed prior to deforestation, thus enabling an accurate assessment of the changes that occurred for each carbon stock stratum. The table below shows the statistics for the proxy areas. The average relative annual deforestation rate is 9.32% per year for forest and 8.66% per year for savannah.

Table 8. The average relative annual deforestation rate for the proxy areas

Floresta					Savana				
Proxy	Desmt.	Area pre-desmat	Anos desmat	D%	Proxy	Desmt.	Area pre-desmat	Anos desmat	D%
px_1	543	559	10	9,71%	px_1	422	440	10	9,58%
px_2	460	486	10	9,47%	px_2	883	1.037	10	8,52%
px_3	112	113	10	9,93%	px_3	121	122	10	9,93%
px_4	358	428	10	8,36%	px_4	237	336	10	7,04%
px_5	770	866	10	8,89%	px_5	76	136	9	6,21%
px_6	354	377	10	9,38%	px_6	181	213	10	8,51%
px_7	310	350	10	8,87%	px_7	270	285	10	9,48%
px_8	348	348	10	10,00%	px_8	188	188	10	9,99%
Total				9,32%	Total				8,66%

Probability of deforestation

As these are private properties, the probability of deforestation in all the properties that make up the first instance of the project is 100%, according to section 1.4 of module VMD0006 (page 12).

Risk of abandonment

Under development.

Annual area of deforestation ($AA_{planned,i,t}$)

The table below shows the estimated annual deforestation areas for the project area. The annual deforestation area was calculated by:

$$AA_{planned,i,t} = (A_{planned,i} \times D_{planned,i,t}) \times L - D_i$$

Where:

$AA_{planned,i,t}$: Baseline of planned annual deforestation area, for stratum i, in hectares (ha);

$A_{planned,i}$: Total area of deforestation planned for the baseline period, in hectares;

$D\%_{planned,i,t}$: Annual proportion of deforestation planned for stratum i, year t.

$L - D_i$: Probability of planned deforestation, equal to 100% for private properties;

Table 9. Baseline of the annual area of future deforestation in the project area ($A_{(planned,i,t)}$)

Project year t	Stratum i of the reference region in the project area				Total	
	1- Forest		2- Savannah		annual	cumulative
	$A_{(planned,i,t)}$		$A_{(planned,i,t)}$		$A_{(planned,i,t)}$	
	annual ha	cumulative ha	annual ha	cumulative ha	ha	
2021	841	841	240	240	1.082	1.082
2022	841	1.683	240	481	1.082	2.164
2023	841	2.524	240	721	1.082	3.245
2024	841	3.365	240	962	1.082	4.327
2025	841	4.207	240	1.202	1.082	5.409
2026	841	5.048	240	1.443	1.082	6.491
2027	841	5.889	240	1.683	1.082	7.573
2028	841	6.731	240	1.924	1.082	8.654
2029	841	7.572	240	2.164	1.082	9.736
2030	841	8.413	240	2.404	1.082	10.818
2031	609	9.023	240	2.645	850	11.667
2032	0	9.023	133	2.778	133	11.800
2033	0	9.023	0	2.778	0	11.800
2034	0	9.023	0	2.778	0	11.800
2035	0	9.023	0	2.778	0	11.800
2036	0	9.023	0	2.778	0	11.800
2037	0	9.023	0	2.778	0	11.800
2038	0	9.023	0	2.778	0	11.800
2039	0	9.023	0	2.778	0	11.800
2040	0	9.023	0	2.778	0	11.800

3.1.5 Additionality (VCS, 3.14)

3.1.5.1 Regulatory Surplus (VCS, 3.14)

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

The activities of the Pantanal REDD+ Program are not mandatory by any law, statute or regulation. Besides, the proponent will follow all standards and processes necessary to guarantee and prove the additionality of the activities proposed here.

3.1.5.2 Additionality Methods (VCS, 3.14)

According to VM0007 methodology (item 7) the Pantanal REDD+ Program uses the latest version of the tool referenced in the T-ADD to identify credible land use scenarios and determine the project additionality.

As such, the tool “VT0001 - Tool for Demonstration and Assessment of Additionality in Activities of the VCS Agriculture, Forestry and Other Land Uses (AFOLU) Project” was applied, version 3.0, dated February 1, 2012. The steps used are described below, according to Tool VT0001:

STEP 1: Identification of alternative land use scenarios to the AFOLU project activity

Sub-step 1a. Identify credible alternative land use scenarios to the proposed VCS AFOLU 2.1.1project activity

According to VT0001 tool, the following scenarios were identified for land use in the absence of the Program:

1. *Maintenance of forestry on properties without financial incentives, such as the implementation of an AFOLU VCS-CCB project*

In this scenario, the current land use in the Project Area would remain unchanged, maintaining forest cover (including forest and forested savannah) in the surplus areas of the properties, but without any financial incentive for its maintenance.

In the regional context, although possible, this scenario is the most unlikely to perpetuate. This is due to the direct costs associated with the protection activities of these areas, as well as the opportunity cost related to the economic use of the properties, since the legislation allows deforestation and, as demonstrated in scenario 4, brings great added value to the owner. Furthermore, the maintenance of rural properties requires the annual payment of various fees and taxes, which implies the need to generate some economic revenue under penalty of financial losses or even the loss of the property for agrarian reform programs.

These factors make it highly unlikely that land use as it existed before the project would continue in the absence of financial incentives to sustain excess property areas. Currently, the revenue sources available

to maintain these areas are limited, beyond potential financing derived from carbon, or the conversion of this area to other land uses.

2. Carrying out REDD+ activities without registering as an AFOLU VCS-CCB project

As indicated in scenario 1, maintaining forest cover without financial incentives is possible, but unlikely. However, considering the different profile of owners participating in the Pantanal REDD+ Program, who are focused on the conservation and sustainable use of their properties, this scenario aligns with the activities proposed by the program. These activities aim to ensure the protection of the Project Area and bring benefits to local communities and biodiversity, although there are no financial incentives from sale of verified carbon credits.

The importance of these activities is directly related to the challenges faced by the Pantanal biome and the owners' understanding of the need to improve their properties.

A significant example of these challenges is the frequency of fires. Despite the reduction in fire rates, more than 8,800 hectares were affected by fire in the biome between January and February of 2023⁵⁰. Although fire is a natural part of the Pantanal, many fires are caused by human action, often by farmers seeking to expand pasture areas. This can result in uncontrollable fires, compromising forest cover and affecting other properties in the region. The threat of fire is imminent for the Project Area, and without adequate management and prevention actions, it could result in irreversible damage.

Furthermore, other problems in the region include the retention of employees on farms, their quality of life and the lack of knowledge about profitable alternatives other than livestock farming. Activities that aim to improve safety, food, access to health and education, in addition to offering technical training and promoting communication between employees, are essential to mitigate losses, reduce rural exodus and diversify income. The activities proposed by the Pantanal REDD+ Program also seek to strengthen governance, improve infrastructure, conserve biodiversity and monitor deforestation risks in the region.

However, since these investments are not mandatory and are generally not carried out purely out of altruism, but rather because the owners have extra resources that can be invested in their farms, these activities could hardly be practiced in the regional scenario, making it difficult or preventing the generation of net positive impacts in the region. Therefore, the economic viability of these actions is unfeasible without the addition of revenue resulting from the sale of credits registered in the VCS.

3. Carrying out legal deforestation to expand livestock farming done by the farm owner

Changing land use from forest areas to pasture areas, within private properties, is the most likely scenario to occur when analyzing the history of land use in the Pantanal, given the predominance of livestock farming as the main economic activity in the region. Based on the analysis of secondary databases, as well as surveys of field diagnoses carried out by the Program, it is possible to observe that many of the

⁵⁰ <https://www.campograndenews.com.br/meio-ambiente/mapbiomas-pantanal-foi-o-bioma-menos-afetado-com-o-fogo-em-fevereiro>

properties adjacent to the farms, participating in the Program, already have land use focused on livestock activities, with the suppression of vegetation cover occurring in the last 10 years.

Data from MapBiomas show that there was an expansion of areas classified as pasture when analyzing the last 10 years of land use and coverage in the region where the project is located, with a higher rate of land use conversion carried out in areas located to the south and east of the properties which are part of the Pantanal REDD+ Program.

According to State Agency for Animal and Plant Health Defense (IAGRO)⁵¹, the State of Mato Grosso do Sul has the fourth largest cattle herd in Brazil with approximately 20 million head of cattle, with beef cattle farming (for food predominantly based on pasture) one of the main economic activities in the region.

News related to the Pantanal, such as the one published in March 2023 on the UOL⁵² website, points out that the main cause of loss of native vegetation in the Pantanal is due to conversion to pastures. Therefore, it is understood that livestock farming is the main factor responsible for the demand for legal suppression of existing vegetation in the Pantanal biome, aiming to convert forest cover to pasture areas.

4. Sale of the property to third parties and conversion of land use from forest cover to livestock farming.

Given the growing demand and resulting increase in property prices in the Project Zone, the sale of the property is a significant possibility. This sale can occur in two different ways: (i) after the occurrence of scenario 3, the property has an increase in its added value and the forest area has already been lost; (ii) with the forest areas still intact, the property is transferred to third parties who can carry out the activities described in scenario 3 and, subsequently, profit from real estate speculation in the region.

To support both approaches, it is possible to resort to the “Terra Nua – Exercício 2023” released by the Ministry of Finance⁵³ indicated below and which demonstrate data on land values in the municipalities of Aquidauana, Miranda and Corumbá, in the state of Mato Grosso do Sul.

Table 10. “Terra Nua – Exercício 2023” released by the Ministry of Finance

Municipality	Crop suitability	Crop regular suitability	Crop restricted suitability	Planted pasture	Forestry or natural pasture	Preservation of fauna and flora
Aquidauana	R\$ 12,626.01	R\$ 8,638.85	R\$ 6,645.28	R\$ 5,316.21	R\$ 3,987.17	R\$ 2,658.11

⁵¹ IAGRO – Available at:

<https://www.iagro.ms.gov.br/Geral/pecuaria/#:~:text=Com%20o%20quarto%20maior%20rebanho,predominantemente%20a%20base%20de%20pastagem>

⁵² UOU – Available at: <https://midiamax.uol.com.br/cotidiano/2023/biomas-em-risco-mato-grosso-do-sul-perde-67-milhoes-de-hectares-de-floresta-em-35-anos/>

⁵³ Valores de Terra Nua - Available at: <https://www.gov.br/receita-federal/pt-br/centrais-de-conteudo/publicacoes/documentos-tecnicos/vtn/tabela-vtn-para-divulgacao-2023-3.pdf/view>

Miranda	No information	No information	No information	R\$ 4,650.00	R\$ 3,320.00	R\$ 1,860.00
Corumbá	No information	No information	No information	R\$ 3,191.12	R\$ 2,597.86	R\$ 1,316.17

To make the consideration of this scenario for additionality more plausible, it is necessary to discard approach (i), since it is already directly affected by scenario 3. In other words, it requires deforestation with the aim of opening pastures to increase the value of the land. Therefore, approach (ii) is more applicable in this context.

Additionally, considering what was exposed in scenario 1, where maintaining forest cover without financial incentives is possible, but unlikely, and taking into account that the owners participating in the Pantanal REDD+ Program have a different profile, focused on conservation and sustainable use of their properties, the most plausible design to apply in this context is the sale of the property with the forest surplus areas intact. Although this approach can reduce the initial sale price of the property, in the long term it can lead to real estate speculation in the region.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations

All alternative land use scenarios identified in substep 1a are in full compliance with mandatory legal and regulatory requirements applicable in the state of Mato Grosso do Sul, which is the focal region of this validation. Although illegal deforestation may occur, whether through invasion or fires of unnatural origin, in the area covered by the Program, this analysis will focus exclusively on legally authorized scenarios. Therefore, none of the previously mentioned alternative land use scenarios will be excluded from the Program's additionality analysis.

Sub-step 1c. Selection of the baseline scenario:

Described in detail in Section 3.1 - Application of the Methodology, item 3.1.4 Baseline Scenario.

STEP 2: Investment analysis to determine that the proposed project activity is not the most attractive from an economic or financial point of view of the identified land use scenarios

Under development

STEP 3: Barrier analysis

VCS "VT0001 - Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities" requires Investment Analysis (Step 2) or Barrier Analysis

(Step 3). In this case, until the completion of this documentation, only the Investment Analysis, described in Step 2, was chosen. This definition is still under analysis by the proponent.

STEP 4: Common practice analysis

Under development.

3.1.6 Methodology Deviations (VCS, 3.20)

Not applicable as no methodological deviations were identified until the moment of completion of this documentation.

3.2 Quantification of Estimated GHG Emission Reductions and Removals

3.2.1 Baseline Emissions (VCS, 3.15)

Estimating pre-deforestation carbon stocks

Under development and revision.

The estimation of the pre-deforestation carbon stock calculations was carried out through a field survey using the forest inventory for the forest and forested savannah strata.

The sampling design was defined to achieve a 15% admissible error, considering a 95% confidence level. 44 and 68 100 x 100m plots were inventoried in the Forests and Savannahs, respectively. The uncertainty found in the total biomass in the project area was 9.33% in the Forests and 6.44% in the Savannahs, thus validating the sampling design applied.

Two reservoirs were inventoried in the field (trees with DBH > 10 cm; palms with DBH > 10 cm); two other reservoirs were calculated from the inventoried reservoirs, using standardized factors for dead wood (0.11) and roots (0.22).

The allometric equation of Chave et al. (2014) was used to estimate the biomass of trees and palms. This equation uses diameter, height and specific wood density. The diameter was measured in the field, the height was estimated using an equation also proposed by Chave et al. (2014), and for the specific density of the wood, the general average for tropical trees was used (0.645 g.cm⁻³).

The table below shows the intermediate data used in the emission calculations for the estimated carbon stock in the REDD+ Pantanal Program areas.

Table 11. Pre-deforestation carbon stock of the Pantanal REDD+ Program for the forest and forested savannah stratum

Name:	Forest			Name:	Savannah		
ID _{icl}	1			ID _{icl}	2		
Average carbon stock per hectare							
Cab _{icl}	Cbb _{icl}	Cdw _{icl}	Ctot _{icl}	Cab _{icl}	Cbb _{icl}	Cdw _{icl}	Ctot _{icl}
tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹	tCO ₂ e ha ⁻¹
456,94	107,84	53,90	618,68	276,54	64,06	32,01	372,61
tC ha ⁻¹	tC ha ⁻¹	tC ha ⁻¹	tC ha ⁻¹	tC ha ⁻¹	tC ha ⁻¹	tC ha ⁻¹	tC ha ⁻¹
124,62	29,41	14,70	168,73	75,42	17,47	8,73	101,62

Estimating post-deforestation carbon stocks

Under development and revision.

The calculation of the carbon stock remaining after deforestation (CP, post, i) can be obtained by applying the standard values adopted by the Ministry of the Environment (MMA) and presented in the Brazilian Fourth Communication (Table 23, page 159).

The table below shows the three most likely types of use to be installed in the Pantanal. The pasture values can be considered for two reasons: (1) extensive pasture, for cattle breeding, is the most likely type of use to occur, as it is the main productive activity in the Pantanal plain; and, (2) the BL-UP module (page 31) advises that when faced with different land use possibilities, you can choose the one with the highest carbon stock, this choice is considered conservative.

Table 12. Carbon stock by type of use after deforestation. Data source: The values in Mg C ha⁻¹ were taken from the "Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference

Post-deforestation land use	tC ha ⁻¹	tCO ₂ e ha ⁻¹
Pasture	10	36,67
Agriculture and pasture mosaic	7,57	27,76
Soybean plantations	5,14	18,85

Estimating the baseline in carbon stock changes

Under development and revision.

The change in carbon stocks in the project area was estimated by equation 13 of Module VMD0006:

$$\begin{aligned} \Delta C_{BSL,i,t} = & AA_{planned,i,t} \times (\Delta C_{AB_{tree,i}} - \Delta C_{WP,i} + \Delta C_{AB_{non\ tree,i}} + \Delta C_{LI,i}) + \\ & + \left(\sum_{t=10}^t A_{planned,i,t} \right) \times (\Delta C_{BB_{tree,i}} + \Delta C_{BB_{non\ tree,i}} + \Delta C_{DW,i}) \times \left(\frac{1}{10} \right) + \\ & + \left(\sum_{t=20}^t AA_{planned,i,t} \right) \times (C_{WP100,i} + \Delta C_{SOC,i}) \times \left(\frac{1}{20} \right) \end{aligned}$$

Where:

$\Delta C_{BSL,i,t}$: baseline estimate of the sum of changes in carbon stocks for all terrestrial reservoirs in stratum i and year t, in tCO₂-e;

$AA_{planned,i,t}$: baseline estimate of the area of planned deforestation in the project area, in stratum i and year t, in hectares (ha);

$C_{WP100,i}$: estimate of the average carbon stock of the year in which deforestation occurred converted into forest products, and assumed to remain stored for 100 years, in tCO₂-e/ha;

$\Delta C_{R,i}$: baseline estimate of the change in the average biomass carbon stock of different terrestrial reservoirs in stratum i, in tCO₂-e/ha. The different reservoirs are:

AB_{tree} : above-ground tree biomass; WP : forest products; $AB_{non\ tree}$: non-tree vegetation; LI : litter; BB_{tree} : below-ground tree biomass, DW : above-ground dead biomass e SOC : soil organic carbon.

The change in carbon stock is given by:

$$\Delta C_{R,i} = C_{R_{pre,i}} - C_{R_{post,i}}$$

Where $C_{R_{pre,i}}$ e $C_{R_{post,i}}$ are the baseline estimates of the pre- and post-deforestation stocks of reservoir R, stratum i. The following sections describe the steps for calculating these and the other components needed to obtain $\Delta C_{BSL,i,t}$.

Table 13. Baseline net carbon stock change in all PAIs

Carbon stock changes per initial forest class <i>icl</i>				Carbon stock changes per post-deforestation			Total net carbon stock change of the project area	
ID _{icl} >	1	2	$\Delta C_{BSL,i,t}$	ID _{iz} >	1	$\Delta C_{BSL,i,t}$	$\Delta C_{BSL,i,t}$	
Name>	Forest	Savannah	annual	Name>	Zone 1	annual	annual	cumulative
Project Year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	Project Year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2021	398.049	68.803	466.852	2021	0	0	466.852	466.852
2022	411.657	71.113	482.770	2022	3.967	3.967	478.803	945.655
2023	425.265	73.423	498.687	2023	7.934	7.934	490.753	1.436.409

Carbon stock changes per initial forest class <i>icl</i>				Carbon stock changes per post-deforestation			Total net carbon stock change of the project area	
ID _{icl} >	1	2	ΔC _{BSL,i,t}	ID _{iz} >	1	ΔC _{BSL,i,t}	ΔC _{BSL,i,t}	
Name>	Forest	Savannah	annual	Name>	Zone 1	annual	annual	cumulative
Project Year <i>t</i>	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	Project Year <i>t</i>	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2024	438.872	75.733	514.605	2024	11.901	11.901	502.704	1.939.113
2025	452.480	78.043	530.522	2025	15.868	15.868	514.654	2.453.767
2026	466.087	80.352	546.440	2026	19.835	19.835	526.605	2.980.372
2027	479.695	82.662	562.357	2027	23.801	23.801	538.556	3.518.928
2028	493.302	84.972	578.274	2028	27.768	27.768	550.506	4.069.434
2029	506.910	87.282	594.192	2029	31.735	31.735	562.457	4.631.890
2030	520.517	89.592	610.109	2030	35.702	35.702	574.407	5.206.298
2031	410.666	89.592	500.258	2031	35.702	35.702	464.555	5.670.853
2032	118.713	58.766	177.479	2032	34.851	34.851	142.628	5.813.481
2033	105.105	19.754	124.859	2033	31.371	31.371	93.489	5.906.969
2034	91.498	17.444	108.942	2034	27.404	27.404	81.538	5.988.507
2035	77.890	15.134	93.024	2035	23.437	23.437	69.588	6.058.095
2036	64.282	12.824	77.107	2036	19.470	19.470	57.637	6.115.732
2037	50.675	10.515	61.189	2037	15.503	15.503	45.687	6.161.419
2038	37.067	8.205	45.272	2038	11.536	11.536	33.736	6.195.155
2039	23.460	5.895	29.355	2039	7.569	7.569	21.785	6.216.940
2040	9.852	3.585	13.437	2040	3.602	3.602	9.835	6.226.775

Greenhouse Gas Emissions GHG

No activities that generate GHG emissions are planned for the project. The GHG emissions baseline was considered to be conservatively zero. Therefore, GHG_(BSL-E,i,t)=0.

Baseline net GHG emissions

The net GHG emissions baseline for planned deforestation will be determined based on equation 1 of VMD0006 v.13:

$$\Delta C_{BSL,planned} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{BSL,i,t} + GHG_{BSL-E,i,t})$$

Where:

ΔC_{BSL,planned}: Net greenhouse gas emissions in the baseline of planned deforestation up to year t*

$\Delta C_{BSL,i,t}$: Net changes in carbon stock in all reservoirs in stratum i of the baseline in year t;

$GHG_{BSL E,i,t}$: Greenhouse gas emissions as a result of deforestation activities within the project boundaries in stratum i of the baseline in year t;

i 1, 2, 3,...: M strata

t 1, 2, 3,...: t* years since the projected start of project activity

Table 14. Net greenhouse gas emissions in the baseline from planned deforestation all PAIs (01-07)

Name>	$\Delta C_{BSL,i,t}$		$GHG_{BSL E,i,t}$		$\Delta C_{BSL,planned}$	
ID _{ici} >	annual	cumulative	annual	cumulative	annual	cumulative
Project Year t	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e	tCO ₂ -e
2.021	466.852	466.852	0	0	466.852	466.852
2.022	478.803	945.655	0	0	478.803	945.655
2.023	490.753	1.436.409	0	0	490.753	1.436.409
2.024	502.704	1.939.113	0	0	502.704	1.939.113
2.025	514.654	2.453.767	0	0	514.654	2.453.767
2.026	526.605	2.980.372	0	0	526.605	2.980.372
2.027	538.556	3.518.928	0	0	538.556	3.518.928
2.028	550.506	4.069.434	0	0	550.506	4.069.434
2.029	562.457	4.631.890	0	0	562.457	4.631.890
2.030	574.407	5.206.298	0	0	574.407	5.206.298
2.031	464.555	5.670.853	0	0	464.555	5.670.853
2.032	142.628	5.813.481	0	0	142.628	5.813.481
2.033	93.489	5.906.969	0	0	93.489	5.906.969
2.034	81.538	5.988.507	0	0	81.538	5.988.507
2.035	69.588	6.058.095	0	0	69.588	6.058.095
2.036	57.637	6.115.732	0	0	57.637	6.115.732
2.037	45.687	6.161.419	0	0	45.687	6.161.419
2.038	33.736	6.195.155	0	0	33.736	6.195.155
2.039	21.785	6.216.940	0	0	21.785	6.216.940
2.040	9.835	6.226.775	0	0	9.835	6.226.775

3.2.2 Project Emissions (VCS, 3.15)

The expected net GHG emissions are estimated ex-ante and use equation no. 1 - Figure 3.37 - from the VCS Module VMD0015 (M-REDD), Version 2.3. The ex-ante deforestation projections in the case of the Program assume that no deforestation has yet occurred. The detailed analysis will be described in the final version of this document, which will be delivered to the VVB.

3.2.3 Leakage Emissions (VCS 2.5, 3.2, 3.6, 3.15, 4.3)

For effective monitoring of leakage areas, it's crucial to identify rural properties owned by the same individuals or entities that own properties within the project area. In this context, we conducted a detailed analysis to establish an accurate correspondence between owners and properties. As a result, we delimited 17 rural properties that are owned by the same individuals or entities that own properties included in the project area.

To define the boundaries of properties in the leakage areas, we primarily used the database shared by the owners, data from the Rural Environmental Registry (CAR), complemented by the Rural Property Environmental Regularization System (SIGEF), and the National System of Rural Property Inventory (SNCI) to distinguish project areas from leakage areas, especially when parts of the properties were already integrated into the project. Regarding the delimitation of forests to be monitored, we employed methodologies adapted to the specific characteristics and locations of the properties.

In the Pantanal, forests and savannahs with more than 10% tree cover were identified using MapBiomas in conjunction with the Tropical Tree Cover dataset for the year 2020, following the same methodology applied in the project area. In properties located in the Cerrado biome, we opted for the use of PRODES for the year 2021, which provided a more precise delineation for the leakage monitoring areas.

PRODES (Monitoring Deforestation in Legal Amazon Project by Satellite) is an initiative developed by the Brazilian National Institute for Space Research (INPE) to track and quantify deforestation in Brazil. This program uses high-resolution satellite images to observe and analyze changes in forest cover. In addition to detecting deforested areas, the system also identifies remaining vegetation, allowing for forest cover analysis. In other regions outside the Pantanal and Cerrado, MapBiomas 2021 has proven to be sufficient for properly classifying forests.

We finalized the process by excluding from the monitoring areas analysis those that overlapped with Legal Reserves and Permanent Preservation Areas. Additionally, fragments of forested areas smaller than 0.5 hectares were removed to ensure accuracy in delineating the areas effectively available for monitoring.

Table 15. Properties included in the Leakage Belt area of the REDD+ Pantanal Program for all PAIs (01-07)

ID	Property area	Monitoring area	Source of monitoring data
lk_1	488	46	PRODES
lk_17	8.478	1.555	Mapbiomes and Tropical Tree Cover
lk_18	34.560	6.113	Mapbiomes and Tropical Tree Cover
lk_2	371	189	PRODES
lk_20	25.010	7.794	Mapbiomes and Tropical Tree Cover
lk_21	242	47	Mapbiomas
lk_22	4.184	160	Mapbiomas
lk_23	1.705	1	Mapbiomas
lk_24	1.816	316	Mapbiomas

ID	Property area	Monitoring area	Source of monitoring data
lk_25	794	186	Mapbiomas
lk_3	3.224	521	PRODES
lk_4	1.726	262	PRODES
lk_5	4.727	552	PRODES
lk_6	483	73	PRODES
lk_7	478	96	PRODES
lk_8	2.193	963	PRODES
lk_9	481	36	PRODES
Total	90.961	18.910	

Once these limits have been set, leakage emissions due to the change in activity of forest land that is legally authorized and documented to be converted into non-forest land can be estimated according to the criteria of VMD0009 v1.3.

However, since the deforestation agent has been identified for the Pantanal REDD+ Program and, according to the survey carried out with the owners, the management plans of the other properties have not undergone significant changes as a result of the project, leakage will not be considered in the baseline. Further details on monitoring the leak are described in the Monitoring Plan in Section 3.3.3.

3.2.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals (VCS, 3.15, 4.1)

Total net reduction in GHG emissions

The total net reductions in greenhouse gas emissions from the REDD+ Pantanal Program activity are calculated based on equation 2 of VM0007:

$$NER_{REDD} = \Delta C_{BSL-REDD} - \Delta C_{WPS-REDD} - \Delta C_{LK-REDD}$$

Where:

NER_{REDD} : Total net GHG emission reductions from REDD project activity up to year t^*

$C_{BSL-REDD}$: Net GHG emissions in the REDD baseline scenario up to year t^*

$C_{WPS-REDD}$: Net GHG emissions in the REDD project scenario up to year t^* - from Module M-REDD

$C_{LK-REDD}$: Net GHG emissions due to leakage from REDD project activity up to year t^*

Calculation of the AFOLU buffer account contribution

Following the guidelines indicated in VM0007 v1.6, for this project the buffer is equal to the net GHG emissions in the baseline minus the net emissions in the project case. Leakage emissions are not taken into account in the buffer calculations, and non-CO2 emissions from fossil fuels and fertilizer use have been conservatively excluded from the project scope.

To arrive at the buffer percentage, risk analysis was applied to the project. The risk analysis for the Pantanal REDD+ Program was carried out using the AFOLU Non-Permanence Risk Tool, v4.2, resulting in a buffer of 22%, as indicated in the following spreadsheet:

State the non-permanence risk rating (%)	22%
Has the non-permanence risk report been attached as either an appendix or a separate document?	☒ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO2e, the Long-term Average (LTA).	Not applicable
Has the LTA been updated based on monitored data, if applicable?	Not applicable
State, in tCO2e, the expected total GHG benefit to date.	6.226.775
Is the number of GHG credits issued below the LTA?	Not applicable

Calculation of verified carbon units (VCUs)

To estimate the number of Verified Carbon Units (VCUs) generated from the implementation of the planned project activity (ex-ante), the following equation was applied:

$$VCU_t = Adjusted_NER_{REDD+} - Buffer_{Planned}$$

Where:

VCU_t : Number of carbon units verified in the year $t = t_2 - t_1$ (VCU)

$Adjusted_NER_{REDD+}$: Total net GHG emission reductions from REDD+ project activity adjusted for uncertainty (t CO2e)

$Buffer_{Total}$: Full retention of the permanence risk buffer (t CO2)

Table 16. Total net GHG emission reductions from REDD+ project activity, total buffer account contribution and number of verified carbon units (VCU) (ex-ante)

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
14-October-21 to 31-December-21	466,852	0	0	102,708	466,852	364,145
1-January-2022 to 31-December-2022	478,803	0	0	105,337	478,803	373,466
1-January-2023 to 31-December-2023	490,753	0	0	107,966	490,753	382,788
1-January-2024 to 31-December-2024	502,704	0	0	110,595	502,704	392,109
1-January-2025 to 31-December-2025	514,654	0	0	113,224	514,654	401,430
1-January-2026 to 31-December-2026	526,605	0	0	115,853	526,605	410,752
1-January-2027 to 31-December-2027	538,556	0	0	118,482	538,556	420,073
1-January-2028 to 31-December-2028	550,506	0	0	121,111	550,506	429,395
1-January-2029 to 31-December-2029	562,457	0	0	123,740	562,457	438,716
1-January-2030 to 31-December-2030	574,407	0	0	126,370	574,407	448,038
1-January-2031 to 31-December-2031	464,555	0	0	102,202	464,555	362,353
1-January-2032 to 31-December-2032	142,628	0	0	31,378	142,628	111,250
1-January-2033 to 31-December-2033	93,489	0	0	20,567	93,489	72,921
1-January-2034 to 31-December-2034	81,538	0	0	17,938	81,538	63,600
1-January-2035 to 31-December-2035	69,588	0	0	15,309	69,588	54,278
1-January-2036 to 31-December-2036	57,637	0	0	12,680	57,637	44,957
1-January-2037 to 31-December-2037	45,687	0	0	10,051	45,687	35,635
1-January-2038 to 31-December-2038	33,736	0	0	7,422	33,736	26,314

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCU's (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
1-January-2039 to 31-December-2039	21,785	0	0	4,793	21,785	16,993
1-January-2040 to 31-December-2040	9,835	0	0	2,164	9,835	7,671
Total	6,226,775	0	0	1,369,890	6,226,775	4,856,884

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation (VCS, 3.16)

Under development and revision.

Data / parameter	$\Delta C_{BSL,planned}$
Data unit	tCO ₂ e ha-1
Description	Net greenhouse gas emissions in the baseline resulting from planned deforestation until year t. In other words, the Change in the total carbon stock of the evaluated reservoirs, after planned deforestation.
Source of data	Described in sections 3.2.1
Value applied	Total: 6,226,775
Justification of choice of data or description of measurement methods and procedures applied	Required by BL-PL, LK-ASP and M-REDD modules
Purpose of data	Carry out calculations of the differences between before and after planned deforestation in the Project Area and in the Leakage Belt
Comments	n/a
Data / parameter	$A_{planned,i}$
Data unit	ha

Description	Total area of planned deforestation during the fixed baseline period for stratum i
Source of data	Described in the section 3.1.4
Value applied	Total: 11,800
Justification of choice of data or description of measurement methods and procedures applied	Required by the BL-PL module
Purpose of data	Used in equation 5 of the module BL-PL
Comments	n/a
Data / parameter	$D_{\text{planned}, i, t}$
Data unit	%
Description	Annual proportion of areas that would be deforested in stratum i, at time t.
Source of data	Calculated using maps using GPS coordinates or remote sensing data. Described in section 3.1.4
Value applied	9.3 % p.a. for forest 8.7 % p.a. for forested savannah
Justification of choice of data or description of measurement methods and procedures applied	Required by BL-PL and LK-ASP modules
Purpose of data	Used in equation 5 of the BL-PL module and for calculating leakage emissions
Comments	n/a
Data / parameter	$L-D_i$
Data unit	%

Description	Probability of deforestation in stratum i
Source of data	Described in section 3.1.4
Value applied	100%
Justification of choice of data or description of measurement methods and procedures applied	Required by BL-PL module
Purpose of data	Used in equation 5 of the BL-PL module
Comments	n/a

Data / parameter	$C_{AB,tree}$
Data unit	tCO ₂ e ha ⁻¹
Description	Average carbon stock per hectare in the tree reservoir (DBH > 10 cm), in the project area
Source of data	Calculated by allometric equation (Chave et al. 2014 and Brown 1997) on field measured data
Value applied	Forests: 214.76 Savannah: 129.97
Justification of choice of data or description of measurement methods and procedures applied	Above-ground biomass estimates were made using forest inventory data and allometric equations. Value required in CP-AB module
Purpose of data	Calculation of baseline emissions
Comments	See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>

Data / parameter	$C_{BB_tree, i}$
Data unit	tCO ₂ e ha ⁻¹
Description	Average carbon stock per hectare in the below-ground biomass carbon pool (trees and palms with DBH > 10 cm) in the Project area

Source of data	Calculated by root:shoot ratio
Value applied	Forests: 50.68 Savannah: 30.10
Justification of choice of data or description of measurement methods and procedures applied	Below-ground biomass estimates were made using forest inventory data and the root:shoot ratio (=0.22). Value required in CP-AB module
Purpose of data	Calculation of baseline emissions
Comments	See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>

Data / parameter	C _{AB,non-tree}
Data unit	tCO _{2e} ha ⁻¹
Description	Average carbon stock per hectare in the palm tree reservoir (DBH > 10 cm), in the project area
Source of data	Calculated using an allometric equation (Chave et al. 2014) based on field-measured data
Value applied	Forests: 15.61 Savannah: 6.86
Justification of choice of data or description of measurement methods and procedures applied	Above-ground biomass estimates were made using forest inventory data and allometric equations. Value required in CP-AB module
Purpose of data	Calculation of baseline emissions
Comments	See document: <i>Carbon Stock Inventory of the Pantanal REDD+Program</i>

Data / parameter	C _{DW, i}
Data unit	tCO _{2e} ha ⁻¹
Description	Average carbon stock per hectare in the dead wood carbon pool in the Project area

Source of data	Calculated by deadwood:shoot ratio
Value applied	Forests: 25.34 Savannah: 15.05
Justification of choice of data or description of measurement methods and procedures applied	Below-ground biomass estimates were made using forest inventory data and the dead wood:shoot ratio (=0.11). Value required in the CP-D module
Purpose of data	Calculation of baseline emissions
Comments	See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>

Data / parameter	CF (conversion factor)
Data unit	t C
Description	Biomass (dry weight) to carbon conversion factor
Source of data	Default value suggested by the CP-AB module
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	Value obtained for the forests of the southern Amazon Value required in CP-AB module
Purpose of data	Cálculo das emissões de linha de base
Comments	See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>

Data / parameter	44/12
Data unit	t CO ₂ e
Description	Carbon mass conversion factor for CO ₂ e
Source of data	Scientific literature: 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 AFOLU
Value applied	44/12

Justification of choice of data or description of measurement methods and procedures applied	IPCC default value. Value required in CP-AB module
Purpose of data	Calculation of baseline emissions
Comments	See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>
Data / parameter	Allometric equation = $(0.0673 * (p * (DBH)^2 * H)^{0.976}) / 1000$
Data unit	Mg ha ⁻¹ (the equation was divided by 1000 so that the result would be in Megagrams)
Description	Equation to convert DBH to aboveground dry mass. This equation was applied to trees and palms with DBH > 10.0cm.
Source of data	Equation from Chave et al. (2014): Chave, J., Réjou-Méchain, M., Búrquez, A., Chidumayo, E., Colgan, M. S., Delitti, W. B., ... & Vieilledent, G. (2014). Improved allometric models to estimate the aboveground biomass of tropical trees. <i>Global Change Biology</i> , 20(10), 3177-3190.
Value applied	View field data table (excel spreadsheet supplementary to this report)
Justification of choice of data or description of measurement methods and procedures applied	Equation developed for all types of vegetation. Equation required in the CP-AB module
Purpose of data	Calculation of baseline emissions
Comments	See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>
Data / parameter	Allometric equation = $EXP(0.893 - E + 0.760 \ln(D) - 0.0340[\ln(DBH)]^2)$
Data unit	meters
Description	Equation to estimate height (m) from DBH (cm), applied to trees and palm trees with DBH > 10.0cm.

Source of data	Equation from Chave et al. (2014): Chave, J., Réjou-Méchain, M., Búrquez, A., Chidumayo, E., Colgan, M. S., Delitti, W. B., ... & Vieilledent, G. (2014). Improved allometric models to estimate the aboveground biomass of tropical trees. <i>Global Change Biology</i> , 20(10), 3177-3190.
Value applied	View field data table (excel spreadsheet supplementary to this report)
Justification of choice of data or description of measurement methods and procedures applied	Equation developed for all types of vegetation. Equation required in the CP-AB module
Purpose of data	Calculation of baseline emissions
Comments	The (E) value (stress factor) calculated by Table provided by Chave et al. (2014) See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>
Data / parameter	Stress factor (E)
Data unit	-
Description	The value of the environmental stress factor (E) varies in relation to the covariates Temperature Seasonality (TS), Climate Water Deficit (CWD) and Precipitation Seasonality (PS).
Source of data	Equation from Chave et al. (2014): Chave, J., Réjou-Méchain, M., Búrquez, A., Chidumayo, E., Colgan, M. S., Delitti, W. B., ... & Vieilledent, G. (2014). Improved allometric models to estimate the aboveground biomass of tropical trees. <i>Global Change Biology</i> , 20(10), 3177-3190. The Stress factor (E) is available in Chave et al. (2014) using the R Platform (http://chave.upstlse.fr/pantropical_allometry.htm)
Value applied	Each sample plot had its associated Stress value. View field data table (excel spreadsheet supplementary to this report)

Justification of choice of data or description of measurement methods and procedures applied	Factor necessary to estimate the height of inventoried trees and palm trees; the height was necessary to apply the equation from Chave et al. (2014) to estimate biomass; biomass is required in the CP-AB module
Purpose of data	Calculation of baseline emissions
Comments	The (E) value (stress factor) calculated by Table provided by Chave et al. (2014) See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>
Data / parameter	root:shoot ratio
Data unit	N/A
Description	Root/shoot relation.
Source of data	Average value between Semideciduous Forests and Savannahs taken after analyzing Table 4.4. in IPCC GL AFOLU – The same Table is also shown in the CP-AB module, and in the Brazilian Fourth Communication document (See Table 28, page 156 of the Report: Land Use Sector, Land Use Change and Forests, 2020).
Value applied	Forest: 0.22 Savannah: 0.22
Justification of choice of data or description of measurement methods and procedures applied	There is a lack of equations for forests and savannah. Value required in the CP-AB module; which also guides the use of the IPCC GL AFOLU Table in the absence of equations
Purpose of data	Calculation of baseline emissions
Comments	The below-ground biomass was calculated for each plot and the aerial part biomass constitutes the sum of trees + palm trees. As we are not completely sure of the typology of each plot, using a standardized value of 0.22 (or 22%) for the root:shoot ratio was a conservative decision, both in relation to the possibility of reaching 2.04 in the Wooded Savannahs and in

	relation to possibility of reaching 0.35-0.40 as reported by Salis et al. (2014), or even the 3.34 from Savanah Parque. See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>
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Data / parameter	Deadwood:shoot ratio
Data unit	N/A
Description	Dead wood/live aerial part ratio, suitable for the type of open rainforest; Note that as defined here, the root-to-shoot ratio is applied as plot biomass
Source of data	Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference Report: Land Use Sector, Land Use Change and Forests, 2020 (See Table 28, page 156)
Value applied	Forest: 0.11 Savannahs: 0.11
Justification of choice of data or description of measurement methods and procedures applied	There is a lack of equations for forests and savannahs. Value required in the CP-D module.
Purpose of data	Calculation of baseline emissions
Comments	As we are not completely sure of the typology of our Savannahs, we used 11%, as it is a conservative value, as it is lower than the 14% estimated for wooded savannahs. See document: <i>Carbon Stock Inventory of the Pantanal REDD+ Program</i>

Data / parameter	D
Data unit	g.cm ⁻³
Description	Basic density of wood
Source of data	Chave et al. (2006)
Value applied	0.645.

Justification of choice of data or description of measurement methods and procedures applied	Required by the equation of Chave et al. (2014) to calculate biomass, which is required by the CP-AB module.
Purpose of data	Calculation of wood products inventory
Comments	Average value for all tropical trees

Data / parameter	<i>COMFi</i>
Data unit	(dimensionless)
Description	Combustion factor for stratum i
Source of data	IPCC (2006)
Value applied	Forests: under survey Savannahs: 0.73
Justification of choice of data or description of measurement methods and procedures applied	Value required in the E-BPB module Standard value for tropical rainforest presented in Table 2.6 of the IPCC (2006)- Volume 4, Chapter 2, Appendix 2)-. The combustion factor is a measure of the proportion of fuel that is actually burned, which varies depending on the size and architecture of the fuel load (i.e., a smaller proportion of large, thick fuels such as tree trunks will be burned compared to fine fuels such as grass blades), the moisture content of the fuel, and the type of fire (i.e. intensity and rate of propagation).
Purpose of data	Calculation of baseline emissions Project emissions
Comments	The default value must be updated whenever new guidelines are produced by the IPCC.

Data / parameter	<i>Gg,i</i>
Data unit	g kg ⁻¹ of dry matter burned
Description	Emission factor for stratum i and gas g

Source of data	IPCC (2006)
Value applied	Forests: Gas CH ₄ =4.7 - Gas N ₂ O= 0.26 Savannahs: Gas CH ₄ =2.3 - Gas N ₂ O= 0.21
Justification of choice of data or description of measurement methods and procedures applied	Value required in the E-BPB module Default values in Table 2.5 of the IPCC (2006) - Volume 4, Chapter 2, Appendix 2). Values considered conservative by the IPCC
Purpose of data	Calculation of baseline emissions Project emissions
Comments	The default value must be updated whenever new guidelines are produced by the IPCC.
Data / parameter	<i>GWP_g</i>
Data unit	t CO ₂ /t gas g
Description	Warming potential for gas g
Source of data	Default values in Table 2.5 of IPCC (2007) - Fourth Assessment Report: Climate Change 2007.
Value applied	Forests: Gwp CH ₄ =25- Gwp N ₂ O=298 Savannahs: Gwp CH ₄ =25- Gwp N ₂ O=298
Justification of choice of data or description of measurement methods and procedures applied	Value required in the E-BPB module Values considered conservative by the IPCC
Purpose of data	Calculating emissions from fires
Comments	The default value must be updated whenever new guidelines are produced by the IPCC.
Data / parameter	C _{POST} , planned,
Data unit	MgCO ₂ e ha ⁻¹

Description	Total carbon stock of the evaluated reservoirs, after planned deforestation
Source of data	Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference Report: Land Use Sector, Land Use Change and Forests, 2020”.
Value applied	Pasture: 36.67 Agriculture and pasture mosaic: 27.76 Soybean Planting: 18.85
Justification of choice of data or description of measurement methods and procedures applied	Required by BL-PL and M-REDD modules
Purpose of data	To calculate the differences between before and after planned deforestation
Comments	
Data / parameter	A _{sp}
Data unit	Hectare (ha)
Description	Area and number of sample plots
Source of data	Inventory
Value applied	Forest Stratum: 44 plots of 1ha each Savannahs stratum: 68 plots of 1 ha each
Justification of choice of data or description of measurement methods and procedures applied	Indication of the sample size of the Project stock.
Purpose of data	During monitoring, equations 2, 6 and 14 of the CP-AB module must be used.
Comments	Where carbon stock increases are included, monitoring must occur at least every 5 years

Data / parameter	Ai
Data unit	ha
Description	Forest stratum area i
Source of data	Geographic coordinates and maps of SR
Value applied	Forest: 9,023 Forested Savannah: 2,778
Justification of choice of data or description of measurement methods and procedures applied	Required by the M-REDD module
Purpose of data	Calculation of baseline emissions
Comments	Should be monitored at baseline reassessments

3.3.2 Data and Parameters Monitored (VCS, 3.16)

Under development and revision. Items better detailed in section 3.3.3.

Data / parameter	A _{DefPA,i,u,t}
Data unit	ha
Description	Area of deforestation registered in the project area in stratum i converted to land use in year t
Source of data	Calculated using remote sensing and data available from reliable sources
Description of measurement methods and procedures to be applied	Monitoring the forest component through remote sensing using satellite images and data from reliable sources.
Frequency of monitoring/recording	Annual
Value applied	Under development
Monitoring equipment	Geotechnologies: remote sensing and geographic information systems

QA/QC procedures to be applied	In mapping changes in forest cover and defining land use classes, data obtained at medium spatial resolution (between 10m and 100 m) will be used. Subsequently, to validate and refine the described mapping, data obtained by high-resolution sensors (up to 5 m pixels) will be used. The minimum accuracy of the land use classification map and land cover classification map is 80%
Purpose of data	Calculation of project emissions
Calculation method	N/A
Comments	N/A
Data / parameter	Δ CWPS-REDD
Data unit	t CO ₂ e
Description	Net GHG emissions in the REDD project scenario until year t*
Source of data	Under development
Description of measurement methods and procedures to be applied	Under development
Frequency of monitoring/recording	Annual
Value applied	Under development
Monitoring equipment	Under development
QA/QC procedures to be applied	Under development
Purpose of data	Calculation of project emissions

Calculation method	Calculated based on VMD0015
Comments	N/A
Data / parameter	$\Delta C_{LK-AS,planned}$
Data unit	t CO _{2e}
Description	Net greenhouse gas emissions due to leaks caused by the displacement of project activities that avoid planned deforestation
Source of data	Under development
Description of measurement methods and procedures to be applied	Under development
Frequency of monitoring/recording	Annual
Value applied	Under development
Monitoring equipment	Under development
QA/QC procedures to be applied	Under development
Purpose of data	Leakage calculation
Calculation method	Calculated based on VMD0009
Comments	N/A

3.3.3 Monitoring Plan (VCS, 3.16, 3.20)

The monitoring of the Pantanal REDD+ Program is focused on tracking changes in carbon stocks to estimate actual changes in greenhouse gas (GHG) emissions. To be considered, these changes must

demonstrate an uncertainty of no more than 15% and be statistically significant. Consequently, not all fluctuations in GHG emissions are taken into account.

In this sense, an essential activity in monitoring is to verify the uncertainty and significance of changes. For this, the project can use the guidance, respectively, from the X-UNC module and the CDM T-SIG tool. Additionally, monitoring for activities classified as APD is guided by the BL-PL module, in addition to the monitoring module itself (M-REDD), which is mandatory in REDD-MF projects.

3.3.3.1 Monitoring current changes in carbon stocks and gas emissions within the Project Area

Technical description of the monitoring task

The Program will encompass continuous monitoring of the implementation of REDD+ activities in property areas within the respective PAs, located in the Pantanal plain. During the planning and development phase of the Program, with regard to the baseline to be developed for each instance, the following steps must be carried out (detailed in sections 3.1 and 3.2 above): i. delimitation of areas eligible for the project for each instance, analyzing the historical period and excluding consolidated areas; ii. the stratification of the area eligible for carbon stock measurement by vegetation classes, in order to reduce uncertainties and increase the precision of the results; iii. baseline carbon stock estimates from forest inventory data; iv. determination of historical deforestation rates for APDef scopes, through proxy areas; and v. delimitation of leak areas.

When remeasured, the forest inventory must apply a random sampling process for each of the two defined strata (Forests and Forested Savannah). Following the results obtained so far in the Program, for each stratum the sampling process must be planned with a sampling error of 15% according to the X-UNC module. In addition to the inventory, maps of the project area and the leakage belt must be analyzed and archived, these will serve to define the area considered in the Pantanal REDD+ Program. The maps must be redone at each baseline renewal event, serving for the project's ex-ante calculations.

Changes in carbon stock due to the conversion of forested areas to non-forested areas through planned deforestation will be monitored, as well as changes in carbon stock due to forest fires and other catastrophic events. These changes will be deducted from the Project's baseline scenario in cases where they are significant. Two main activities to cover this monitoring consist of: 1) monitoring deforestation through satellite images; and, 2) improve surveillance within the properties that make up the Program, with the opportunity to verify in the field cases in which deforested areas were not detected by satellite and confirm areas that were detected.

Within the project limits there is no logging or firewood collection, and this can be demonstrated by the forest area within the project area, with the maps and the parameter Ai (project forest area). It is important to highlight that CO₂ reductions in the carbon stock due to the burning of biomass are counted as a change in the carbon stock, and non-CO₂ gases were conservatively excluded as recommended by REDD-MF (see item above on definition of reservoirs).

Data and parameters to be collected

Data/ Parameter	Description	Unit	Source
$\Delta\text{CBSL}_{\text{planned}}$	Baseline net gas emissions from planned deforestation up to year t	MgCO ₂ e	Calculations are made according to equations 1 and 13 of the BL-PL module, using $A_{\text{planned}, i}$
$A_{\text{burn}, i, t}$	Burned area in stratum i in year t	ha (hectare)	SR and GIS with field validation
$A_{\text{DistPA}, i, t}$	Impacted area by natural disturbances in stratum i, in year t	ha (hectare)	SR images associated with field verification
$A_{\text{DegW}, i}$	Area potentially impacted by the degradation process in stratum i	ha (hectare)	Terrain Verification and GIS
$A_{\text{P}, i}$	Total area of degradation sample plots in stratum i	ha (hectare)	Terrain Verification and GIS
CDegW, i, t	Biomass carbon from trees cut and removed through illegal logging and the degradation process of firewood and charcoal extraction from plots measured in stratum i in year t	MgCO ₂ e	Calculated from the forest inventory
$E_{\text{biomassBurn}}, i, t$	Greenhouse gas emissions due to biomass burning in stratum i in year t	Mg CO ₂ e	Calculated
$\Delta\text{CWPS-REDD}$	Net GHG emissions in the project scenario	MgCO ₂ e	Total net emissions

Overview of Data Collection Procedures

The mapping of forest cover in the Project area is done using satellite images with a spatial resolution of 30 meters or higher; with few clouds; captured close to the dry season, with appropriate radiometric quality; the images must be classified to identify the categories of forest, non-forest vegetation, and deforestation; the analysis must involve the Pre-processing; Interpretation and classification; and, Post-processing stages; each of these phases must be well explained in the PD.

These images must be from qualified and scientifically recognized sources, such as PRODES and DETER, developed by INPE, MapBiomass, developed by a collaborative network formed by NGOs, universities, and technology startups, among other qualified and recognized sources that may be evaluated to meet the

requirements of data quality and accuracy (90% accuracy). These maps are part of the baseline, and the latter two are also part of the monitoring.

For the project area, net emissions are calculated using Equation 1 of the M-REDD module ($\Delta C_{WPS-REDD}$),

$$\Delta C_{WPS-REDD} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{P,DefPA,i,t} + \Delta C_{P,Deg,i,t} + \Delta C_{P,DistPA,i,t} + GHG_{P-E,i,t} - \Delta C_{P,Enh,i,t})$$

Equation 1 of the M-REDD

module

To achieve this goal, the first step is to verify the land use change in the project area, using the images mentioned above. The calculation of these emissions requires monitoring the emissions resulting from different disturbance categories, referred to as stratum "q" in the M-REDD module, which are:

- 1) deforestation that occurs in the project area ($\Delta C_{P,DefPA,i,t}$);
- 2) degradation that occurs in the project area ($\Delta C_{P,Deg,i,t}$);
- 3) natural disturbances that occur in the project area ($\Delta C_{P,DistPA,i,t}$);
- 4) emissions of greenhouse gases as a result of deforestation and degradation activities within the project area ($GHG_{P-E,i,t}$);
- 5) forest growth and sequestration during the project in areas projected to be deforested in the baseline ($\Delta C_{P,Enh,i,t}$).

This last value ($\Delta C_{P,Enh,i,t}$) can be conservatively assumed to be zero, according to the M-REDD module. The other values will be detailed below for the entire project area, following the guidelines of the M-REDD module, as well as the specific calculations for planned deforestation areas, following the guidelines of the BL-PL module.

Monitoring deforestation in the project area ($\Delta C_{P,DefPA,i,t}$)

The net change in carbon stock as a result of deforestation is equal to the deforested area multiplied by the emission per unit area; this formula is presented in equation 3 of the M-REDD module, as shown below:

$$\Delta C_{P,DefPA,i,t} = \sum_{u=1}^U (A_{DefPA,u,i,t} * \Delta C_{pools,P,Def,u,i,t})$$

Equation 3 of the M-REDD module

Therefore, the analysis phase of land use change maps is important to define ($A_{DefPA,u,i,t}$), which is the deforestation area recorded in stratum i converted to land use u in the year, within the project area. Additionally, the net changes in carbon stock in all reservoirs in the case of land use in the project ($\Delta C_{pools,Def,,u,i,t}$) are also required, which can be calculated using equation 5 of the M-REDD module, as shown below:

$$\Delta C_{pools,Def,i,t} = C_{BSL,i} - C_{P,post,i} - C_{WP,i}$$

Equation 5 of the M-REDD module

Equation 5 of the M-REDD module considers the carbon stock (em CO₂e ha⁻¹) in all reservoirs found in the baseline (CBSL, i,), originating from the forest inventory, the post-deforestation stock (CP, post, i), considering the land use type "u" (which is the type of use to which the area would be converted if deforested), and the carbon stock sequestered in timber products from logging (CWP, i,). Since there is no logging in the Pantanal REDD+ Program, this value can be considered zero.

To calculate the remaining carbon stock after deforestation (CP, post, i), standard values adopted by the Ministry of the Environment (MMA) and presented in the Brazilian Fourth Communication (Table 23, page 159) can be applied. Table 17 presents three types of land use most likely to be installed in the Pantanal. Pasture values can be considered for two reasons: (1) extensive pastures for cattle ranching are the most likely type of use to occur, as it is the main productive activity in the Pantanal plain; and (2) the BL-UP module (page 31) guides that when faced with different land use possibilities, one can choose the one with the highest carbon stock, this choice is considered conservative.

Table 17. Carbon stock by land use after deforestation. Data source: Values in Mg C ha⁻¹ were extracted from the "Fourth National Communication on Brazil's Biennial Update Report to the United Nations Framework Convention on Climate Change – Reference Report: Land Use Sector, Land Use Change and Forests, 2020"; and the values in Mg CO₂e ha⁻¹ were calculated using the factor 44/12.

Land use after deforestation	Mg C ha ⁻¹	Mg CO ₂ e ha ⁻¹
Pasture	10	36,67
Agriculture and pasture mosaic	7,57	27,76
Soybean plantation	5,14	18,85

Monitoring of degradation that occurs within the project area ($\Delta C_{P,Deg,i,t}$)

In general, methods using remote sensing technology are not able to directly measure the changes in degradation that may occur in the area, but they do have some ability to identify forest strata that have undergone a change in biomass, which helps in monitoring. The calculation can be done using Equation 7 of the M-REDD module, as shown below:

$$\Delta C_{P,Deg,i,t} = \Delta C_{P,DegW,i,t} + \Delta C_{P,SelLog,i,t}$$

Equation 7 of the M-REDD module

This equation considers two types of changes:

1) ($\Delta C_{P,DegW,i,t}$) – the net change in carbon stock as a result of degradation through illegal logging for timber or firewood and charcoal extraction in the project area in stratum i in year t; Mg CO₂-e

2) ($\Delta CP, SelLog, i, t$) – the net change in carbon stock as a result of degradation originating from selective logging in forest management areas certified by FSC, considering stratum i in year t ; ; Mg CO₂-e. Since this possibility does not exist in the area, this value can be considered zero.

To calculate $\Delta CP, DegW, i, t$ the first step is to determine if there is potential for illegal activities related to timber, firewood, and charcoal extraction. According to the area report, these activities do not seem to be significant in the Pantanal REDD+ Program area. Therefore, this change can be assumed to be zero.

Ideally, throughout the project's existence, these activities can be monitored every 2 years using good remote sensing images, such as those from DETER, associated with field verification. The Participatory Rural Appraisal (PRA) can also be used, conducted every 2 or 3 years with area residents. If degradation is detected, it is necessary to identify the area potentially subject to degradation and monitor carbon changes through equation 8 of the M-REDD module.

$$\Delta C_{P, DegW, i, t} = A_{DegW, i} * \frac{C_{DegW, i, t}}{AP_i}$$

Equation 8 of the M-REDD module

Monitoring of natural disturbances

Emissions originating from forest degradation caused by natural disturbances must be estimated. Natural disturbances arise from tectonic activities (such as earthquakes, landslides, volcanoes), extreme weather conditions (such as hurricanes), droughts, or forest fires. These emissions can be omitted if they are considered insignificant through the T-SIG Tool. Also, a risk assessment can be performed.

The net change in carbon stock as a result of the natural disturbance is equal to the disturbed area multiplied by the emission per unit area.

When the disturbance event occurs ex post in the project area, the disturbed area must be defined and mapped as in the monitoring of ARRL,forest,t and ARRD,unplanned,hrp.

For planned deforestation, the sum of $ADistPA, q, i, t$ will be equal to the overlap area between the disturbed area and the sum of the planned deforestation area in the project area ($D\%planned, i, t * A_{planned, i}$), summed in the year the disturbance occurred.

The Net change in carbon stock as a result of natural disturbance in the project area in stratum i in year t ($\Delta CP, DistPA, i, t$) is calculated using equation 20 of the same module, which considers all classes (or strata) of disturbance:

$$\Delta C_{P, DistPA, i, t} = \sum_{q=1}^Q (A_{DistPA, q, i, t} * \Delta C_{P, Dist, q, i, t})$$

Equation 20 of the M-REDD module

This equation 20 requires the area impacted by natural disturbance in each class or stratum ("q") of disturbance ($ADistPA, q, i, t$) and the net change in carbon stock as a result of these disturbance classes

($\Delta C_{P,Dist,q,i,t}$). Where fire occurs, $\Delta C_{P,Dist,q,i,t}$ is similar to the burned area $A_{burnPA,q,i,t}$ and no other disturbance should be considered in that burned area.

These total changes resulting from each disturbance class ($\Delta C_{P,Dist,q,i,t}$) are calculated by equation 23 of the M-REDD module:

$$\Delta C_{P,Dist,q,i,t} = C_{BSL,i} - C_{P,Dist,q,i} - C_{WP,q,i}$$

Equation 23 of the M-REDD module

The value of $C_{BSL,i}$ corresponds to the carbon stock (in CO₂e ha⁻¹) in all reservoirs found in the baseline, originating from the forest inventory. It is subtracted from the values of $C_{P,Dist,q,i}$ and $C_{WP,Dist,q,i}$, which represent the stocks after the disturbance, CWP for timber products, which can be considered null as already presented above, and $C_{P,Dist}$ for the carbon stock in different reservoirs accounted for in the baseline (for its calculation, equation 24 of the M-REDD module can be used). If there are burns, the amount of carbon stock after the burn can be considered as zero. In equation 24 of the M-REDD module, only the stocks included in the baseline that were trees ($DAP \geq 10cm$), palms ($DAP \geq 10cm$), dead wood, and roots (below ground) should be considered.

$$C_{P,Dist,q,i} = C_{AB_tree,i} + C_{BB_tree,i} + C_{AB_non-tree,i} + C_{BB_non-tree,i} + C_{DW,i} + C_{LI,i} + C_{SOC,i}$$

Equation 24 of the M-REDD module

Monitoring of GHG emissions

Significant non-CO₂ greenhouse gas emissions occurring within the project boundaries must be assessed; however, these emissions must also be significant, and for this, the T-SIG tool must be used.

Emissions are calculated by summing the emissions originating from burns, fossil fuel use, and nitrogen compounds (such as fertilizers). The latter two were excluded from the Pantanal REDD+ Program, resulting in only emissions from burns.

In the event of a fire (forest fire), the E-BPB module should be used to calculate the emissions of gases; the equation 1 of this module is:

$$E_{biomassburn,i,t} = \sum_{g=1}^G \left(\left((A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i}) \times 10^{-3} \right) \times GWP_g \right)$$

Equation 1 of the E-BPB module

Where:

$E_{biomassburn,i,t}$ = Greenhouse gas emissions from biomass burning in stratum i in year t for each GHG (CH₄, N₂O) (t CO₂e) - CO₂ is not accounted for here, as it is already accounted for in the reservoir stock.

$A_{burn,i,t}$ = Burned area in stratum i in year t (ha)

$B_{i,t}$ = Average aboveground biomass stock before burning in stratum i , year t (t d.m.ha⁻¹)

$COMF_i$ = Combustion factor for stratum i (dimensionless)

$G_{g,i}$ = Emission factor for stratum i and gas g (g kg⁻¹ of burned dry matter)

GWP_g = Global warming potential for gas g (t CO₂/t gas g)

$g = 1, 2, 3 \dots$ G greenhouse gases, including carbon dioxide, methane, and nitrous oxide (unitless)

$i = 1, 2, 3, \dots M$ strata (unitless)

$t = 1, 2, 3, \dots t^*$ time elapsed since the start of the project activity (years)

The aboveground biomass values are derived from the forest inventory, and the values of the factors required in the equation above are presented in Table 18:

Table 18. Standardized values for calculating emissions from fires.

Parameters	Value Forests Stationary/ Cerradão	Value Savannahs	Data source
$COMF$ – Combustion Factor	0,5*	0,73**	IPCC (2006)
G_g – Emission Factor	G CH ₄ =4,7 G N ₂ O= 0,26	G CH ₄ =2,3 G N ₂ O= 0,21	IPCC (2006), volume 4, chapter 2, table 2.5
GWP_g^{***} – Warming Potential	Gwp CH ₄ =25 Gwp N ₂ O= 298	Gwp CH ₄ =25 Gwp N ₂ O= 298	

* In the IPCC (2006), there is no specific value for Dry Forests, so the same value used for Tropical Forests was applied; **Value for Tropical Savannahs; ***alue updated in the IPCC Errata. Available at: <https://www.ipcc.ch/site/assets/uploads/2018/05/ar4-wg1-errata.pdf>

Monitoring of emissions from planned deforestation (as per the BL-PL module)

For planned deforestation, it is necessary to pre-establish how much of the planned deforestation will be avoided during the project's development ($A_{planned, i}$), the annual proportion (or rate) of areas that would be deforested ($D\%_{planned, i}$), and explain how the land would be used if the planned deforestation were to occur. Equation 5 of the BL-PL module helps calculate the annual area of planned deforestation:

$$AA_{planned, i, t} = (A_{planned, i} * D\%_{planned, i, t}) * L - D_i$$

Equation 5 of the BL-PL module

Where:

$AA_{planned,i,t}$ = Annual area of planned deforestation in the baseline for stratum i at time t; ha

$D\%_{planejado,i,t}$ = Projected annual proportion of lands to be deforested in stratum i during year t. If the actual annual proportion is known and documented (e.g., 25% per year for 4 years), it can be used; %

$A_{planned,i}$ Total area of planned deforestation during the reference period for stratum i; ha

$L-Di$ = Deforestation likelihood for stratum i; %

The value of $L-Di$ can be considered 100%, as the BL-PL module establishes that in areas of planned deforestation that are not under government control and not zoned for deforestation). This is the case of the Pantanal REDD+ Program.

Changes in carbon stocks and greenhouse gas emissions are calculated using Equation 1 of the BL-PL module:

$$\Delta C_{BSL,planned} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{BSL,i,t} + GHG_{BSL-E,i,t})$$

Equation 1 of the BL-PL module

Where:

$\Delta C_{BSL,planned}$ = Net greenhouse gas emissions in the baseline from planned deforestation until year t^* ; Mg CO₂e

$\Delta C_{BSL,i,t}$ = Net changes in carbon stock across all reservoirs in baseline stratum i in year t; t CO₂e

$GHG_{BSL,E}$ = Greenhouse gas emissions resulting from deforestation activities within the project area in baseline stratum i in year t; t CO₂e ano⁻¹

$I = 1, 2, 3, \dots M$ strata.

$t = 1, 2, 3, \dots t^*$ years elapsed since the planned start of the project activity.

The value de $\Delta C_{BSL,i,t}$ is achieved using equation 13 of the BL-PL module:

$$\begin{aligned} \Delta C_{BSL,i,t} = & AA_{planned,i,t} * (\Delta C_{AB_{tree,i}} - \Delta C_{WP,i} + \Delta C_{AB_{non-tree,i}} + \Delta C_{LI,i}) \\ & + \left(\sum_{t=10}^t A_{planned,i,t} \right) * (\Delta C_{BB_{tree,i}} + \Delta C_{BB_{non-tree,i}} + \Delta C_{DW,i}) * \left(\frac{1}{10} \right) \\ & + \left(\sum_{t=20}^t AA_{unplanned,i,t} \right) * (C_{WP100,i} + \Delta C_{SOC,i}) * \left(\frac{1}{20} \right) \end{aligned}$$

Equation 13 of the BL-PL module

The differences in reservoirs $\Delta C_{Reservatório}$ can be calculated together, as shown above in the item on "Deforestation Monitoring," and can also use the values for post-deforestation land uses.

The value of $GHG_{BSL, E}$ is achieved with equation 15 of the BL-PL module; it is worth noting that only emissions from burned biomass are considered:

$$GHG_{BSL, E, i, t} = E_{FC, i, t} + E_{BiomassBurn, i, t} + N_2 O_{direct-N, i, t}$$

Equation 15 of the BL-PL module

3.3.3.2 Monitoring the Leakage Belt Area

Technical description of the monitoring task

Once the project activities are installed, actions for the conservation of forest areas are also installed, and it is expected that such activities will reduce deforestation in the area. However, the threats for deforestation to occur continue, and the actions of deforestation-causing agents may simply migrate to surrounding areas and continue deforestation in those other areas to which they migrated. That is, deforestation moves from an area now 'with conservation actions' to an area 'without conservation actions'. This migration of areas is called 'deforestation displacement'. This typically occurs due to deforestation prevention activities installed in the project area, so deforestation displacement is a consequence of project installation, and this consequence must be monitored. Thus, the project must install an area, outside the project area, where deforestation must be monitored, this area is called the 'leakage belt'.

It may occur that in the Pantanal REDD+ Program, the leakage belt does not need to be monitored. This is allowed when the specific deforestation agent is identified and the management plans and/or land use designations of the other areas in the Program expansion region, but under the same deforestation agents (which must be identified by location), have not materially changed as a result of the project (for example, when there are no new concessions for logging, increased deforestation of forests for agricultural production, or increased use of fertilizers to increase agricultural yields).

Similarly, leakage does not need to be accounted for when it can be demonstrated that it is not related to the project and responds to factors beyond the control of the project proponents (for example, markets, policies, etc.). The project proponent must provide verifiable information demonstrating this situation, such as official documents, market analyses, peer-reviewed literature, and other relevant sources, as well as an assessment showing how and to what extent this affected leakage emissions.

Data and parameters to be collected.

Data / Parameter	Description	Unit	Source
$A_{planned, I, OP}$	Total area of planned deforestation outside the project area, during the reference period for stratum i; ha	ha (hectare)	Remote sensing images

$D\%_{\text{planejado},i,t,OP}$	Projected annual proportion of lands to be deforested, outside the project area, in stratum i during year t.	%	Calculated
$\Delta C_{LK-AS,planned}$	Net greenhouse gas emissions due to leakage in APDef projects until year t		Calculated
$LKA_{planned,i,t}$	Leakage belt area in stratum i in year t.	ha (hectare)	Calculated
$NewR_{i,t}$	New deforestation in stratum i in year t, carried out by the same agents of planned deforestation, in locations where there is no leakage.	ha (hectare)	Calculated

Overview of Data Collection Procedures

The net greenhouse gas emissions due to leakage in APDef projects must be calculated according to Equation 1 of module LK-ASP:

$$\Delta C_{LK-AS,planned} = \sum_{t=1}^{t^*} \sum_{i=1}^M (LKA_{planned,i,t} \times \Delta C_{BSL,i}) + GHG_{LK,E,i,t}$$

Equation 1 of module LK-ASP

Where:

$\Delta C_{LK-AS,planned}$ = net greenhouse gas emissions due to leakage in APDef projects until year t (Mg CO₂e).

$LKA_{planned,i,t}$ = leakage belt area in stratum i in year t (ha).

$\Delta C_{BSL,i}$ = net changes in carbon stock in all reservoirs in stratum i (Mg CO₂e ha⁻¹), in the baseline.

$GHG_{LK,E,i,t}$ = greenhouse gas emissions due to leakage in stratum i in year t (Mg CO₂e)

i 1, 2, 3, ... M strata (unitless)

t 1, 2, 3, ... t* time elapsed since the beginning of the project activity (years).

In this equation, the carbon stock (em CO₂e ha⁻¹) in the reservoir set is considered, and its calculation was shown above using Equation 5 of module M-REDD, where the value from the forest inventory (or baseline, CBSL, i,) is subtracted from the post-deforestation stock (CP, post, i), considering the land use type "u" (which is the land use type to which the area would be converted if deforested).

When estimating the total deforestation area in the project area and in the leakage belt areas, it is possible to monitor possible displacements (leakage) to properties that do not participate in the Program. The predicted deforestation within the project boundaries is then subtracted from the total deforestation.

This subtraction gives the expected deforestation if no leakage occurs; or the value will be higher than the expected deforestation when leakage occurs, and the difference will be the leaked deforestation area.

To calculate the leakage belt area in stratum i in year t ($LKA_{planned,i,t}$), two steps are necessary:

1. The new deforestation in stratum i in year t , carried out by the same agents of planned deforestation, in locations where there is no leakage ($NewR_{i,t}$) must be calculated; this value represents the average number of hectares deforested per year in all concessions, considering the same agents of deforestation. For this, the formula of Equation 4 of module LK-ASP must be used:

$$NewR_{i,t} = (D\%_{planned,i,t,OP} \times A_{planned,i,OP})$$

Equation 4 of module LK-ASP

To obtain this calculation, the guidance provided in Module BL-PL, Part 1.2 (Planned Deforestation Area $A_{planned,i}$) can be used to identify all areas that can be converted to non-forest lands outside the project boundaries during the baseline period ($A_{planned,i,t,OP}$). Apply Section 1.3 (Planned Deforestation Rate $D\%_{planned,i,t}$) of Module BL-PL to estimate the deforestation rate (deforested area per year, ha year⁻¹) in such areas ($D\%_{planned,i,t,OP}$). Note that the designation OP refers to areas outside the project area.

2. Calculate $LKA_{planned,i,t}$ using Equation 6 of module LK-ASP:

$$LKA_{planned,i,t} = A_{defLK,i,t} - NewR_{i,t}$$

Equation 6 of module LK-ASP

The deforestation area recorded in the leakage belt area in stratum i converted to land use u in year t ($A_{defLK,i,t}$) was mentioned in the parameters regarding project installation, as this needs to be defined during project installation; therefore, we emphasize that in the parameters mentioned above (for emissions in the leakage belt area) this parameter was mentioned for the second time.

Finally, to monitor emissions of other greenhouse gases, the formulas are the same as those used to calculate emissions in the project area, obviously replacing the area with the leakage belt area to be used.

3.3.3.3 Monitoring the ex-post net change in carbon stock and GHG emissions

According to VM0007, ex-post calculations of net anthropogenic greenhouse gas emissions reductions should follow the guidelines of module M-REDD; and M-REDD stipulates that all calculations can be used ex-ante and ex-post. Therefore, all items presented above for monitoring current changes should also be monitored ex-post.

3.3.3.4 Reassessment of baseline projections

The baselines will be reassessed over time in the Program, as guided by VM0007, because the agents, drivers, and underlying causes of deforestation change dynamically. However, this update should not be

frequent or unexpected, as it can create serious uncertainties in the market. The baseline will be updated every 10 years and in a planned manner. Two activities are necessary in updating the baseline:

1. Update information on changes in land use and land cover from the baseline.
2. Update the carbon component of the baseline.

3.3.3.5 Quality control and assurance procedure

During the implementation of the Pantanal REDD+ Program, all activities will be continuously monitored and evaluated by the proponent, allowing for a continuous feedback process to incorporate improvements in the procedures applied to all types of monitored data, consequently ensuring better quality to the Program.

Maps showing changes in carbon stock due to the conversion of forests into non-forest areas by unplanned and planned deforestation will undergo an analysis of overall accuracy and the kappa index obtained from a confusion matrix, such as the Congalton and Green matrix (1999). At least 100 points will be randomly distributed in relation to the area analyzed through a geographic information system. Validation will be performed using high-resolution satellite images and/or field-collected data. The minimum mapping accuracy, according to REDD-MF, for each class or category in the land use and land cover map should be 90%. In addition to satellite images, field checks in areas where forest conversion to non-forest areas is identified will be conducted.

3.3.4 Dissemination of Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

The Climate Monitoring Plan and all its results will be publicly disclosed on the VERRA registration page and on the Biofílica Ambipar Environment website.

For stakeholders without internet access, a printed copy can be provided at strategic points (farm headquarters and partner offices). Information about the Program will also be disseminated via email and WhatsApp (to a pre-established list of stakeholders), and on social media platforms such as Instagram and LinkedIn of the project proponent.

In this way, it is expected that all necessary efforts to provide full transparency and distribute key information and documents of the Program will be achieved.

3.4 Optional Criterion: Climate Change Adaptation Benefits

The Pantanal REDD+ Program is applying for Gold Level for climate due to its significant support in helping communities and/or biodiversity adapt to the impacts of climate change through strategies aligned with the activities planned by the Program.

3.4.1 Regional Climate Change Scenarios (CCB, GL1.1)

Human activities, mainly through greenhouse gas emissions, have caused global warming, with the global surface temperature reaching a value 1.1 °C higher between 2011-2020 than in the period 1850-1900. Global greenhouse gas emissions continued to rise, with historical and ongoing unequal contributions resulting from unsustainable energy use, land use and land-use change, lifestyles, and consumption and production patterns between regions, countries, and within them, and among individuals (IPCC, 2023⁵⁴).

There have been widespread and rapid changes in the atmosphere, ocean, cryosphere, and biosphere. Human-induced climate change is already affecting many climate and weather extremes in all regions of the world. This has resulted in widespread adverse impacts, and nature- and people-related losses and damages. Vulnerable communities that have historically contributed least to the current climate change are disproportionately affected (IPCC, 2023).

The Summary Report of the Sixth IPCC Assessment (AR6) highlights the rapid emission reductions necessary to achieve intermediate climate goals - reducing greenhouse gas emissions by 43% by 2030 and 60% by 2035, to net-zero emissions by mid-century, and to avoid global temperatures exceeding the dangerous 1.5 °C tipping point. Exceeding the 1.5 °C limit, even temporarily, will cause even more severe and often irreversible impacts - from the extinction of local species to the drying of salt marshes and loss of human lives due to heat stress. Restricting the exceedance of the 1.5 °C limit, in timing and magnitude, will be essential to ensure a safe and habitable future, as well as to keep warming as close as possible or below 1.5 °C. However, the document recognizes that current policies are off track to achieve these goals, despite the variety of economic solutions available (WWF, 2023⁵⁵).

From unprecedented floods to record heatwaves, the impacts of the climate crisis are more evident than ever. Rising temperatures, more severe and frequent extreme weather events, and irregular rainfall are driving biodiversity loss, affecting food prices, undermining livelihoods, and have been associated with increasing social and gender inequalities and large-scale displacement (UNDP, 2023⁵⁶).

Regarding Brazil, a study conducted by USAID⁵⁷ shows that in the Pantanal region, recent increases in the interannual variability of floods and droughts threaten local species adapted to seasonal floods, the broader ecosystem, and humans who depend on these natural resources. It is evident that climate change is affecting all regions of the planet, and the Pantanal is no exception. These climate changes have been affecting the biome in various ways, including changes in geological processes, increased risk of fires, and impacts on biodiversity.

Furthermore, it can be said that the ongoing climatic variability in the Pantanal has been modulating the magnitude of river discharge from year to year in the region, as well as the sediment load from highlands

⁵⁴ Relatório Síntese, IPCC. Disponível em: https://www.gov.br/mcti/pt-br/acompanhe-o-mcti/sirene/publicacoes/relatorios-do-ipcc/arquivos/pdf/copy_of_IPCC_Longer_Report_2023_Portugues.pdf

⁵⁵ WWF. Disponível em: <https://www.wwf.org.br/?85141>

⁵⁶ UNDP. Disponível em: https://www.undp.org/latin-america/publications/guidance-note-climate-peace-and-security-latin-america-and-caribbean?gad_source=1&gclid=EAlaIQobChMI2JiIrnaggMVvyqtBh1eIQKBEAAYAAEglwg_D_BwE

⁵⁷ USAID, CLIMATE RISK PROFILE BRAZIL Disponível em: https://www.climateintelinks.org/sites/default/files/asset/document/2018-April-30_USAID_CadmusCISF_Climate-Risk-Profile-Brazil.pdf

to plains. This also affects groundwater recharge, as well as the sinking and accommodation of river channels, according to the *Jornal da Unesp*⁵⁸. Also, concerning trend analysis published in the *Unesp* magazine, researchers suggest that the intensity of summer precipitation and the number of dry days in autumn/winter due to climate change have been consistently increasing. Thus, there is more rain and a higher sediment load in the rainy seasons and a greater water deficit in the dry seasons.

Another highlight is the fire in the biome, this element is also being pressured by climate change. According to researchers from *Unesp*⁵⁹, just as with rain, fire also has a dynamic of longer temporal cycles, in addition to the annual variation itself. However, with the average global temperature increase, aligned with the impacts of environmental destruction in the Amazon and deforestation due to agriculture occurring in the headwaters and springs of rivers that drain into the Pantanal (most of them in the Cerrado), this dynamic has been altered. In other words, human degrading actions on ecosystems are certainly factors that multiply the effects of climate changes in certain annual or multiannual periods.

If climate management continues as it is, the Pantanal, which has already lost 29% of its water surface and flooded fields between the flood of 1988 and the last one in 2018, may be dry in 70 years⁶⁰. Therefore, it is crucial to adopt firefighting structures, integrated fire policies, and, mainly, for Brazil to address its major conservation challenges.

Figure 17 below shows the historical average temperature in Mato Grosso do Sul, with a clear warming trend from 1951 to 2020. Analyzing climate models, it is noticed that in most of them there is a trend of increasing average temperatures in the coming years. Figure 18 illustrates this trend up to 2099. Thus, analyzing historical data and future models of temperature and precipitation, it is evident that climate changes are occurring and causing alterations in the environment.

⁵⁸Jornal da Unesp | A caixa d'água pantaneira, disponível em: <https://jornal.unesp.br/2023/03/15/a-caixa-dagua-pantaneira/>

⁵⁹Jornal da Unesp | A caixa d'água pantaneira, disponível em: <https://jornal.unesp.br/2023/03/15/a-caixa-dagua-pantaneira/>

⁶⁰Fapesp na mídia. Disponível em: <https://namidia.fapesp.br/cientistas-dizem-que-se-gestao-do-clima-continuar-como-esta-o-pantanal-deixara-de-existir-mato-grosso/352600>

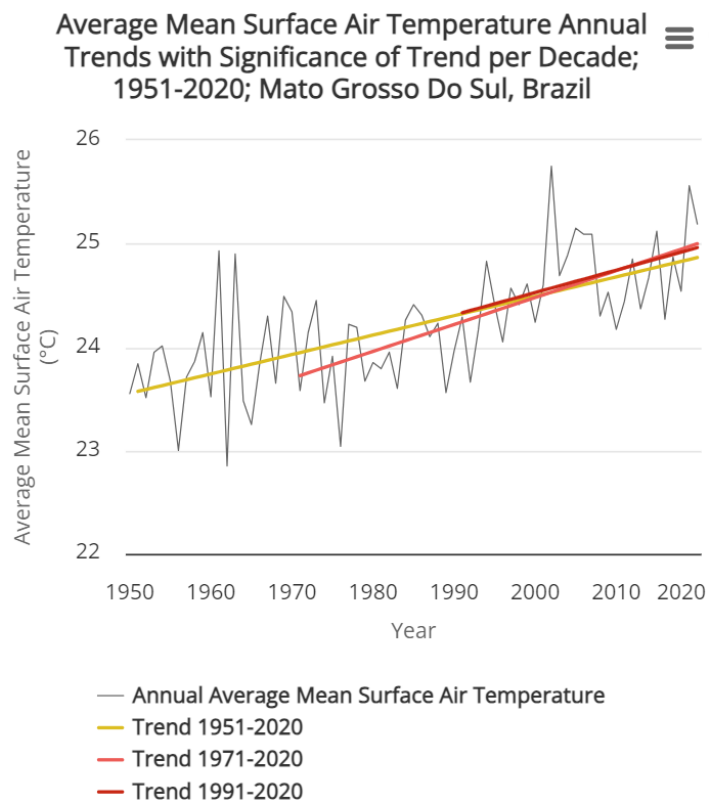


Figure 17: Average air surface temperature graph. Source: Climate Change Knowledge Portal.

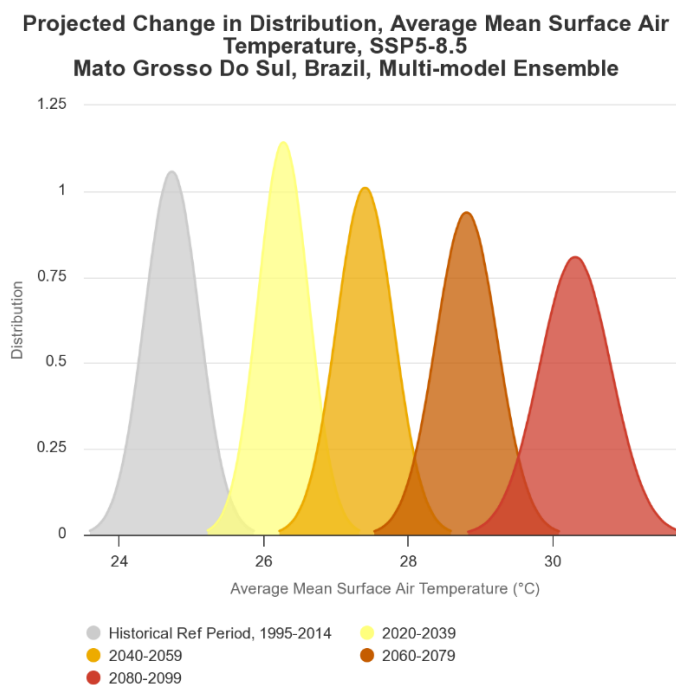


Figure 18: Projected average air surface temperature. Source: Climate Change Knowledge Portal.

The loss of biodiversity, intense droughts, changes in ecosystems, and changes in the dynamics of livestock and agriculture are possible changes in the local land use scenario due to these climate change scenarios.

3.4.2 Climate Change Impacts (CCB, GL1.2)

The IPCC's AR6 report⁶¹ presents evidence of global warming already underway. The evidence is: the unprecedented concentration of CO₂ in the last 2 million years; the unprecedented retreat of glaciers in the last 2 thousand years; the last decade has been warmer than any period in the last 123,000 years; sea levels have risen faster than in any century in the last 3,000 years; summer ice cover in the Arctic is the lowest it has been in a thousand years; ocean warming faster than at any time since the ice age; and ocean acidification reached the highest level in the last 26 thousand years.

According to the same report, each fraction of a degree increase in temperature intensifies climate risks such as biodiversity loss, water stress, reduced food security, increased number of fires, more extreme heat events, sea level rise, floods, and coral reef decline.

On a global scale, climate change affects the well-being of communities in various aspects such as health, food and water security, resource availability, and extreme weather events, putting various populations in vulnerable situations. According to the World Health Organization (WHO), climate change increases the incidence of food, water, and vector-borne diseases (mosquitoes). Infectious diseases such as leishmaniasis, malaria, dengue, and other arboviruses are more sensitive to these changes, as reported by the Oswaldo Cruz Foundation (Fiocruz⁶²). In addition to these, hepatitis A is also a concern, as the virus causing the disease can be transmitted through the consumption of contaminated water and food, especially in areas lacking basic sanitation or frequently affected by floods.

When we say that food security is affected by climate change, in practical terms, it means that food production can be hindered by high temperatures, desertification, water stress, and other processes resulting from climate change (Adapta Clima - MMA⁶³). This can lead to a decrease in food availability and an increase in food price volatility. Moreover, climate change can also affect the nutritional quality of food. For example, the increase in carbon dioxide in the atmosphere can decrease the content of essential nutrients in many crops (ALPINO et al., 2022⁶⁴).

Regarding biodiversity, the Living Planet Report⁶⁵ released in 2020 by the WWF network indicates that monitored populations of mammals, birds, amphibians, reptiles, and fish declined by an average of two-thirds between 1970 and 2016. According to them, the main direct driver of biodiversity loss in terrestrial systems in recent decades has been changes in land use, mainly the conversion of untouched native

⁶¹ IPCC's AR6 report. Available at: <https://www.unep.org/pt-br/resources/relatorios/sexta-relatorio-de-avaliacao-do-ipcc-mudanca-climatica-2022>

⁶² Portal Fiocruz. Disponível em: <https://portal.fiocruz.br/mudancas-climaticas>

⁶³ Adapta Clima. Disponível em: <http://adaptaclima.mma.gov.br/seguranca-alimentar-e-nutricional-no-contexto-da-mudanca-do-clima>

⁶⁴ ALPINO et al., 2022. Disponível em: <https://www.scielo.br/j/csc/a/Rdr4LGpjWwGfmkgxMs6pLSL/#>

⁶⁵ WWF 2020. Disponível em: https://wwfbr.awsassets.panda.org/downloads/lpr_pt_2020_v2.pdf

habitats into agricultural systems. Climate change is not yet the main factor in global biodiversity loss, but over the next few decades, it is expected to become as or more important than other factors.

However, predicted climate changes are likely to have an impact on biodiversity conservation by reducing the amount of water available in the ecosystem during the dry season, decreasing the amount of habitat available for species. This can result in ecosystem degradation around remaining water sources, threatening species survival. More irregular rainfall can also compromise biodiversity by affecting fire regimes, as hotter and longer dry seasons can increase the occurrence of fires.

In addition to the references, the Socioeconomic and Environmental Diagnosis sought evidence that would bring perceptions from local stakeholders regarding some issues related to climate change, through field diagnosis. Below are presented these perceptions and concerns raised for the project zone.

- Loss of biodiversity (fauna and flora).
- Forest degradation.
- Increase in extreme climate events (droughts and floods).
- Increase in air temperature, consequently increasing the number of forest fires and health problems for the local population.
- Changes in water availability.
- Loss of productivity.
- Loss of economic activities related to non-timber forest products.
- Increase in the number of cases of tropical diseases.
- Alteration of the well-being of the population.
- Decrease in food production and availability, and an increase in food price volatility.

Therefore, climate change has a profound impact on biodiversity and the well-being of populations, so it is crucial that measures are taken to mitigate negative effects. Therefore, the Pantanal REDD+ Program actively works to mitigate climate change.

3.4.3 Measures Needed and Designed for Adaptation (CCB, GL1.3)

Although certain negative impacts of climate change are uncontrollable and inevitable, and beyond the control of the proponent, the Pantanal REDD+ Program aims to increase the resilience of the community and biodiversity, adopting various project activities as adaptation options.

By having activities that reduce greenhouse gas emissions, the Pantanal REDD+ Program also works to help communities and biodiversity adapt to the likely impacts of climate change. Maintaining standing

forests helps reduce the amount of carbon dioxide in the atmosphere, as well as providing habitats for wildlife.

The benefits of the Program seek to reduce the vulnerability of local inhabitants to climate change through actions to protect the habitat in which they are inserted and promote sustainable actions in rural areas. Part of this strategy is the implementation of additional and alternative sources of income and the diversification of Payments for Environmental Services (PES) in forest and forested savannah areas, aiming to guarantee the economic viability of the sustainable practices adopted. Furthermore, the Pantanal REDD+ Program will work on the educational axis, as education is vitally important for adapting to climate change, offering opportunities and solutions to the local population.

Concern with the well-being and improvement of the quality of life of employees and owners is an essential component to achieve the goals established by the Pantanal REDD+ Program. Thus, actions were planned with the aim of improving safety, food quality, access to health and education, technical training and communication processes between farm employees.

Firefighting is also an adaptation strategy promoted by the Program. It is known that higher temperatures can increase the incidence of fires, therefore, promoting firefighting activities will bring benefits to the community and biodiversity.

Additionally, the Pantanal REDD+ Program develops activities that directly and indirectly help maintain valuable services for the ecosystem and adapt to climate change, such as hydrological regulation, pollination, pest control, and the expansion of habitat connectivity in different types of environments and climates.

4 COMMUNITY

4.1 Without-Project Community Scenario

4.1.1 Descriptions of Communities at Project Start (CCB, CM1.1)

The socio-economic and environmental diagnosis of the Pantanal REDD+ Program was carried out by the specialized team of Restaura Consultoria Ambiental e Treinamentos LTDA between October 2023 and January 2024, and the method of characterization and identification of urban and rural populations was based on secondary and primary data obtained through field surveys, combined with the systematization of previous studies and surveys, as well as available information. The analysis of the Social, Economic, Environmental, and Cultural Dimensions includes elements such as historical and current occupation, population density, number of groups and social actors, levels of education, and access to education and health services. These factors are essential for a comprehensive understanding of the reality of the municipalities in question. As of the completion of this documentation, this survey is in the final stage of consolidation, so here are presented the intermediate data found.

The first indicator presented is the FIRJAN Municipal Development Index (IFDM acronym in Portuguese), highlighting the importance of this composite indicator created in 2008, which addresses, with equal weighting, the areas of Employment & Income, Education, and Health, consolidating the level of local socio-economic development into a single number.

To facilitate interpretation, four reference concepts for IFDM were established: low stage of development (0.0 to 0.4), regular development (0.4 to 0.6), moderate development (0.6 to 0.8), and high stage of development (0.8 to 1.0). The analysis of these data provides a comprehensive understanding of the social panorama of the municipalities, providing essential subsidies for the design of strategies, policies, and specific projects to meet the needs of local communities within the scope of the Pantanal REDD+ Program.

The analysis of the data presented in Table 19, regarding IFDM for the municipalities of Aquidauana, Corumbá, and Miranda, gains even more scope when considering both the state ranking of the state of Mato Grosso do Sul, composed of 79 municipalities, and the national ranking of Brazil, composed of 5,568 municipalities. The results, referring to the year 2016 and published in 2018, provide a more complete understanding of the positioning of these localities at the regional and national levels, highlighting opportunities for improvement.

Table 19. FIRJAN Municipal Development Index (IFDM) for the municipalities of Aquidauana, Corumbá and Miranda, % IFDM Employment & Income, % IFDM Health, % General IFDM Ranking - National and % General IFDM Ranking - State for 2016

Municipality	IFDM (2016)	IFDM Employment & Income (2016)	IFDM Education (2016)	IFDM Health (2016)	General IFDM Ranking - National (2016)	General IFDM Ranking - State (2016)
Aquidauana	0.64%	0.47%	0.72%	0.74%	3442 °	63°
Corumbá	0.65%	0.47%	0.72%	0.77%	3207 °	60°
Miranda	0.69%	0.56%	0.70%	0.79%	2444 °	45°

Source: FIRJAN Municipal Development Index (2016)

A analysis of the results, considering both the state and national rankings, provides a preliminary view of the development of the municipalities in a broader context in Brazil. This evaluation not only highlights specific areas of opportunity for each locality but also guides the implementation of strategic interventions within the scope of the Pantanal REDD+ Program, aiming at promoting sustainable development at the state and national levels.

Other information regarding population census, historical and current occupation, economic development and urban changes, levels of education, access to education services, access to health services, and expectations of population growth in the municipalities of Miranda, Aquidauana, and Corumbá were gathered and are present in the socioeconomic results report.

Looking specifically at the area of the Pantanal REDD+ Program, 137 questionnaires were applied, identifying a population composed of approximately 65% male and 35% female individuals, with 61% falling within the "adult" age group, 32% "young", and 7% "elderly", and classified according to their declared race as 56% "mixed race", 21% "white", 15% "black", 7% "indigenous," and 1% "yellow." The field study did not identify traditional communities such as indigenous, riverside dwellers, or quilombola communities living in the project area. Due to being private properties, the area is inhabited by farm workers engaged in livestock farming, tourism, and other general services. Below are some data collected on the properties belonging to the Program.

C2-P01-Y24: represented by 5 respondents, there is a predominance of young age group, with 80% of participants falling within the 15 to 29 age range. Ethnic diversity is notable, with 60% of the population self-identifying as mixed race, 20% as white, and 20% as black. Male predominance (60%) and unanimous identification as heterosexuals provide an initial overview of the social composition of this community, where the majority are in stable relationships (80%).

C4-P01-Y24: composed of 23 respondents, there is an average age of 32 years, with significant representation of the young age group (65.22%). Ethnic diversity is evident, with the majority being mixed race (56.52%), followed by white (17.39%), black (21.74%), and a presence of yellow workers (4.35%). Sexual orientation shows a majority of heterosexuals (91.30%), although there is a notable presence of homosexual workers (8.70%). The variety of marital statuses reflects the heterogeneity of this community.

C5-P01-Y24: represented by 18 respondents, presents a diversified age distribution, with the majority falling within the adult age range (73.47%). The ethnic predominance is mixed race (72.22%), followed by white (5.56%), black (16.67%), and indigenous (5.56%). The gender composition is predominantly male (83.33%), and all participants identify as heterosexual, with the majority being single (33.33%).

C7-P01-Y24: composed of 2 respondents, stands out for the ethnic composition equality (100% mixed race) and an average age of 56 years. The adult age range (30 to 59 years) represents 68.42% of the population. Gender balance is evident, with 50% male and 50% female respondents, all heterosexual and married.

C1-P01-Y24: with 8 respondents, there is an ethnic distribution among mixed race (50%) and white (50%) workers. The average age is 42 years, with the adult age range representing the majority (87.50%). Male gender predominance (75%), and sexual orientation is predominantly heterosexual (100%), with a diversity of marital statuses.

C3-P01-Y24: with 6 respondents, stands out for the self-declaration of all as mixed race, mostly male (83.33%). The average age is 50.83 years, with the adult age range representing the majority (83.33%). Sexual orientation is predominantly heterosexual (100%), and regarding marital status, there is a diversity, with singles (50%), in a stable relationship (33.33%), and divorced (16.67%).

C6-P01-Y24: the data reveal a diverse and representative community, composed of 52 respondents. In terms of declared color, the farm harbors a varied ethnic-racial mix, with 11.54% indigenous, 23.08% white, 17.31% black, and 48.08% mixed race. In terms of gender, there is equity, with 50% women and 50% men. Sexual orientation also demonstrates diversity, with 86.54% heterosexual, 11.54% homosexual, and 1.92% bisexual. Regarding marital status, the community presents a significant distribution, with 19.23% married, 3.85% divorced, 67.31% single, 7.69% in a stable union, and 1.92% widowed.

Salary Income:

The salary distribution of workers on the researched rural properties reveals a significant concentration in the intermediate salary ranges, represented by ranges ranging from R\$ 1,501 to R\$ 5,000, totaling 68% of the total respondents. This suggests a considerable supply of work in these salary ranges, possibly reflecting the demand for specific skills and qualifications. On the other hand, approximately 17% of respondents are in the salary range between R\$ 1,300 and R\$ 1,500, indicating a supply of work in a more modest pay range, possibly requiring a different level of education and technical skills.

Family Composition:

We observe that most respondents (43.80%) have families composed of more than 3 to 5 individuals, indicating a significant presence of larger families on the properties. Next, about 24.82% of respondents report having families composed of 2 individuals, while 27.01% have families of only 1 individual, suggesting a considerable proportion of small families or individuals living alone. On the other hand, a small portion of respondents (4.38%) indicate having families with more than 6 people, pointing to the presence of extended families in some rural properties.

When analyzing housing arrangements, it is observed that 45 respondents (approximately 32.85%) live with colleagues in farm dormitories, while 53 respondents (about 38.69%) live with their spouse and/or children. Eleven respondents (approximately 8.03%) share housing with parents and/or other relatives, and 26 respondents (about 18.98%) live alone. Two respondents did not provide information about who they live with. These data highlight the different types of residential arrangements existing in Pantanal's rural properties, demonstrating the diversity of family structures and the variety of family and social relationships present in this context.

Education Levels:

The education level of rural workers plays a crucial role in understanding social dynamics and community development. A detailed analysis of the educational characteristics of respondents in various farms provides valuable insights for the implementation of educational policies and targeted social programs. In this context, we will address the education levels of workers in different farms, highlighting the particularities of each property.

C2-P01-Y24: We observe a variety in the education levels of the workers. The majority (60%) have incomplete high school education, while 20% have incomplete elementary education and another 20% have completed high school.

C4-P01-Y24: In terms of education, the Farm presents a remarkable diversity in the educational levels of the workers. From completed elementary education (21.74%) to post-graduate studies (8.70%), the property reflects a wide range of backgrounds. This heterogeneity suggests an educationally versatile community.

C5-P01-Y24: Most workers have incomplete elementary education (61.11%). A small percentage are literate (16.67%), while another 16.67% have incomplete high school education, and 5.56% have completed high school.

C7-P01-Y24: The education level is limited, with all respondents (100%) having only incomplete elementary education.

C1-P01-Y24: Most workers have incomplete elementary education (37.50%). There is a notable presence of literate individuals (37.50%), while 12.50% have completed higher education and another 12.50% are enrolled in higher education.

C3-P01-Y24: It reveals a population with a high literacy rate (66.67%), and the distribution in education levels includes incomplete elementary education (16.67%) and incomplete high school education (16.67%). This educational diversity contributes to the richness of the community.

C6-P01-Y24: among the 52 individuals surveyed, only 3.85% of the population is classified as literate, while 23.08% have incomplete elementary education and 1.92% have completed elementary education. It is worth noting that 28.85% have completed high school, while 15.38% have incomplete high school. Additionally, a considerable contingent of 13.46% have completed higher education, 3.85% are currently enrolled in higher education, and 7.69% are pursuing post-graduate studies. The presence of individuals with incomplete higher education (1.92%) and those pursuing post-graduate studies (7.69%) indicates a

continuous pursuit of qualification and academic improvement. This educational heterogeneity in the C6-P01-Y24 community suggests a wide range of knowledge and skills, promoting an environment conducive to the exchange of experiences and the joint development of initiatives aimed at the diversified activities of the property.

Access to Health Services

The field research reveals a challenging scenario regarding the health of workers on rural properties. Although there are occasional vaccination campaigns and collective efforts that include various services such as social assistance, education, document updates, occupational safety guidance, and fire prevention actions involving institutions like the Fire Department, Prevfogo, and SOS Pantanal, significant obstacles still persist. The lack of specific healthcare stands out as a concern, highlighting the logistical difficulties faced by workers in accessing healthcare services regularly.

During the Covid-19 pandemic, properties implemented preventive measures, including vaccinations for all workers. These actions were fundamental in mitigating the risks associated with the virus's spread, demonstrating a proactive response from property owners and managers. However, the lack of regular healthcare services remains a gap to be addressed, underscoring the need for more comprehensive strategies to address health issues in this context.

In cases of work accidents or encounters with venomous animals, the research shows that property owners immediately transport workers to urban areas, using means of transportation such as cars or even airplanes. This transportation aims to ensure that workers have access to specialized care in the public health system. However, this practice also highlights the logistical challenges faced in rural areas, where rapid access to healthcare often requires extraordinary measures.

Access to Communication Channels

Communication plays an important role in the lives of Pantanal farm residents, being a vital element for daily routines, activity coordination, and emotional well-being. The main communication methods adopted reveal an efficient adaptation to the particularities of the rural environment. Communication radios emerge as an indispensable tool in the farm's daily activities, facilitating efficient coordination among workers, even in extensive and remote areas.

In addition, the internet stands out as a multifunctional resource. Internally, it provides a platform for communication among workers, contributing to operational efficiency. Externally, it connects farm residents with urban administrators, enabling discussions and resolutions of work-related issues. The internet also plays an essential role in communicating with families, offering a means of contact that promotes peace of mind and emotional stability during periods when workers remain isolated on the properties.

The research also reveals that Pantanal farm residents have access to modern technologies and equipment in their work. Tractors, boats, cars, computers, and the internet are common elements that drive efficiency in activities. The presence of technologies such as monitoring cameras, firefighting

engines, and specific applications demonstrates a commitment to innovation to improve safety and productivity.

It is important to note that, even in the face of challenging rural conditions, farm residents are integrated into a diverse range of technologies, showing that modernization is a growing reality. This combination of communication channels and modern technologies not only optimizes daily operations but also highlights the resilience and adaptability of these communities in the face of contemporary demands.

The social dimension of the populations on the farms under study was thoroughly examined through a detailed analysis of data collected via structured questionnaires. The careful standardization of this information allowed for a consistent comparative analysis, revealing important nuances about the social profile of each community.

Based on the data analysis, we can understand in a more contextualized manner the specific characteristics of each property, respecting their particularities and enabling the implementation of more effective and sensitive measures to the identified needs.

4.1.2 Interactions between Communities and Community Groups (VCS, 3.19; CCB, CM1.1)

The dynamics of interaction among workers on Pantanal farms reflect a wealth of ethnic, gender, and marital diversity, shaping social relationships within these communities. Farms, aware of the importance of harmonious coexistence, provide comfortable housing conditions, creating an environment conducive to interaction among employees.

Among the researched farms, C6-P01-Y24, C4-P01-Y24, C5-P01-Y24, and C1-P01-Y24 stand out, as they go beyond the work environment by promoting celebratory events such as traditional June festivals and year-end celebrations. These initiatives not only strengthen bonds among workers but also contribute to building a sense of community and belonging.

Beyond festive events, landowners organize tours and exchanges between farms, enabling the exchange of technical knowledge and the improvement of agricultural practices. This connectivity between properties not only expands the workers' technical repertoire but also fosters a collaborative spirit among rural communities.

The interaction also extends to communication channels, playing a vital role in daily life. Connectivity via communication radios alerts about crucial issues such as the start of fires, wildlife sightings, road and waterway conditions, as well as informing about livestock movements and passage of groups. This communication network becomes essential for safety and efficiency in daily operations.

The research also evidenced significant interaction with fire prevention institutions in the Pantanal, notably with the Fire Department. The conduct of training sessions on the properties highlights the collaboration between workers and authorities, promoting awareness and preparedness to face fire-related challenges in the region.

Preliminary data reveals that the interaction among social groups and actors on Pantanal farms transcends work boundaries, extending to social events, exchanges between properties, and collaboration with prevention institutions. This dynamic strengthens community ties, contributing to sustainable development and resilience of rural communities in the Pantanal.

4.1.3 High Conservation Values (CCB, CM1.2)

According to the field research, the outlook for the region's future is diverse, reflecting the concerns and hopes of the interviewees. A crucial point highlighted is the urgent need for environmental conservation to ensure sustainable development. It is often emphasized that without this conservation, the environment necessary for future generations will be at risk. Many interviewees emphasize the importance of conserving fauna and flora, recognizing their essential contribution to ecosystem maintenance and to prevent species extinction.

On the other hand, there is a concern about the dichotomy between the growth of the tourism sector and the losses in agricultural activity. However, there are optimistic views, such as the potential increase in tourism and livestock farming, the strengthening of the hotel sector, and expectations of investments in education and improvements in quality of life.

Given this diversity of perspectives, the team responsible for the socio-environmental survey of the Pantanal REDD+ Program is compiling the necessary information for the adequate delineation of the HCVs related to the well-being of the community in the project area. Based on the different contexts that occur in the Pantanal, it is being identified whether the areas that provide critical ecosystem services, which are fundamental for the subsistence of communities, and those that are critical for the traditional cultural identity of the communities, will be classified as HCVs for the region of operation of the Program. Although this consolidation is not yet complete, this Section will be integrated into the final version of the documentation to be submitted to the VVB.

4.1.4 Without-Project Scenario: Community (CCB, CM1.3)

In the scenario without the project, the expectation is the continuation of the conditions found during the social baseline, where a low level of education, lack of access to health and education services, low professional qualification, and maintenance of the economy based on daily work with livestock or tourism are observed.

The non-occurrence of the Program would result in the worsening of socio-economic problems for people who would be positively impacted by the project's activities, such as:

- Absence of training in conservation and environmental awareness.
- Reduction in job generation.
- Lack of work, studies, and scientific research focused on the proposed strategies for climate, community, and biodiversity.

- Loss of forested areas due to fire, compromising the health and well-being of the population.
- Communities with limited access to public policies and services.
- Perpetuation of high turnover of employees on properties.
- Lack of institutional articulation and partnerships can result in fragility in responding to socio-environmental challenges and lack of support for sustainable development initiatives.
- Absence of diversification of income sources and investments in local infrastructure can keep communities in a cycle of economic and social stagnation, limiting opportunities for growth and development.
- Lack of training in first aid, rescue plans, and infrastructure maintenance can increase communities' vulnerability to natural disasters and health emergencies.
- Lack of access to quality internet systems and adequate educational infrastructure can exclude communities from education, employment, and information access opportunities.
- Absence of alternative systems for water treatment and soil and vegetation conservation practices can lead to the loss of natural resources and environmental imbalance, as well as health problems.
- Lack of awareness and education on environmental issues, such as illegal hunting, predation of livestock by wild animals, and the human-wildlife relationship, can result in conflicts between local communities and environmental authorities.

Such a scenario could also lead to rural exodus, meaning that residents might move to cities, where there is a risk of marginalization due to the low absorption capacity of labor in the region. In summary, the lack of access to the quality-of-life improvement activities proposed by the REDD+ Program could contribute to the perpetuation of poverty, environmental degradation, and vulnerability of local communities.

4.2 Net Positive Community Impacts

4.2.1 Expected Community Impacts (CCB, CM2.1)

The Pantanal REDD+ Program is concluding the final phase of consolidating the results of the socio-environmental diagnosis conducted on the properties. In this regard, the identified impacts are being presented in an expanded form to the main stakeholders. In the final version of the documentation, the impacts will be divided by groups aligned with the findings.

Community group	Stakeholders of the Pantanal REDD+ Program
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Impact(s)	- Promoting economic growth, creating job opportunities, and training the workforce. - Implementing effective measures for conserving natural resources by reducing local deforestation, actively preserving biodiversity, and substantially reducing emissions of pollutants that cause diseases in the population. - Stimulating creativity and innovation, driving technological development, and encouraging local scientific research. - Significant improvement in health conditions, ensuring universal access to essential basic services, and strengthening infrastructure to provide a better quality of life for the populations in the project's area of operation. - Investing in education improvement, actively promoting equal opportunities, and strengthening social cohesion to create a fairer and more inclusive society. - Preserving and actively promoting the rich cultural heritage of the Pantanal region and strengthening local identity.
Type of benefit/cost/risk	Direct tangible benefits
Change in well-being	- Boost local growth, creating jobs and training the workforce for key sectors in the region. - Conservation measures preserve biodiversity and reduce emissions, improving air quality and reducing diseases. - Incentive for technologies adapted to the Pantanal, while local research expands the understanding of ecosystems and supports sustainable decisions. - Improvement in access to basic services in the region. - Capacity building for the population to fully participate in the economy and society, promoting a fairer and more inclusive society. - Strengthening of the identity and pride of the Pantanal population.

4.2.2 Negative Community Impact Mitigation (VCS, 3.19; CCB, CM2.2)

The activities proposed by the Pantanal REDD+ Program do not provide, nor do they plan to provide, negative impacts on the well-being of local communities. As a forest conservation project, the Program

does not have activities harmful to the community and the environment. Additionally, all planned mitigation measures for identified risks to the well-being of communities are described in previous sections of this document.

4.2.3 Net Positive Community Well-Being (VCS, 3.19; CCB, CM2.3, GL1.4)

The activities of the Pantanal REDD+ Program were designed to provide several benefits to the local community. To achieve this, stakeholders of the project were consulted in order to tailor the proposed activities according to their real needs. Meetings were held with employees and farm owners to identify the main "issues" from their perspective, so that the activities of the Pantanal REDD+ Program could be designed as accurately as possible.

Among the positive impacts foreseen for the local community, the program plans to promote the quality of life and well-being of employees and farm owners participating in the Pantanal REDD+ Program. This will be achieved through activities aimed at improving safety, food quality, facilitating access to healthcare and education, enhancing technical training, and improving communication among farm employees.

It's important to note that these themes are adaptable by property, as some of the demands identified throughout the process of surveying local "issues" were very specific to each property and region. Therefore, some activities may be tailored according to the annual plan that will be developed between the proponent and the partners.

Considering the planned activities both in the general context of the region where the properties are located and specific issues identified for each location, the following activities aimed at improving the well-being and quality of life of the local community stand out:

- Training for technical capacity building of employees and their families.
- Support for educational improvements in the region.
- Implementation of diversified productive systems, favoring a more sustainable and dynamic economy.
- Implementation of vegetable gardens on farms, ensuring fresher and higher quality food.
- Support for itinerant medical care campaigns.
- Development and execution of an emergency rescue plan.
- Implementation of higher-quality internet systems.
- Alternatives for water treatment.
- Conversations and workshops addressing themes of adaptation and preparation for possible future scenarios resulting from climate change, aiming to increase resilience.

- Investment in the maintenance of internal farm roads, contributing to safer mobility and access to the properties.

4.2.4 High Conservation Values Protected (CCB, CM2.4)

Even though the HCVs of the Pantanal REDD+ Program are in the final stage of consolidation, it is understood by the proponent that the activities proposed by the Pantanal REDD+ Program do not provide, nor do they plan to provide, negative impacts on the identified HCVs.

As a forest conservation project, the Program does not have activities harmful to the community and the environment. Additionally, all planned mitigating measures for the identified risks to the well-being of communities are described in earlier sections of this document.

4.3 Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.1)

For the Pantanal REDD+ Program, negative impacts on stakeholders are not expected or are unlikely. On the other hand, positive impacts of the Program can be observed, aligning with the objectives of these stakeholders. The following are the anticipated positive impacts for each stakeholder:

NGOs and Other Social Movements:

- Contribute to achieving environmental agendas and commitments.
- Encourage the development of public policies.
- Expand local governance and improve it.
- Encourage the production and dissemination of knowledge.
- Encourage participation, representation, active voice, and political articulation.

Other Public Institutions:

- Contribute to achieving environmental agendas.
- Promote increased job creation.
- Expand local governance and improve it.
- Disseminate environmental knowledge aimed at mitigating environmental risks.

Research and Extension Institutions:

- Promote the development of research to contribute to professional training, advancement of science, biodiversity conservation, and socioeconomic development.

It is worth noting that all local communities, as well as other actors residing in the project area, whether participating in project activities or not, will benefit from all positive impacts related to the maintenance and conservation of forest cover and biodiversity. All communities and other actors will benefit from sustainable development, as well as opportunities generated by Program activities, improving quality of life and well-being.

As indicated above, negative impacts from these activities are not expected; however, they should be identified so that the Pantanal REDD+ Program can implement actions for their prevention and mitigation. Thus, potential negative impacts are identified as follows:

- Lack of engagement from communities and other stakeholders in Program activities and other partnerships.
- Failure in communicating Program actions and in addressing possible conflicts arising from the implementation and management of activities.
- Competition regarding the allocation of community members' time, time spent in meetings with government agencies and other institutions versus time allocated for livestock farming or tourism.

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

As mentioned above, no negative impacts on other stakeholders are expected or are unlikely to occur in this project.

However, considering the potential impacts identified in Section 4.3.1, the Program proposes a mitigating measure, which is the implementation of participatory strategies in the design of the activity and in the decision-making regarding the most appropriate timing and structure of interaction, with the joint construction of the agenda minimizing the overlap of activities, as has been done.

Additionally, a conflict resolution procedure has been structured, and if it is not effective, it will be readapted throughout the Program's lifecycle for the best forms of communication and conflict resolution.

4.3.3 Net Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

As described and detailed in Sections 4.3.1 and 4.3.2, the activities proposed by the Pantanal REDD+ Program are not expected to generate negative impacts on the well-being of other local actor groups.

The activities to be carried out in relation to the surrounding communities are mainly based on articulation with government agencies and other local institutions precisely to promote improved living conditions, greater access to public policies, and local quality of life. The Program's objectives aim only for impacts that promote inclusion and well-being for communities and other involved parties.

4.4 Community Impact Monitoring

4.4.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The Community Monitoring Plan focuses on monitoring activities covered by the areas of "Promotion of Sustainable Actions in Rural Areas" and "Sustainable Development and Quality of Life". For each proposed action in this theme, specific methodologies and indicators are directed. Below is a brief description of the main actions listed for the Program.

Prospecting new properties: involves registering the number of new areas included in the Program during each monitored period, aligned with information from Biofílica Ambipar Environment's New Business Team.

Monitoring Indicators

- Number of new areas included in the Program

Implementation of institutional articulation/partnerships with public agencies, private entities, and associations promoting sustainable actions in the rural Pantanal: through the articulation carried out by Biofílica Ambipar Environment, with its respective responsible areas, the partnerships established aligned with the activities of the Program will be regularly monitored. Additionally, together with the participating property owners, their official involvement in associations promoting sustainable actions in the rural Pantanal will be monitored.

Monitoring Indicators

- Number of institutional partnerships established
- Number of property owners participating in associations promoting sustainable actions in the rural Pantanal.

Implementation of alternative income sources on the properties: through the strategic planning outlined with each participating property of the Program, the number of alternative income sources implemented with the Program's incentive will be counted, involving both PSA models and sustainable management of forest areas.

Monitoring Indicators

- Number of Payment for Environmental Services (PSA) models implemented on the properties

- Number of alternative income sources implemented on the properties

Conducting training sessions, talks, and workshops for employees and their families on various topics:

after establishing the strategic planning of the properties, it will be crucial to register and monitor the participation of employees in these events, especially those focused on topics different from the activities present on the properties. Similarly, given the need to hold talks/workshops on climate change, the participation of employees should also be monitored. Both records should specify the involvement of women and young people in each activity.

Monitoring Indicators

- Frequency of training sessions conducted
- Participation in training sessions conducted
- Number of talks/workshops on climate change held
- Participation in talks/workshops on climate change held
- Number of women and young participants in the activities

Implementation of actions focused on improving the quality of life of rural workers: This line of action is broader within the properties, even though the main beneficiaries are the employees. Initially, it will be necessary to conduct a survey with local educational institutions, social organizations, and authorities to identify and record all initiatives aimed at improving educational infrastructure in the region. This may include school construction, provision of educational materials, teacher training, among others. The specific actions of each property in this line will depend on the strategic planning outlined by the partners.

A second theme involves supporting medical assistance campaigns held on the properties, either through mobile clinics, partnerships with local health institutions, or other medical assistance programs. Through the records of these actions, such as campaign reports, participant records, and beneficiary feedback, it will be possible to assess the impact of the actions.

Focusing on the internal context of the property, the existence and implementation status of emergency rescue plans should be surveyed with property managers. Based on this, a review of documents, such as security plans, training registration processes, among other actions, can be carried out.

In this activity, it is crucial that all obtained records specify the involvement of women and young people in each activity.

Monitoring Indicators

- Number of actions focused on improving the educational infrastructure of the region
- Number of medical assistance campaigns supported on the properties
- Number of emergency rescue plans implemented and/or improved on the properties
- Number of women and young residents on the properties

Implementation of actions focused on improving property infrastructure: once the strategic planning of the properties is completed, monitoring of the activities defined in this theme will begin with a survey directly with the owners about which sustainable practices, as well as which infrastructure improvements, have been implemented on the property. This survey will be based on all documentary analysis (such as records and reports).

Additionally, remote sensing tools and georeferencing technology may be used to visually map and monitor changes in properties over time. Direct feedback from beneficiaries of sustainable practices and improvements in physical infrastructure may be collected through interviews, focus groups, or satisfaction surveys, in total alignment with the communication tools outlined for the program.

Monitoring Indicators

- Number of sustainable practices implemented on Properties
- Number of actions to improve physical infrastructure on properties

It should be noted that the finalization of this plan will depend on the completion of the data assessment for the Project Area. Therefore, these pieces of information may undergo some changes until the final version of the documentation. However, updated data will be made available for validation by the VVB.

4.4.2 Monitoring Plan Dissemination (CCB, CM4.3)

The Climate Monitoring Plan and all its results will be publicly disclosed on the VERRA registry page and on the Biofíllica Ambipar Environment website.

For stakeholders without internet access, a printed copy may be provided at strategic points (farm headquarters and partner offices). Information about the Program will also be disseminated via email and WhatsApp (using a pre-established contact list of stakeholders), and on social media platforms such as Instagram and LinkedIn, of the project proponent.

In this way, it is expected that all necessary efforts to provide full transparency and distribute key information and documents of the Program will be achieved.

4.5 Optional Criterion: Exceptional Community Benefits

Not applicable. The Pantanal REDD+ Program is not intended to be validated at the Community Gold Level.

4.5.1 Exceptional Community Criteria (CCB, GL2.1)

Not applicable

4.5.2 Short-term and Long-term Community Benefits (CCB, GL2.2)

Not applicable

4.5.3 Community Participation Risks (CCB, GL2.3)

Not applicable

4.5.4 Marginalized and/or Vulnerable Community Groups (CCB, GL2.4)

Not applicable

4.5.5 Net Impacts on Women (CCB, GL2.5)

Not applicable

4.5.6 Benefit Sharing Mechanisms (CCB, GL2.6)

Not applicable

4.5.7 Benefits, Costs, and Risks Communication (VCS, 3.18; CCB, GL2.7)

Not applicable

4.5.8 Governance and Implementation Structures (CCB, GL2.8)

Not applicable

4.5.9 Smallholders/Community Members Capacity Development (CCB, GL2.9)

Not applicable

5 BIODIVERSITY

5.1 Without-Project Biodiversity Scenario

5.1.1 Existing Conditions (VCS, 3.19; CCB, B1.1)

The biodiversity assessment of the Pantanal REDD+ Program was conducted by the specialized team of Restaura Consultoria Ambiental e Treinamentos LTDA between October 2023 and January 2024, based on primary and secondary data collection, to support project activities and understand the dynamics of the region, as well as to map existing conditions. Until the completion of this documentation, this survey is in the final stage of consolidation, so here are the intermediate data found.

The Pantanal is a biome widely known for being the largest flooded plain in the world, but it is more than that; it is a mosaic containing flooded and non-flooded areas, deciduous forest, savannah (cerrado), riparian forests, pastures, temporary areas, permanently aquatic areas, and characteristic monodominant formations. It is a biodiversity reservoir with organisms adapted to the cycles of drought, flooding, and fire, requiring special attention for its conservation and restoration, especially with the catastrophic deforestation and fires that have recently occurred (Embrapa⁶⁶).

While the Pantanal is known as the largest flooded plain on the planet, the biome is also a transition region between the Amazon and the Cerrado, and it harbors species from both areas. Therefore, the vegetation is a mosaic influenced by the Amazon, the Cerrado, and the Chaco.

The region covered by the project is rich and hosts a diverse fauna, comprising, so far, 712 identified species, distributed among 158 families and 55 orders. Regarding the avifauna, 342 native species have been inventoried, distributed among 62 families and 26 orders. The families with the highest species richness are Tyrannidae (n=42), Thraupidae (n=33), Furnariidae (n=18), Psittacidae (n=18), and Accipitridae (n=16).

Regarding the mammalian fauna, 90 native species have been inventoried, distributed among 24 families and 10 orders. The families with the highest species richness include Cricetidae (n=19), Didelphidae (n=18), and Felidae (n=8 species). Considering the herpetofauna, which includes amphibians and reptiles, 115 species have been surveyed, belonging to 29 families and 4 orders. The richest families were Dipsadidae (n=30), Hylidae (n=15), and Leptodactylidae (n=14).

The ichthyofauna was represented by 121 species, distributed among 30 families and 8 orders. The highest species richness was found in the Characidae family (n=29), followed by Cichlidae (n=13) and Serrasalminae (n=10). Finally, for the insect fauna, 44 species have been inventoried, distributed among 13 families and seven orders. So far, the families with the highest species richness are Formicidae (n=14), Apidae (n=9), and Acrididae (n=6).

⁶⁶ Bioma Pantanal, Ciência e Cultura – Embrapa. Disponível em:
<https://ainfo.cnptia.embrapa.br/digital/bitstream/doc/1158955/1/Bioma-Pantanal-ciencia-cultura.pdf>

The main pressures and threats to fauna in this region come from deforestation for the introduction of grass species to favor traditional extensive livestock farming, and fires (Alho & Sabino, 2011⁶⁷; Tomas et al. 2021⁶⁸). Both lead to fragmentation, habitat loss, and reduction in the availability of natural prey or plants used for food, shelter, or breeding sites.

Regarding flora, the vegetation is classified into different phytosociological classes: Cerrado and Cerradão (wooded/forested savannah), semi-deciduous seasonal forest, capões, monodominant patches, flooded fields with native and cultivated pasture, bays, and salinas, riparian forest, and murundus fields. Phytosociological classes in the Pantanal originate from the influence of different factors such as the duration and frequency of flooding and burning, soil types, relief, rainfall, topography, and others (see Vila da Silva et al. 2022). Over time, plant species in the Pantanal have developed specific adaptations (morphological, physiological, and anatomical) to survive and thrive in the adverse conditions of this wetland. These adaptations, such as tolerance to flooding and waterlogging, contribute to the diversity of vegetation types (Bogarín et al. 2023). Understanding the interaction of these factors helps explain the rich mosaic of vegetation types found in the Brazilian Pantanal. The dynamic nature of the ecosystem, with its seasonal floods and diverse habitats, contributes to the high biological diversity of the region. Specifically, each evaluated area is composed of different phytosociological classes (Table 20).

Table 20. Phytosociological classes classified in each evaluation area.

Assessment areas	Phytophysiognomies
C2-P01-Y24	Cerrado e cerradão (savannah), semi-deciduous forest, forest on the latitudinal elevations of the terrain (capões in Portuguese), monodominant patches (C. americana, B. cynodifolia, S. haemospermum), flooded fields.
C5-P02-Y24	Forest on the latitudinal elevations of the terrain (capões in Portuguese), formações monodominantes (Vochysia divergens e Leptobalanus parvifolius),
C4-P01-Y24	Forest on the latitudinal elevations of the terrain (capões in Portuguese), flooded fields, native and cultivated pasture, bays and salinas, riparian forest, semi-deciduous seasonal forest.
C5-P01-Y24	Cerrado e cerradão (savannah), flooded fields, native pasture, semi-deciduous seasonal forest, murundus fields.
C7-P01-Y24 e C7-P02-Y24	Cerrado, Semi-deciduous seasonal forest on ridges, native pasture, monodominant patches (canjiqueiral).

⁶⁷ ALHO, C. J. R.; SABINO, J. A conservation agenda for the Pantanal's biodiversity. Brazilian Journal of Biology, v. 71, p. 327-335, 2011.

⁶⁸ TOMAS, Walfrido Moraes et al. Distance sampling surveys reveal 17 million vertebrates directly killed by the 2020's wildfires in the Pantanal, Brazil. Scientific Reports, v. 11, n. 1, p. 23547, 2021.

Assessment areas	Phytophysionomies
C1-P01-Y24	Riparian forest, Cerrado (wooded/forested savannah), flooded fields, semi-deciduous seasonal forest, forest on the latitudinal elevations of the terrain (capões in Portuguese), monodominant patches (carandazal, paratudal, acurizal).
C3-P01-Y24	Riparian forest, flooded fields, forest on the latitudinal elevations of the terrain (capões in Portuguese).
C6-P01-Y24	Riparian forest, flooded fields, swamps, monodominant “carandazal”, wooded/forested savannah (Cerradão).

Here are some photographic records collected in 2023 during field studies.



Figure 19 C5-P01-Y24 Photo a) Hyacinth Macaw (*Arara-azul*); b) Marsh Deer (*Cervo-do-pantanal*); c) Capybara (*Capybara*); d) Southern Screamer (*Tochã-cinzenta*).



Figure 20. C2-P01-Y24 Photo a) Bare-faced Curassow (*Mutum-de-penacho*); b) plumbeous ibis (*Curicaca-real*)



Figure 21. C4-P01-Y24 a) Red-and-green Macaw (*Arara-vermelha* in Portuguese); b) Jabiru Stork (*Tuiuiú* in Portuguese); c) Maria-faceira (*Syrgma sibilatrix*); d) Pantanal Caiman (*Jacaré-do-pantanal* in Portuguese).

As a biome, the Pantanal is classified as a seasonally flooded tropical savannah. It is a biome influenced by several biogeographical provinces, including the Cerrado, Amazon, and Atlantic Forest in Brazil, as well as the Paraguayan and Bolivian Chaco, and the Chiquitano forests of Bolivia. (Junk et al. 2006)⁶⁹.

In the avifauna, this influence is perceived due to the restricted distribution of some species such as the Chaco Chachalaca (*aracuã* in Portuguese) (*Ortalis canicollis*), the Striped Cuckoo (*rapazinho-do-Chaco* in Portuguese) (*Nystalus striatipectus*), the Nanday Parakeet (*periquito-de-cabeça-preta* in Portuguese) (*Aratinga nenday*), and the Black-bellied Antwren (*chororó-do-Pantanal* in Portuguese) (*Cercomacra melanaria*).

Regarding the mammalian fauna, the influence of neighboring biomes such as the Cerrado is noticeable in the presence of the Giant Anteater (tamanduá-bandeira in Portuguese) (*Myrmecophaga tridactyla*) ou White-lipped Peccary (queixada) (*Tayassu pecari*). The occurrence of small mammals is greatly influenced by the Amazon and Atlantic Forest (species of the family Cricetidae), while the Chaco biome influences the occurrence of the Chaco Mouse Opossum (*catita* in Portuguese) (*Cryptomys chacoensis*), and the Azara's Night Monkey (*macaco-da-noite*) (*Aotus azarae*) (Hannibal et al. 2017)⁷⁰.

For the herpetofauna, the influence of the Chaco is observed in the presence of the anurans *Scinax fuscomarginatus* and *Leptodactylus chaquensis*, in the lizards of the genus *Cercosaura*, and in the snakes of the genera *Chironius*, *Xenodon*, and *Phalotris* (Jansen, 2009)⁷¹. The ichthyofauna is mainly influenced by the Amazon region, with emphasis on the genera *Hypostomus* and *Moenkhausia* (Ribeiro et al. 2013)⁷².

As this is a preliminary survey, below are listed the rare, threatened, or endangered species identified so far. Once the diagnostic analyses are completed, this information will be updated.

Species and habitat	Justification of why the project will not have a negative impact on the habitats of rare, threatened, or endangered species.
Mutum-de-penacho - <i>Crax fasciolata</i> Arara-azul - <i>Anodorhynchus hyacinthinus</i> Cervo-do-pantanal - <i>Blastocerus dichotomus</i> Queixada - <i>Tayassu pecari</i> Anta - <i>Tapirus terrestris</i>	The fauna species listed here are threatened to some degree according to the International Union for Conservation of Nature (IUCN). The Pantanal REDD+ Program will not negatively affect the habitats of rare, threatened, or endangered species, as the planned activities for this REDD+ project focus on research, conservation, and monitoring of forest and savannah habitats in the Pantanal biome. Positive impacts are expected from the project due to the proposed activities, generating net

69 JUNK, W. J. et al. Biodiversity and its conservation in the Pantanal of Mato Grosso, Brazil. Aquatic Sciences, v. 68, p. 278-309, 2006.

70 HANNIBAL, W. et al. Biogeography and conservation of non-volant mammals from the Urucum Mountains: a Chiquitano dry forest ecoregion in western Brazil. Mammalia, v. 81, n. 2, p. 169-180, 2017.

71 JANSEN, M. The herpetofauna of selected ecoregions in Bolivia: studies on taxonomy, diversity, and biogeography, with special reference to the Chiquitano Region. 2009. Tese de Doutorado. Universitätsbibliothek Johann Christian Senckenberg.

72 RIBEIRO, A. C. et al. Distributions and phylogeographic data of rheophilic freshwater fishes provide evidences on the geographic extension of a central-brazilian amazonian palaeoplateau in the area of the present day Pantanal Wetland. Neotropical Ichthyology, v. 11, p. 319-326, 2013.

Tamanduá-bandeira - <i>Myrmecophaga tridactyla</i> Macaco prego- <i>Sapajus cay</i> Ariranha - <i>Pteronura brasiliensis</i> Tapeti - <i>Sylvilagus brasiliensis</i> Cobra-cega - <i>Epictia clinorostris</i> Lagarto-ápodo - <i>Bachia bresslaui</i> Cobra-cega - <i>Epictia clinorostris</i>	benefits to biodiversity that would likely not occur in the absence of the project.
<i>Neptunia pubescens</i> Benth. <i>Ipomoea subrevoluta</i> Choisy <i>Mezilaurus vanderwerffii</i> F.M.Alves & Baitello <i>Trichilia stellato-tomentosa</i> Kuntze <i>Dulacia egleri</i> (Bastos) Sleumer <i>Coccoloba rigida</i> Meisn.	The flora species listed here are threatened to some degree according to the IUCN. The Pantanal REDD+ Program will not negatively affect rare, threatened, or endangered species, as the planned activities aim to maintain the standing forest. Thus, project activities focused on firefighting and deforestation prevention are highlighted. Positive impacts are expected from the project due to the proposed activities, generating net benefits to biodiversity that would likely not occur in the absence of the project.

5.1.2 High Conservation Values (CCB, B1.2)

Through the global document of the HCV Resource Network, the classification of High Conservation Value (HCV) is in the finalization process, which is nothing more than "a biological, ecological, social, or cultural value of exceptional or critical importance." For this work, the first three values (HCV 1, 2, and 3) are being considered, which correspond to biodiversity criteria. Although this consolidation is not yet complete, this section will be integrated into the final version of the documentation to be delivered to VVB. So far, the following possible attributes have been listed:

High conservation value	HCV 1. Areas containing globally, regionally or nationally significant concentrations of biodiversity values (e.g. endemism, endangered species, refugia).
Qualifying attribute	From the vegetation diagnosis, 918 species were recorded in different phytophysognomies of the Pantanal.
Focal area	Under Development
High conservation value	HCV 2. Globally, regionally or nationally significant large landscape-level areas where viable populations of most if not

	all naturally occurring species exist in natural patterns of distribution and abundance.
Qualifying attribute	Under Development
Focal area	Under Development

High conservation value	HCV 3. Areas that are in or contain rare, threatened or endangered ecosystems.
Qualifying attribute	Under Development
Focal area	Under Development

5.1.3 Without-project Scenario: Biodiversity (CCB, B1.3)

The Living Planet Report released in 2020 by the WWF network indicates that monitored populations of mammals, birds, amphibians, reptiles, and fish decreased by an average of two-thirds between 1970 and 2016. According to them, the main direct driver of biodiversity loss in terrestrial systems in recent decades has been changes in land use, mainly the conversion of untouched native habitats.

Biodiversity loss is not just an environmental problem. It also affects development, the economy, global security, ethics, and morality. Furthermore, it is a matter of self-preservation. Biodiversity plays a crucial role in providing food, fibers, water, energy, medicines, and other genetic resources; and is vital for climate regulation, water quality, pollution, pollination services, and flood and storm control. Moreover, nature sustains all dimensions of human health, as well as offering intangible contributions, such as inspiration and learning; physical and psychological experiences; and the formation of identity. These elements are fundamental to the quality of life and cultural integrity of humans (WWF, 2020⁷³).

For the Pantanal REDD+ Program, the most likely land use scenario without the project is the conversion of forest into pasture for livestock. Baseline forest loss projections in a scenario without the project indicate a total area of 11 thousand hectares deforested in the first 10 years in the current project area. Thus, deforestation has a significant negative impact on biodiversity, both fauna and flora. Furthermore, it is worth noting that the Pantanal is a unique ecosystem that harbors diverse species of plants and animals, some of which are endemic. In this context, the impact of land use change in the absence of the project is clear.

⁷³ WWF 2020. Available in: https://wwfbr.awsassets.panda.org/downloads/lpr_pt_2020_v2.pdf

5.2 Net Positive Biodiversity Impacts

5.2.1 Expected Biodiversity Changes (VCS, 3.19; CCB, B2.1)

The expectations for changes in biodiversity as a result of the Program were estimated using the Theory of Change method, which is more clearly defined in Section 2.1.17. With this definition in mind, the cause-and-effect link between the planned activities for the project, the actions taken, and the expected short-, medium-, and long-term results becomes clearer.

All activities were established with the aim of conserving and maintaining native forest vegetation in the project area. Regarding biodiversity, it is understood that the most significant transformations, compared to the scenario without the project implementation, are linked to the conservation of natural habitats. This allows for the preservation or growth of the variety of flora and fauna species, which would be at risk due to changes in land use.

Therefore, it is evident that safeguarding the forest through the Program's actions and the REDD+ mechanism will result in beneficial changes in biodiversity as expected. Furthermore, the biodiversity monitoring plan, being a crucial tool, will allow for the verification and investigation of possible changes that may adversely affect biological diversity. This will provide a clearer understanding of the population dynamics of species, including endemic and vulnerable ones, as well as the conflicts resulting from human interaction with nature.

In summary, the expected changes in Project biodiversity are:

- Maintenance and possible addition of species identified in the scenario without the Project.
- Ensuring the conservation of habitats and species in the Pantanal.
- Reducing illegal activities through Project activities.
- Reducing losses caused by forest fires through Project activities.
- Increasing awareness and sensitization of partner owners, collaborators, neighbors, and local communities regarding environmental issues, reducing hunting pressure.
- Contribution to the conservation of threatened and/or endemic species.
- Reduction of threat levels according to the IUCN Red List.
- Promoting an ecologically balanced environment.
- Maintaining landscape connectivity.

The changes in key elements of biodiversity are described in the tables below.

Biodiversity element	Flora
Estimated change	Positive change due to activities reducing deforestation and forest degradation, resulting in the maintenance and/or increase of flora species.
Justification of change	The project will promote the maintenance of native forests through activities aimed at containing the expansion of deforestation. This objective will be achieved by conducting frequent monitoring in the Project Area and the Leakage Belt using deforestation alerts via remote sensing. Another point that contributes to the maintenance of this biodiversity element is the implementation of patrimonial surveillance if necessary for the property. Monitoring of pasture areas is also planned to ensure they do not encroach upon the Project Area and Leakage Belt. Additionally, the set of activities related to fire prevention and mitigation is indispensable when it comes to flora conservation.

Biodiversity element	Fauna
Estimated change	Positive change resulting from activities aimed at reducing deforestation and forest degradation, leading to habitat conservation.
Justification of change	The activities designed for the Program are aimed at conserving wildlife habitats. These include remote monitoring, patrimonial surveillance, firefighting, and mitigation, as well as environmental education actions aimed at raising awareness about the consequences and damages of illegal hunting and fishing. These actions have been devised based on a previously conducted socio-environmental diagnosis and will be implemented in the Program to ensure success in achieving the conservation objectives throughout the project longevity.

Biodiversity element	Fauna and Flora
Estimated change	Positive change resulting in biodiversity conservation in the project zone.
Justification of change	Considering the vast richness of biodiversity present in the Pantanal biome, actions such as continuous monitoring of fauna and flora and monitoring of endangered species classified as High Conservation Value Attributes (HCVAs) stand



out. These actions aim to understand ecosystem dynamics and identify any significant changes. Additionally, local scientific research is essential to expand knowledge and promote species conservation, crucial for achieving the objectives of the Pantanal REDD+ Program.

5.2.2 Mitigation Measures (VCS, 3.19; CCB, B2.3)

Considering the rich biodiversity present in the Pantanal biome, the Program aims to preserve areas belonging to this region by avoiding deforestation and, consequently, maintaining the habitat of the species present. Continuous monitoring of fauna and flora and monitoring of endangered species classified as High Conservation Value Attributes (HCVAs) are planned with the goal of understanding ecosystem dynamics and detecting any significant changes. Additionally, initiatives are scheduled to promote scientific research in the region to expand knowledge and favor species preservation. Simultaneously, within the context of each property, awareness campaigns with employees and local communities about the impacts of illegal hunting and fishing on Pantanal biodiversity will be recommended. The objective of these joint actions is to ensure the maintenance of the unique biodiversity of the Pantanal, safeguarding its species and ecosystems.

In this regard, no significant negative impacts on biodiversity are foreseen as a result of project activities. All Program activities have been designed and structured to act as mitigating measures for the main impacts and threats to biodiversity, both internal and external, identified in the Socioeconomic and Environmental Diagnostics studies. Thus, the proposed measures will be more effective and meaningful, resulting in relevant and beneficial changes for the region, its surroundings, and biodiversity.

5.2.3 Net Positive Biodiversity Impacts (CCB, B2.2, GL1.4)

Considering that project activities directly contribute to the maintenance of forest cover and species diversity in the project zone, the projected net impacts on biodiversity in the project zone will be positive compared to conditions in the land use scenario without the project.

The Program will implement direct measures for the study, conservation, and monitoring of flora and fauna species categorized as near-threatened, vulnerable, threatened, and endemic by the IUCN. Additionally, High Conservation Value Areas (HCVAs) listed in section 5.1.2 above will be monitored.

Regarding species with some degree of threat, such as the Ariranha or Giant Otter in English (*Pteronura brasiliensis*) and Tapeti or Brazilian Cottontail Rabbit in English (*Sylvilagus brasiliensis*) their habitats will be protected, with ample forest cover available. The jaguar (onça-pintada in Portuguese), tapir (anta), giant anteater (tamanduá-bandeira), capuchin monkey (macaco prego), collared peccary (queixada), and marsh deer (cervo-do-pantanal), among other important species, will be protected from illegal hunting, forest degradation, and fragmentation.

5.2.4 High Conservation Values Protected (CCB, B2.4)

The activities carried out by the Pantanal REDD+ Program throughout its duration will generate positive impacts on areas of high conservation value, as they include measures to maintain biodiversity-related attributes, such as forest landscape connectivity and the conservation of species of fauna and flora, including those threatened with extinction.

Therefore, no negative effects related to biodiversity are anticipated from the project's proposed activities on High Conservation Value Areas.

5.2.5 Species Used (VCS, 3.19; CCB, B2.5, B2.6)

As a REDD+ APD project, the activities of the Pantanal REDD+ Program do not involve the use of species, including non-native ones.

5.2.6 Invasive Species (VCS, 3.19; CCB, B2.5)

As a REDD+ APD project, the activities of the Pantanal REDD+ Program do not include the use of invasive species; thus, the population of any invasive species will not increase as a result of the project.

5.2.7 GMO Exclusion (CCB, B2.7)

The Pantanal REDD+ Program ensures that genetically modified organisms (GMOs) will not be used in its activities, as greenhouse gas emissions reduction will be achieved through avoided planned deforestation.

5.2.8 Inputs Justification (VCS, 3.19; CCB, B2.8)

The Pantanal REDD+ Program ensures that fertilizers, chemical pesticides, and biological control agents will not be used in the Project Area, as greenhouse gas emissions reduction will be achieved through avoided planned deforestation.

5.2.9 Waste Products (VCS, 3.19; CCB, B2.9)

Although some properties participating in the Pantanal REDD+ Program have actions to minimize waste generation, involving identification/classification, packaging, temporary storage, and transportation to

the appropriate destination, mitigating the effects of pollution caused by waste in the environment, for the Program's activities, no additional waste generation beyond those commonly generated is anticipated (mainly organic waste (food scraps) or solids such as metal, paper/cardboard, glass, and plastic).

Nevertheless, it is a priority for the proponent to constantly encourage its partners to implement good waste management practices, encouraging the maintenance and improvement of these practices, as well as incentivizing those who have not yet implemented them. This work is closely aligned with the training activities proposed by the Program. It is important to emphasize that as the activities are implemented, the identification, classification, and management of their waste will be enhanced.

5.3 Offsite Biodiversity Impacts

5.3.1 Negative Offsite Biodiversity Impacts (CCB, B3.1) and Mitigation Measures (CCB, B3.2)

As this is a conservation project with activities from the Pantanal REDD+ Program focused on fauna and flora monitoring, firefighting, environmental education, and scientific research promotion, there are no anticipated negative effects outside the Project Zone resulting from these actions.

During the Program's monitoring process, if impacts are identified, the proponent will mobilize with stakeholders to mitigate any potential negative impact, collectively articulating adaptive management measures through external partnerships with public organizations, private sector, civil society, and research institutions.

5.3.2 Net Offsite Biodiversity Benefits (VCS, 3.19; CCB, B3.3)

In general, the net impacts on biodiversity in the Project Zone will be positive compared to the scenario without the project, as well as outside the Project Zone. Therefore, at this time, no mitigation actions have been outlined. The project has significant positive impacts on biodiversity, including the conservation of large areas for threatened and vulnerable species. With these activities, the surrounding areas of the Pantanal REDD+ Program will also benefit.

If negative effects on biodiversity occur outside the Project Zone, mitigating measures will be outlined and planned by the proponents, considering stakeholder engagement, promoting adaptive management, and ensuring the desired benefits for biodiversity.

5.4 Biodiversity Impact Monitoring

5.4.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

The Biodiversity Monitoring Plan focuses on monitoring activities covered by the "Biodiversity Conservation" area of operation. For each proposed action in this theme, specific methodologies and indicators are directed. Below is a brief description of the main actions listed for the Program.

Fauna and Flora Monitoring: will be conducted through periodic inventories of local biodiversity, supplemented using additional tools such as camera traps in strategic areas. Species occurrence data and biodiversity records are analyzed and recorded in specific databases.

Monitoring indicators:

- Number of identified species of fauna and flora
- Frequency of sightings of key species
- Biological diversity indices in monitored areas

Encouragement of local scientific research: will be carried out through the establishment of partnerships with research institutions and universities, as well as the organization of scientific events and workshops for the dissemination of results and integration of the local scientific community.

Monitoring indicators:

- Number of approved and conducted scientific research projects
- Scientific publications resulting from the research
- Participation of the local community in research projects

Threatened species monitoring: aligned with fauna and flora monitoring, regular monitoring of the areas where these species occur will be conducted using specific technologies. These data are used to assess the conservation status of the species and implement appropriate protection measures.

Monitoring indicators:

- Number of sightings of threatened species
- Population trends of threatened species

Awareness about illegal hunting and fishing: promoted through lectures, workshops, and educational activities for employees and the local community, in addition to the production and distribution of educational materials. There is a possibility of partnerships with enforcement and security agencies being established to assist in combating these illegal practices.

Monitoring Indicators:

- Participation in awareness campaigns

- Number of complaints registered on farms regarding illegal activities

Awareness about cattle predation by jaguars: awareness events will be held with cattle ranchers participating in the Program, where information about preventive and compensatory measures is shared.

Monitoring Indicators:

- Participation in awareness campaigns
- Adoption of cattle protection measures

It is worth noting that the consolidation of this plan will depend on the finalization of the data evaluation for the Project Area. Therefore, this information may undergo some changes until the final version of the documentation. In any case, updated data will be made available for validation by the VVB.

5.4.2 Biodiversity Monitoring Plan Dissemination (CCB, B4.3)

The Climate Monitoring Plan and all its results will be publicly disclosed on the VERRA registry page and on the Biofílica Ambipar Environment website.

For stakeholders without internet access, a printed copy can be provided at strategic points (farm headquarters and partner offices). Information about the Program will also be disseminated via email and WhatsApp (to a pre-established list of stakeholders), and on social media platforms such as Instagram and LinkedIn of the project proponent.

In this way, it is expected that all necessary efforts to provide full transparency and distribute key information and documents of the Program will be achieved.

5.5 Optional Criterion: Exceptional Biodiversity Benefits

The Pantanal REDD+ Program is seeking Gold-level Biodiversity Status, as the Project Zone includes an area of high priority for biodiversity conservation that meets the criteria of vulnerability or uniqueness, identifying trigger species.

5.5.1 High Biodiversity Conservation Priority Status (CCB, GL3.1)

In the survey conducted by Restaura Consultoria Ambiental e Treinamentos LTDA, which is currently in the final consolidation phase, there were 25 species with distribution restricted to the Pantanal in Brazil, totaling 3 species of herpetofauna, 8 of ichthyofauna, 4 species of mastofauna, as well as 10 species of avifauna, which are detailed below.

Avifauna: As rare species, we highlight the tiriba-fogo or Deville's Parakeet in English (*Pyrrhura devillei*), the rapazinho-do-Chaco (*Nystalus striatipectus*), e o pica-pau louro or Pale-crested Woodpecker in English (*Celeus lugubris*). Out of the 342 species surveyed so far, 9 are classified under some degree of threat according to the IUCN (2023).

Mastofauna: As rare species, we highlight the small cats (*Leopardus colocolo*, *Leopardus geoffroyi*, *Leopardus tigrinus* e *Leopardus wiedii*), the cachorro-vinagre or Bush Dog in English (*Speothos venaticus*), which naturally occur in low densities. Also rare are the tatu-canastra or Giant Armadillo (*Priodontes maximus*) and the tatu-bola or Three-banded Armadillo in English (*Tolypeutes matacus*). Out of the 90 mammal species surveyed so far, 26 are classified under some degree of threat according to the IUCN (2023).

Herpetofauna: As rare species, we highlight animals of the genus *Amphisbaena* and the víbora-do-Pantanal or Paraguayan Lancehead in English (*Dracaena paraguayensis*). ut of the 115 species surveyed, 4 are classified under some degree of threat according to the IUCN (2023).

Ichthyofauna: Among the ichthyofauna, 121 species were identified, of which 2 are threatened according to the IUCN (2023). As a rare species, we highlight the catfish *Farlowella isbruckeri*.

Entomofauna: As a rare species, we highlight the ant *Atta saltensis*, whose status has not yet been assessed by the IUCN. Among the 44 species surveyed so far, none are under any degree of threat, according to the IUCN classification (2023).

Table 21. Threatened species according to the IUCN Red List of Threatened Species.

BIRDS		
IUCN Threat Categories	Common Name	Scientific Name
Vulnerable (VU)	Mutum-de-penacho	<i>Crax fasciolata</i>
	Arara-azul	<i>Anodorhynchus hyacinthinus</i>
MAMMALS		
IUCN Threat Categories	Common Name	Scientific Name
Vulnerable (VU)	Cervo-do-pantanal	<i>Blastocerus dichotomus</i>
	Queixada	<i>Tayassu pecari</i>
	Anta	<i>Tapirus terrestris</i>
	Tamanduá-bandeira	<i>Myrmecophaga tridactyla</i>
	Macaco prego	<i>Sapajus cay</i>
Endangered (EN)	Ariranha	<i>Pteronura brasiliensis</i>

	Tapeti	<i>Sylvilagus brasiliensis</i>
REPTILES AND AMPHIBIANS		
IUCN Threat Categories	Common Name	Scientific Name
Vulnerable (VU)	cobra-cega	<i>Epictia clinorostris</i>
	Lagarto-ápodo	<i>Bachia bresslaui</i>
	Cobra-cega	<i>Epictia clinorostris</i>
FLORA		
IUCN Threat Categories	Scientific Name	
Vulnerable (VU)	<i>Neptunia pubescens</i> Benth.	
	<i>Ipomoea subrevoluta</i> Choisy	
	<i>Mezilaurus vanderwerffii</i> F.M.Alves & Baitello	
	<i>Trichilia stellato-tomentosa</i> Kuntze	
Endangered (EN)	<i>Dulacia egleri</i> (Bastos) Sleumer	
	<i>Coccoloba rigida</i> Meisn.	

It is worth noting that, since the data indicated here are in the final phase of evaluation with the responsible team, these pieces of information may undergo some changes until the final version of the documentation. Nevertheless, the updated data, as well as all evidence from primary and secondary surveys conducted, will be made available for validation by the VVB.

5.5.2 Trigger Species Population Trends (CCB, GL3.2, GL3.3)

This section is under development, review, and finalization. Below are listed species, currently classified as 'Endangered (EN)' by the IUCN (2023). However, the analyses of population trends at the beginning of the project, as well as scenarios with and without the implementation of the Program, are still being consolidated."

Trigger species	Fauna: Ariranha (<i>Pteronura brasiliensis</i>)
Population trend at start of project	Under development
Without-project scenario	Under development

With-project scenario	Under development
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Trigger species	Fauna: Tapeti (<i>Sylvilagus brasiliensis</i>)
Population trend at start of project	Under development
Without-project scenario	Under development
With-project scenario	Under development

Trigger species	Flora: <i>Dulacia egleri</i> (Bastos) Sleumer
Population trend at start of project	Under development
Without-project scenario	Under development
With-project scenario	Under development

Trigger species	Flora: <i>Coccoloba rigida</i> Meisn.
Population trend at start of project	Under development
Without-project scenario	Under development
With-project scenario	Under development

APPENDIX 1: STAKEHOLDER DESCRIPTION TABLE

Under development and review

Type	Stakeholder	Rights, interest, and overall relevance to the project
"Community"	Rural Property Owners Partners of the Pantanal REDD+ Program	Rights: Holders of credit rights, responsible for investments, development, and implementation of the REDD+ Project. Interests: Ensuring the inclusion of workers in project activities. Relevance: High - Decisive participation in the region's operations and implementation of socio-economic development activities.
	Workers of the Rural Properties	Rights: Beneficiaries of the social activities outlined in the project. Interests: Participating in project activities, accessing training and capacity-building opportunities that enable improvements in their living conditions. Relevance: High - Important for the success of social activities, deforestation control, and development of sustainable activities.
Public environmental institutions	SEMADESC - Secretaria de Estado de Meio Ambiente, Desenvolvimento, Ciência, Tecnologia e Inovação (State Secretariat of Environment,	Rights: Conduct environmental inspection and regulation in the Pantanal; Research, Development, and Innovation for Pantanal sustainability.

Type	Stakeholder	Rights, interest, and overall relevance to the project
	Development, Science, Technology, and Innovation)	Interests: Promote environmental preservation, monitor the impact of REDD+ Program activities on Pantanal biodiversity and ecosystems.
	Embrapa Pantanal	Relevance: High - Essential to ensure compliance with environmental regulations and guarantee the conservation of Pantanal ecosystems.
Private Institutions	Sebrae Mato Grosso do Sul e Sebrae Mato Grosso - Pró Pantanal – Programa de Apoio à Recuperação Econômica do Bioma Pantanal (Pantanal Biome Economic Recovery Support Program)	Rights: Facilitators, potential partners of the activities Interests: Rural Professional Training and Social Promotion; Sustainable development in tourism, creative economy, and agribusiness sectors.
	Senar - Serviço Nacional de Aprendizagem Rural. (National Rural Learning Service)	Relevance: Low - they can assist in implementing activities, but initially, they are not crucial.
Non-Governmental Organizations	Acaia Pantanal	Rights: Facilitators, potential partners of the activities
	Ampara Animal	Interests: Fundraising and maintaining their operations; Bringing families and communities closer to participating in project activities, strengthening social integration; Generating knowledge; Promoting biodiversity conservation and improving the quality of life of communities; Generating positive impact.
	Comitiva Esperança	
	Ecoa - Ecologia e Ação	
	Grupo de Resgate de Animais em Desastres	
	Instituto de Conservação de Animais Silvestres (ICAS)	Relevance: Medium - Important partners in the region due to their established local credibility.

Type	Stakeholder	Rights, interest, and overall relevance to the project
	Instituto Homem Pantaneiro (IHP)	
	Instituto Mamede de Pesquisa Ambiental e Ecoturismo	
	Fundação Neotrópica do Brasil	
	Instituto Onçafari	
	Instituto Arara Azul	
	SOS Pantanal	
	WCS Brasil	
	WWF	
Associations and Unions	ABPO – Associação Brasileira de Produtores Orgânicos (Brazilian Association of Organic Producers)	Rights: Facilitators, potential partners in the Program (encouraging the inclusion of new areas) Interests: Representatives of the interests of Pantanal rural landowners Relevance: High - Important partners in the region due to their established local credibility and to support the expansion of the Program.
	ASPNP - Associação Sul-Mato-Grossense de Produtores de Novilho Precoce (Mato Grosso do Sul Association of Early Steer Producers)	
	UNIPAN - União dos Pantaneiros da Nhecolândia (Union of Pantaneiros of Nhecolândia)	
	Sindicato Rural de Anastácio (Rural Union of Anastácio)	

Type	Stakeholder	Rights, interest, and overall relevance to the project
	Sindicato Rural de Aquidauana (Rural Union of Aquidauana)	
	Sindicato Rural de Corumbá (Rural Union of Corumbá)	
	Sindicato Rural de Coxim (Rural Union of Coxim)	
	Sindicato Rural de Miranda e Bodoquena (Rural Union of Miranda and Bodoquena)	
	Sindicato Rural de Porto Murtinho (Rural Union of Porto Murtinho)	
	Sindicato Rural de Rio Verde de Mato Grosso (Rural Union of Rio Verde de Mato Grosso)	
Educational and Research Institutions	UFMS - Universidade Federal de Mato Grosso do Sul (Federal University of Mato Grosso do Sul)	Rights: Collaborate with studies and research regarding project interventions and other management activities, and their impacts. Interests: Produce and disseminate knowledge, develop and publish scientific papers, support with the deployment of researchers. Relevance: Medium - Important for technological knowledge and cooperation with the municipality, supporting clear-cut control.
	UEMS - Universidade Estadual de Mato Grosso do Sul (State University of Mato Grosso do Sul)	
	IFMS - Instituto Federal de MS (Federal Institute of Mato Grosso do Sul)	

APPENDIX 2: COMMERCIALY SENSITIVE INFORMATION

Under development and review

Section	Information	Justification
2.1.9 e 2.5.6	Land ownership documents and legal status of properties participating in the Program	Under development
2.1.9 e 2.5.6	Agreements signed between the parties involved	Under development
2.1.22 e 2.4.5	Financial performance spreadsheet of the Project and other related documents	Under development
2.1.22 e 2.4.5	Financial Statements of Biofílica Ambipar Environment	Under development
2.1.19, 2.3.5 e 2.3.10	Minutes of meetings between Biofílica Ambipar Environment and the parties involved in the Program	Under development
2.1.19, 2.3.5 e 2.3.10	Attendance list with personal information of Program participants	Under development